



SKG16.1 Control Theory and Applications

Skill Group 16.1

Operations Management

The PETRONAS Group adopts zero tolerance against all forms of bribery and corruption. We abide by the PETRONAS Code of Conduct and Business Ethics (CoBE) & Anti-Bribery and Corruption (ABC) Manual, guided by our Shared Values and Statement of Purpose.

Revision History

Rev	Date	Prepared By	Notes
13	May 16	Prakash K.K.	Updated Concepts of Control
13A	Feb 17	Andriny Rusman	Minor Corrections
14	Nov 17	Andriny Rusman	Inclusion of Retrieval Practice
15	Feb 18	Andriny Rusman	Incorporation of exercises within the training material
16	Oct 19	Andriny Rusman	Updated ILC and improved explanations
17	Mar24	Andriny Rusman	Revamp on template and improved explanation
20a	Jul24	Andriny Rusman	Updated to 5 layers of control

Before we begin . . .

- Introduction of trainer & participants
- Typical daily schedule (time we begin, break & end)
 - Start at 9am
 - Morning break at about 1030am
 - Lunch around 1230 noon
 - Afternoon break around 315pm
 - Stop at around 430pm
- We will have a short quiz to see current understanding that you have on Process Controls.

Quiz Time



- <https://forms.office.com/r/1aHtKg4dHR>
- If you don't know the answer, please select 'Don't know'

Programme Objectives

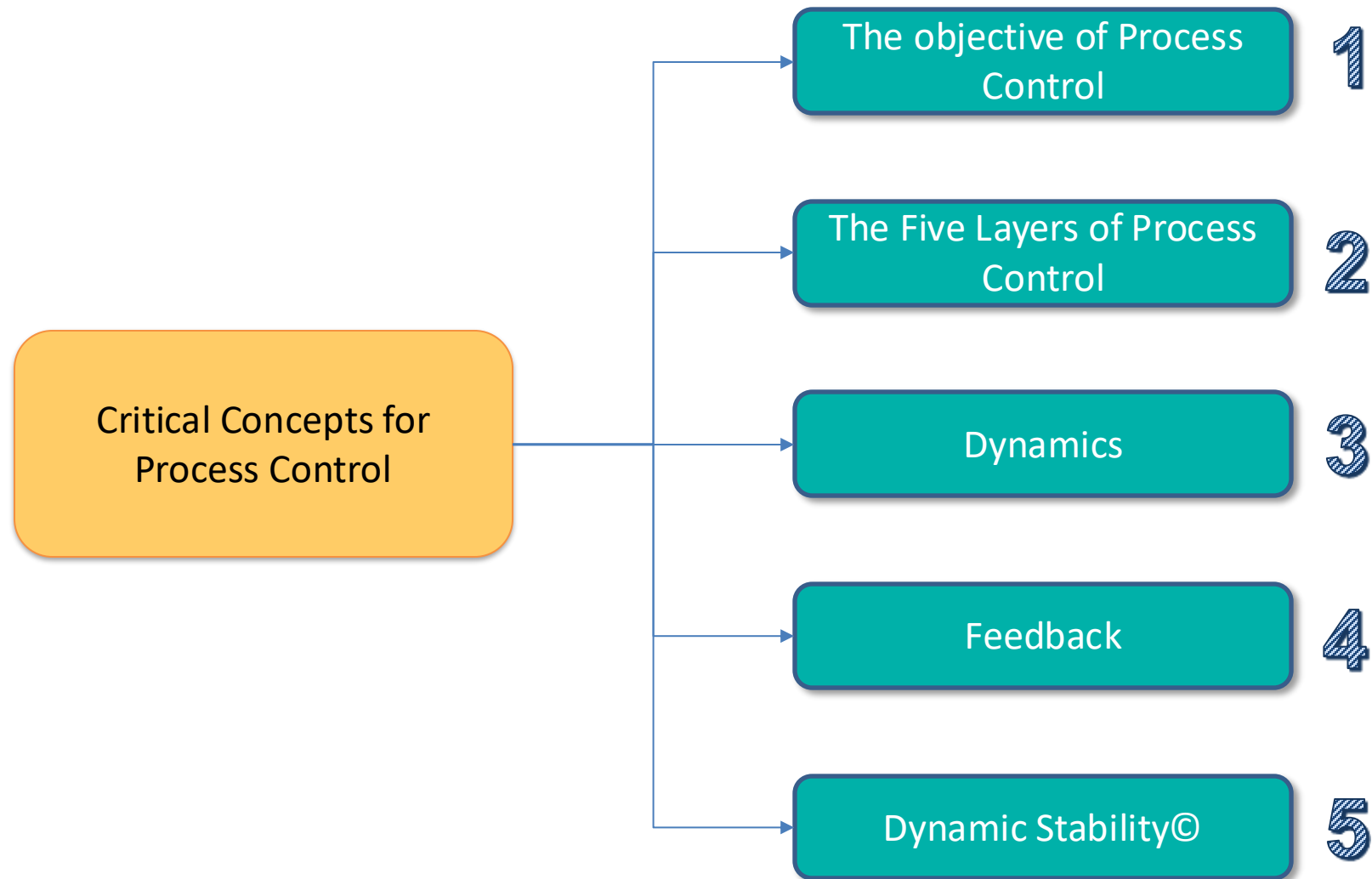
1. Understand the concepts of Process Control as applied in PETRONAS (Process Control Definition, Five Layers of Control and Dynamic Stability[©])
2. Be able to explain Process Dynamics/Behaviour and their criticality in control
3. Be able to explain the “3 + 2 Steps to Understanding Base Layer Control”
4. Understand the concepts behind Tuning and Tuning Techniques
5. Understand the different forms of Intermediate Layer controls (Cascade, Split Range, Override and Ratio Control Loops)
6. Understand the process requirement for Advanced Regulatory Controls (such as combustion, compression and distillation)
7. Aware of the concepts of APC (MPC) and Property Inference

Contents

- Process Control Overview
 - PETRONAS Concepts of Process Control
- Layer 1 – Base Layer Controls
- Layer 2 – Intermediate Layer Controls
- Layer 3 – Advanced Regulatory Controls
- Layer 4 – Advanced Process Controls

PETRONAS Process Control Concepts

Important Concepts in PETRONAS Process Control

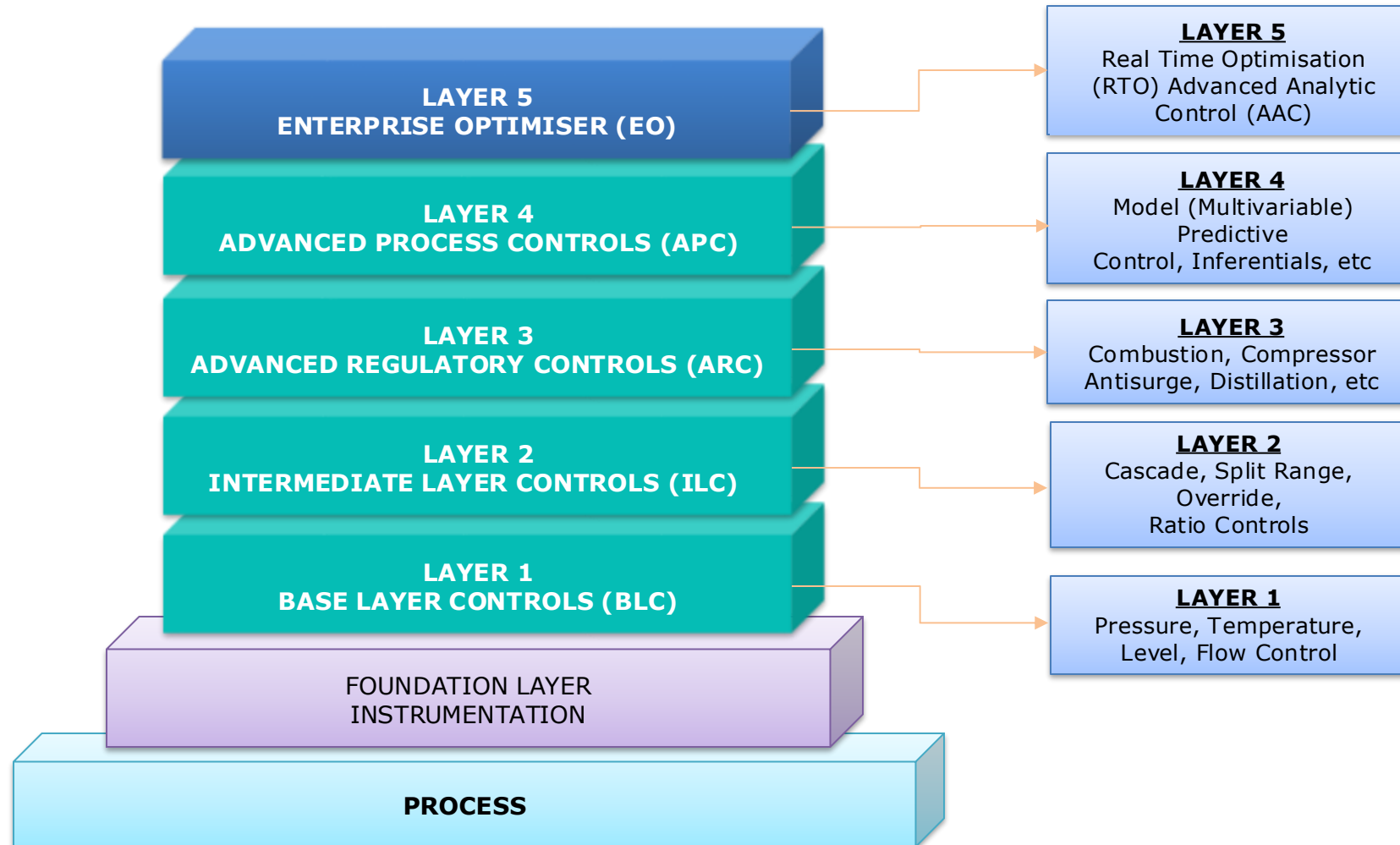


Concept 1 : Objective of Process Control

The field of Process Control refers to the selection and application of the 'PETRONAS Five Control Layers', with the objective of:

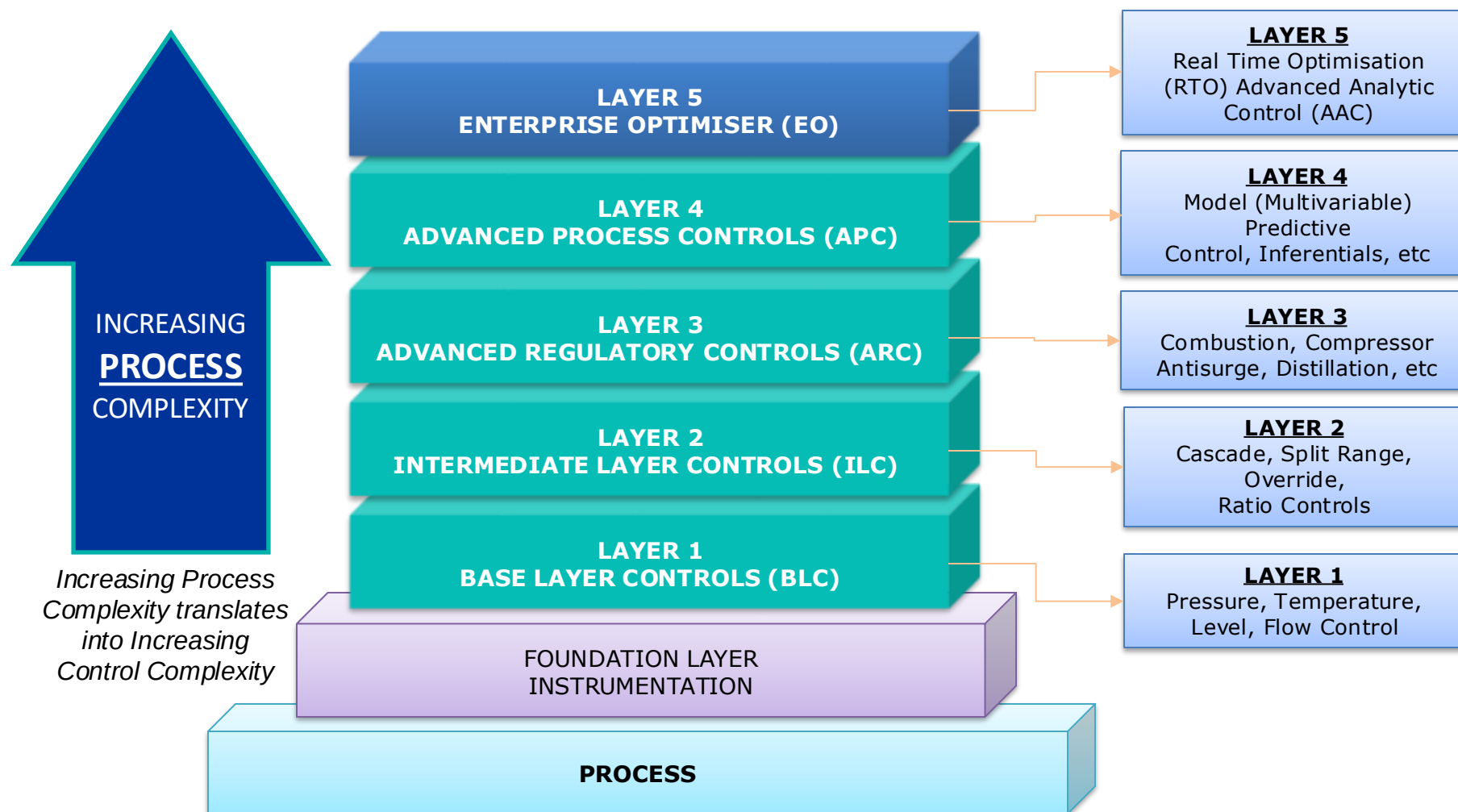
- achieving Dynamic Stability®
- improving product specifications,
- increasing yield and throughput,
- and increasing efficiency.

Concept 2: The PETRONAS Five Control Layers

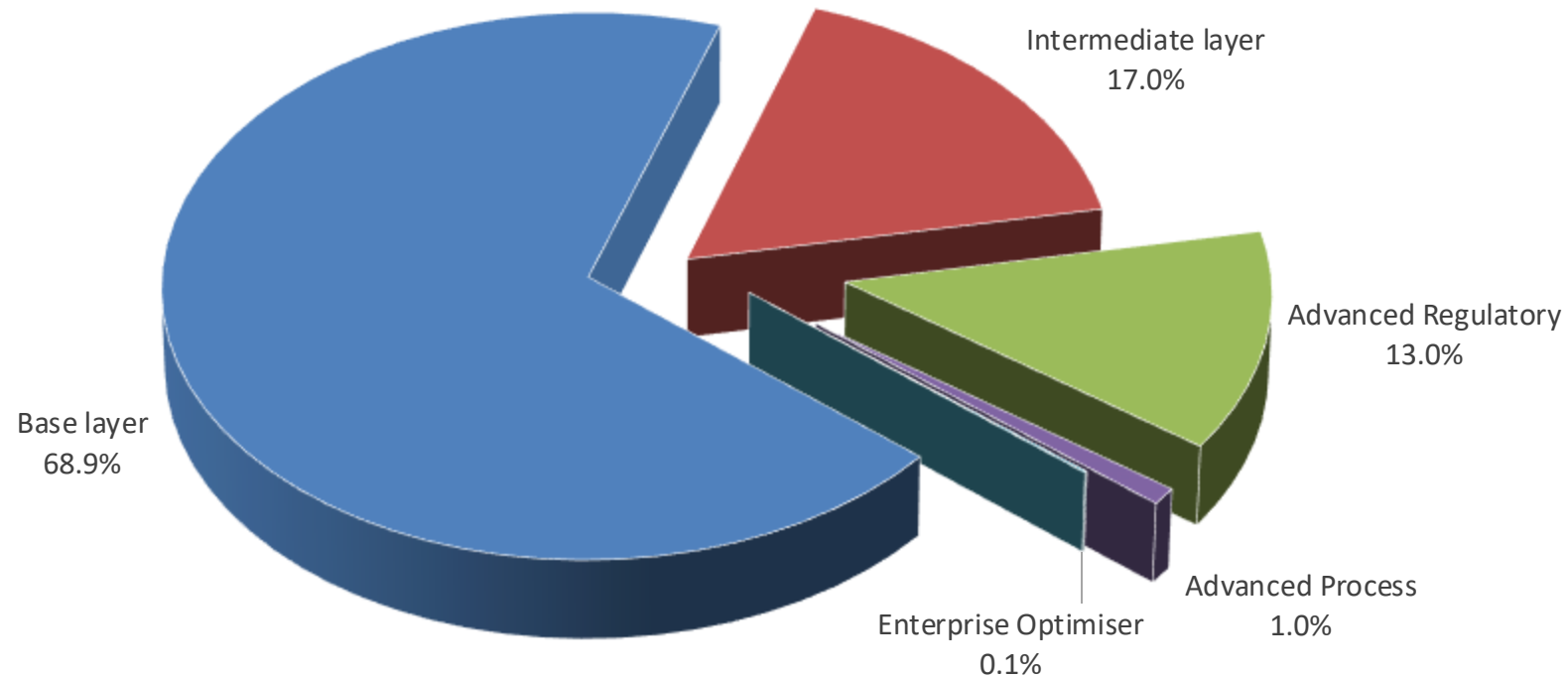


Concept 2: Five Control Layers

Why Layer the Controls?

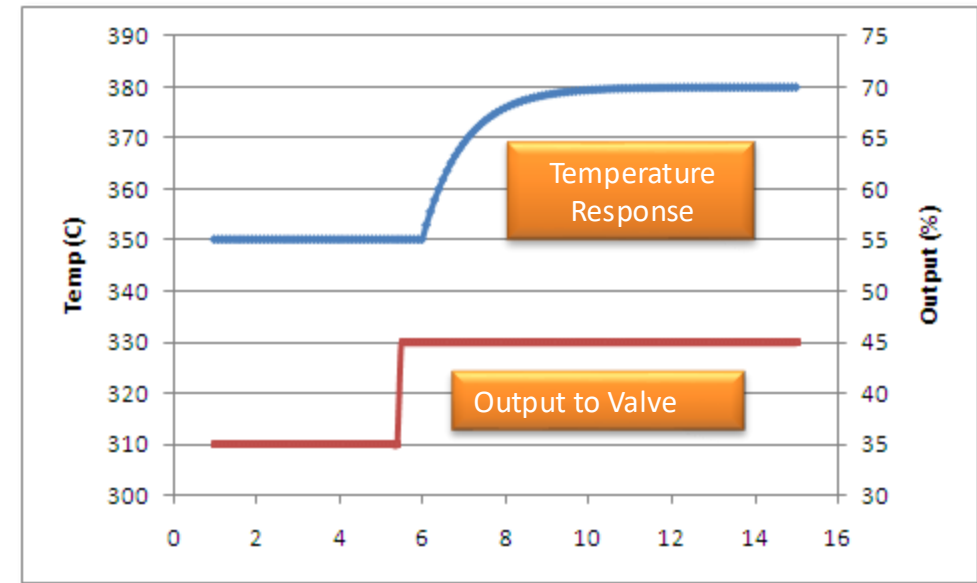
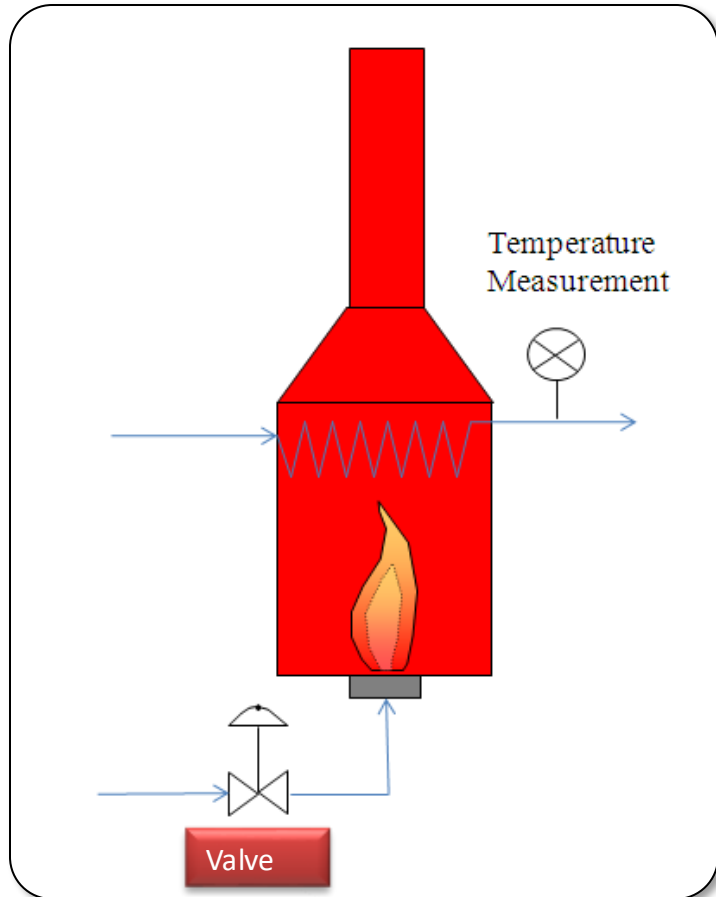


Concept 2: The Five Layers of Control Typical Distribution



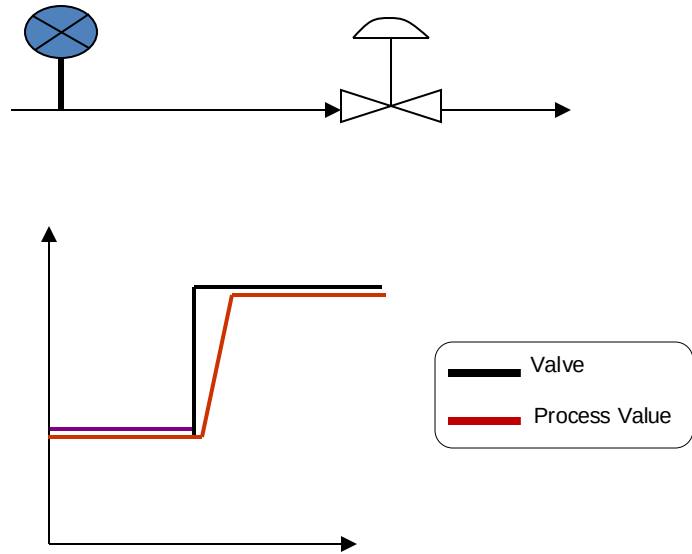
Source: Controller Distribution Analysis on various plants from 2004 to 2017

Concept 3 : Dynamics!



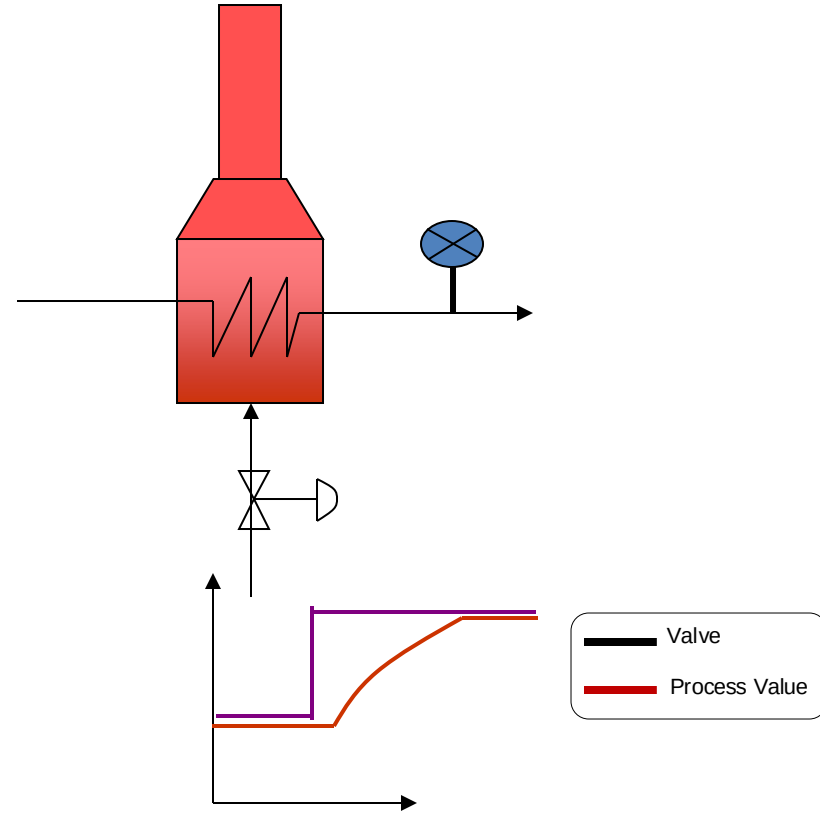
- i. Process response is NOT instantaneous for most processes!
- ii. It varies with the type of process/parameter.
- iii. Understanding dynamics allows better control of the process.

Concept 3 : Dynamics



Process Reaction Curve

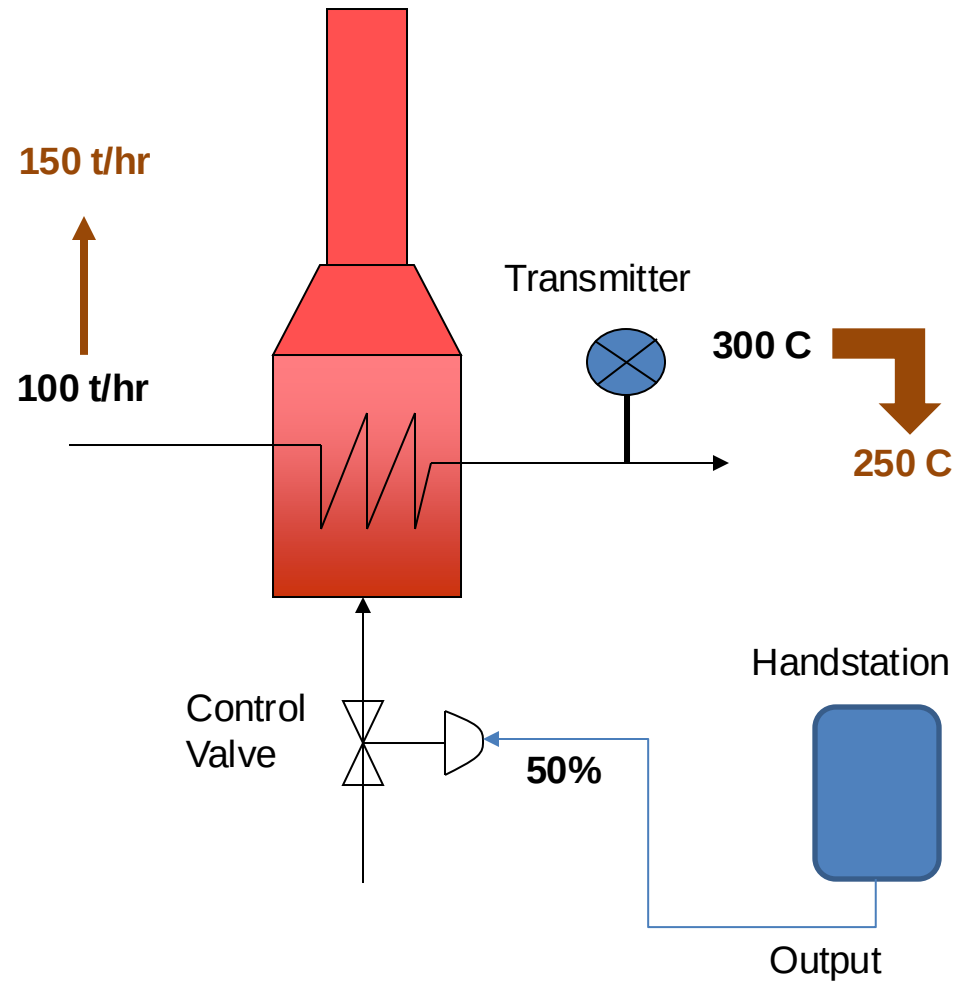
Flow



Process Reaction Curve

Temperature

Concept 4 : FEEDBACK



Situation :

- i) A process furnace is running in a stable condition with a feed of 100t/hr and an outlet temperature of 250C
- ii) The output to the fuel control valve is at 50%

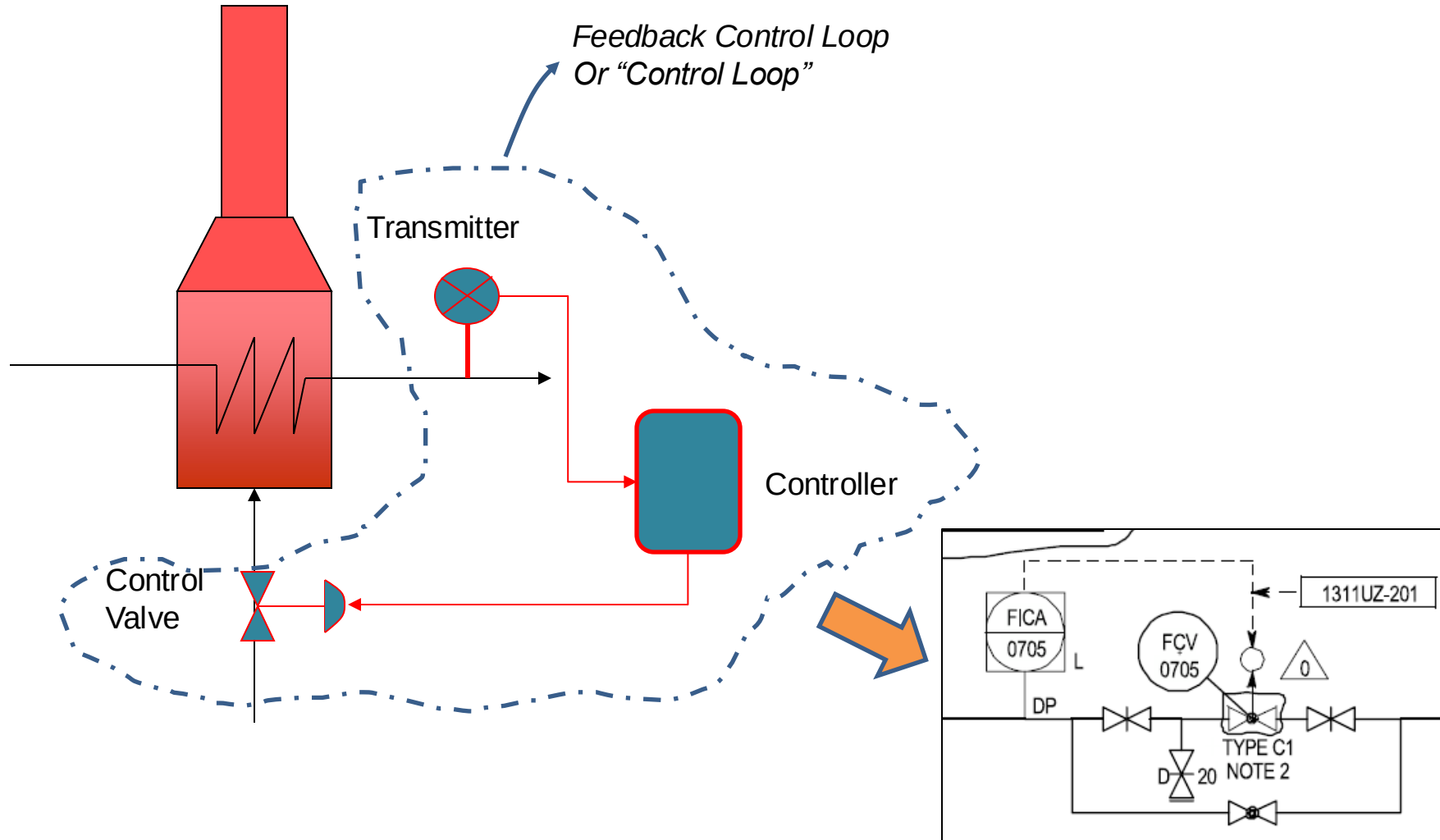
Transition

- i) The process undergoes a transition from 100t/hr to 150t/hr.
- ii) This reduces the exit temperature from 300C to 250C
- iii) Assume there is only a HC (handstation)

Query :

- i) What should you do to the output?
- ii) By how much?
- iii) How do we know when to stop changing the output?

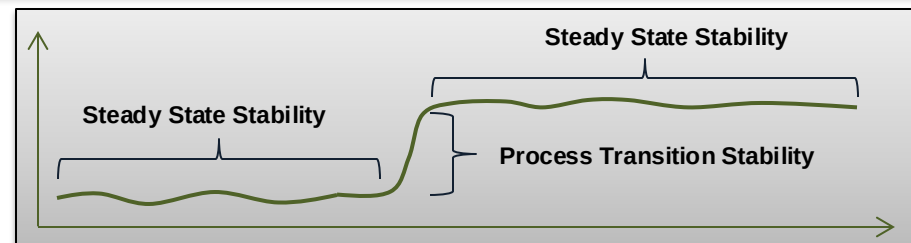
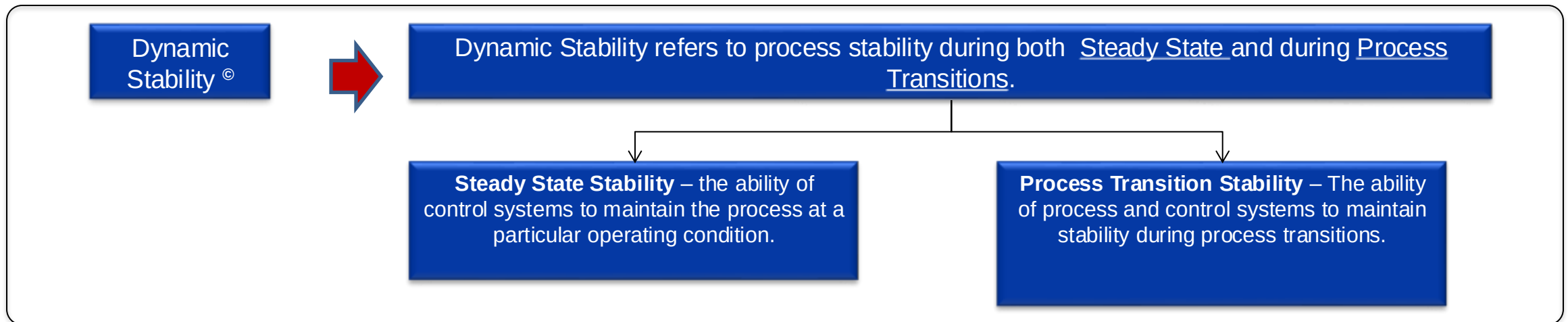
Concept 4 : FEEDBACK



Concept 5 : Dynamic Stability[©]

The field of Process Control refers to the selection and application of the PETRONAS Four Control Layers, with the objective of:

- achieving **Dynamic Stability[©]**
- improving product specifications,
- Increasing yield and throughput,
- and increasing efficiency.



Retrieval Practice

1. There are 4 key objectives of control. One of these objectives is however a pre-requisite for all the others. Which is this objective?
 - a) Enhance product specification
 - b) Improve monetization
 - c) Improve stability
 - d) Enhance efficiency

2. There are ____ Layers of Control:
 - a) 3
 - b) 4
 - c) 5
 - d) 6

3. The Layers of Control were developed primarily to address
 - a) Increasingly complex configurations
 - b) Increasingly complex control systems
 - c) Increasingly complex processes
 - d) Increasingly complex DCS

Retrieval Practice

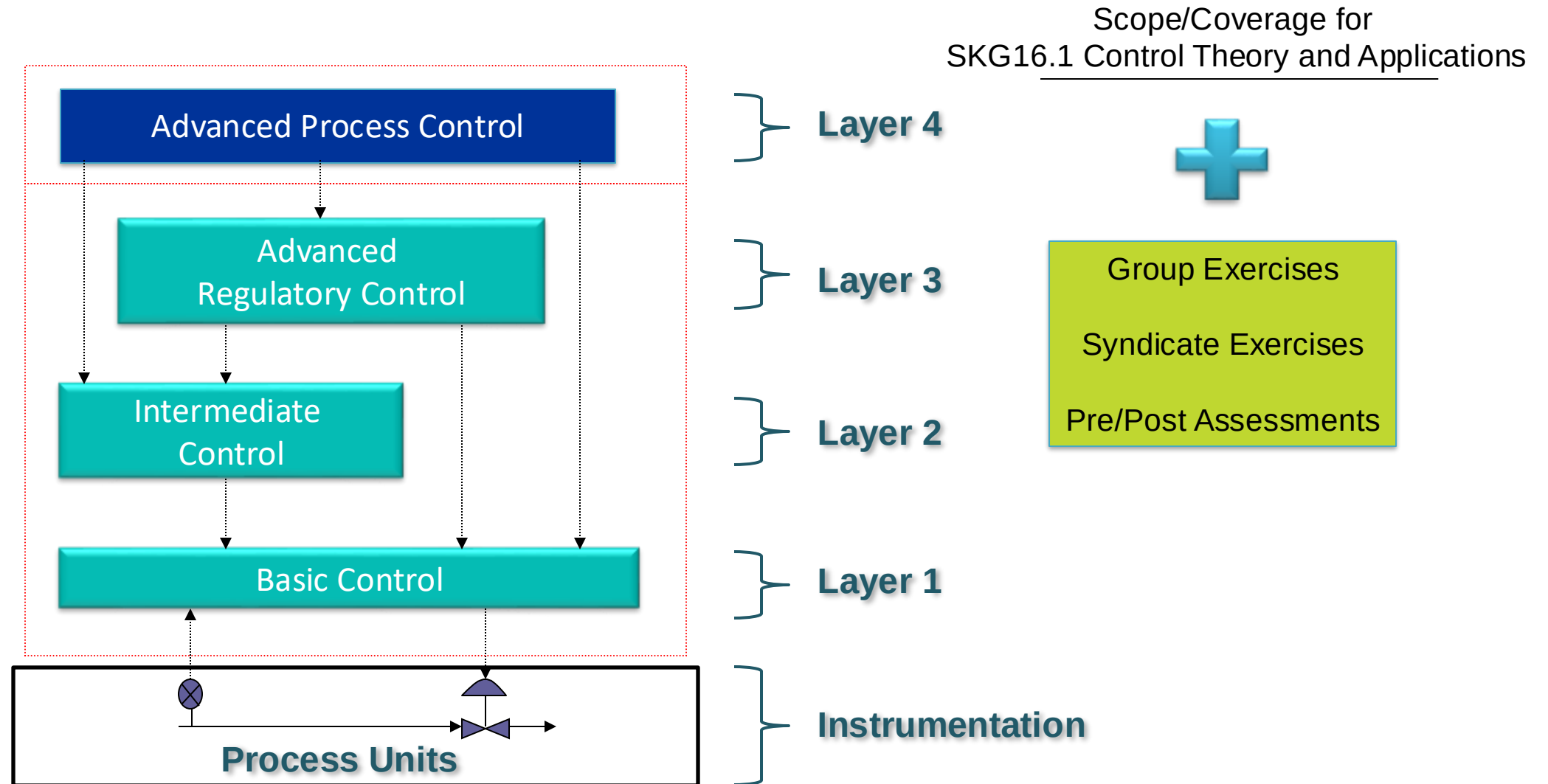
4. Dynamic Stability was coined to explain differences in Dynamic Behaviour
 - a) Between a process disturbance and PV changes
 - b) Between a process disturbance and Setpoint changes
 - c) Between a process disturbance and valve change
 - d) Between process disturbances only

5. Feedback is explained by one of the following sentences:
 - a) Feedback is obtained from the valve to manipulate it correctly
 - b) Feedback is obtained from the process to manipulate the transmitter by the right amount
 - c) Feedback is obtained from the process to manipulate the process by the right amount
 - d) Feedback is obtained from the process to manipulate the temperature correctly

Summary

- Process Control is applied to achieve **Dynamic Stability** and to achieve high levels of **Optimisation**.
- The application of Process Control is through the **PETRONAS 5 Control Layers**
- The objective of applying the higher layers of control for **Improved Plant Stability and Profitability**

The scope of this Control Theory and Applications course



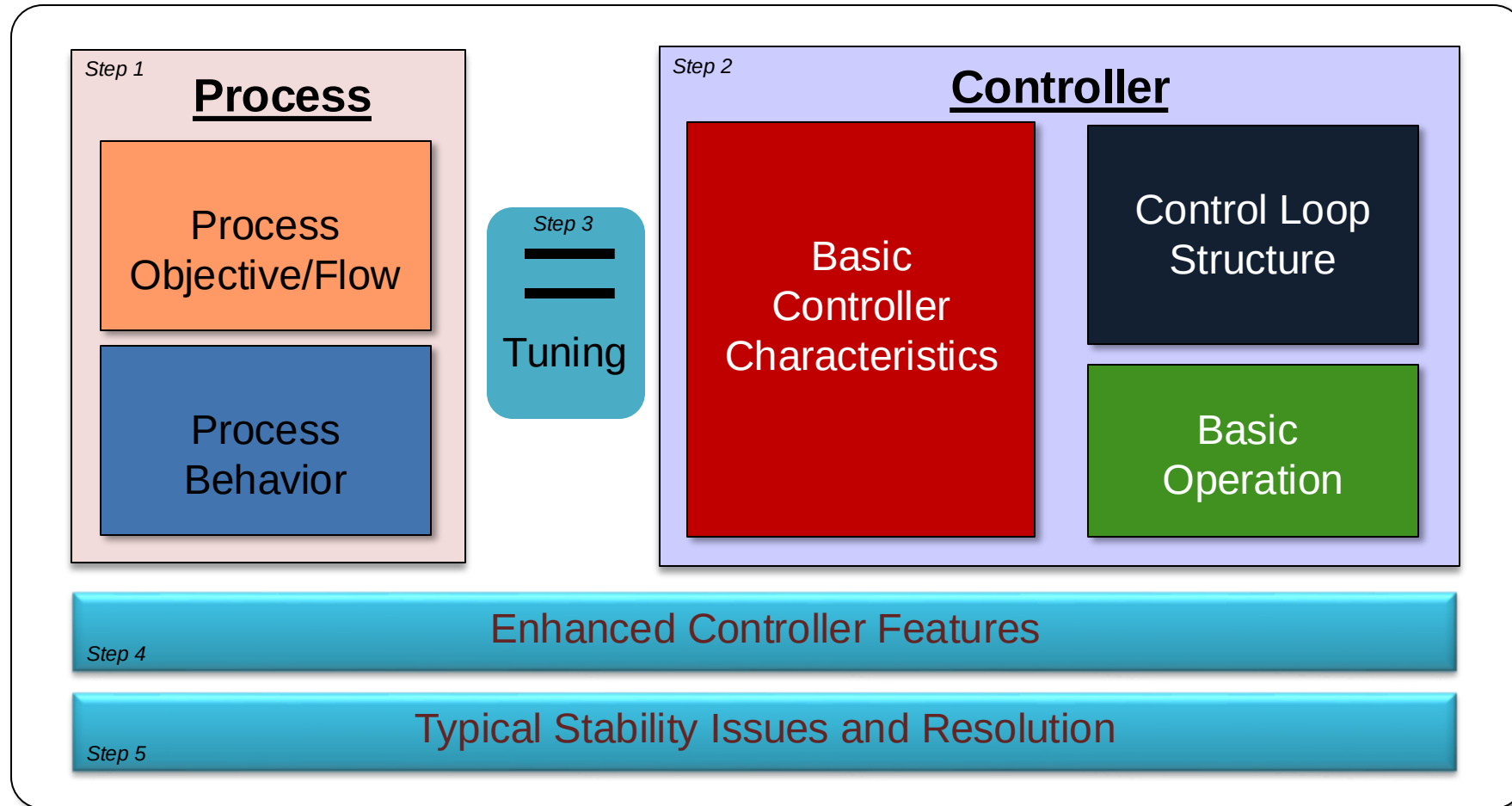
LAYER 1

Base Layer Controls

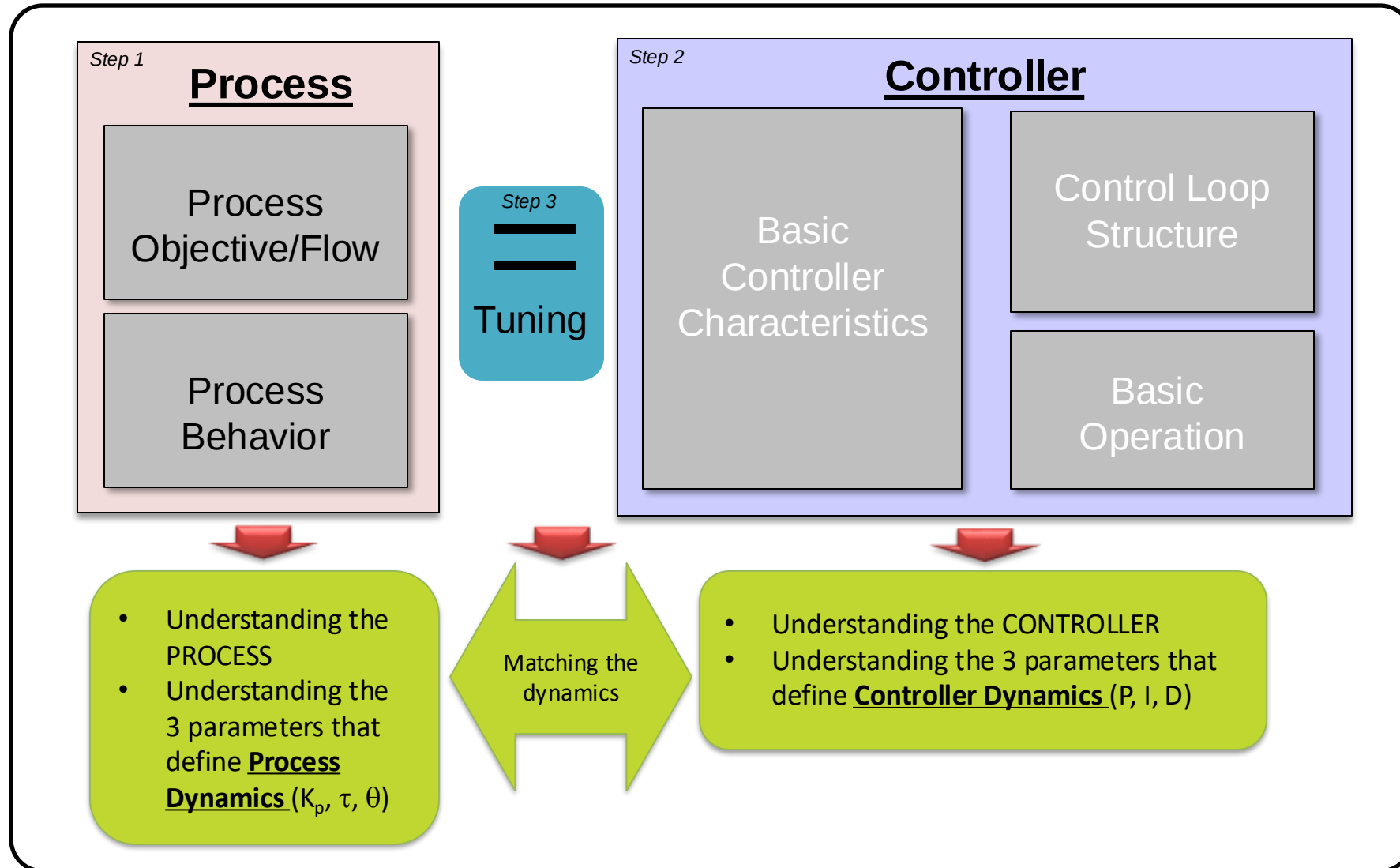
Base Layer Controls

- The basic means of control of any process facility is via a single loop control
- Other more complex controls are built on the top of this Basic Single Loop Control Layer.
- Thus, the basic control layer in a facility is known as the “**Base Layer Control**” or “**BLC**”

The “3 + 2 Steps” Approach for Understanding BLC



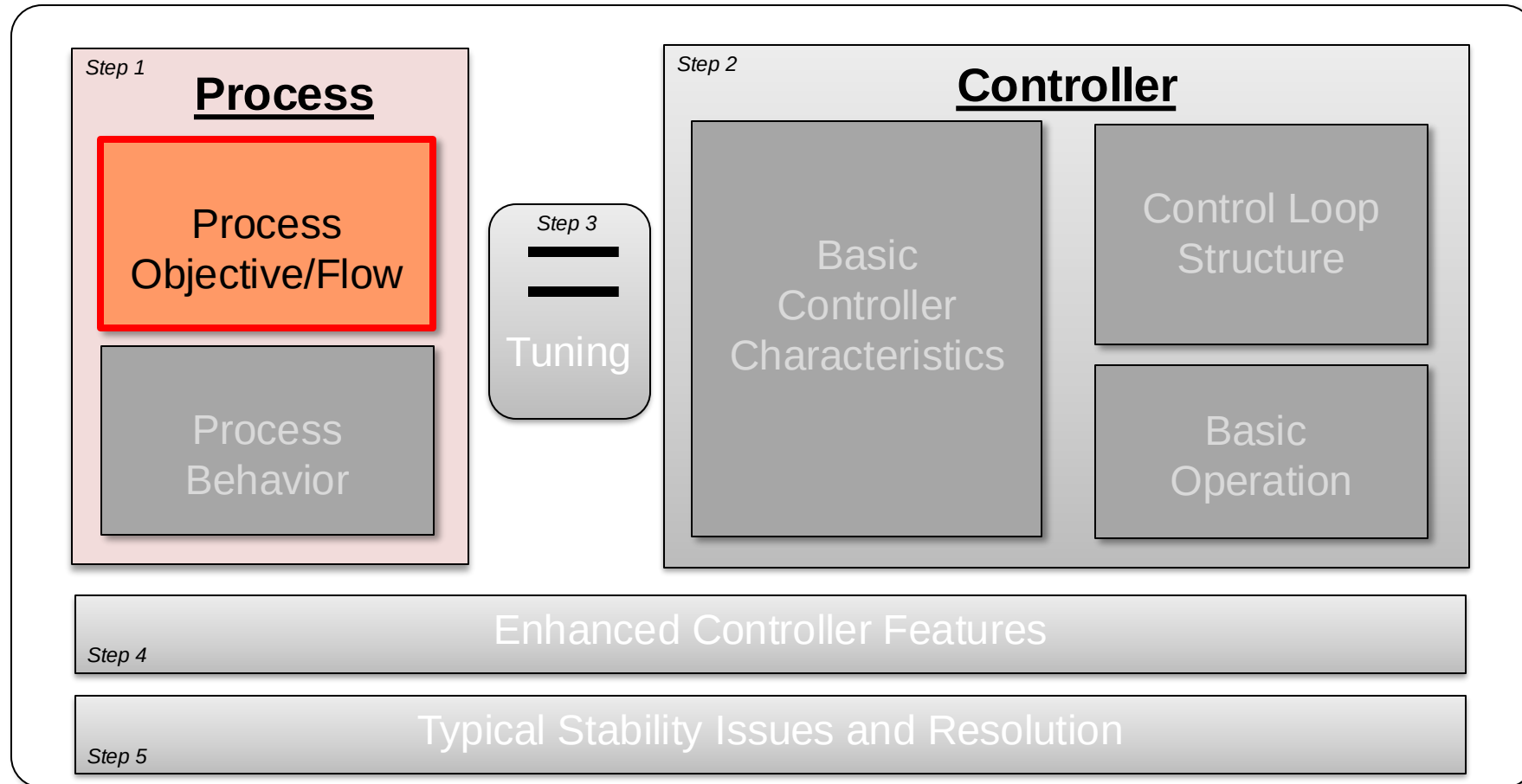
The 3 Core steps for understanding BLC



Base Layer Control

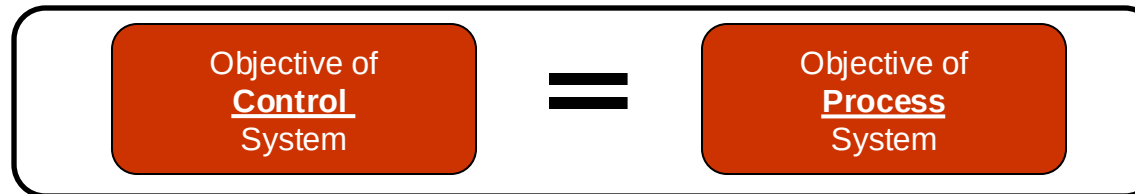
Step 1 - Understanding the Process

Process Objective/Flow



The criticality of Process Knowledge

- The objective of Process Control is firstly to achieve a **dynamically stable process**. A “dynamically stable” process is stable under both conditions of:
 - *Process disturbances*
 - *Transition operations*
- In order to meet this Process Control objective, it is CRITICAL to
 - Understand the **objective** of the process
 - Understand the **behaviour** of the process
- This will allow the matching of the Control System to the Process System

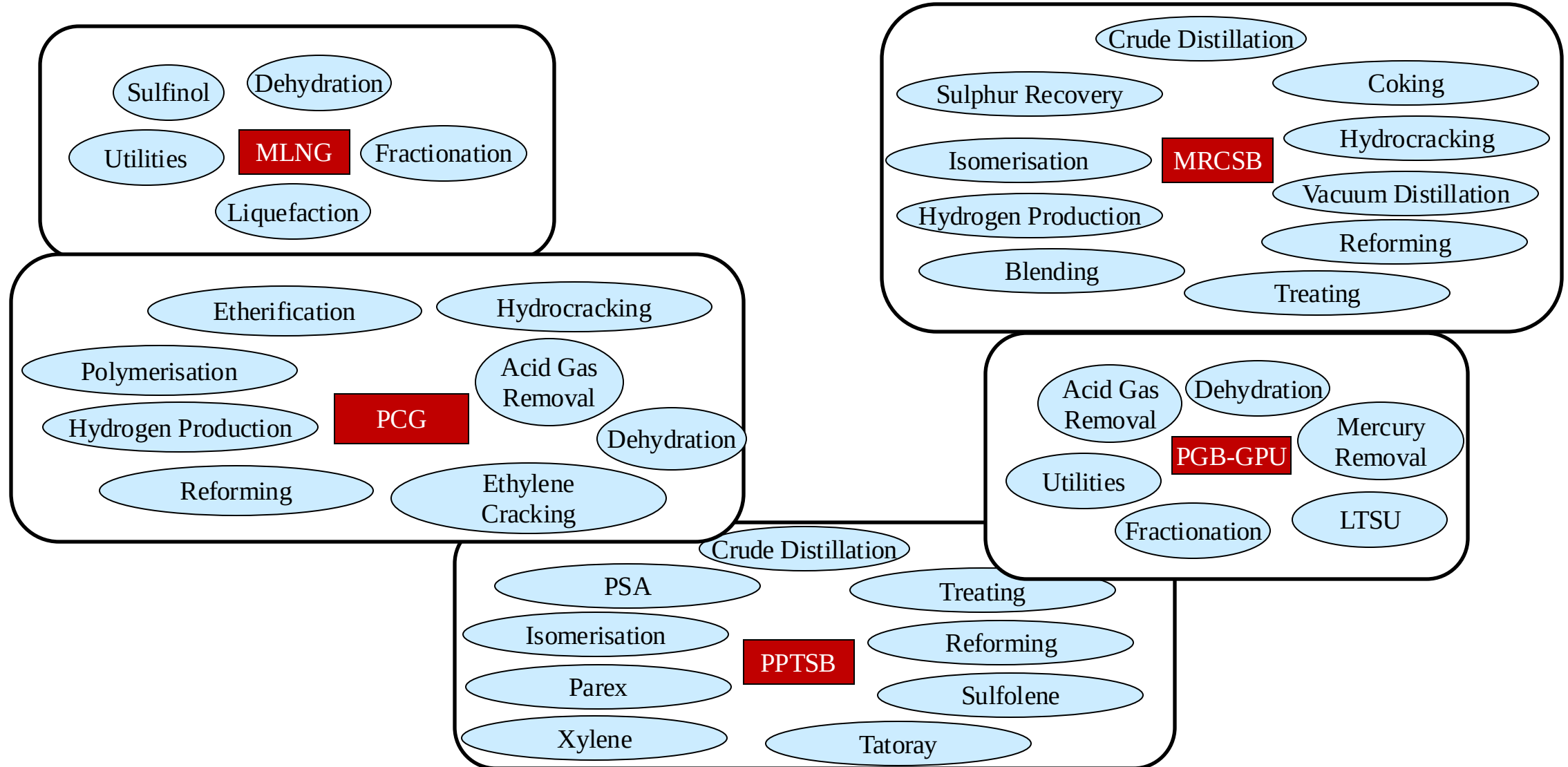


Understanding the Process

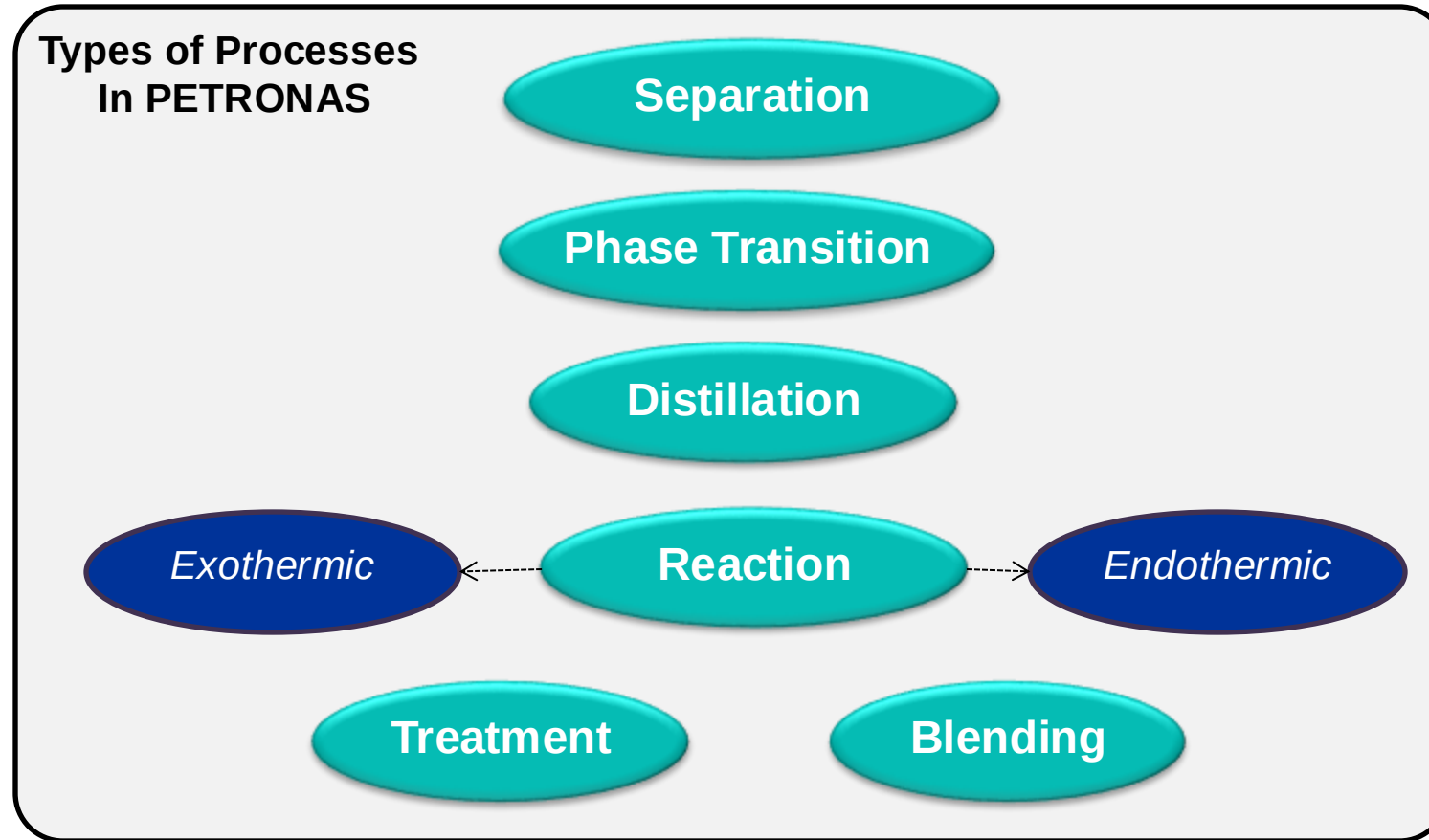
Understanding the Process implies knowledge (at the very least) of the following:

- Process Objective
- Process Equipment
- Process Flow
- Process parameters during steady state operation AND
- Process behaviour during a dynamic response of a process
- And preferably the governing economics of the plant/unit

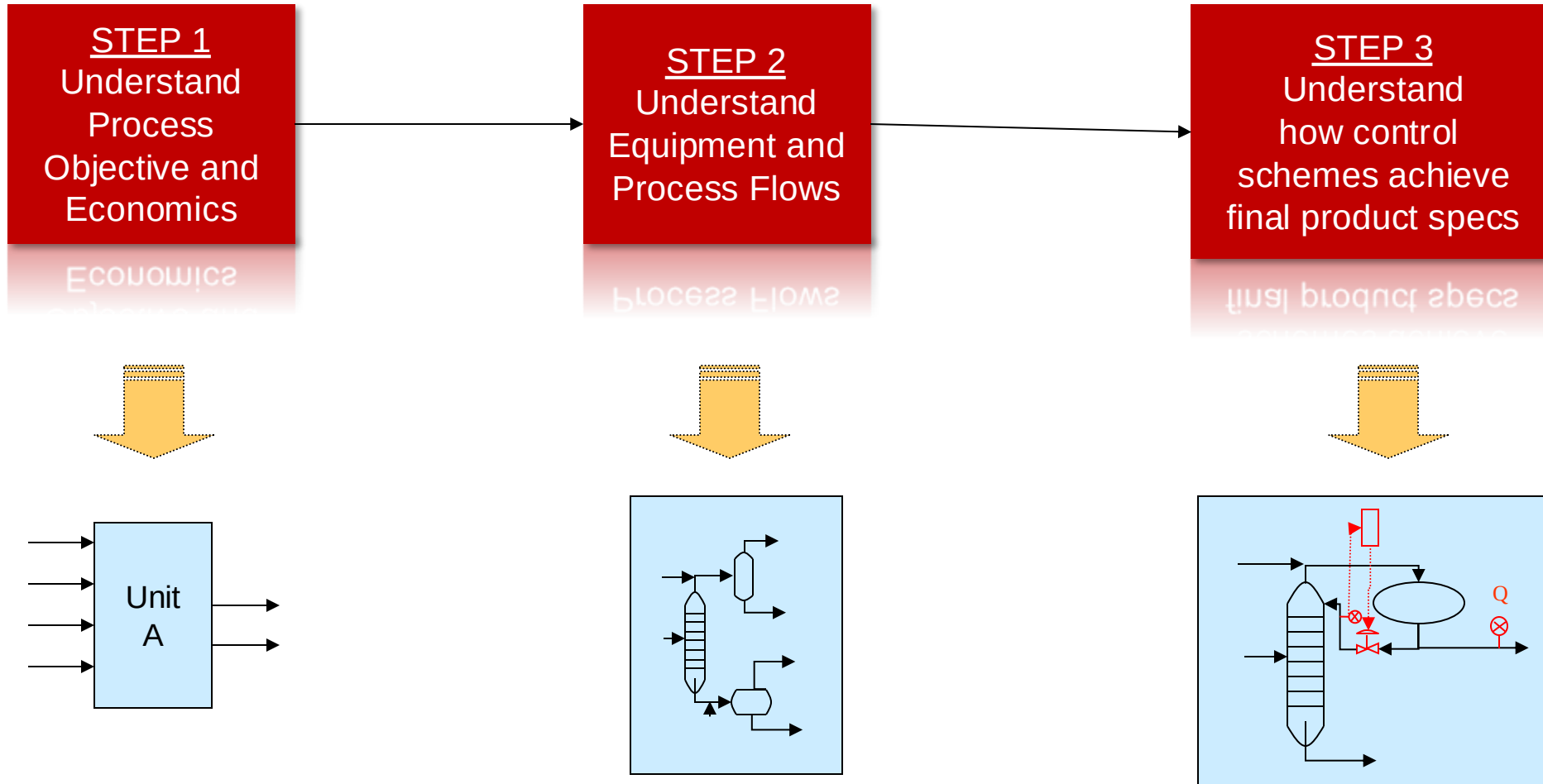
Process Knowledge in PETRONAS



Overview of Basic Process types



An approach to Process Understanding



Objective of Control Systems

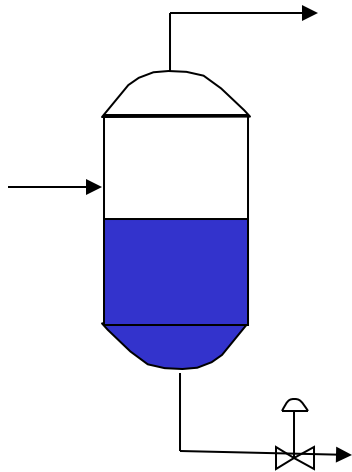
The objective of the control system is to provide a means to control a process at a desired value (typically known as setpoint or set value).

- If the desired value changes or the process itself changes, the control system must be able to bring the process back to the desired value in a quick and stable manner.

Group Exercise #1

Control Scheme Design

Unit OP1

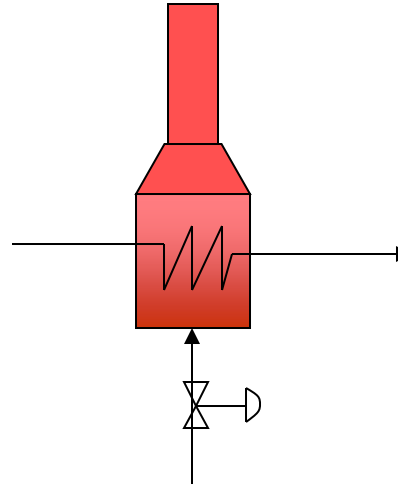


Key Process Variable =

Failure to control PV,
PV crosses high limit, impact =

Failure to control PV
PV crosses low limit, impact =

Unit OP2

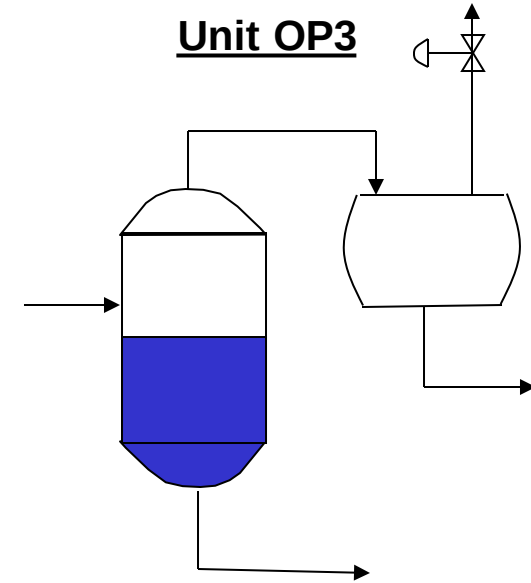


Key Process Variable =

Failure to control PV,
PV crosses high limit, impact =

Failure to control PV
PV crosses low limit, impact =

Unit OP3

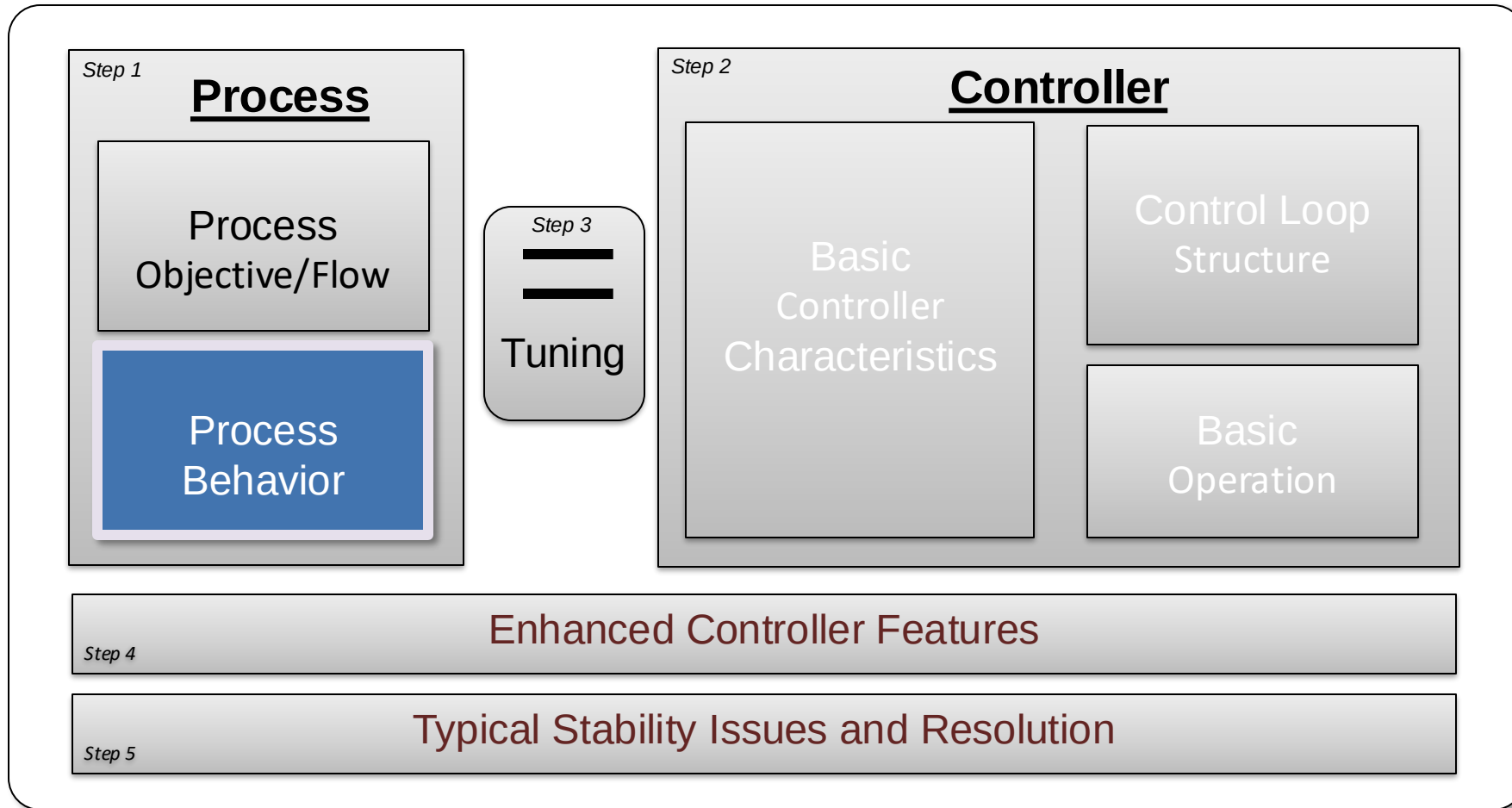


Key Process Variable =

Failure to control PV,
PV crosses high limit, impact =

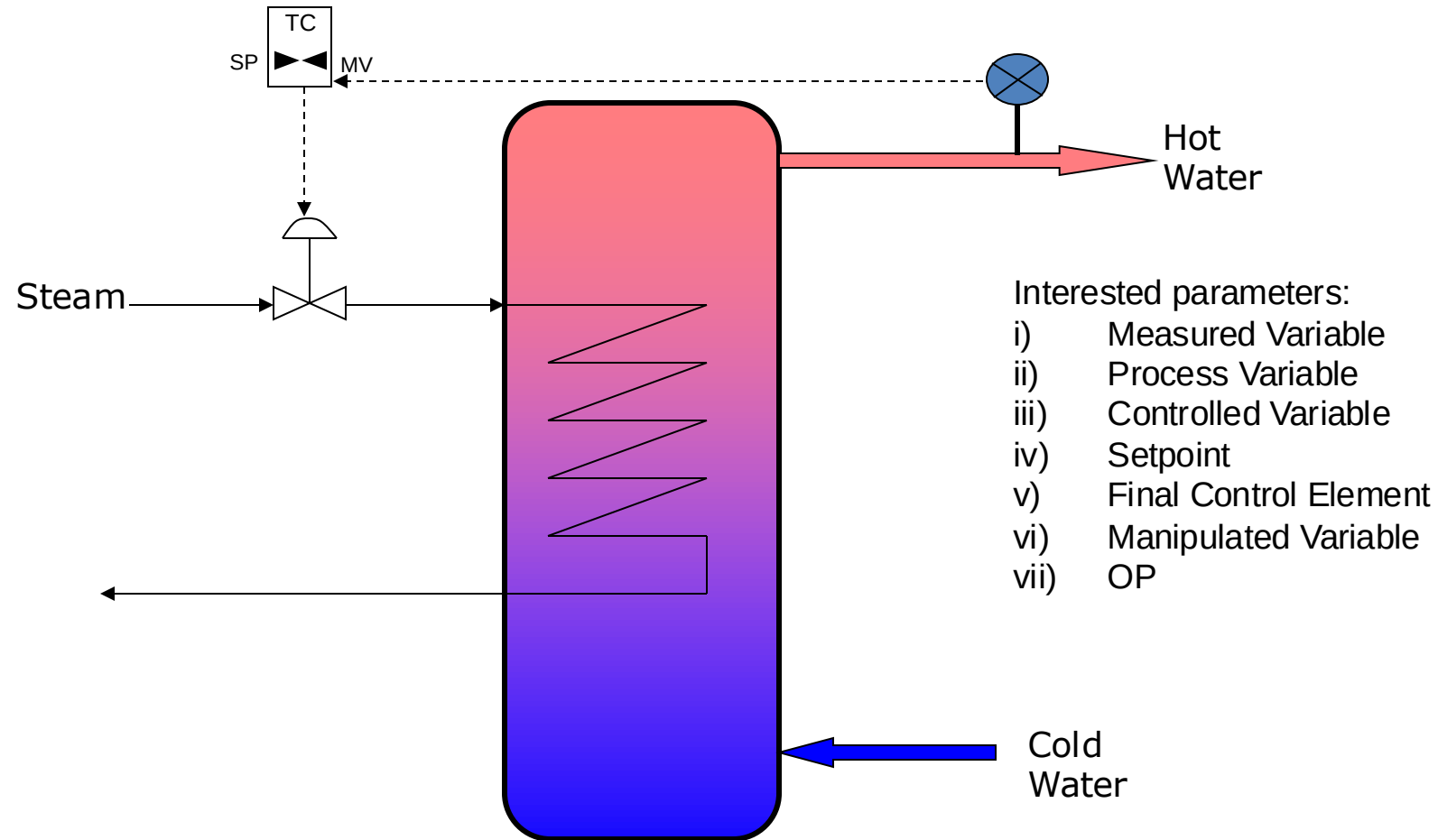
Failure to control PV
PV crosses low limit, impact =

Scope of Base Layer Controls

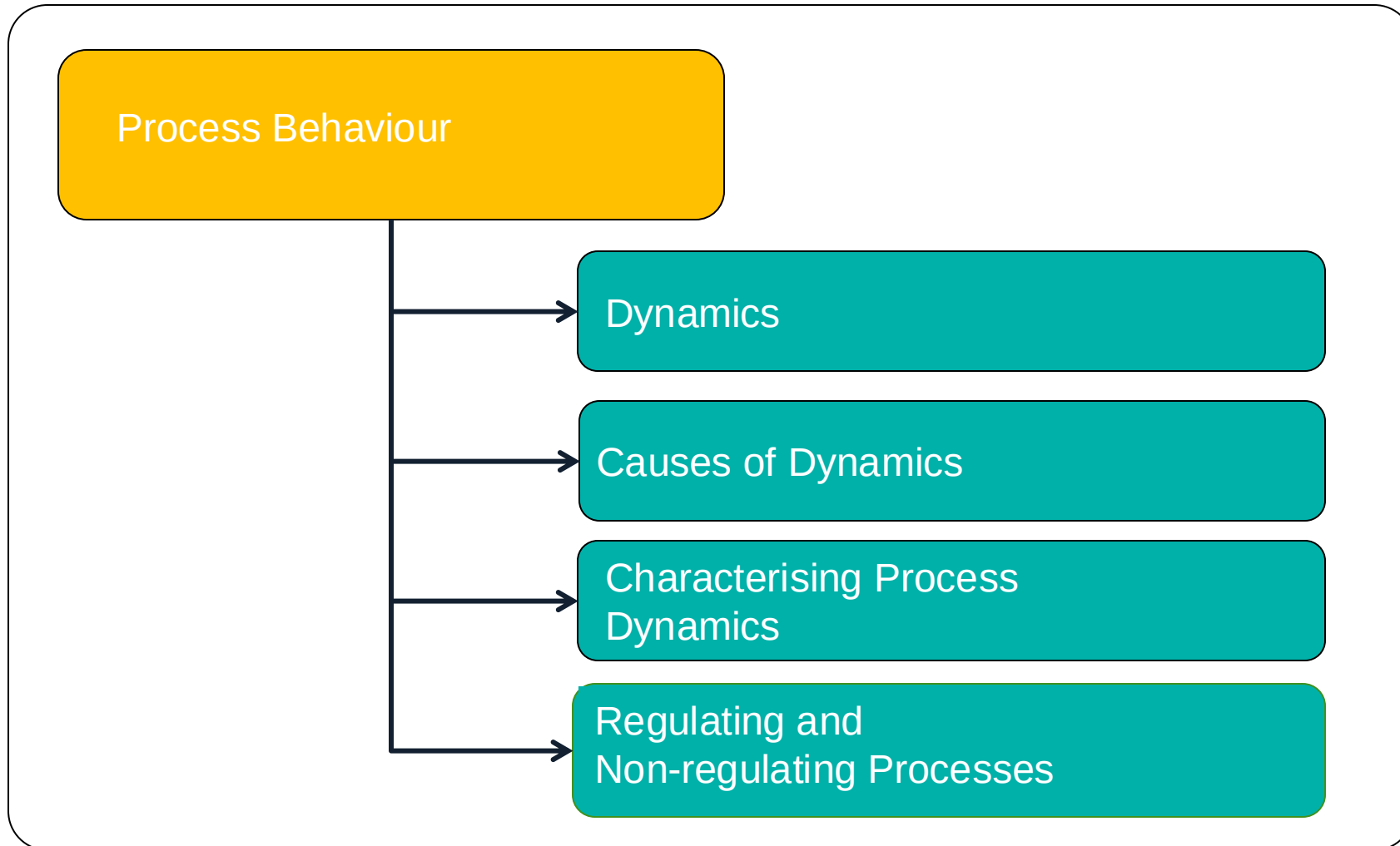


Process Fundamentals

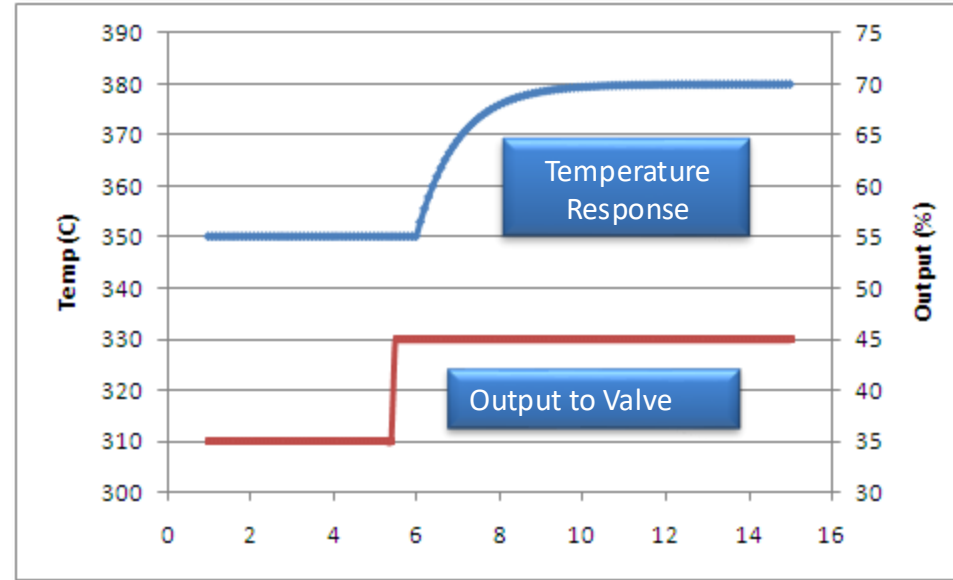
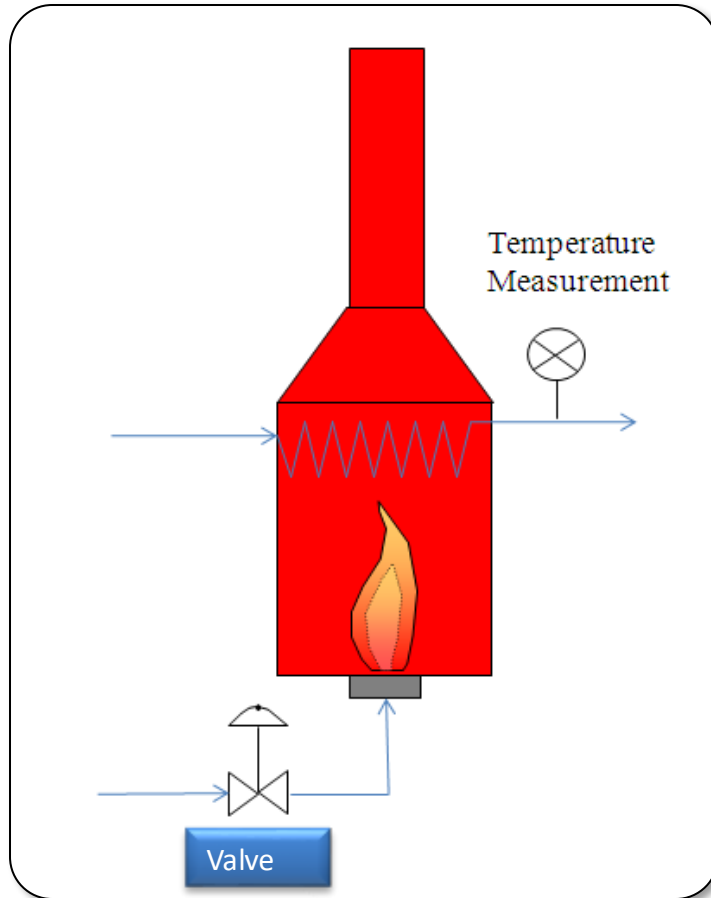
- Terminologies



Process Behaviour - Concepts



Dynamics!



- i. Process response is NOT instantaneous!
- ii. It varies with the type of process/parameter.
- iii. Understanding dynamics allows better control of the process.

Characteristics of Process Dynamics

Characteristics of the dynamic nature of a process:

- Process Capacitance/Resistance
- Process Deadtime

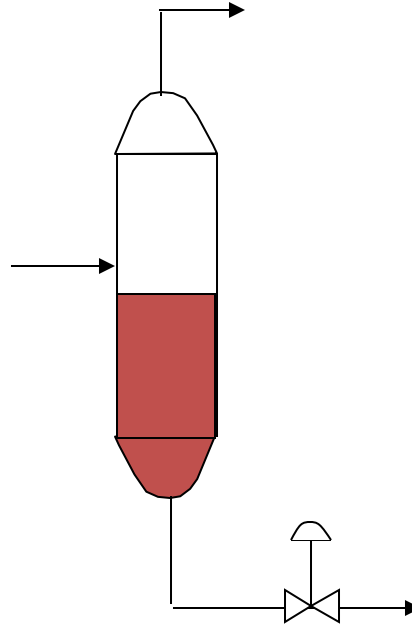
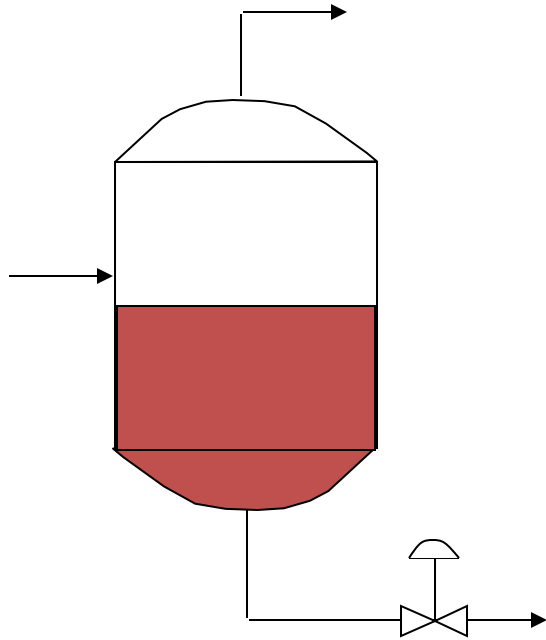
Process Capacitance/Resistance

It represents a system's:

- Ability to absorb or accumulate energy or material
- Resistance to change of mass or energy stored in the system.

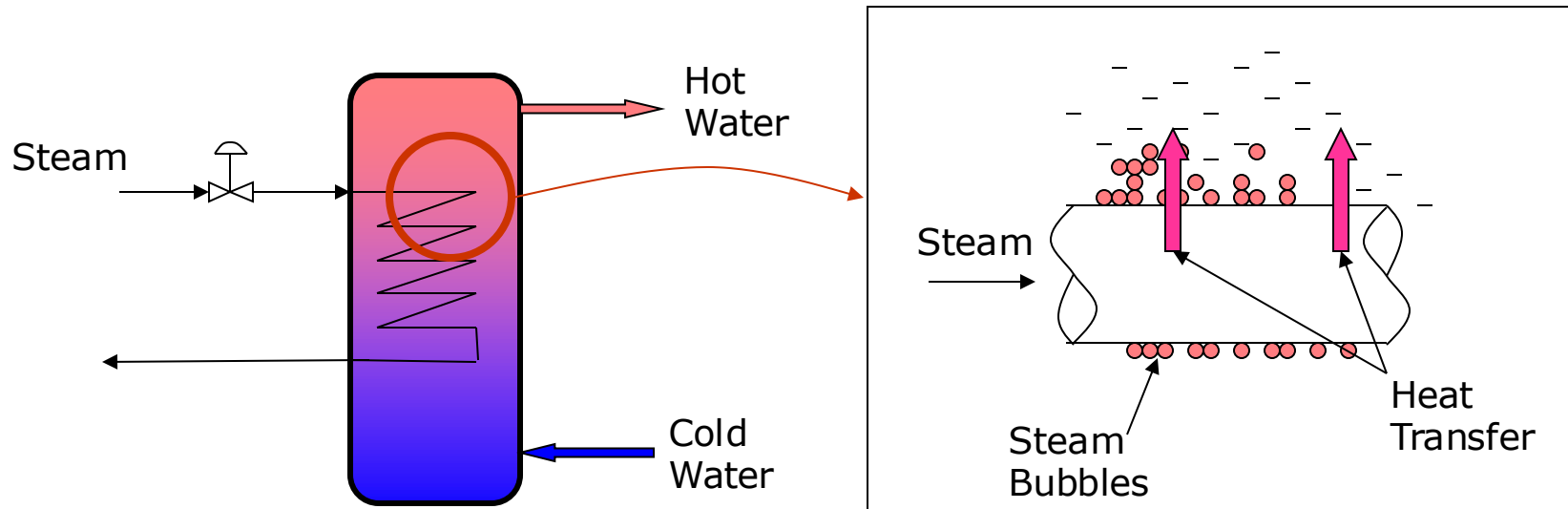
Capacitance

- Vessel



Valve sizes are identical.
Pressures upstream and downstream of
control valve are identical for both
vessels.
Which level will drop faster?

Resistance



The resistance of the process is its basic capability to resist the transfer of energy.

Examples include heat resistivity of the pipe or heat capacity of the fluid.

The resistance of the process is its basic capability to resist the transfer of energy. Examples include heat resistivity of the pipe or heat capacity of the fluid.

Metal	BTU/hr F Ft	W/(m K)	Notes
Copper (at 300 C)	213	369	
Aluminium (at approx 90C)	124	215	
Aluminium Bronze (at 20C)	44	76	
Carbon steel (max 0.5 % C at 20C)	31	54	
Carbon steel (max 1.5 % C at 20C)	21	36	
Titanium (at 20C)	11	19	range 7 till 45 W (m K)
Stainless steel (at 20C)	7	12	range till 23 W (m K)

Thermal conductivities of various metals (source : www.engineeringtoolbox.com)

Remember!

Process dynamics are **dependent** on

Process design &

- *Once design is completed, process dynamics is largely fixed*

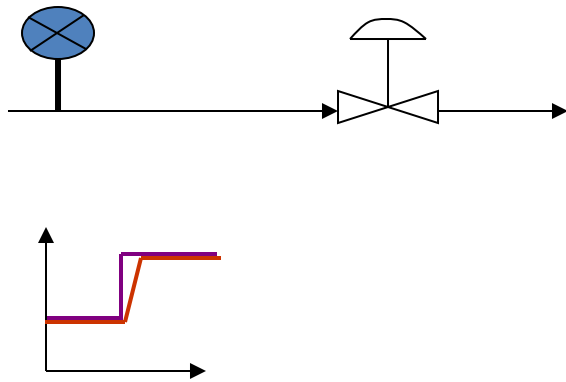
Operating conditions

- *different operating modes and conditions (for instance running the plant at 100% and running the plant at 50%) can result in different dynamics)*

Implication

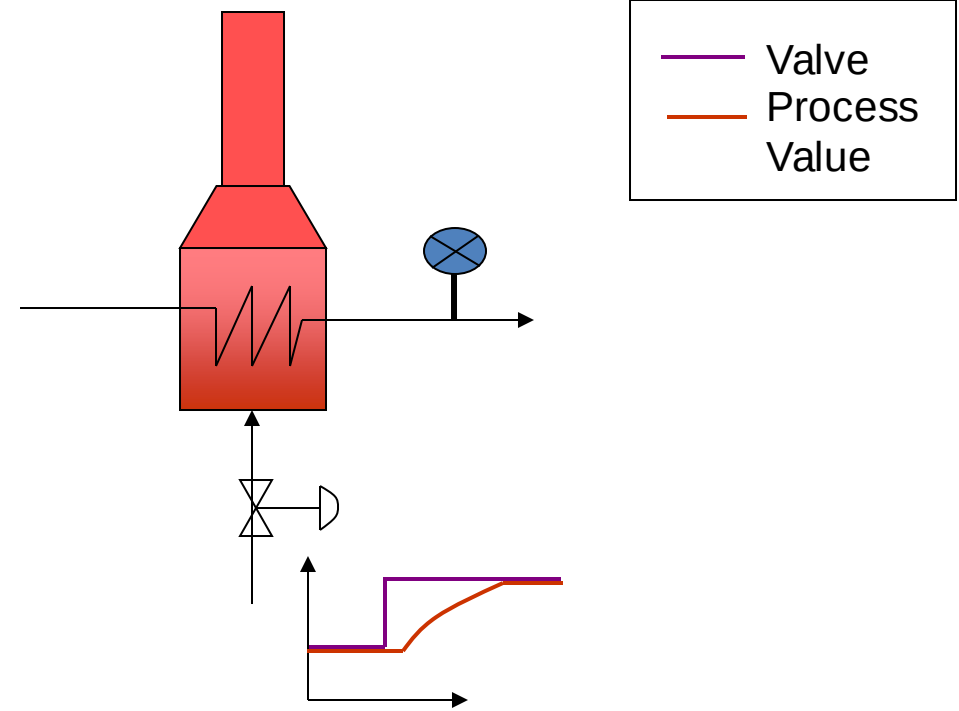
Since Process Dynamics cannot be changed, stable automatic control is achieved through manipulation of Controller Dynamics.

Action of 4 key parameters



Process Reaction Curve

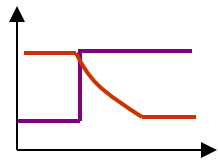
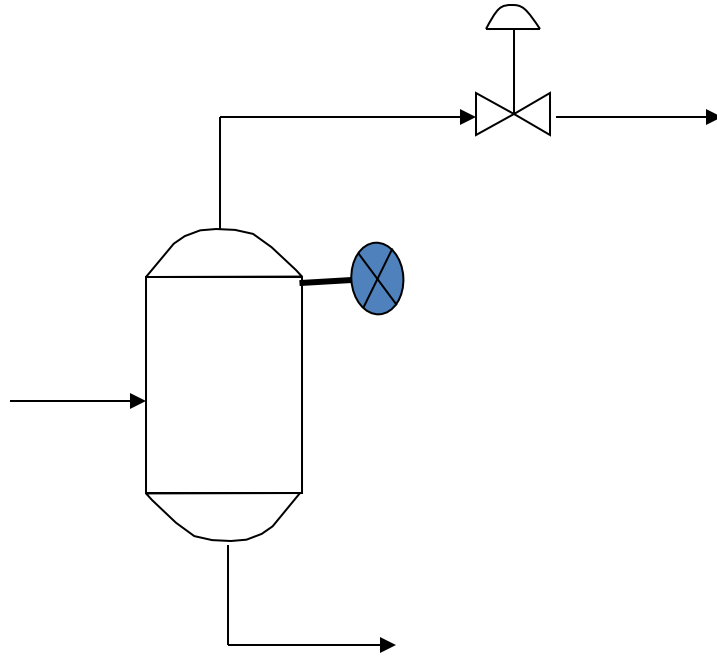
Flow



Process Reaction Curve

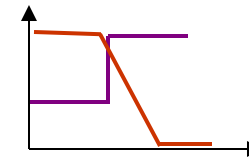
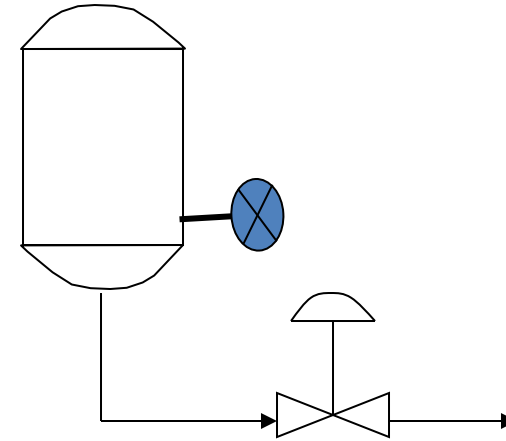
Temperature

Action of 4 key parameters (2)



Process Reaction Curve

Pressure

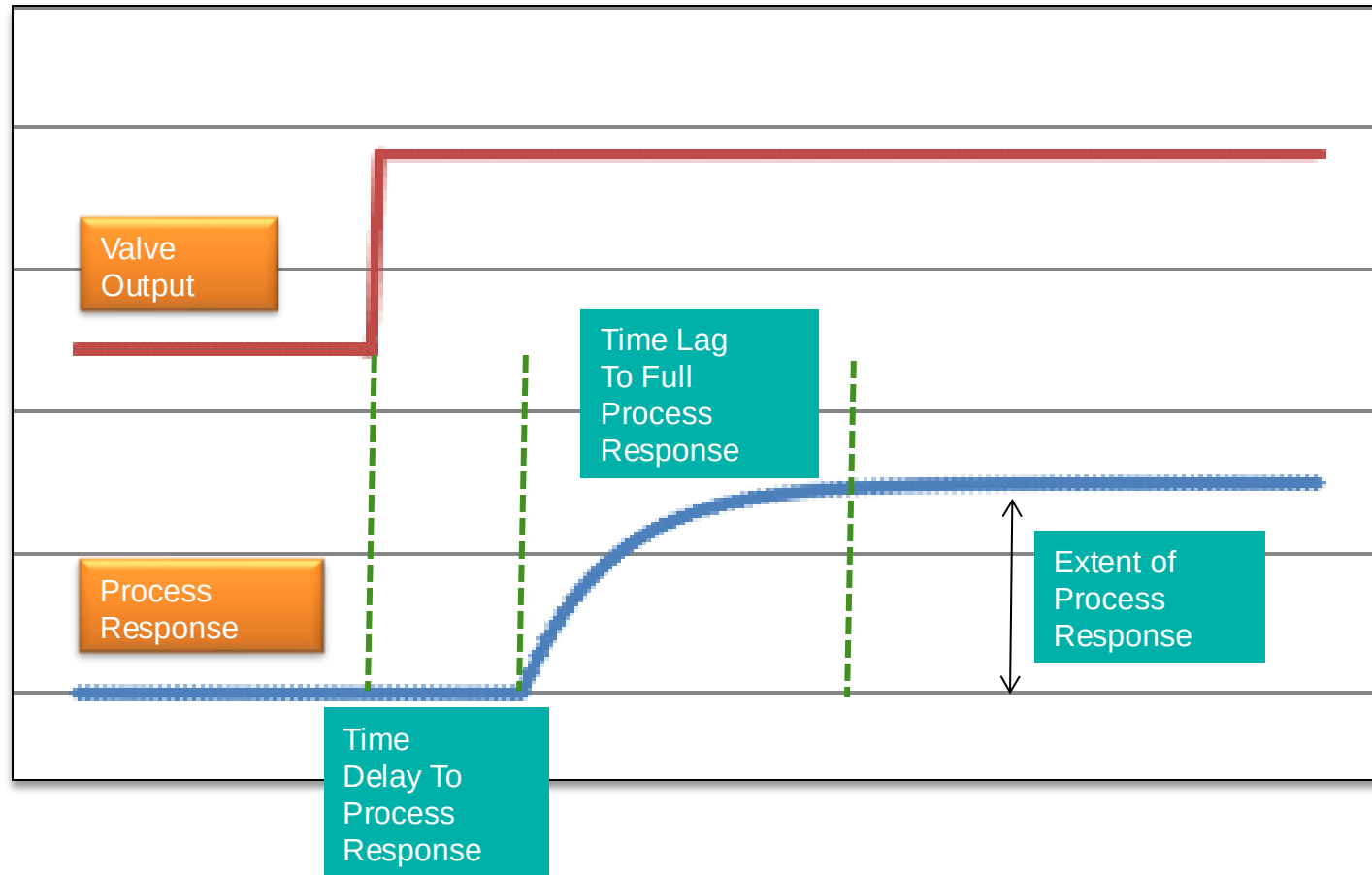


Process Reaction Curve

Level

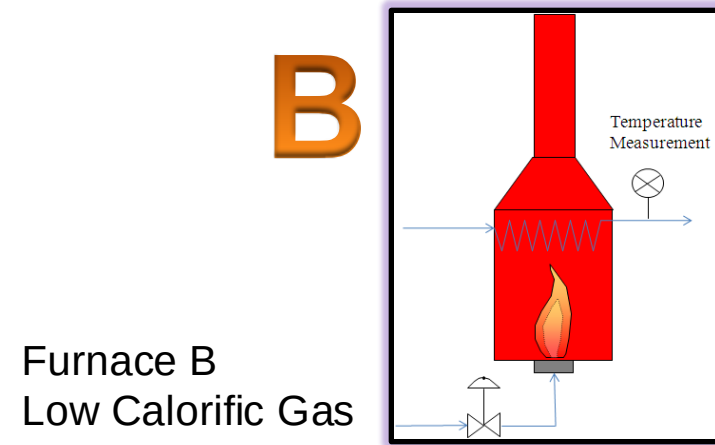
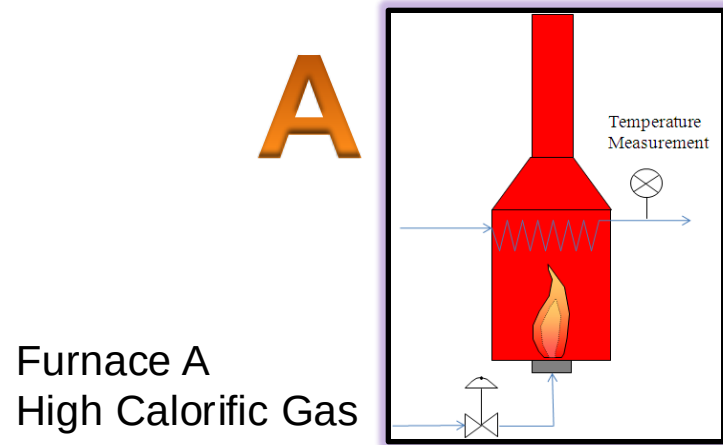
Characterising Process Dynamics

Three parameters are used for characterising Process Dynamics



Process Gain

Let's compare 2 furnaces running on fuel gas of different calorific value, i.e. Furnace A running on a higher calorific value compared to furnace B.

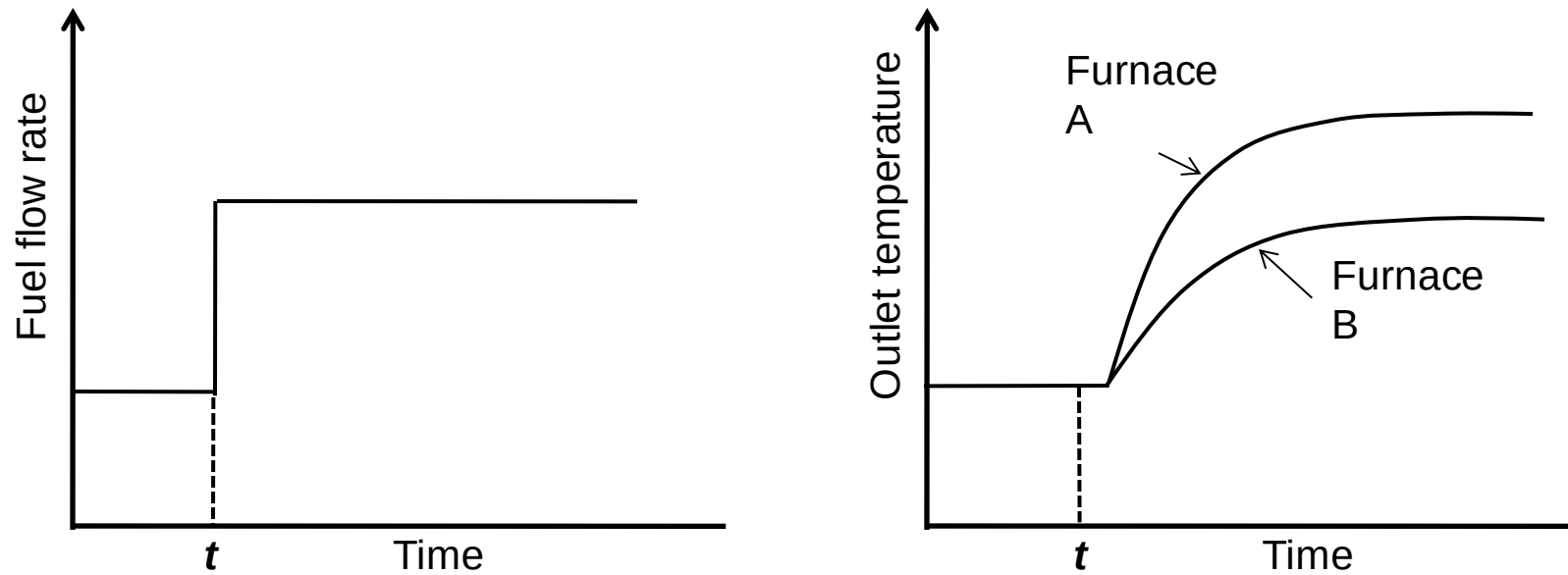


Definitions :

Calorific Value - the amount of heat produced by complete combustion of a fuel (units : kJ/kg)

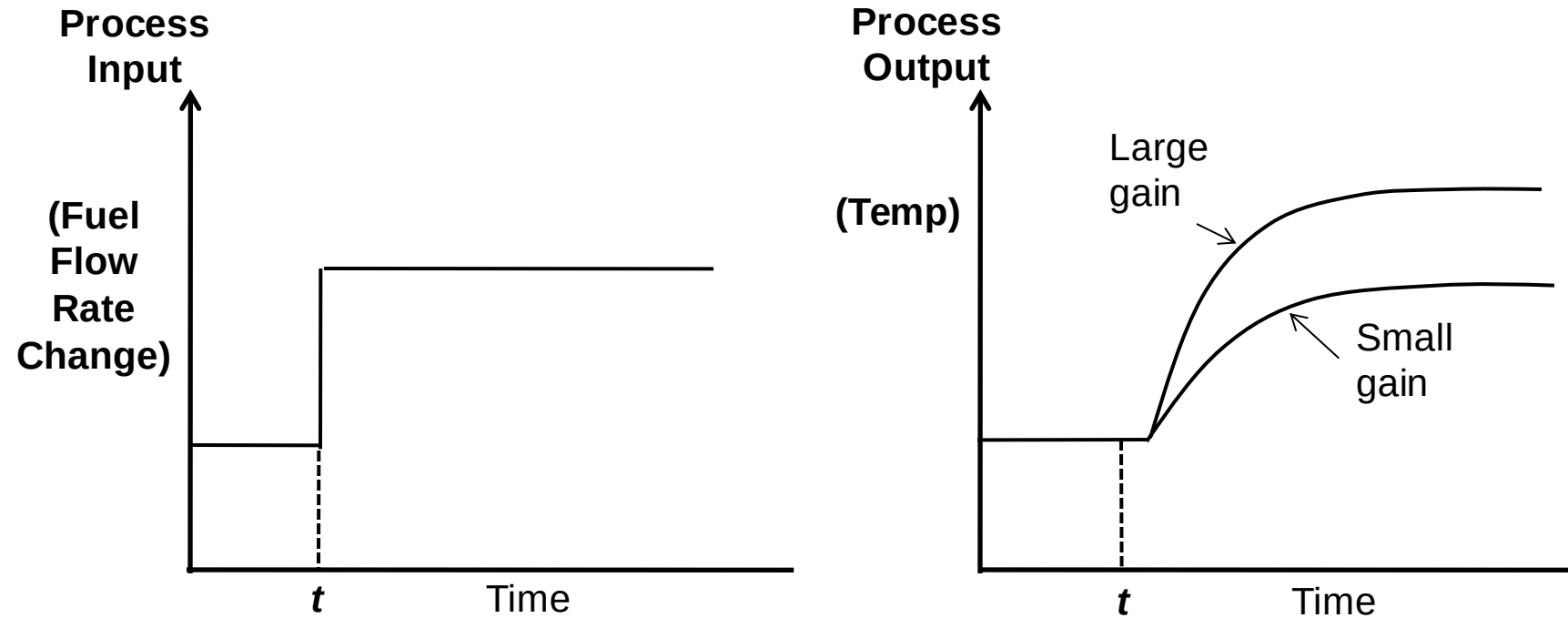
Process Gain

Upon a same step increase of fuel in both furnaces, the response on the outlet temperature was observed as follows:



Process Gain

As such, a process with a higher gain will produce a higher change in the output.



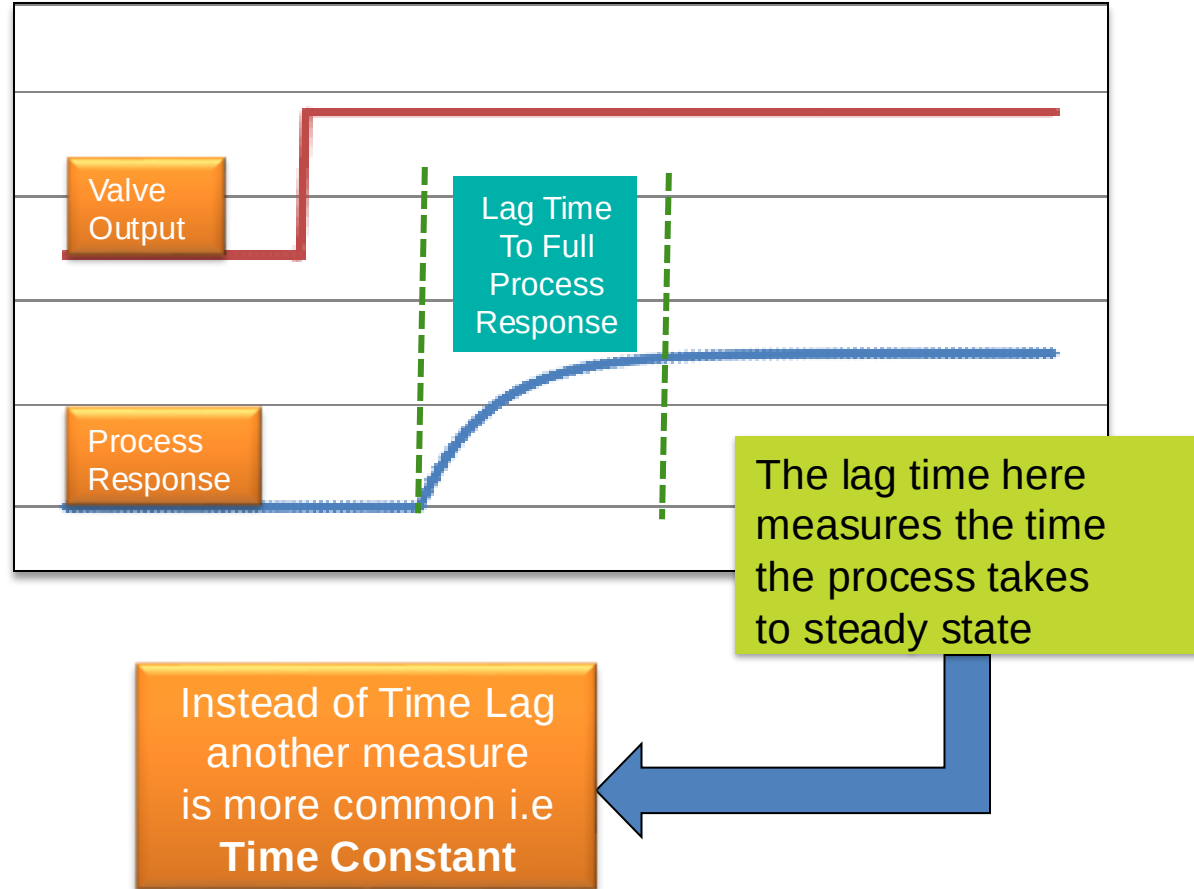
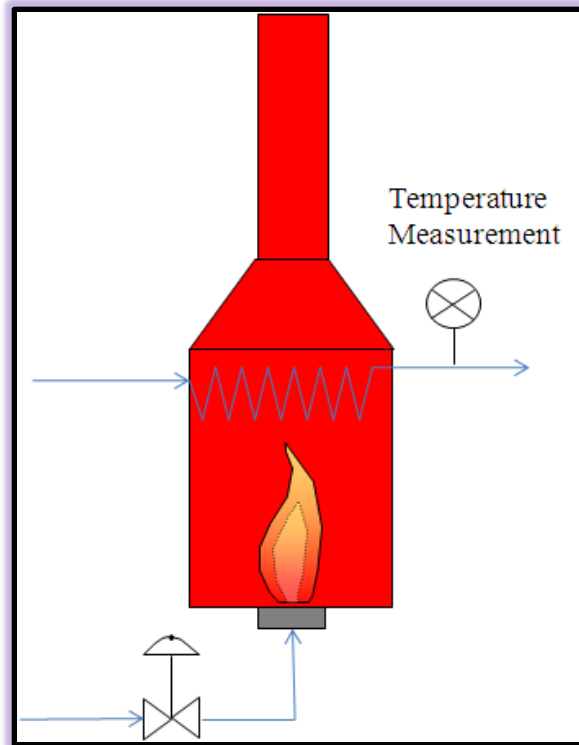
Process Gain

Process gain is the ratio of the change in output to a change in input.

The gain term is essentially dimensionless (*meaning no units*)

$$\text{Process Gain} = \frac{\text{Change in Output}}{\text{Change in Input}}$$

Lag time and Time Constant

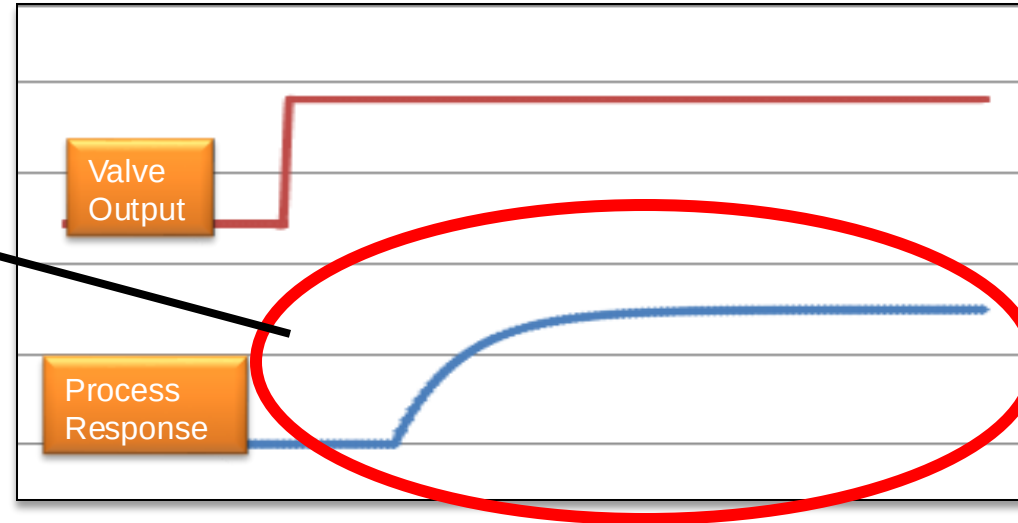


What is time constant?

Process behaviour here can be defined by an equation below:

$$\text{Process Response} = [1 - \exp(-t/\text{Tau})]$$

Where Tau is the Process **TIME CONSTANT**



Tau = 1 min

Time	Process Response (%)
0	0.0%
1	63.2%
2	86.5%
3	95.0%
4	98.2%
5	99.3%
6	99.8%
7	99.9%

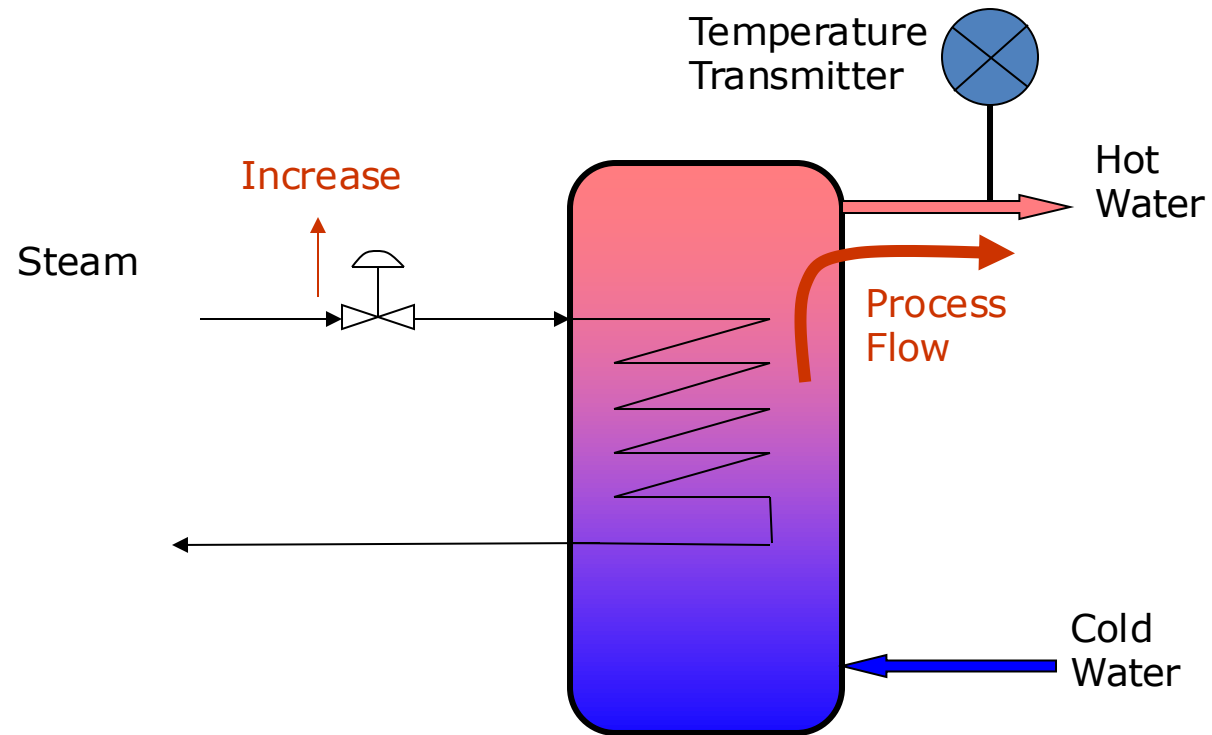
Key Conclusions:

- Process reaction time can be characterised by a **Time Constant**.
- The **Time Constant**, represents the time it takes for the process to reach 63.2 % of its final value

Note

i) The above is for a first order process

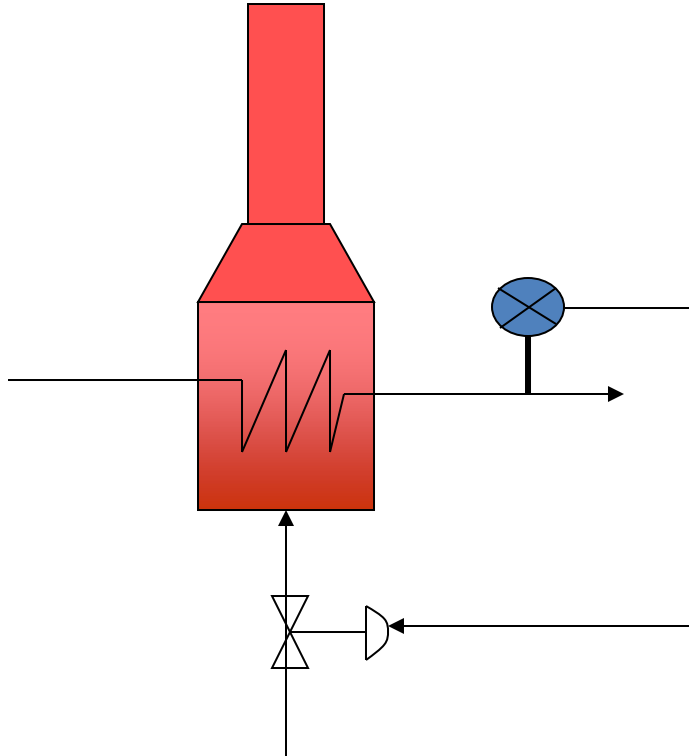
Deadtime



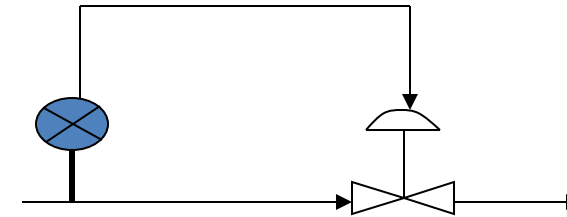
“**Dead time** is defined as the time delay between change in the control signal and the beginning of its effect on the measured value of the controlled variable.”

Lingo : Also known as transportation lag or velocity lag.

Comparison of Deadtime and Time constant



Deadtime (temp) : Significant/Insignificant?
Time Constant (temp) : Fast/Slow?

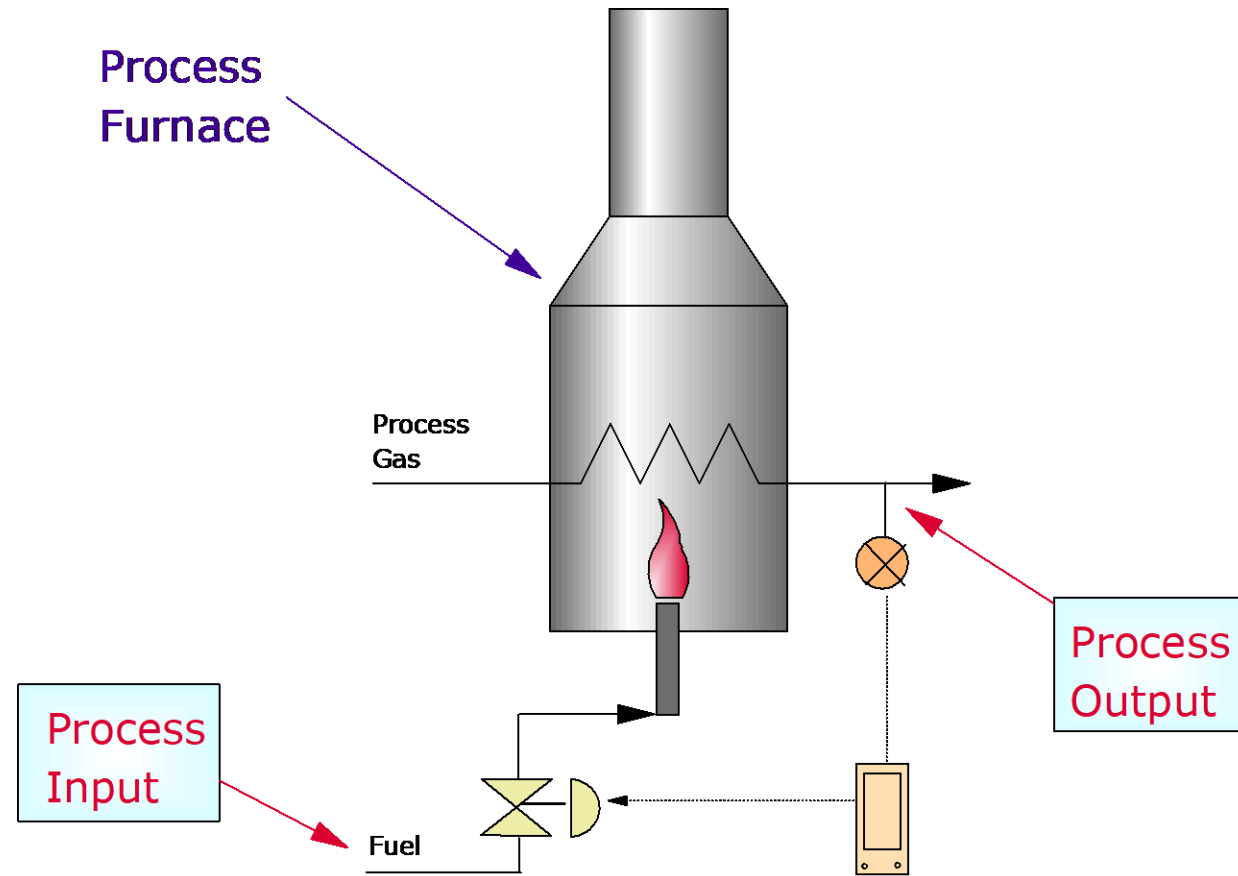


Deadtime(flow) : Significant/Insignificant?
Time Constant (flow) : Fast/Slow?

How to identify Process Dynamics

- Open loop test
- 2 methods to obtain process dynamics parameters:
 - Steepest slope method
 - 2-point method (*not discussed in this course*)

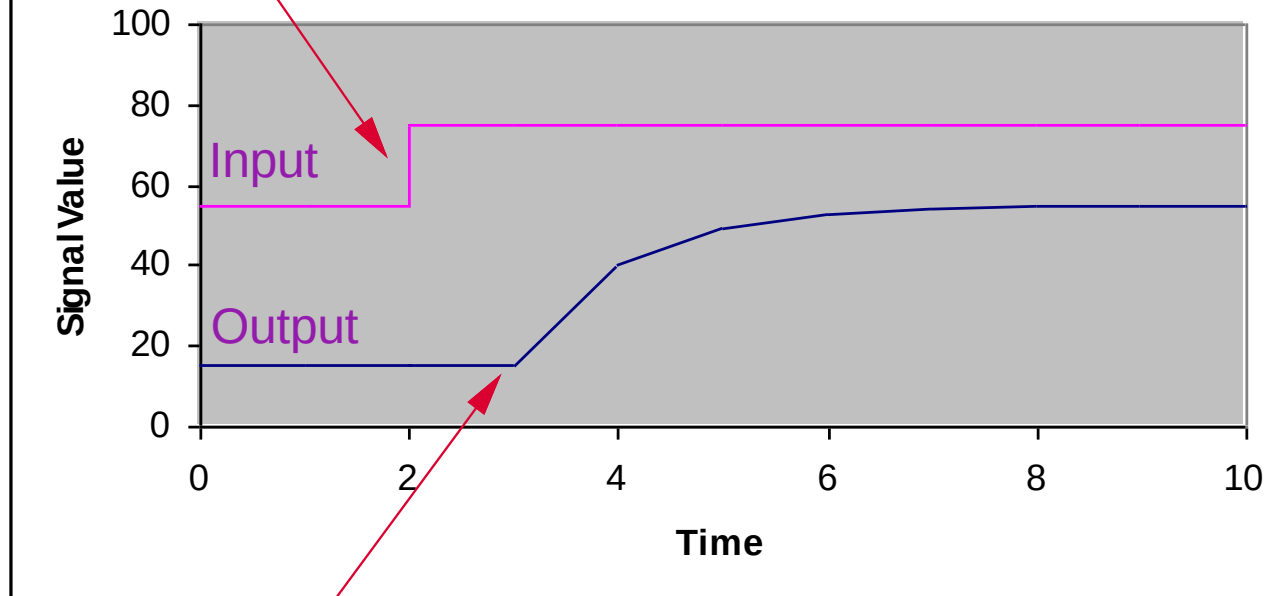
Process Example



Step Response

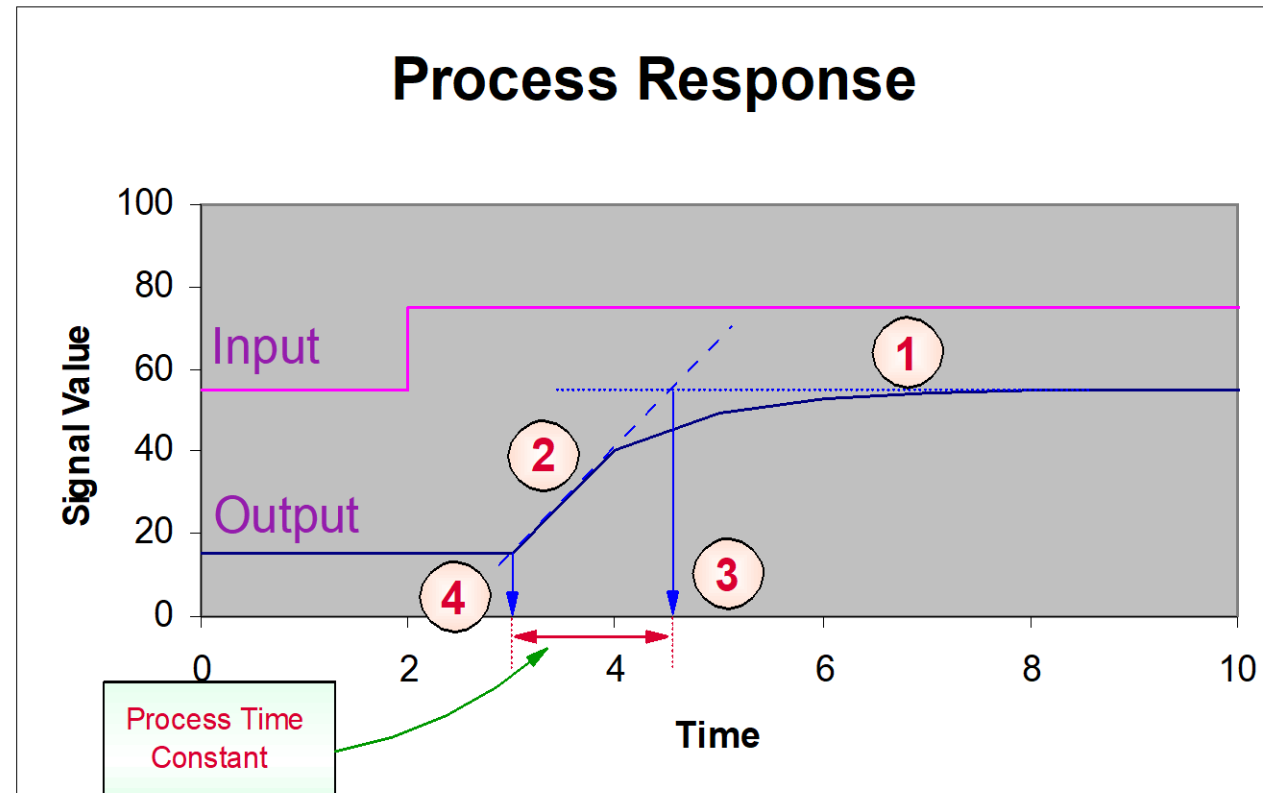
Step Change in Process Input

Process Response



Process Reaction curve

Steepest Slope Method



Step 1 : Draw a line back from the maximum output of the process

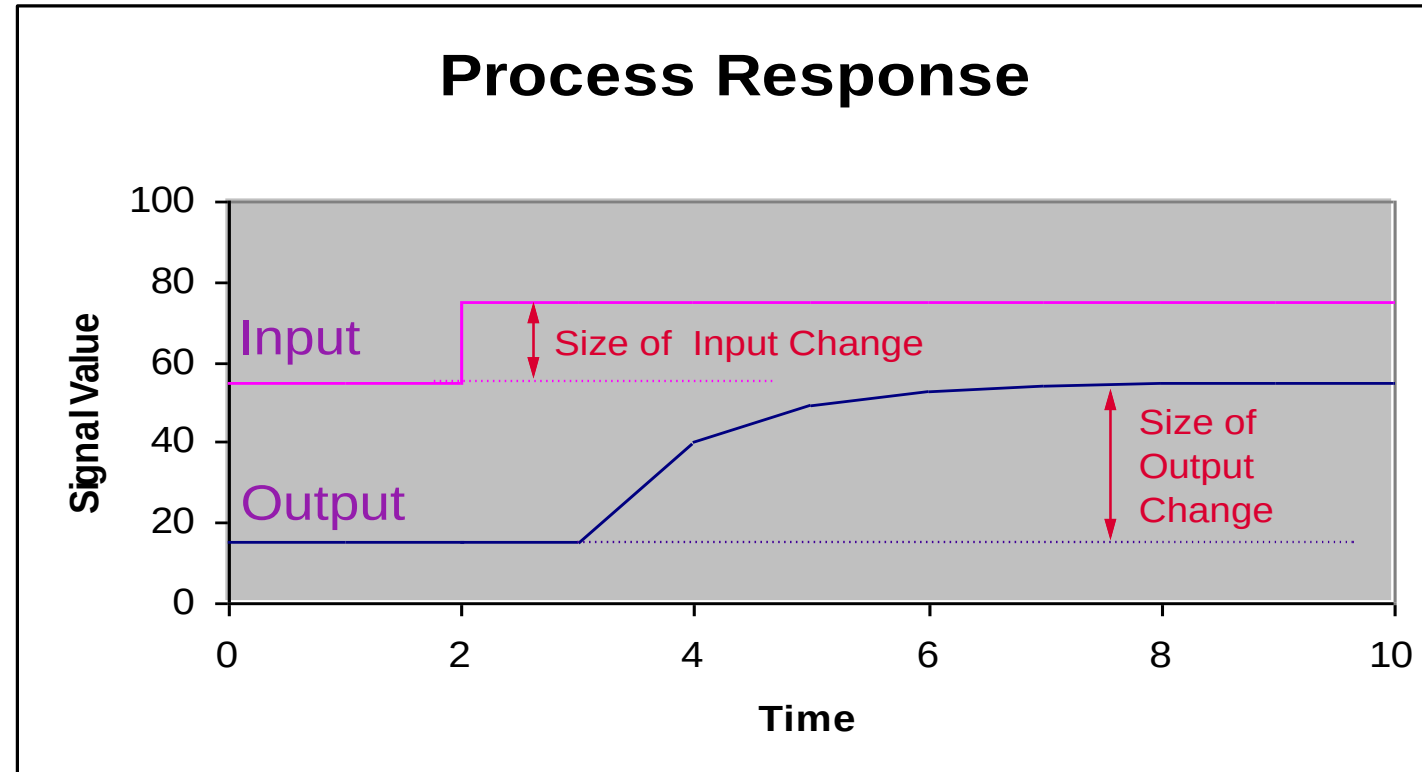
Step 2 : Draw the steepest tangent to the process output curve

Step 3 : Draw a line from the intersection of the lines of step 1 and the tangent, to the x-scale

Step 4 : Draw a line from the intersection of the tangent with the process curve at the other end, to the x-scale

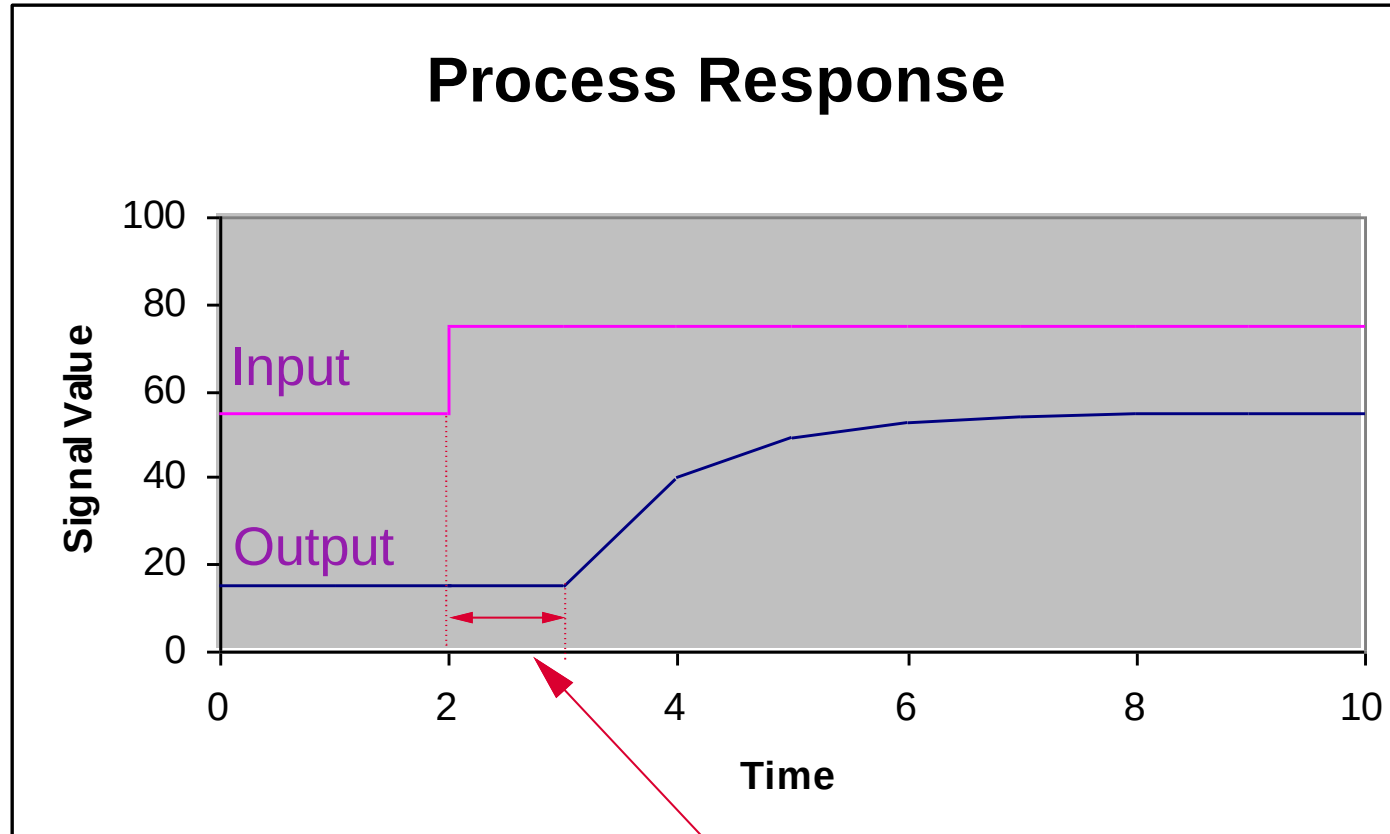
The distance between the intersections on the x-scale (from steps 3 & 4) is the Process Time Constant

Process Gain



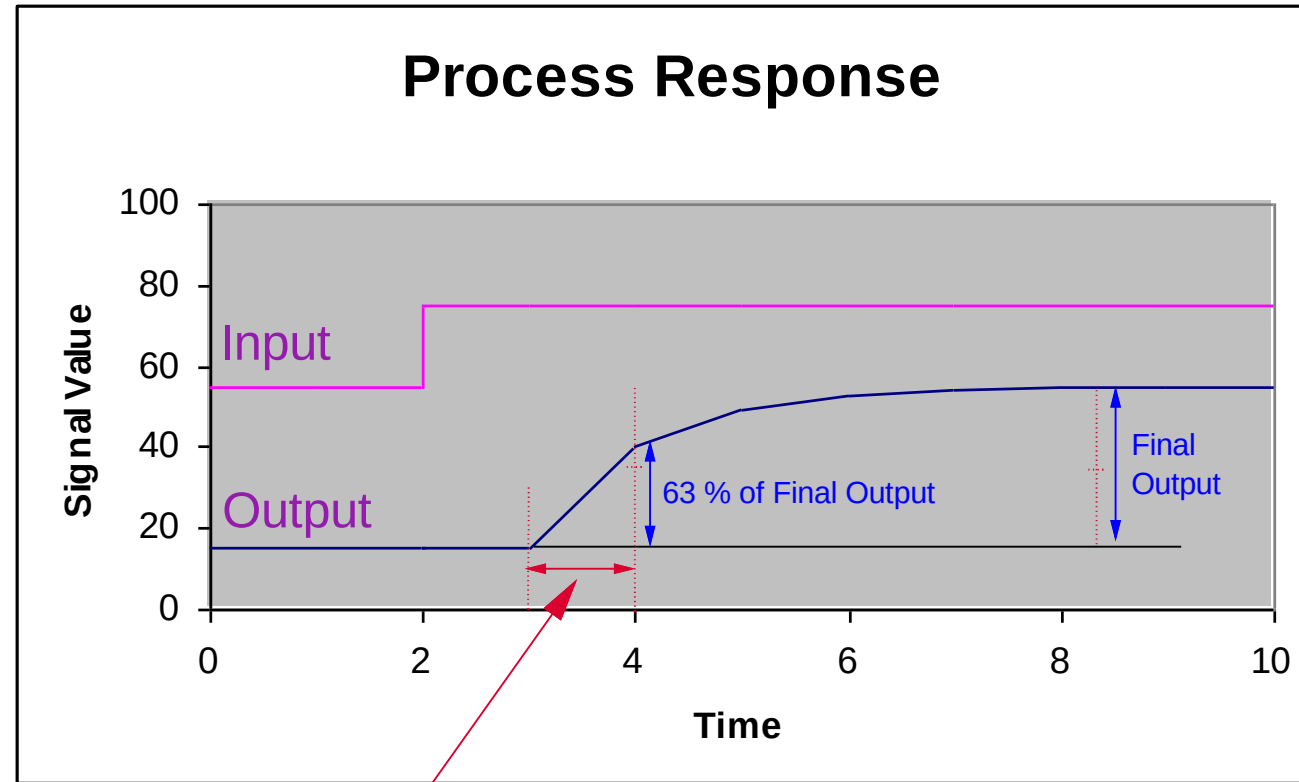
$$\text{Process Gain} = \frac{\text{Size of Output Change}}{\text{Size of Input Change}}$$

Deadtime



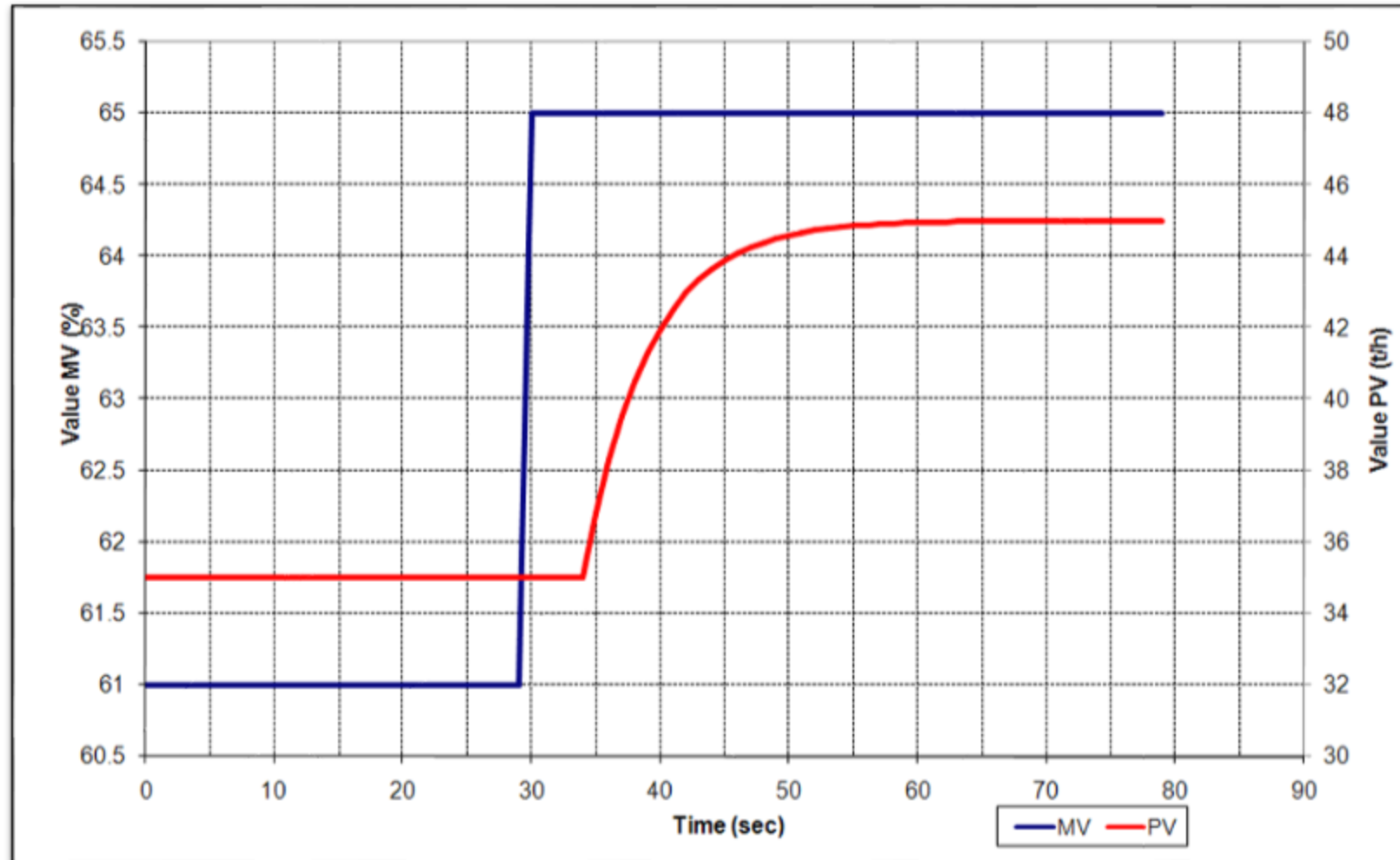
Time taken for process
to respond, i.e. Deadtime, θ

Time Constant



Time taken for process (Output) to reach
63 % of its final value, i.e. Process Time Constant, τ

Exercise: Calculate Process Dynamics



Find K_p , t and Deadtime. PV is ranged from 0 to 100 t/h.

Self-Regulating/Non-Self-Regulating

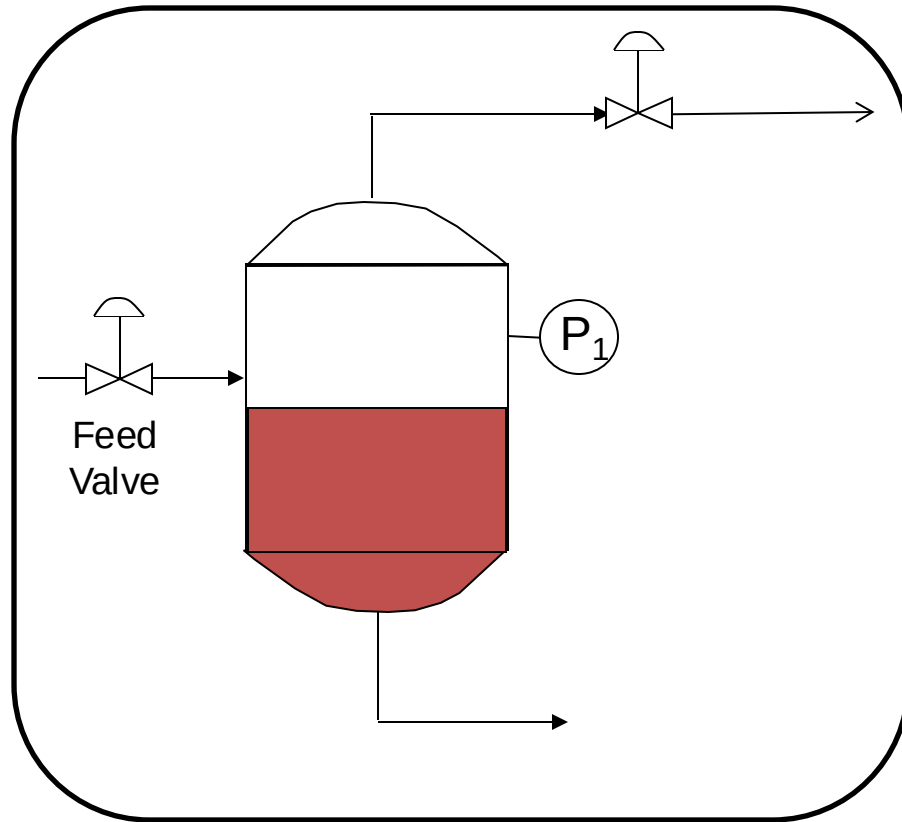
Self-regulating

- The ability of the process to reach a new equilibrium on an input change
 - Most processes exhibit this

Not self-regulating

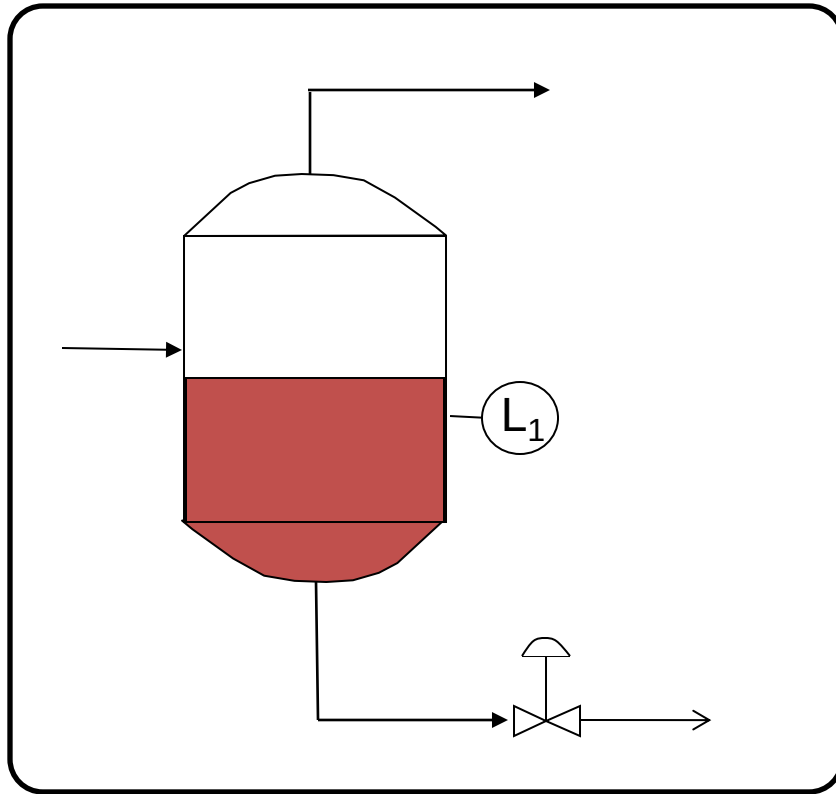
- Process does not reach equilibrium on an input change
 - Level and some temperatures

Self-regulating



- i) Consider a Propane vessel at a stable pressure of 10 barg
- ii) All valves are on manual.
- iii) What will happen to the pressure if the **feed valve** is opened by 10%? And why?

Nonself-regulating



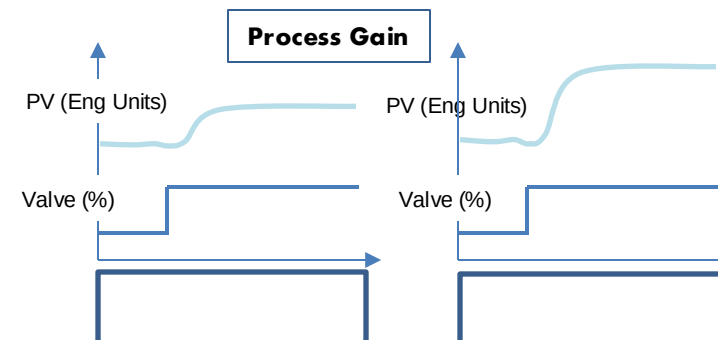
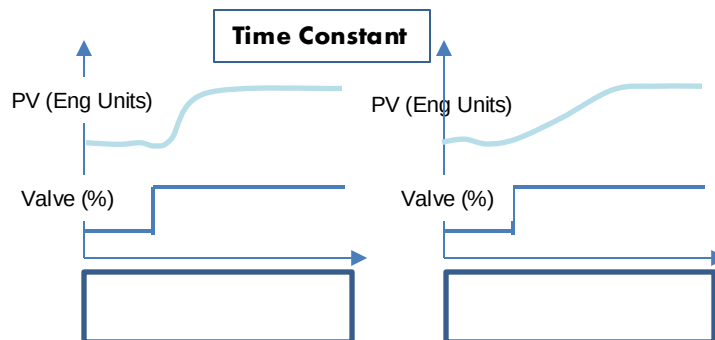
- i) Consider a pressurized (8 barg) butane vessel with a stable level
- ii) What happens to the level if the bottom discharge valve is further opened by 10%? And why?

Retrieval Practice

1. In the “3+2 Step” approach to understanding Base Layer Control (or BLC), how many critical steps are there to understand Base Layer Control?
 - a) 1 Key Step (understanding the Process)
 - b) 2 Key Steps (understanding the Process and the Controller)
 - c) 3 Key Steps (understanding the Process, the Controller and Tuning)
 - d) 4 Key Steps (understanding the Process, the Controller, Tuning and Enhanced Features)
2. What fundamental activity are we doing when we tune a controller?
 - a) Stabilise the controller
 - b) Stabilise the process
 - c) Match the Process Dynamics to the Controller Dynamics
 - d) Match the Controller Dynamics to the Process Dynamics
3. Process Understanding is critical in Process Control primarily to
 - a) Understand the Process
 - b) Understand the Controller
 - c) Understand why the Process needs to be stable
 - d) Understand why the Controller Relationships are critical

Retrieval Practice (..2)

4. Tick which of the following statements are true (can be more than 1)
- The dynamics of the process cannot be changed by a controller
 - The dynamics of the process is determined by the Operating Point
 - The dynamics of the process is also determined by material used in the process and volume present
 - The dynamics of the process is affected strongly by PID tuning
5. You have been asked to explain the difference between “Process Gain” and “Time Constant”. You will use an example if a valve output is changed from 30% to 40%. In this case
- “Process Gain” represents _____ the Process will react (how much/how much time)
 - “Time Constant” represents _____ the Process will take to react (how much/how much time)
6. Write down which of the processes below have a high/low gain and which has a short/long time constant.



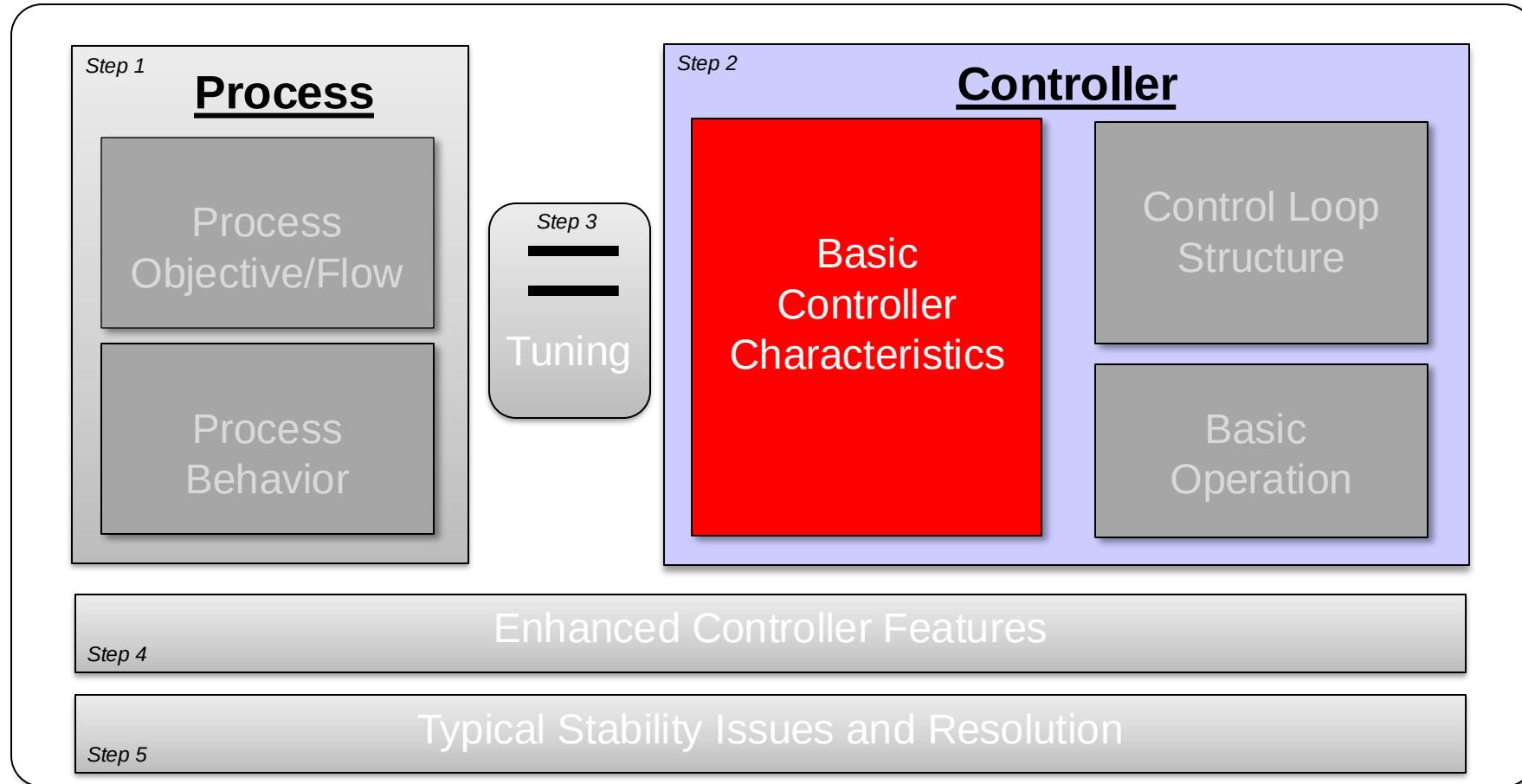
Key Take Aways - Process Knowledge

- 2 types of processes:
 - Self-regulating
 - Non self-regulating
- Self-regulating process dynamics/ characteristics
 - $\text{ProcessGain} = \frac{\text{Size of Output Change}}{\text{Size of Input Change}}$
 - Time constant: Time taken for process (Output) to reach 63 % of its final value
 - Deadtime: Time taken for process to respond
- Method of identifying process dynamics:
 - Open loop test

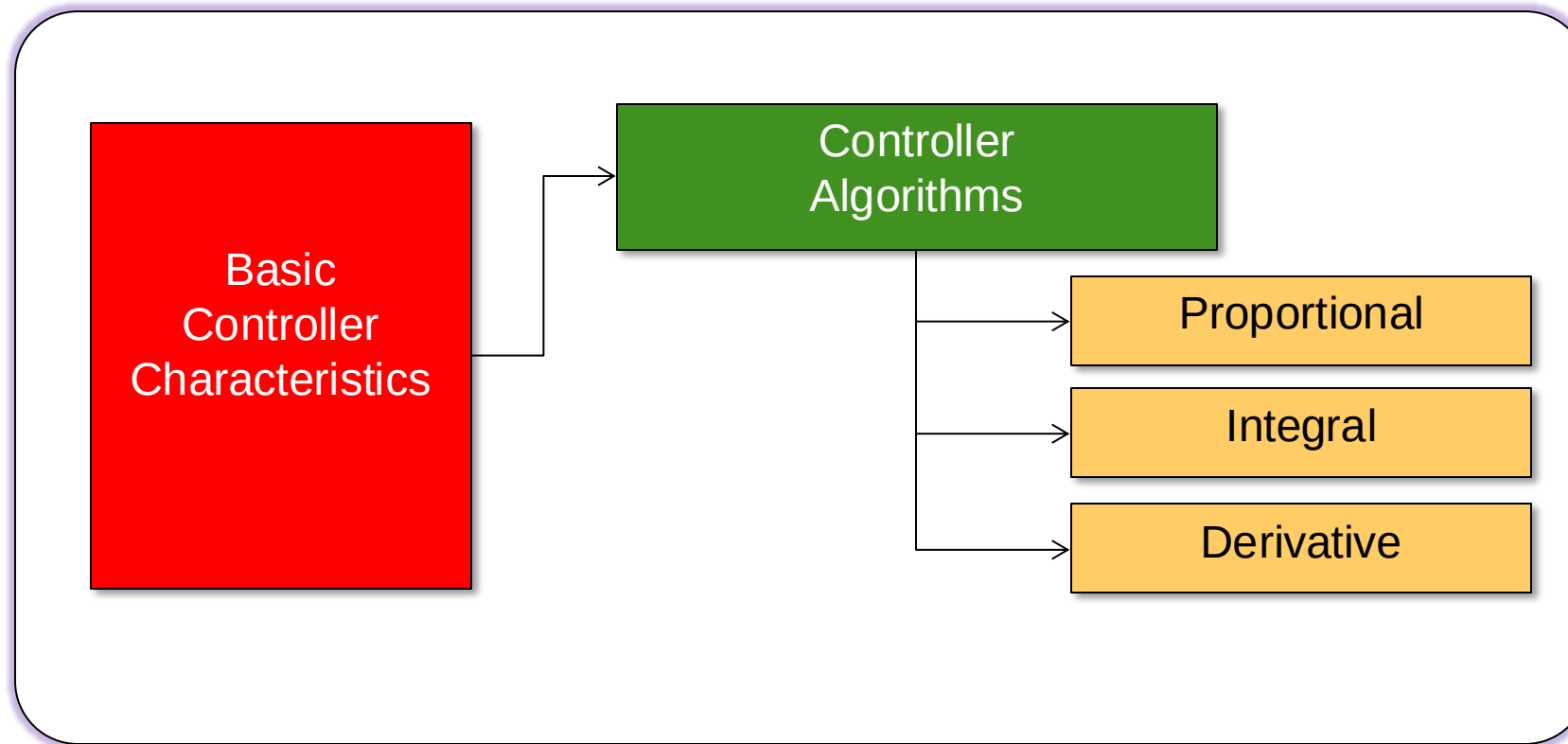
Base Layer Control

Step 2 - Understanding the Controller

Basic Controller Characteristics

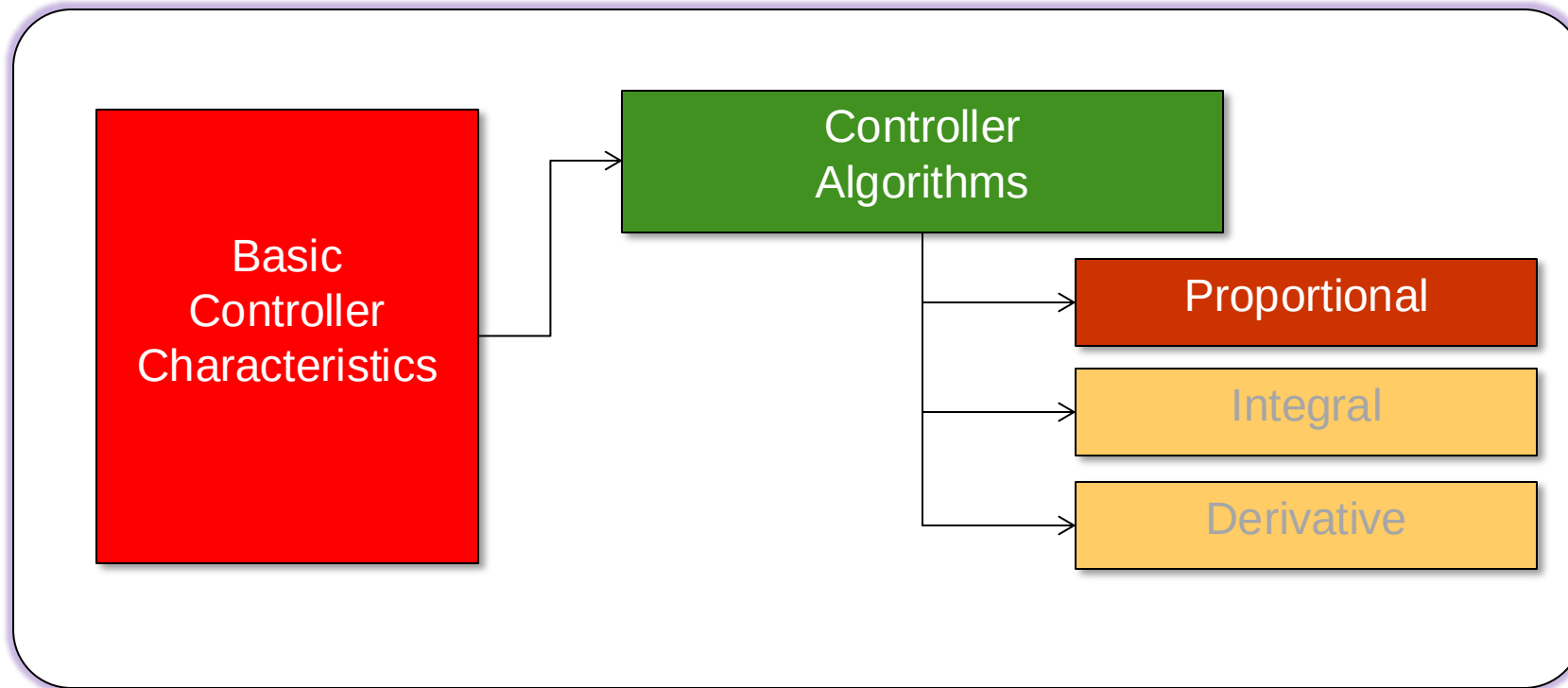


Basic Controller Characteristics

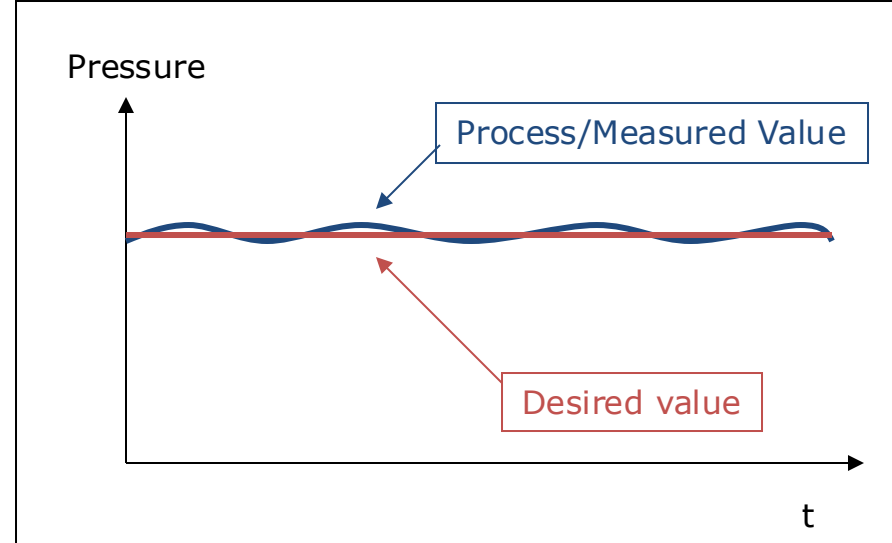
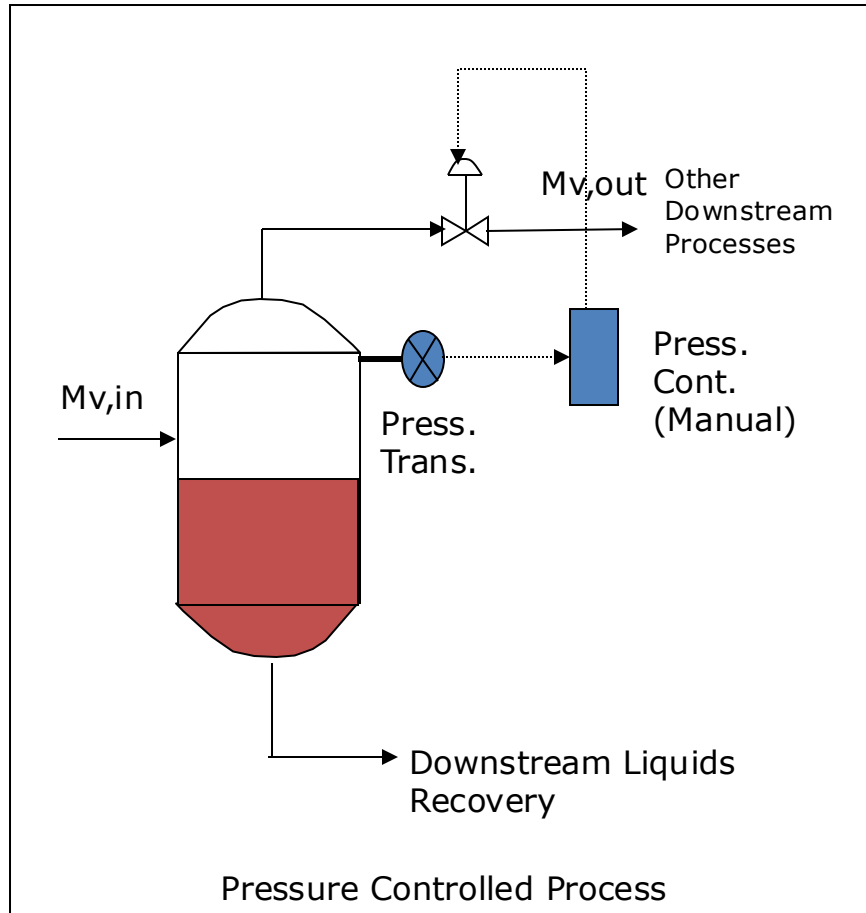


Controller Algorithms

Proportional Action



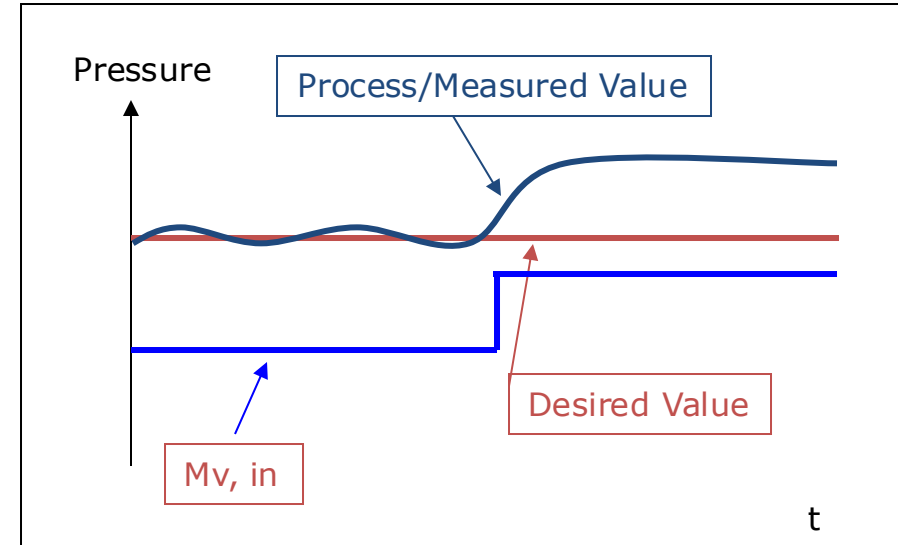
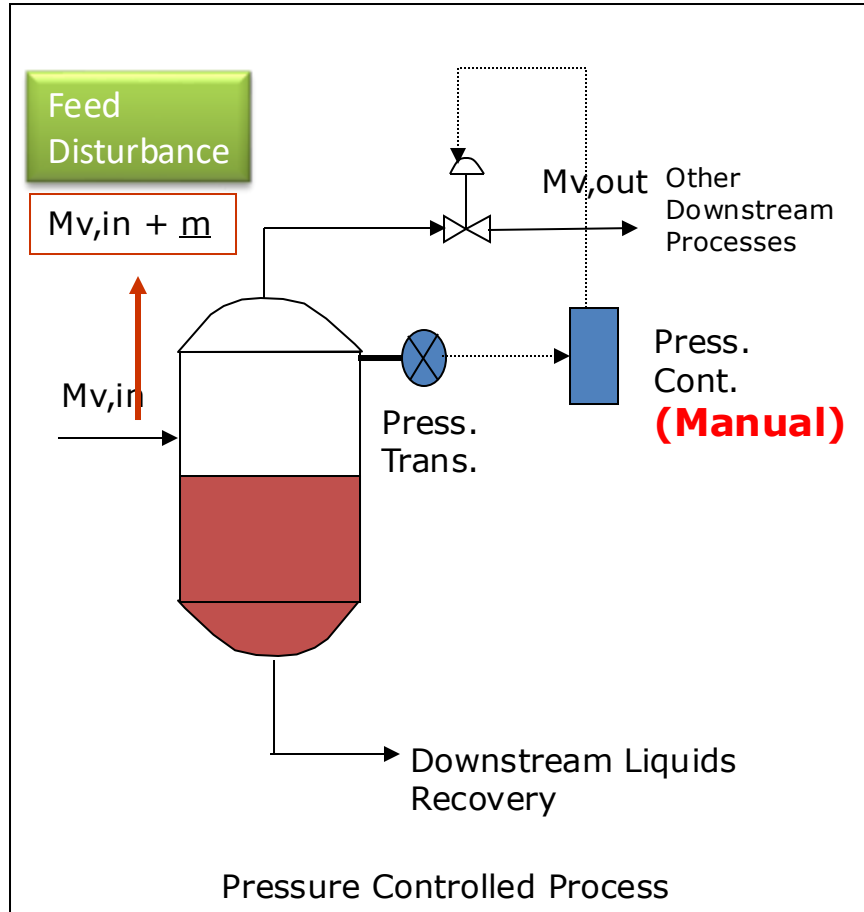
Error



Notes:

- i) Controller in Manual
- ii) $Mv_{in} = Mv_{out}$

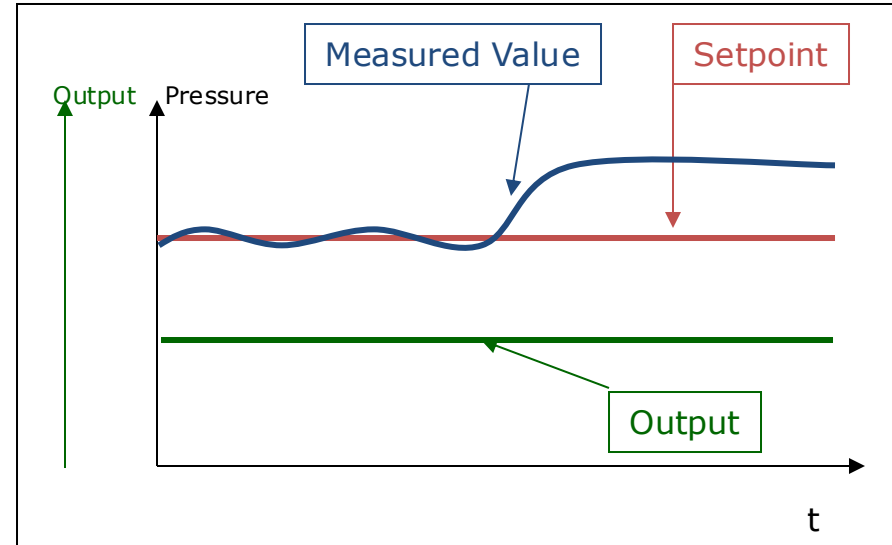
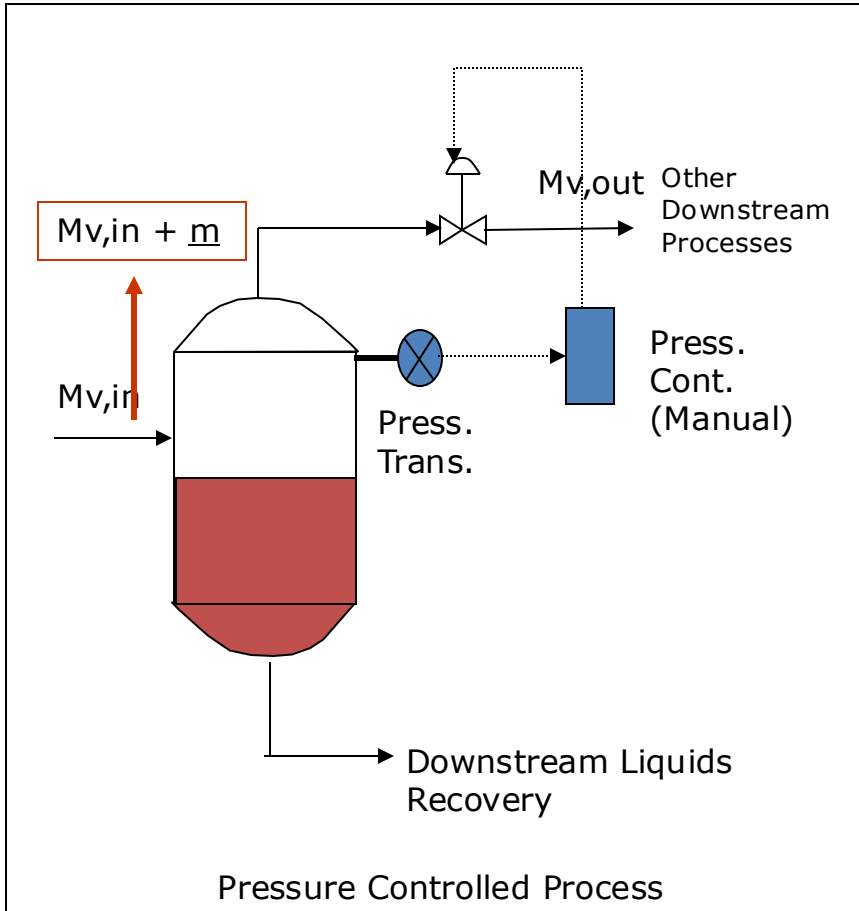
Error (2)



Notes:

- i) Controller in Manual
- ii) $Mv_{in} > Mv_{out}$
- iii) Error = Setpoint - Process Variable

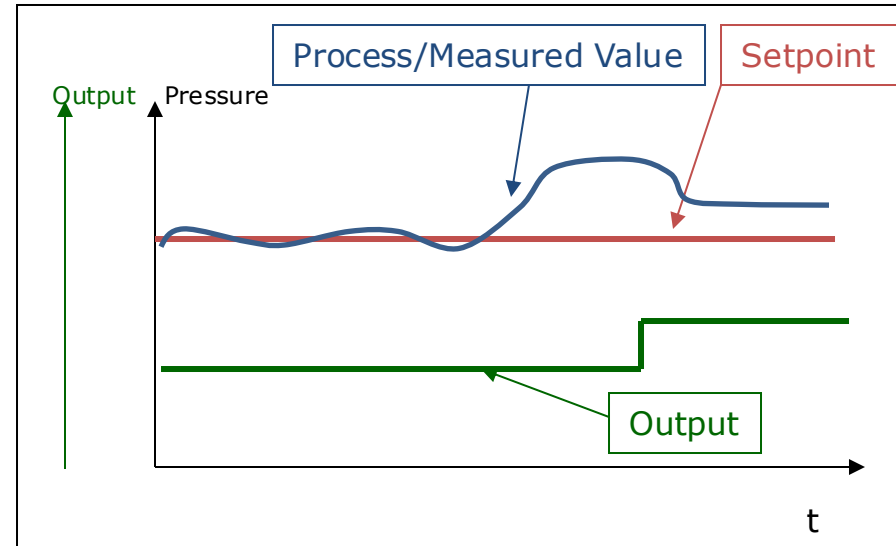
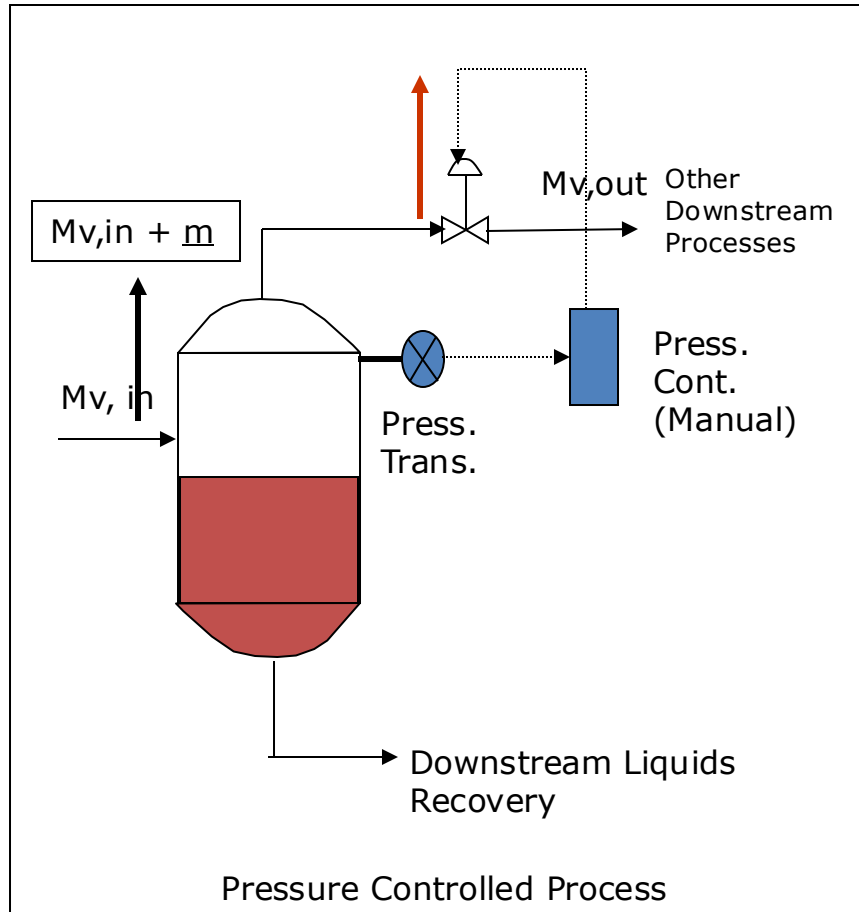
Removal of Error



Notes:

i) Controller in Manual

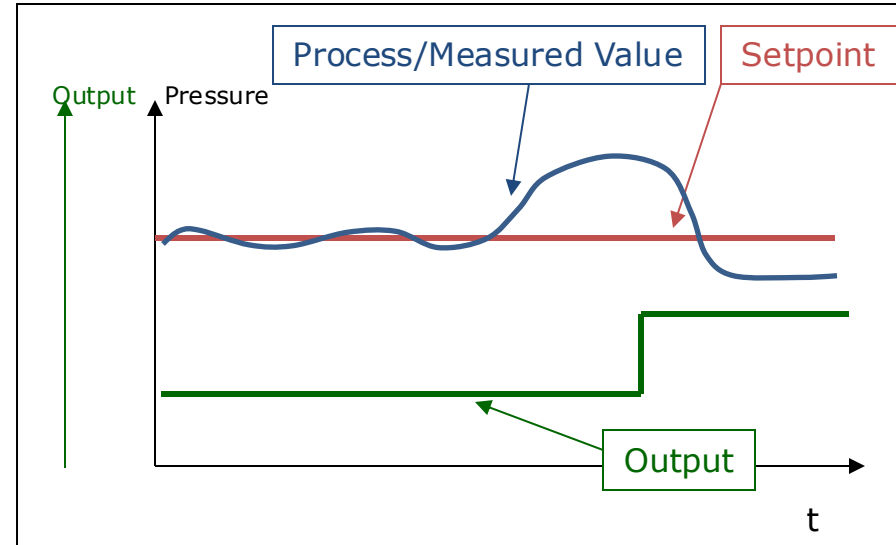
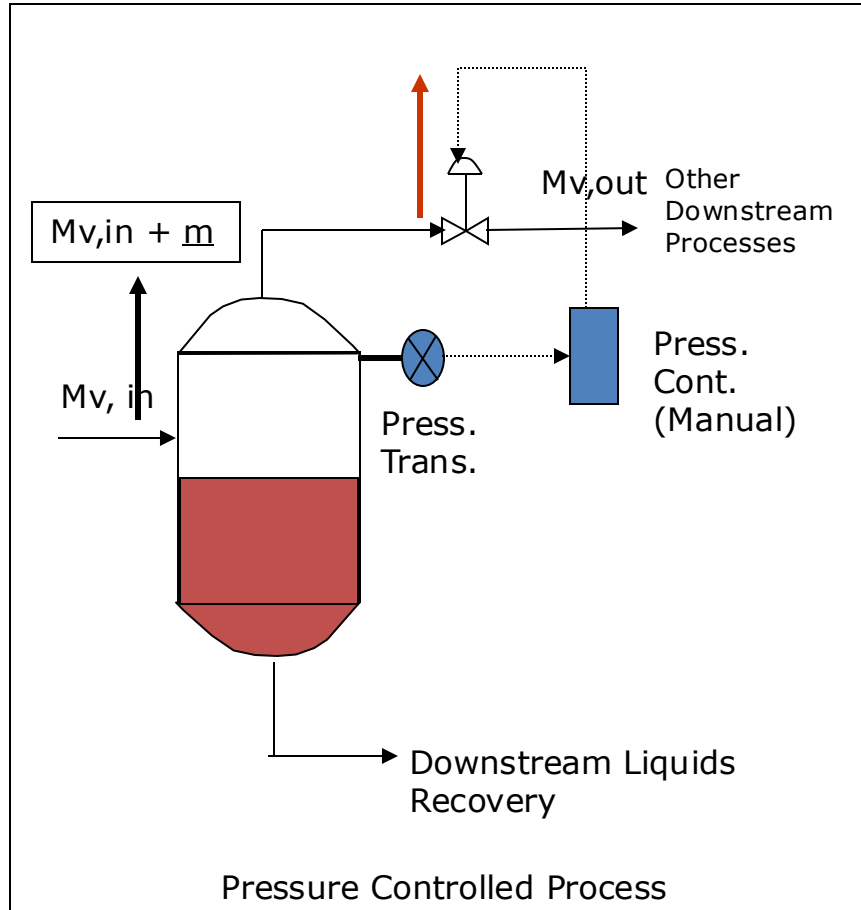
Removal of Error – too little



Notes:

- i) Controller in Manual
- ii) Output increased
- iii) The increase in output has to be proportional to the size of the error.

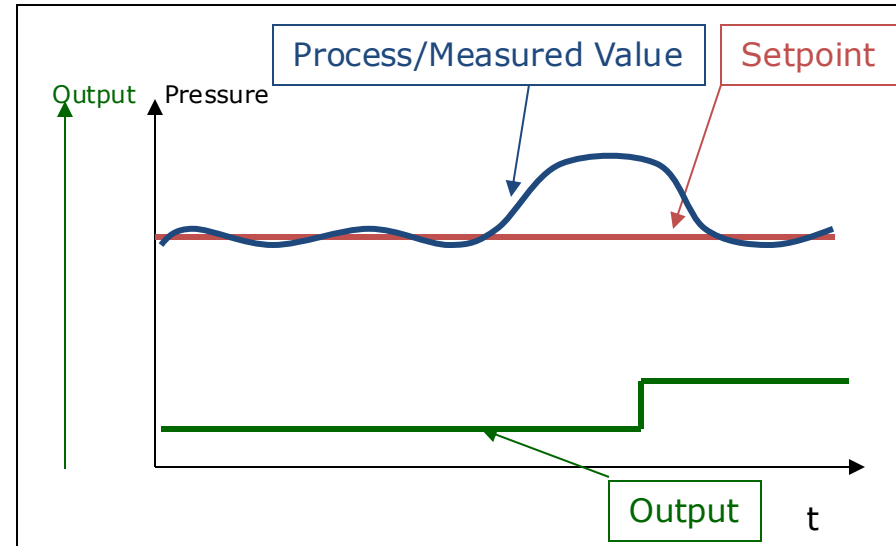
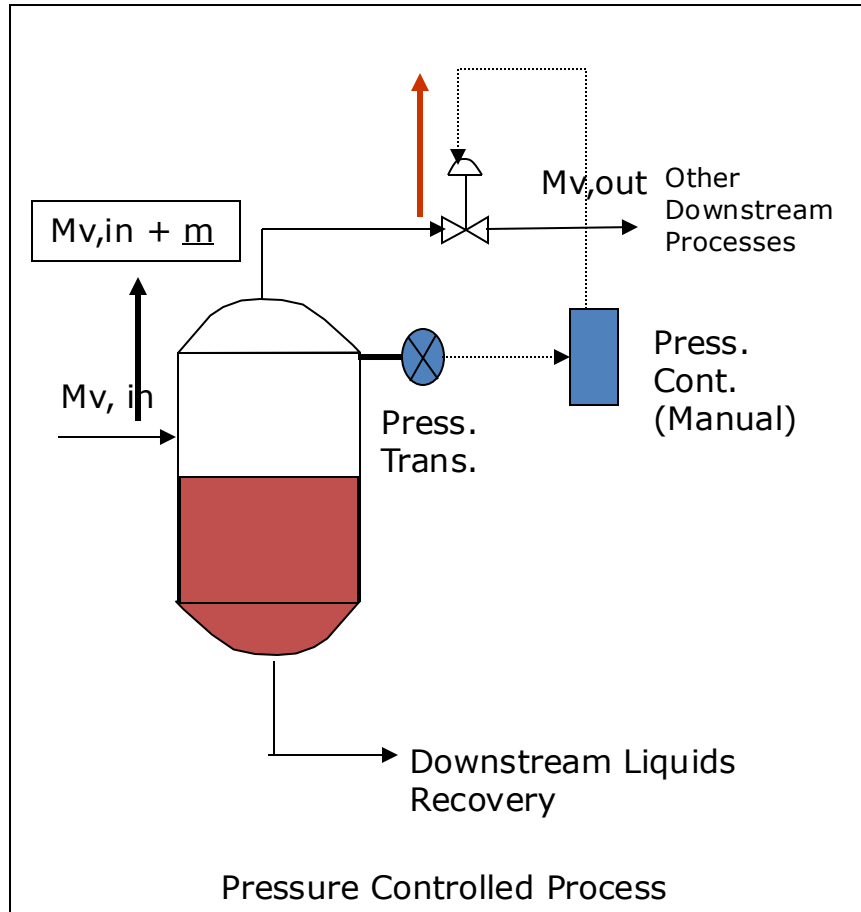
Removal of Error – too much



Notes:

- i) Controller in Manual
- ii) Output increased
- iii) The increase in output has to be proportional to the size of the error.

Removal of Error – Just right!



Notes:

- i) Controller in Manual
- ii) Output increased
- iii) The increase in output has to be proportional to the size of the error.

The change in output must be “proportional” to the size of the error

The Removal of Error

Conclusion

- To bring the pressure back down, there is a need to change the output of the controller
- The size of the change has to be in proportion to the size of the error.
 - Definition of Proportional Action or P-action

Proportional Action

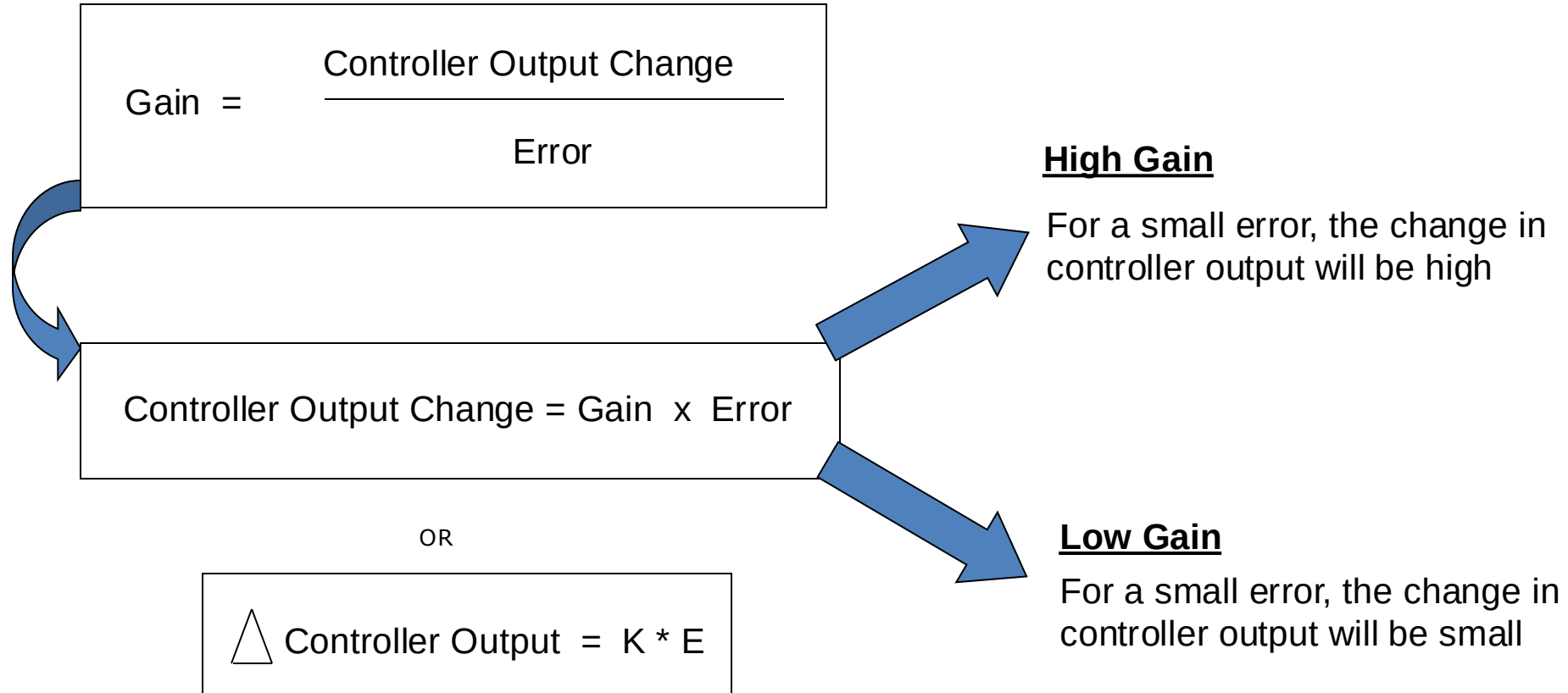
Definition of Proportional Action

- The action of a controller to change its output in proportion to the size of the error generated.

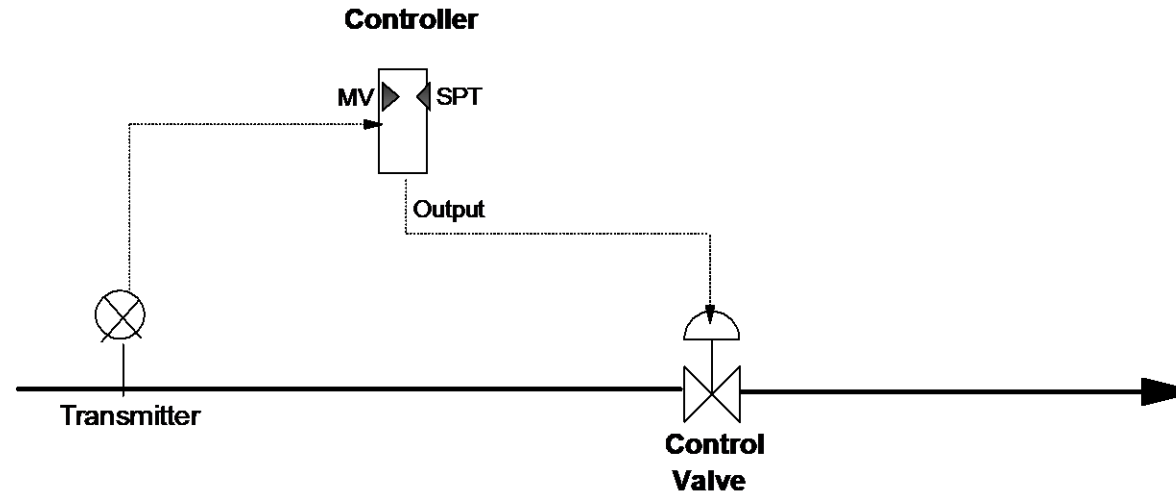
Definition of Proportional Gain

- The variation of the output of the controller per unit error is called the proportional gain.

Definition



Question (1) for Gain



Process:

- i) Transmitter measures a flow between 0 to 100 t/d of process fluid.
- ii) Control Valve operates typically from 0 ~ 100 % opening
- iii) Normal load is at 50 t/d and valve opening is at 50 %
- iv) Gain of Controller is 1.
- v) Controller is a proportional only controller
- vi) No SPT change, just eliminate error

What will be the final controller output if the Process Value increases to 70 t/d?

Worked Example

The 4 Step Process

1. Error = Setpoint - Process/Measured value

➤ $\text{Error} = (50 - 70) \text{ t/d} = -20 \text{ t/d}$

2. Error (%) = Error / Full range * 100

➤ $\text{Error (\%)} = (-20 \text{ t/d}) / 100 \text{ t/d} * 100 = -20 \%$

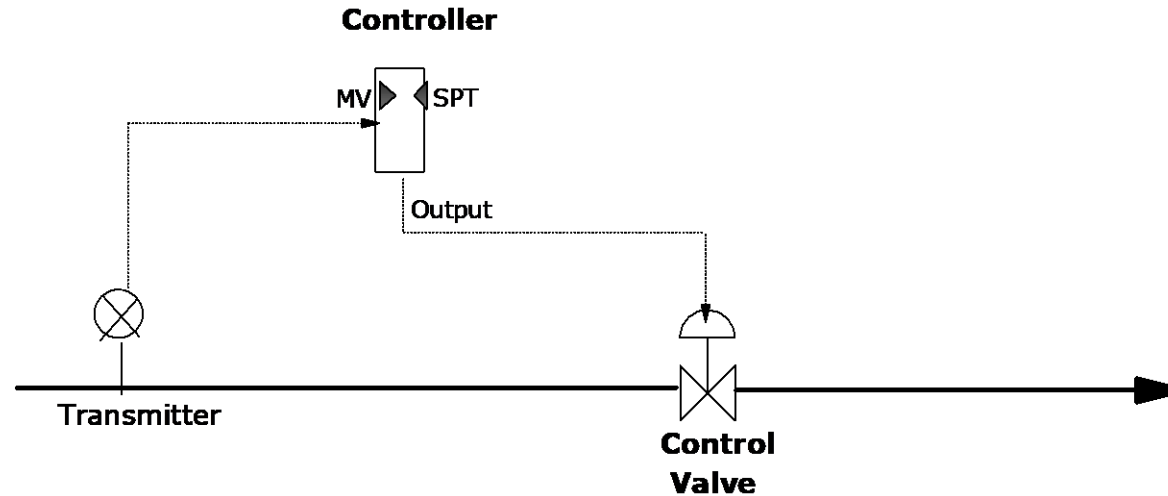
3. Change in output (%) = Error (%) * Gain

➤ $\text{Change in output} = -20 \% * 1 = -20 \%$

4. Final output = Original o/p + Change in o/p

➤ $\text{Final Output} = 50 \% + (-20\%) = 30 \%$

Exercise



Process:

- i) Transmitter measures a flow between **0 to 1000 t/d** of process fluid.
- ii) Control Valve operates typically from 0 ~ 100 % opening
- iii) Normal load is at 500 t/d and valve opening is at 50 %
- iv) Gain of Controller is 1.5.
- v) Controller is a proportional only controller
- vi) No SPT change, just eliminate error

What will be the final controller output for both the cases below?

Case (1) - Process Value increases to 700 t/d,

Case (2) – Process Value drops to 200 t/d

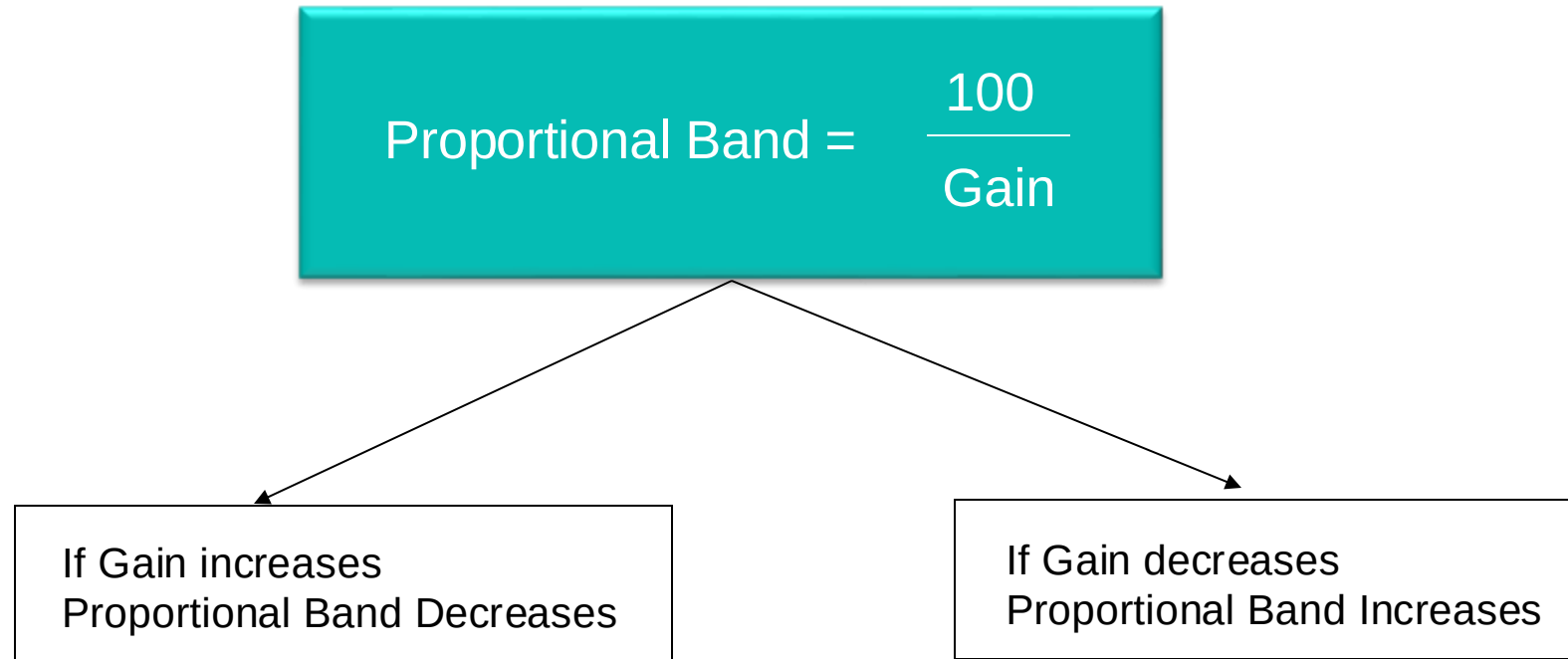
Understanding Proportional Band

Units used by DCS Vendors

PID Settings	Yokogawa	Honeywell	Foxboro
P	PB (%)	Gain (K)	PB (%)
I	Seconds/repeat	Minutes/repeat	Minutes/repeat
D	Seconds	Minutes	Minutes

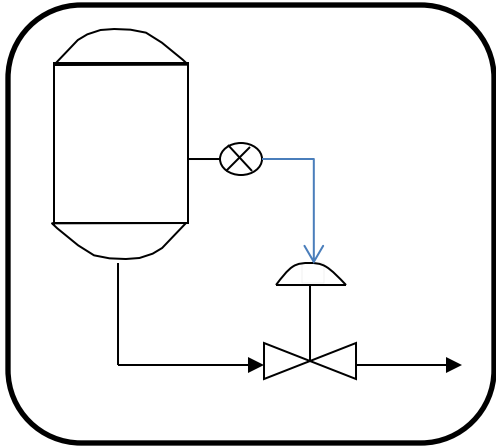
What is Proportional Band?

Relationship between Gain and Proportional Band

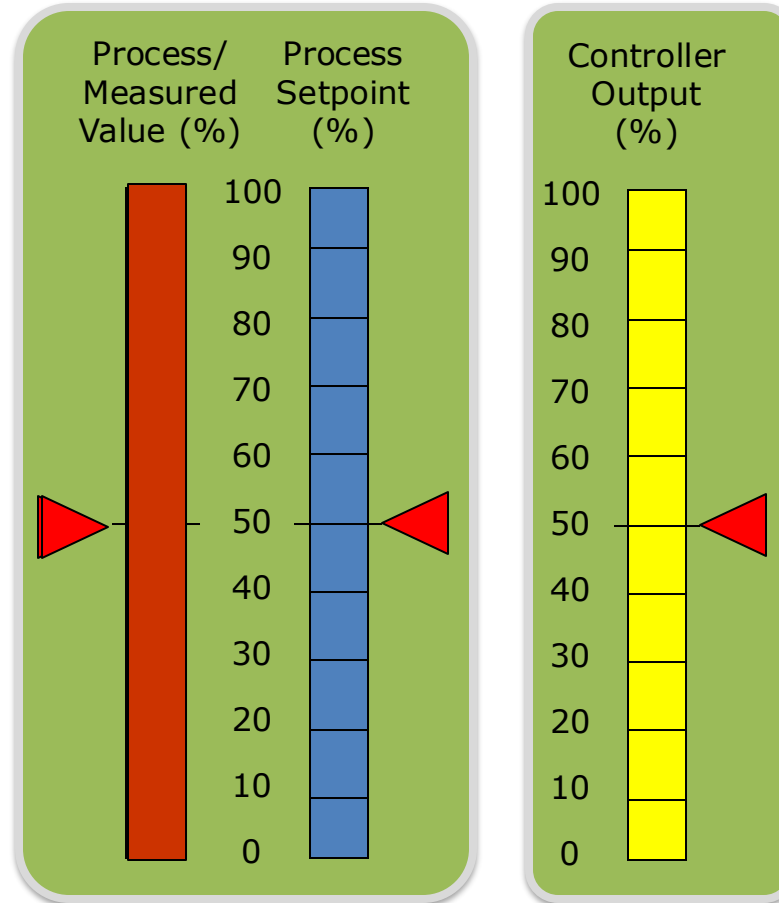


Lingo : PB, Pband, Propband

Worked Example – Pband 1a



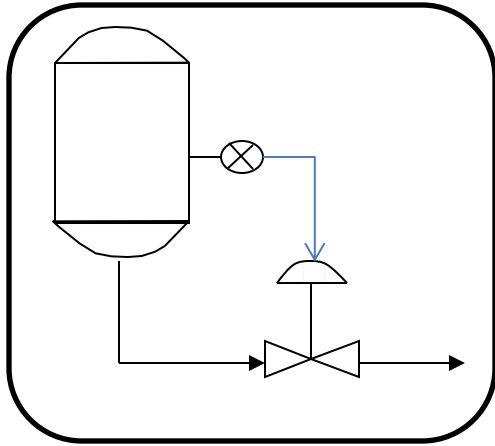
Controller Faceplate



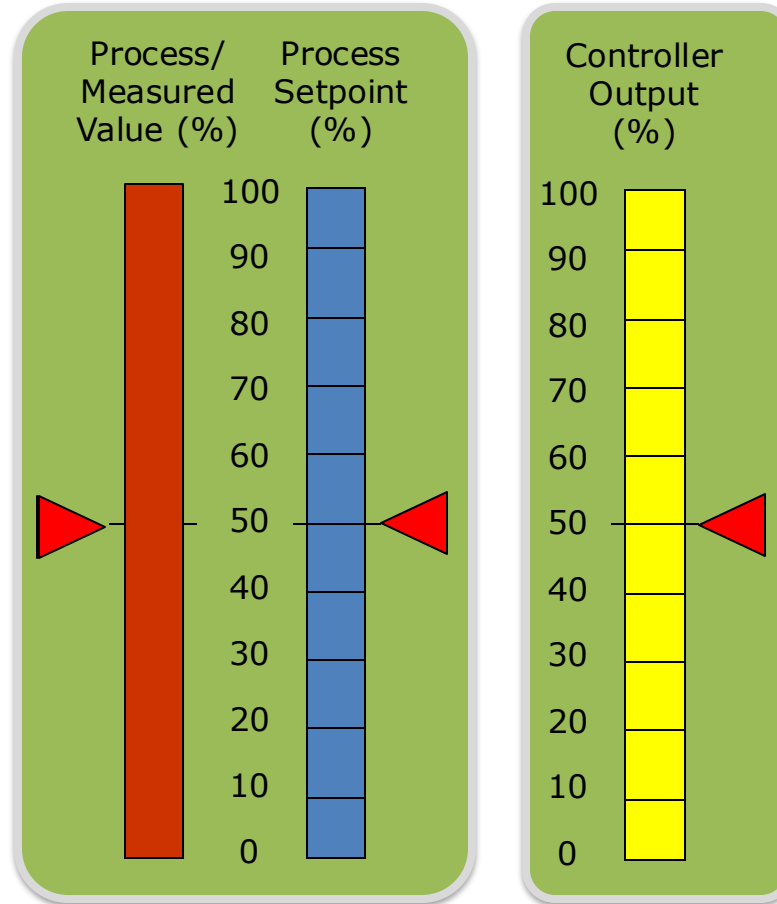
Controller Gain = 1

What is the Proportional Band?

Worked Example – Pband 1b



Controller
Faceplate

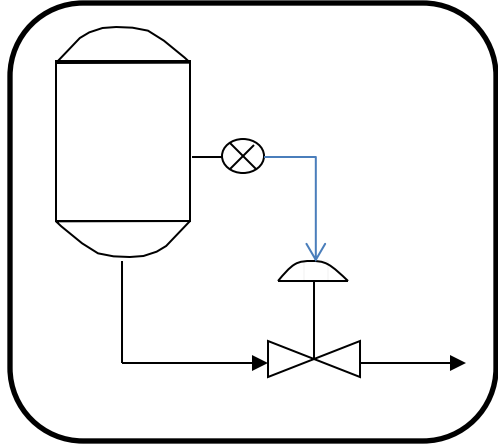


Controller Gain = 1

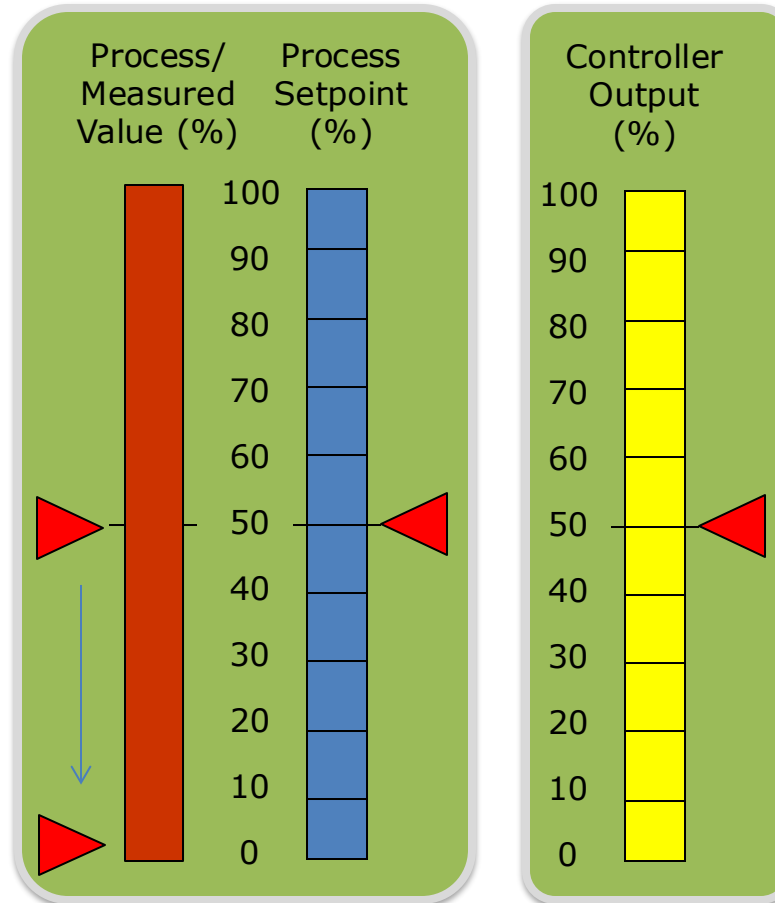
What is the Proportional Band?

$$\begin{aligned}\text{Pband} &= 100/\text{Gain} \\ &= 100/1 \\ &= 100\end{aligned}$$

Worked Example – Pband 2a



Controller Faceplate

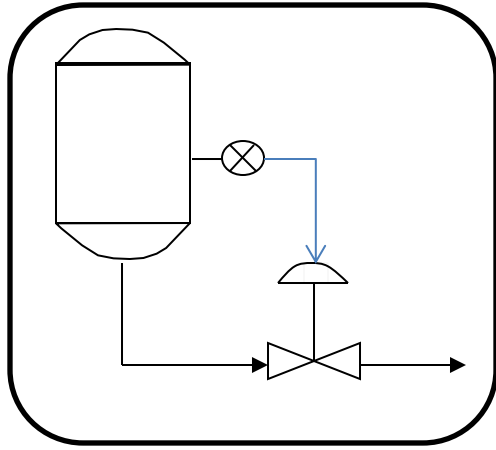


Controller Gain = 1

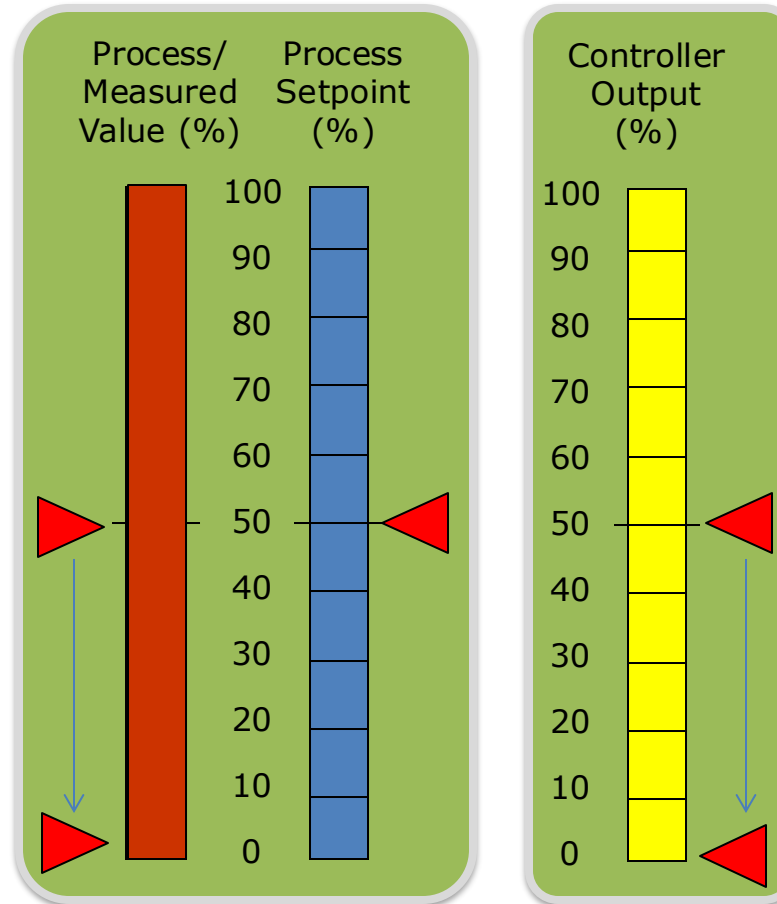
What should the controller do (output)?

What happens if level (measured value) drops to zero percent?

Worked Example – Pband 2b



Controller Faceplate



Controller Gain = 1

What should the controller do (output)?

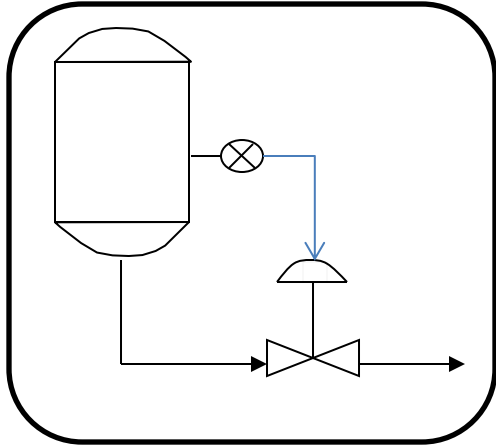
As PV increases, the OP must increase to return the level, this implies direct acting.

What happens if level (measured value) drops to zero percent?

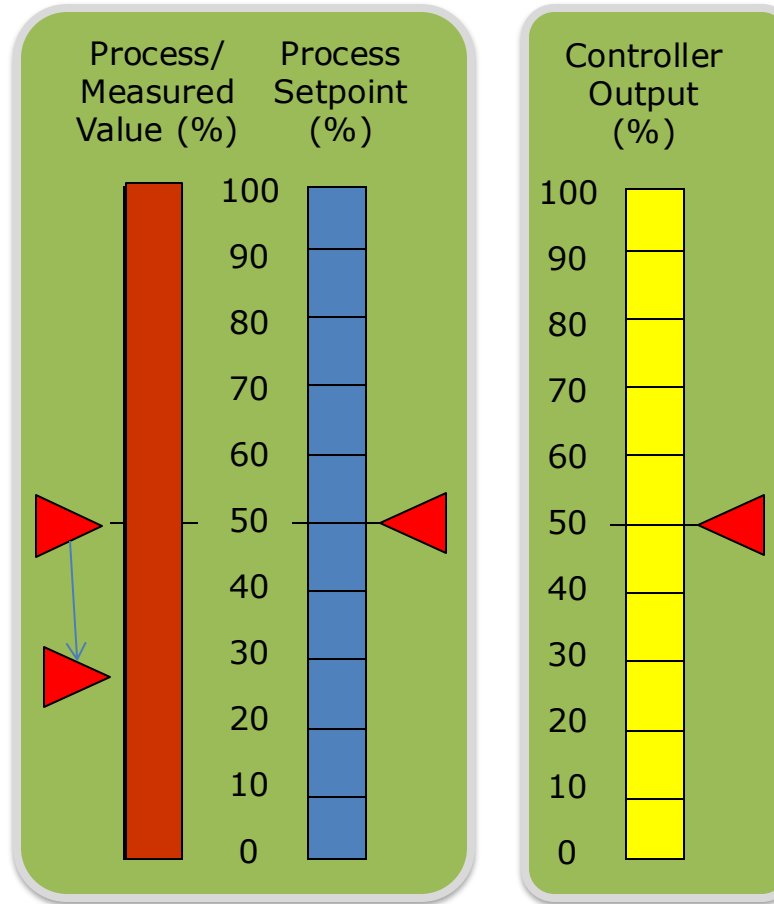
PV = 0, SP = 50%

- Error = PV – SPT = 50%
- Error % = 50%
- Change in output = error % x gain = -50% x 1 = (-50%)
- Final Op = 0%

Worked Example - Pband 3a



Controller Faceplate



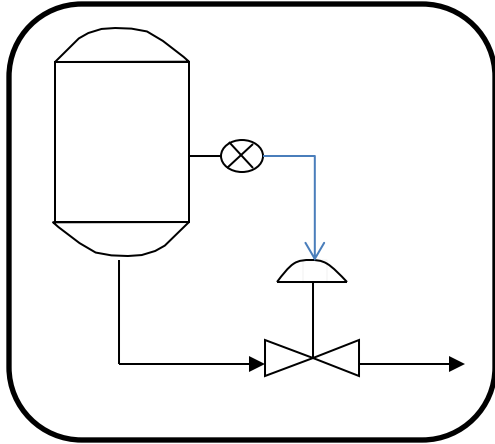
Controller Gain = 2

What is the Pband?

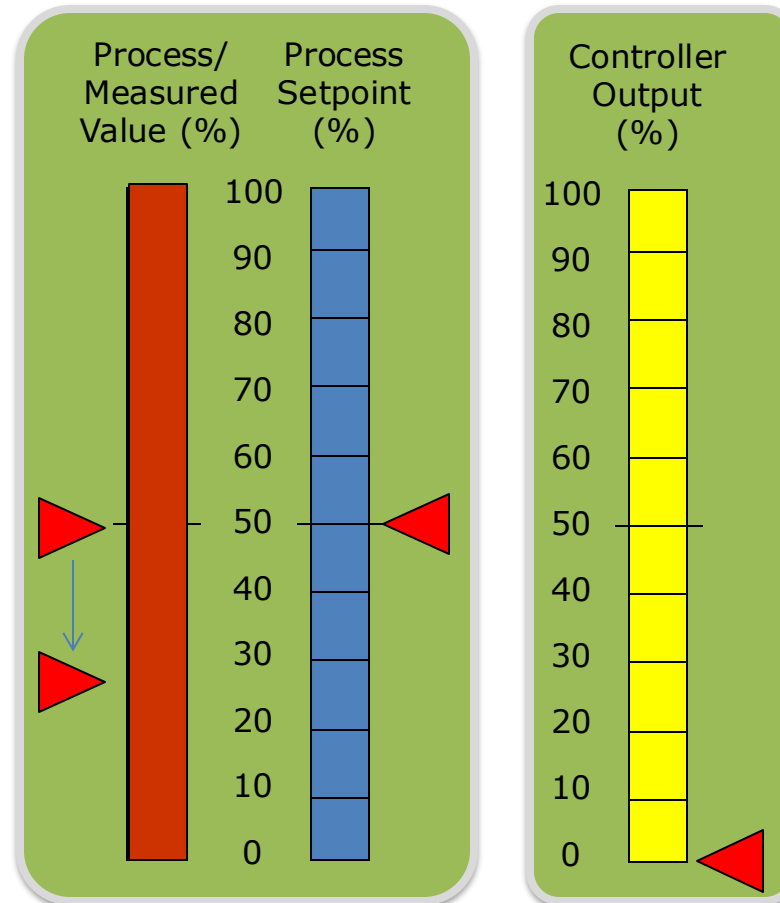
What happens to the output if level (measured value) **drops to 25%**?

Will there be any change to the controller output if level drops below 25%?

Worked Example - Pband 3b



Controller Faceplate



Controller Gain = 2

What is the Pband?

$$\text{Pband} = 100/G = 50\%$$

What happens to the output if level (measured value) **drops to 25%**?

- i) Error = 25%
- ii) Error % = 25 %
- iii) Change in OP = 50%
- iv) Final Op = 0%

Conclusion : Decreasing the Proportional band increases the sensitivity of the controller.

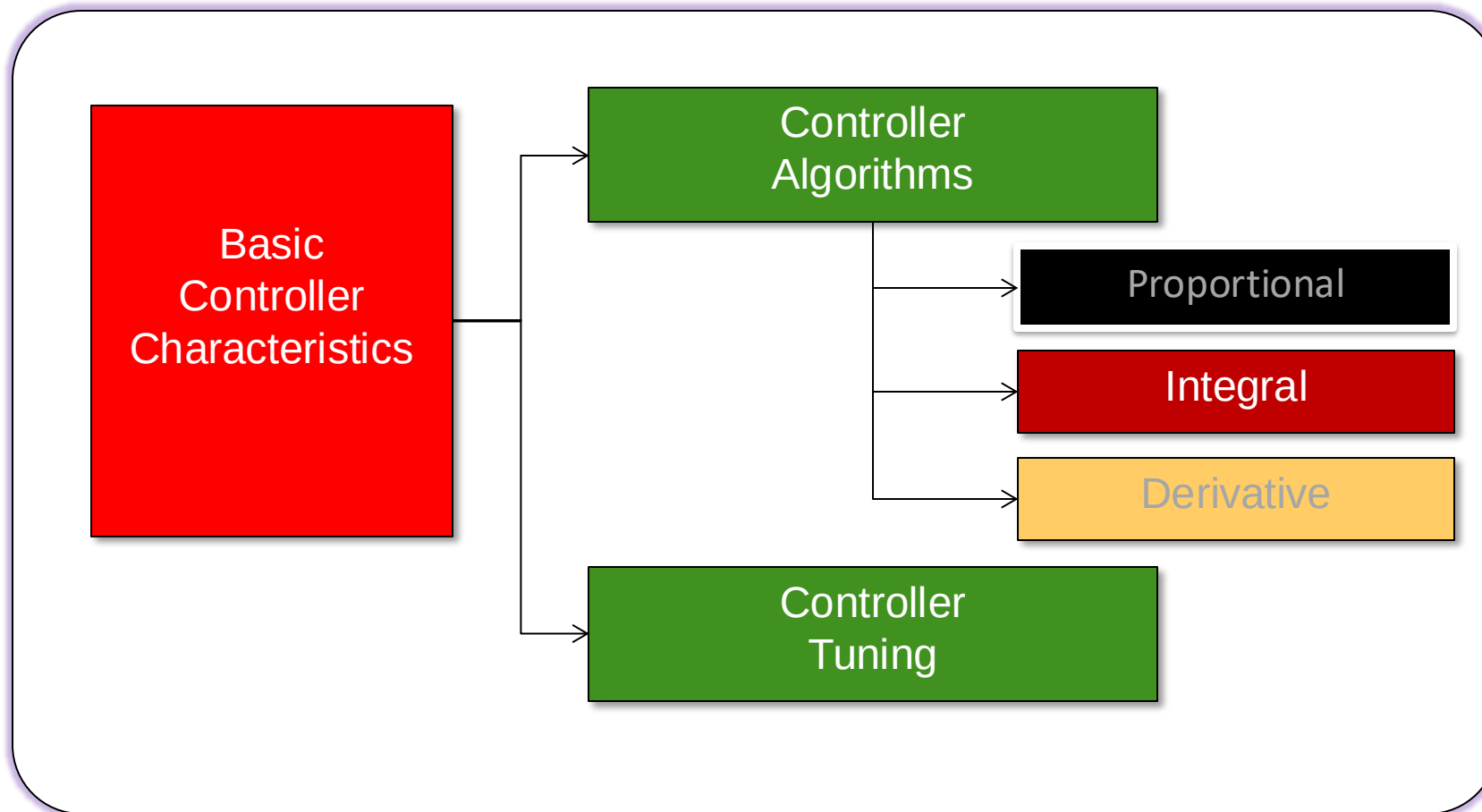
Conclusions for PBand

The relationship between Controller gain and Proportional Band is inverse

- Increasing the gain is the same as decreasing the proportional band

Controller Algorithms

Integral Action



Integral Action

For an integral only controller:



As long as there is an ERROR, I-Action will CONTINUOUSLY change the output of the controller.

Note that if ERROR is not removed, I-Action will eventually cause the controller to be SATURATED.

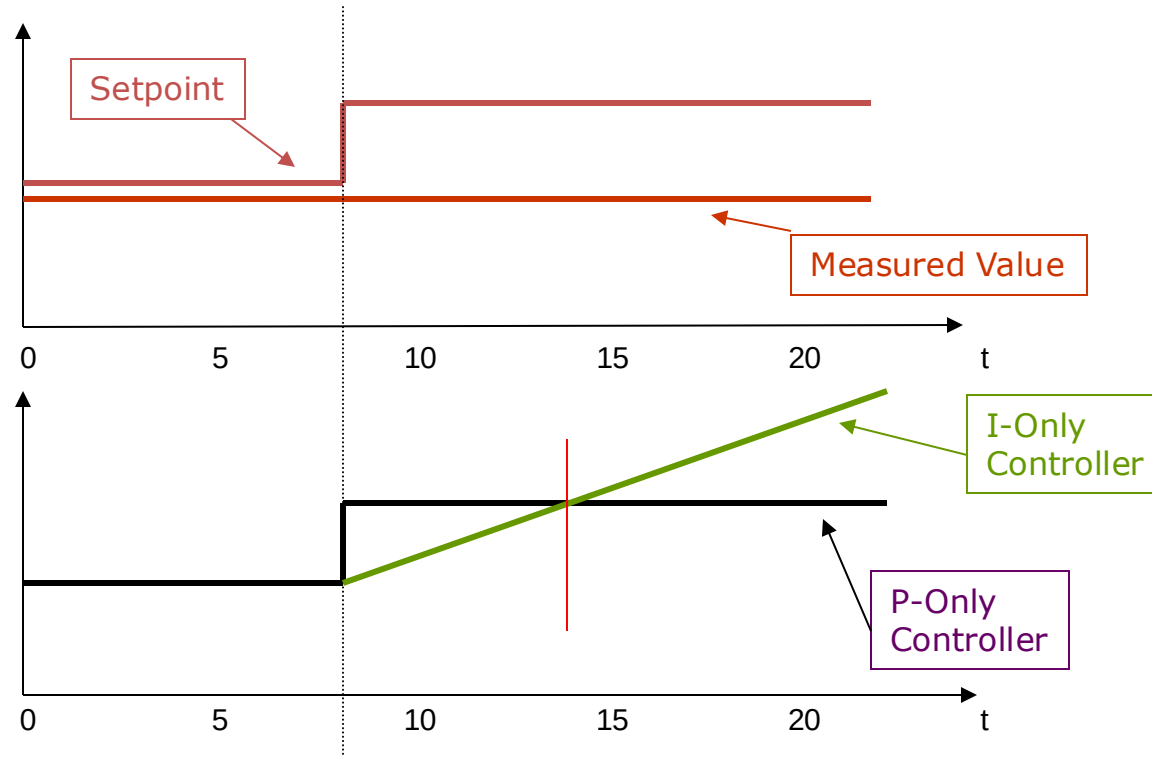
Size of Integral Action

The amount of integral action is indicated by the “integral action time”

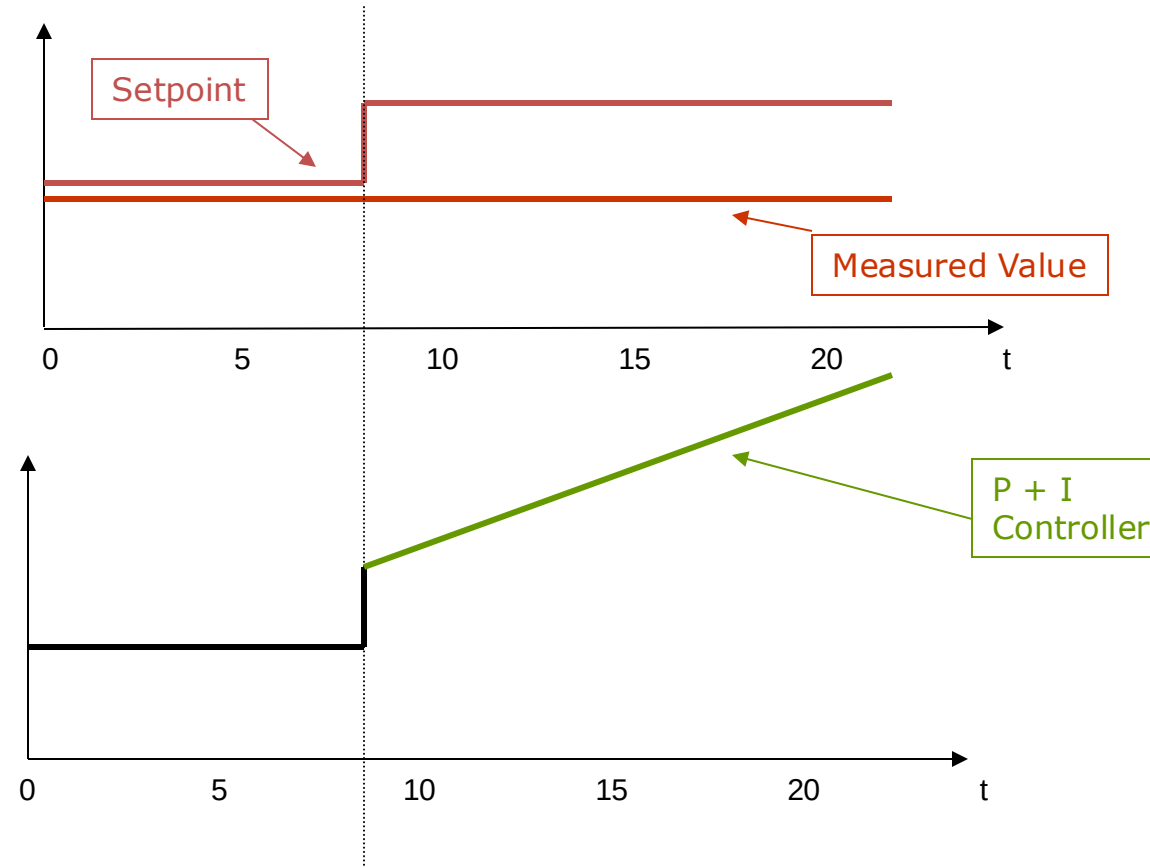
Integral Action Time

- Time needed to bring about the same change in the controller output as was initially changed by the P-Action

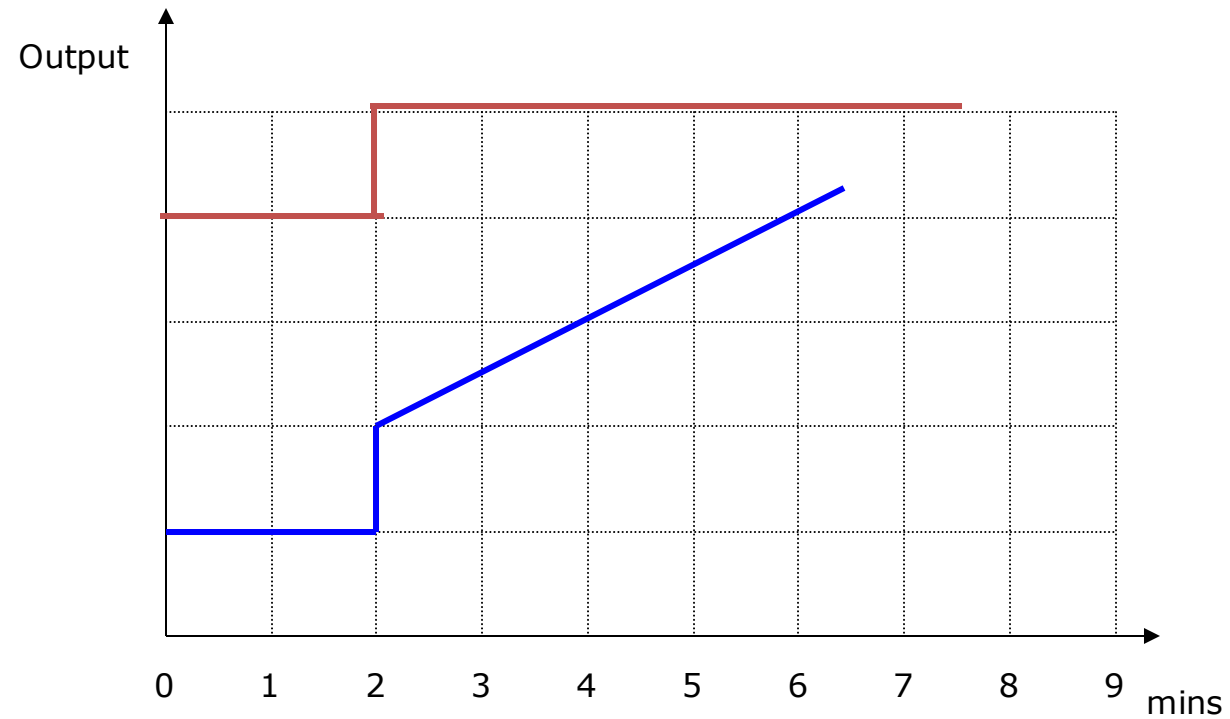
Integral Action Comparison



Combining P & I Action



Integral Time



Integral Time = ? minutes

Equations

For I-Only
Controller
(Form A)

$$\text{Change in Output} = \frac{\text{Gain}}{\text{Integral Time}} \int (\text{error}) dt$$

For P+I
Controller

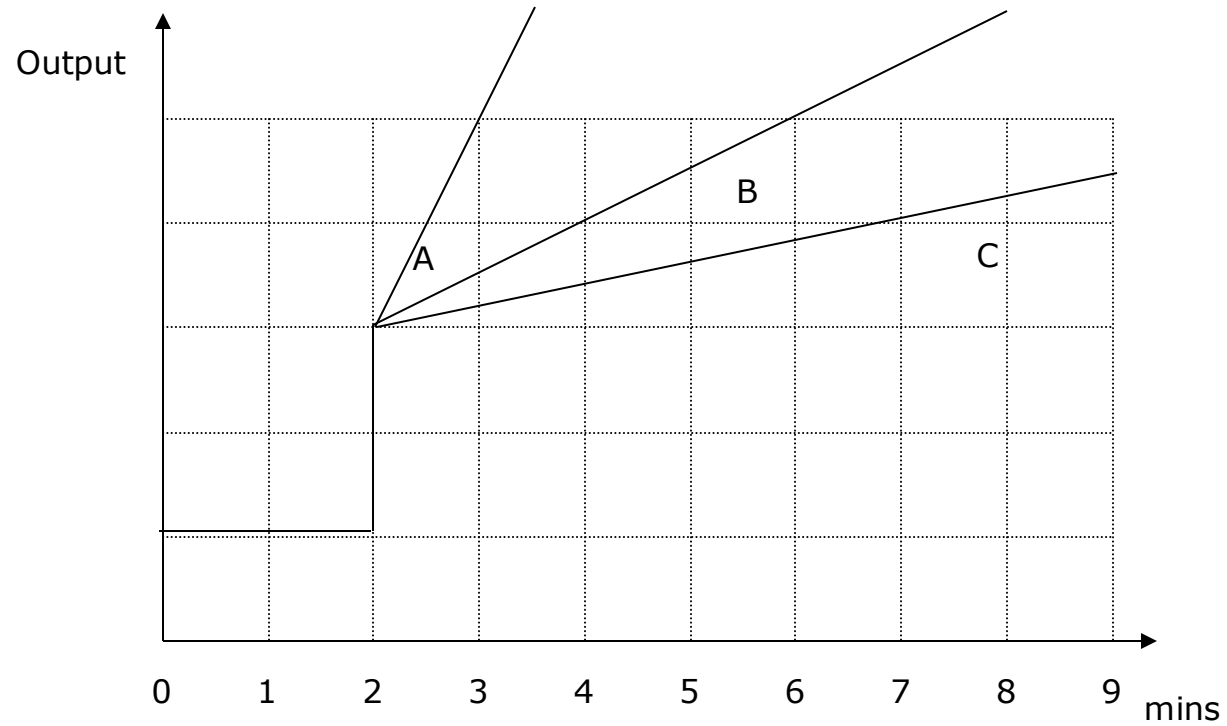
$$\text{Change in Output} = (\text{Gain} * \text{Error}) + \frac{\text{Gain}}{\text{Integral Time}} \int (\text{error}) dt$$

Impact of Change in Integral Time

As integral time is decreased:

- The time taken to repeat the output is reduced
- Therefore, the action by the integral term is stronger

Integral Time



Determine the Integral Times.

Alternative Units for Integral Time

– Reset Rate

Another way of defining integral time is by using the term “Reset Rate”

- The units for Reset Rate is “repeats per minute”
- This is exact opposite of Integral time and is related to Integral time by the following equation:

$$\text{Reset Rate} = \frac{1}{\text{Integral Time}}$$

Note:
Choice between Integral time and
Reset rate is primarily set by the
DCS vendor

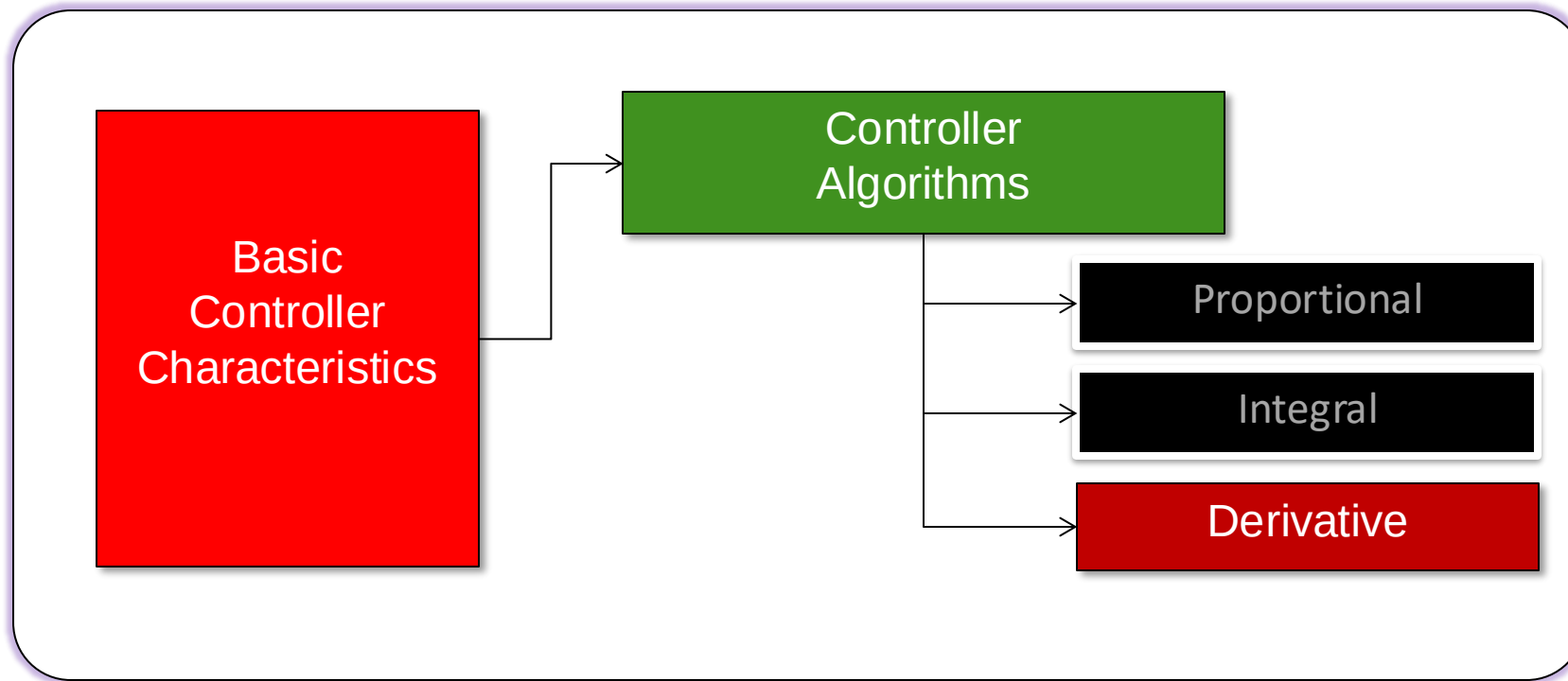
Equation (for Reset Rate)

For I-Only
Controller
(Form B)

$$\text{Change in Output} = \text{Gain} * \text{Reset Rate} \int (\text{error}) dt$$

Controller Algorithms

Derivative Action

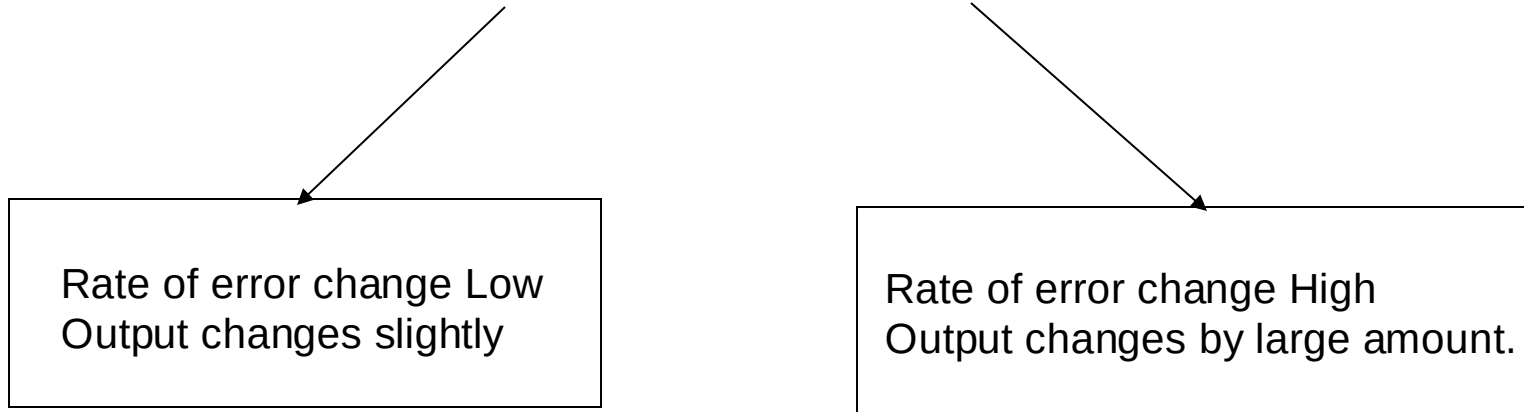


Usage of Derivative Action

- Derivative action becomes necessary as the size of the processing equipment increases.
 - This results in the increase process lag & deadtime.
- As a result, for large processes, it is not good enough to respond to an error when it is already present.
- The purpose of derivative action is to anticipate process errors and take corrective action in advance.
 - Derivative action speeds up the controller action by taking the action now, rather than later when the full error is already present

Derivative Action

With derivative action, the output of a controller is proportional to the rate of change of error.



Derivative Units

- The unit for derivative setting is the derivative time (T_d)
- Derivative time
 - Upon a change in error, the size of the correction will be equal to the size of correction the proportional action would have made T_d time later.
- Therefore, the larger the T_d , the bigger the size of correction.

Lingo : D-time, D-action, Pre-act

Equation

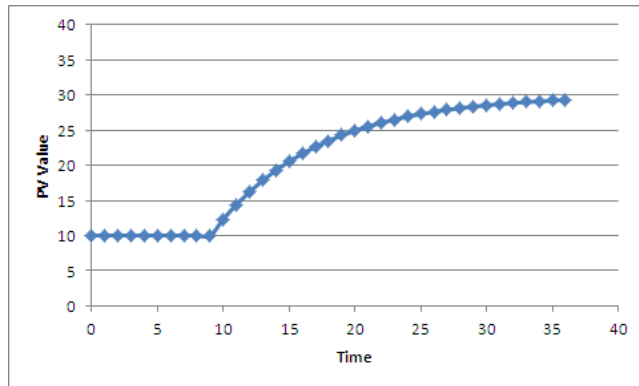
$$\text{Derivative Term} = \text{Gain (K)} * \text{Derivative Time, Td} * \frac{d(\text{error})}{dt}$$

Rate of Change
Of error

Derivative Action

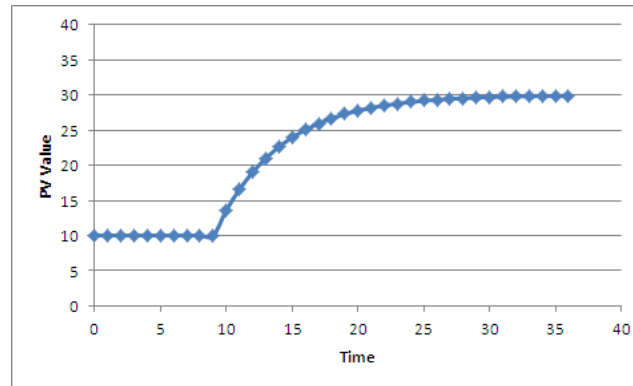
Process Disturbance Scenarios

Small



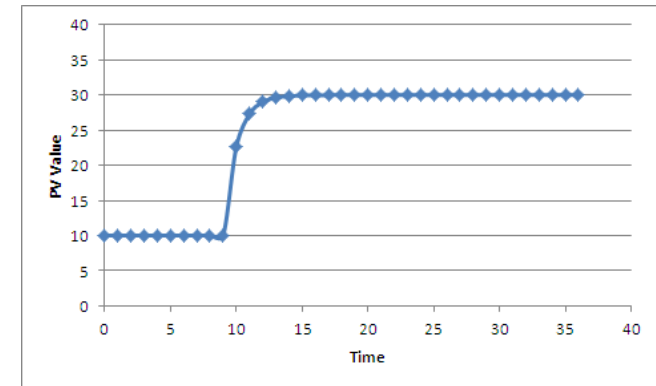
$$\frac{dE}{dt}_{\max} = 2.1$$

Medium



$$\frac{dE}{dt}_{\max} = 3.6$$

Significant



$$\frac{dE}{dt}_{\max} = 12.6$$

- Implications
 - Small amounts of derivative can counter the effect of fast process disturbances as the dE/dt (or dPV/dt) term increases in proportion to the size of the disturbance and can have an immediate stabilising effect on output
 - Good for processes that suffer from upstream (and uncontrollable) disturbances

Derivative Action Notes

- Long Derivative Time
 - Large amount of derivative action
- Note that since the term is dependent on the rate of change, it can make controllers very fast acting
 - It is therefore important to NOT utilise this function where process noise is present as this will be amplified and will lead to instability
- When a setpoint change occurs, the rate of change of error explodes as the time taken to carry out the setpoint change is almost zero.
 - To prevent this, most control vendors inhibit the derivative term on a setpoint change

The PID Equation

$$\text{Output} = K * E + K * \frac{1}{T_i} \int E dt + K * T_d * \frac{dE}{dt}$$



*Proportional
Action*

(P-Term)

*Integral
Action*

(I-Term)

*Derivative
Action*

(D-Term)

Note:

- i) *This is the default equation in most textbooks*
- ii) *There is an issue with the implementation of this equation in DCS and this is associated with the D-Action. This will be covered in a later section.*

Other Typical Design Notes

Basic controller requirements:

1. Bumpless transfer capability
 - When controller is put to Manual, require the setpoint to track the current PV
 - This is to prevent a bump when controller put back to Auto due to the error that exists if setpoint not tracking the PV
2. Bad input/output (PV failure, loss of connection to valve)
 - Normally PV is configured to remain as last good value
 - Similarly for control valve, valve will remain at its last good position if lost connection to controller

Retrieval Practice

1. The P-Action or Proportional Action of the controller is represented by one of the following statements:
 - a) The change in output of the controller is based on what happens to the Output (against the PV)
 - b) The change in output of the controller based on what happens to the PV (against the setpoint)

2. For the P-Action, what is the parameter that is first calculated to compute the output:
 - a) The Process Variable
 - b) The Setpoint
 - c) The difference between the Setpoint and Process Variable
 - d) The difference between the Output and Setpoint

3. How would you ideally set the P-action
 - a) Set it such that the Setpoint will not change
 - b) Such that the output change is sufficient to bring the PV back to setpoint
 - c) Set it such that the output change occurs by a fixed amount
 - d) Set it such that the error continues to increase

Retrieval Practice (2)

4. The I-Action or Integral Action of the controller is necessary because
 - a) It is not possible for the Setpoint to change without an Integral action
 - b) It is not possible for the PV to change without an integral action
 - c) It is unlikely for the gain to exactly bring the PV to the Setpoint
 - d) It is unlikely for the gain to exactly bring the Setpoint to the PV

 5. What is the difference between the tuning parameters for P-Action and I-action
 - a) Decreasing Gain increases aggressiveness, Increasing Integral action decreases aggressiveness
 - b) Decreasing PB increases aggressiveness, Increasing integral time increases aggressiveness
 - c) Decreasing Gain increases aggressiveness, increasing Integral time increases aggressiveness
 - d) Decreasing PB increases aggressiveness, increasing integral time decreases aggressiveness

 6. For a particular heater, there are oftentimes that the feed is disturbed daily by 10%. Every two days, the feed is disturbed by 20% and every three days by 30%. This causes large variations in the outlet temperature. You have been tasked to improve the PI outlet temperature controller. The equipment vendor tells you to apply D-action. Would you agree to use D and why?
-
-

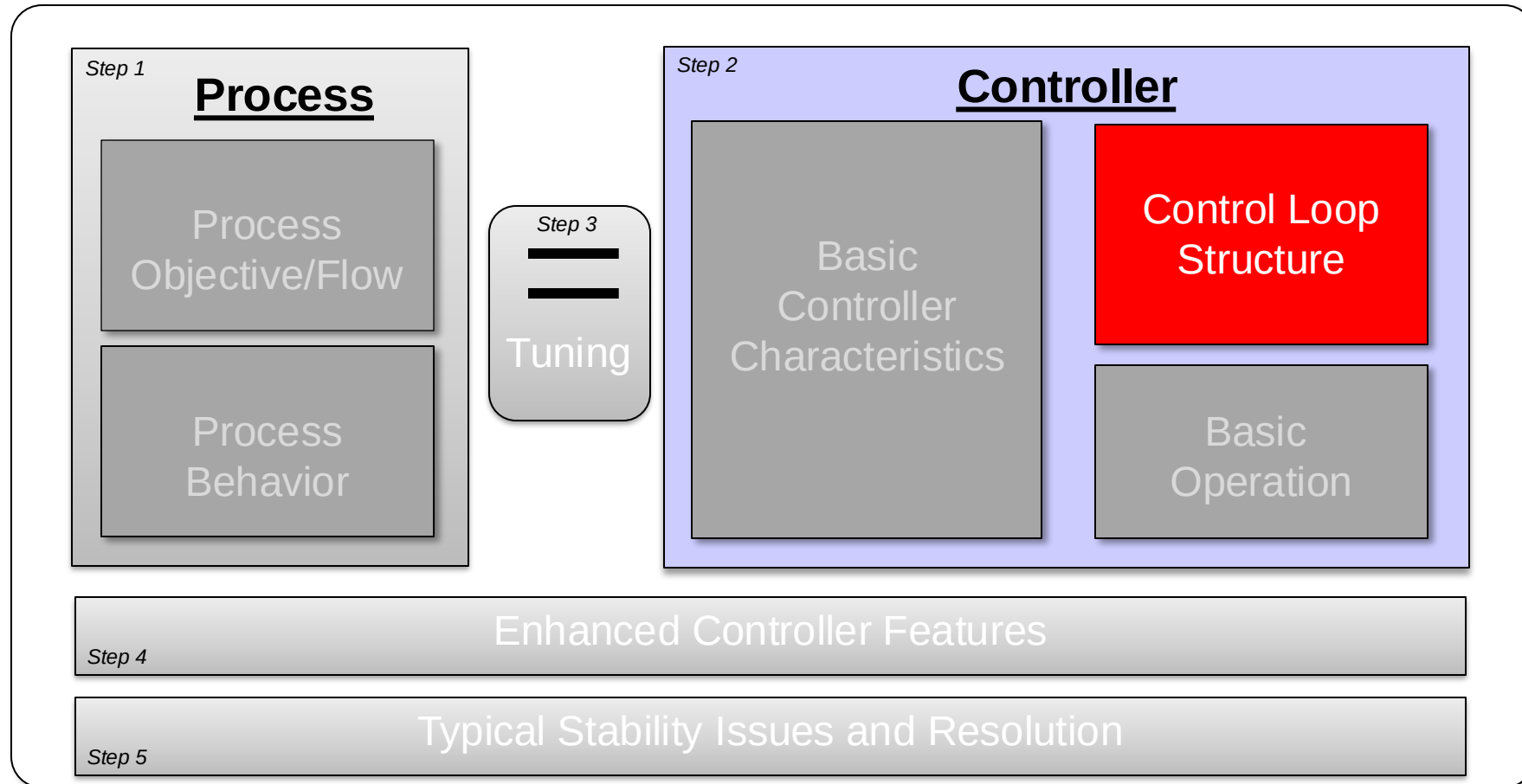
Key Take Aways

- Proportional Action
 - Change in output a error
 - $\text{Controller Output Change} = \text{Gain} \times \text{Error}$
 - $\text{PB} = 100/\text{Gain}$
- Integral Action
 - Used to eliminate offset
 - Rate of change of output a error
 - $\text{Reset Rate} = 1/\text{Integral Time}$
- Derivative Action
 - Used to speed up controller
 - Output of controller a rate of change of error.
- Ensure proper units for T_i & T_d
 - Mins/repeat or Seconds/repeat

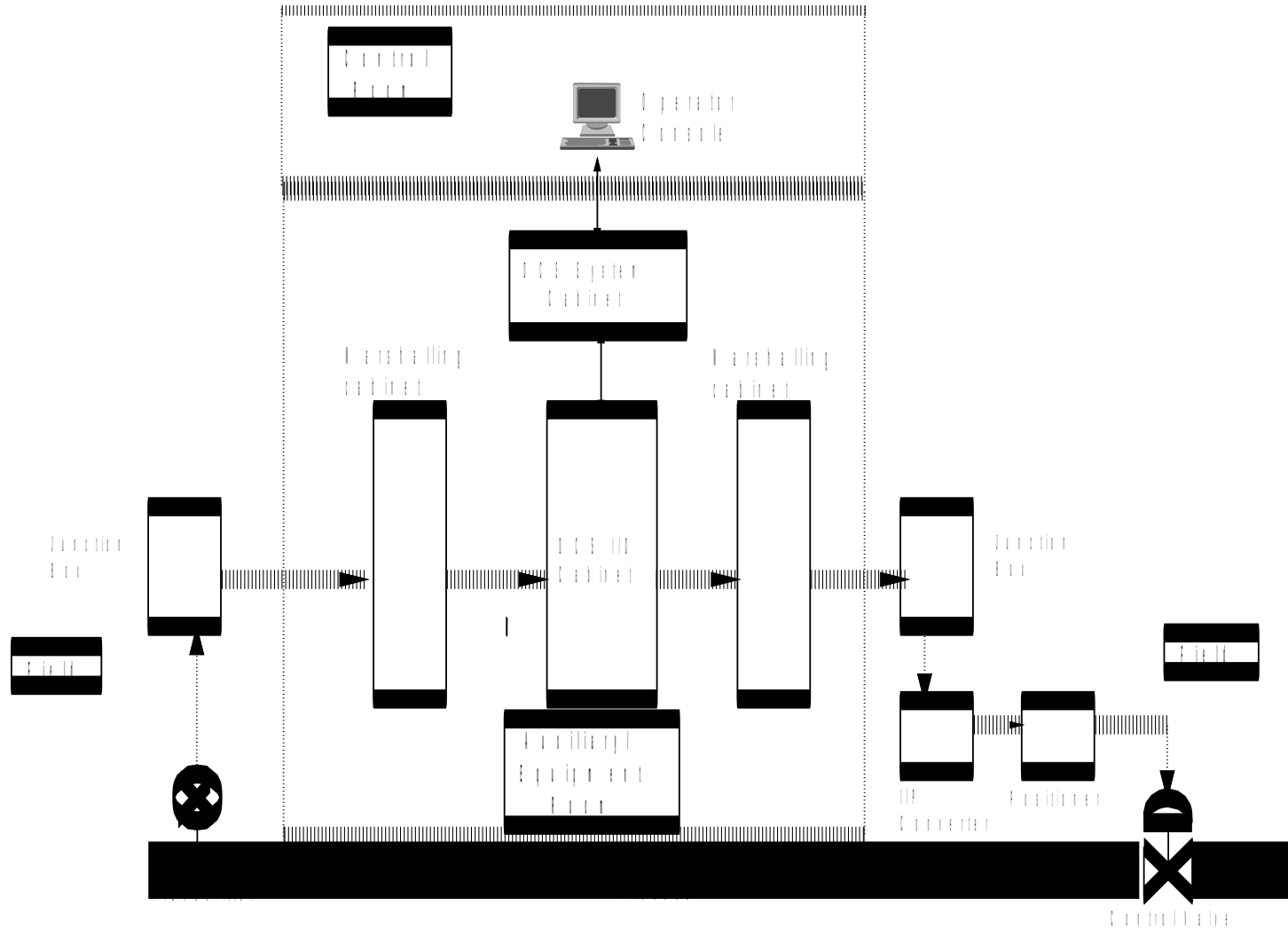
Base Layer Control

Step 2 - Understanding the Controller

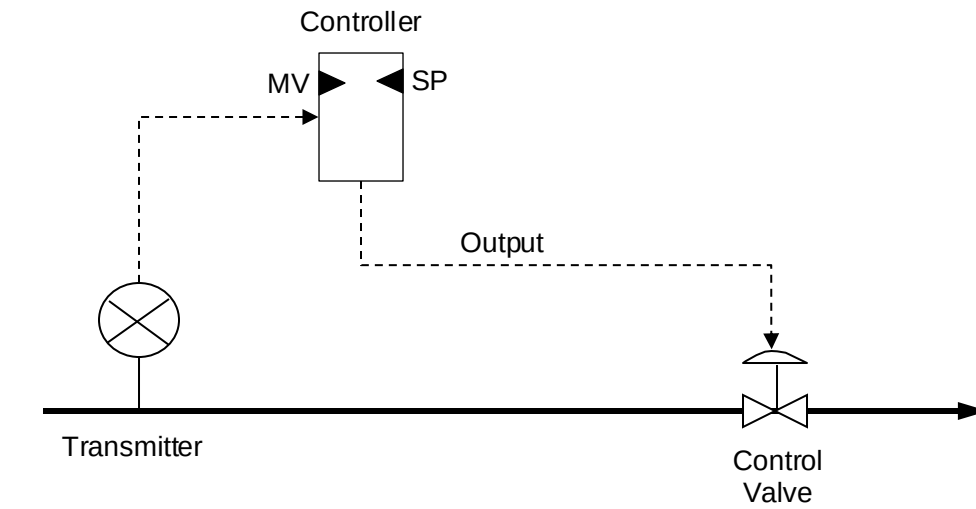
Control Loop Structure



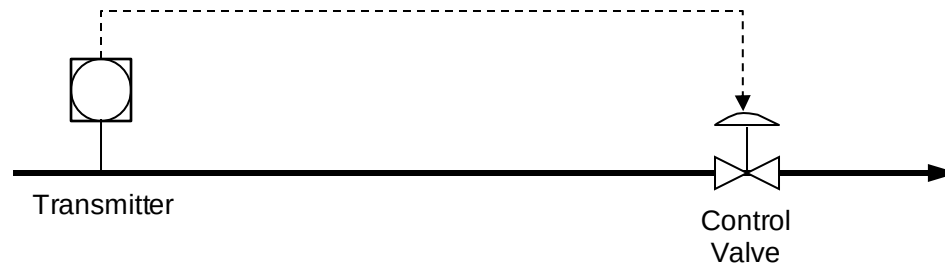
Typical Instrumentation Configuration for a single Control Loop



Simplified Structures



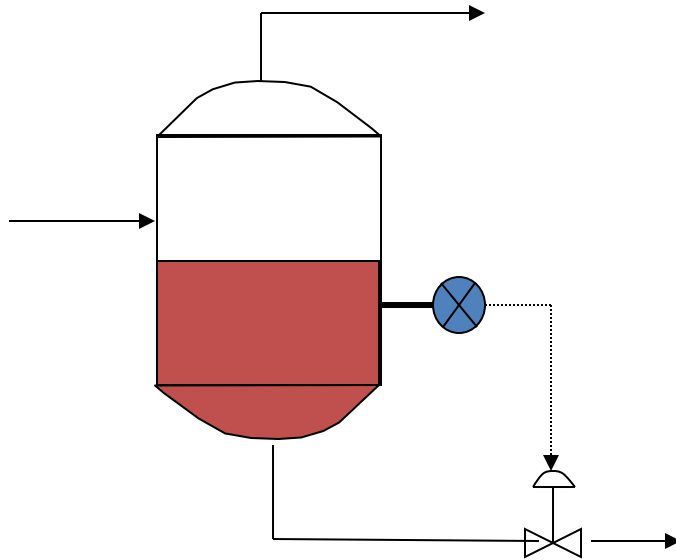
Simplified
Diagram



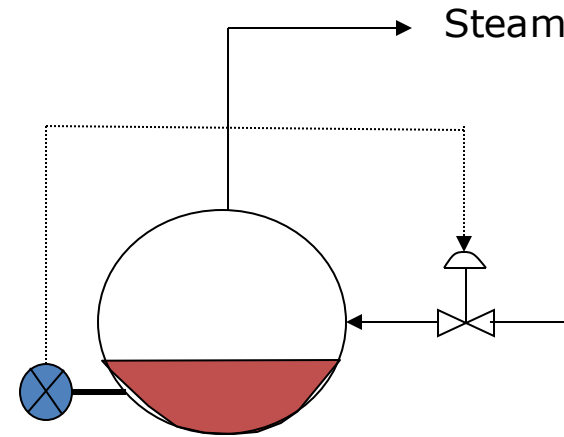
Typical for
P&IDs

Flow Control Loop

Level Control



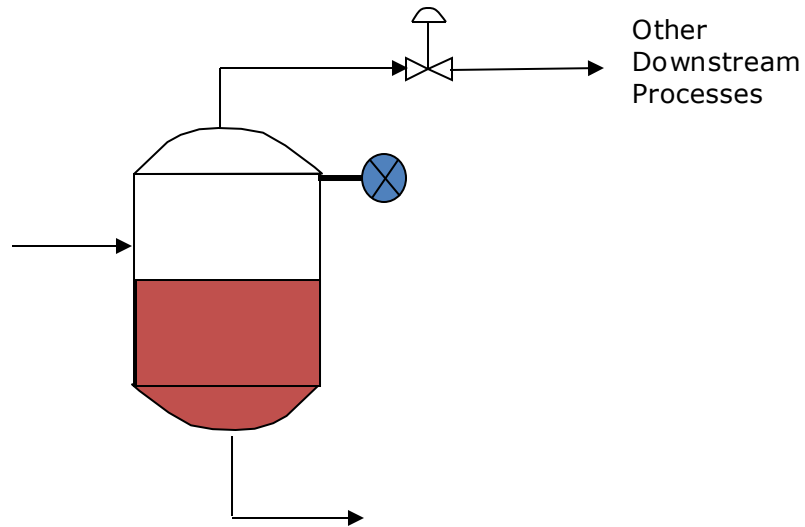
Level on Vessel/Column Outlet
Eg : Most liquid filled equipment



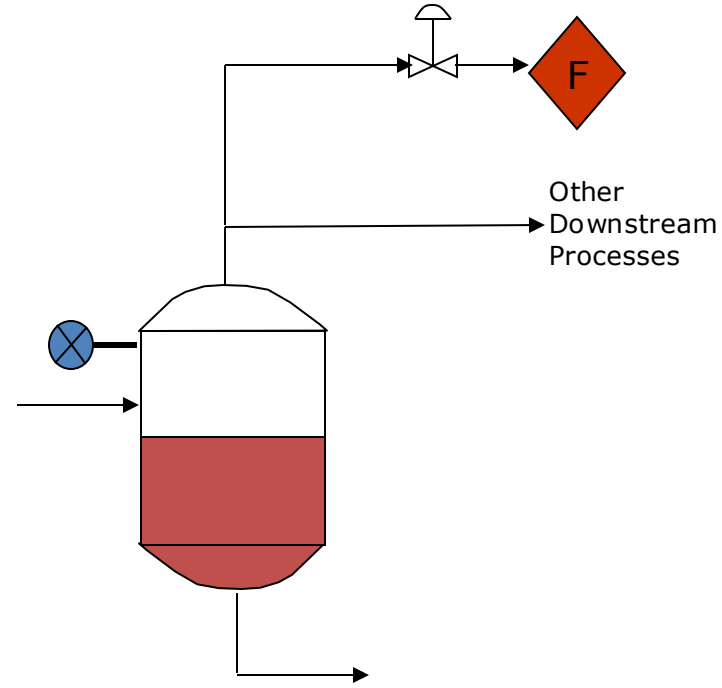
Level on Vessel Inlet
Eg : Boiler Drum

Pressure Control

How would you design a single controller for these processes?



Normal Load Control



Abnormal Load Control

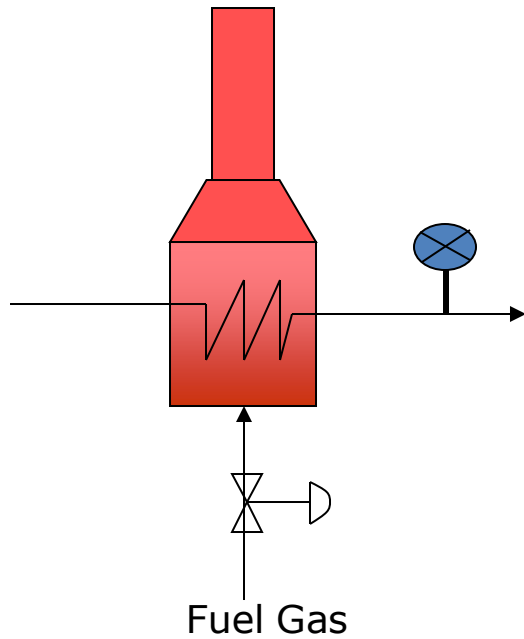
Legend



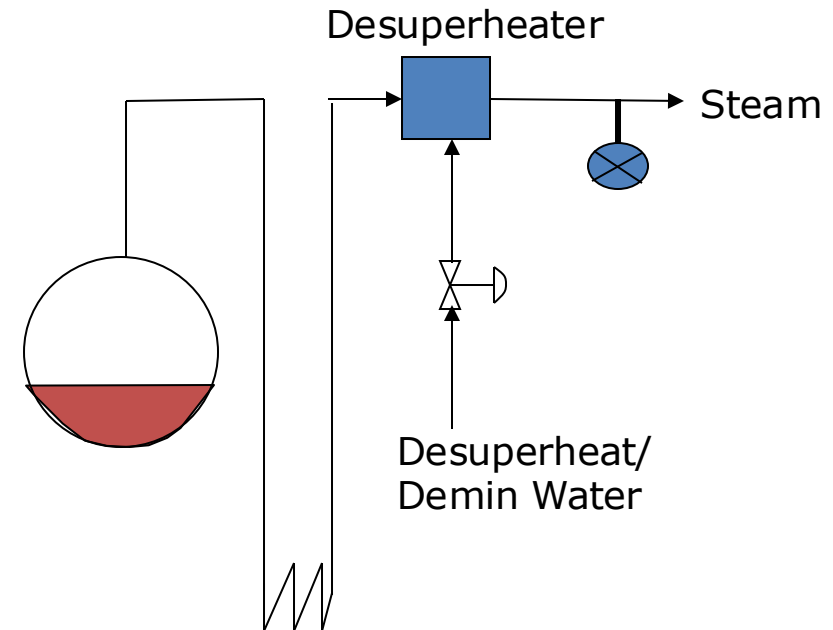
: Flare Header

Temperature Control

How would you design a single controller for these processes?

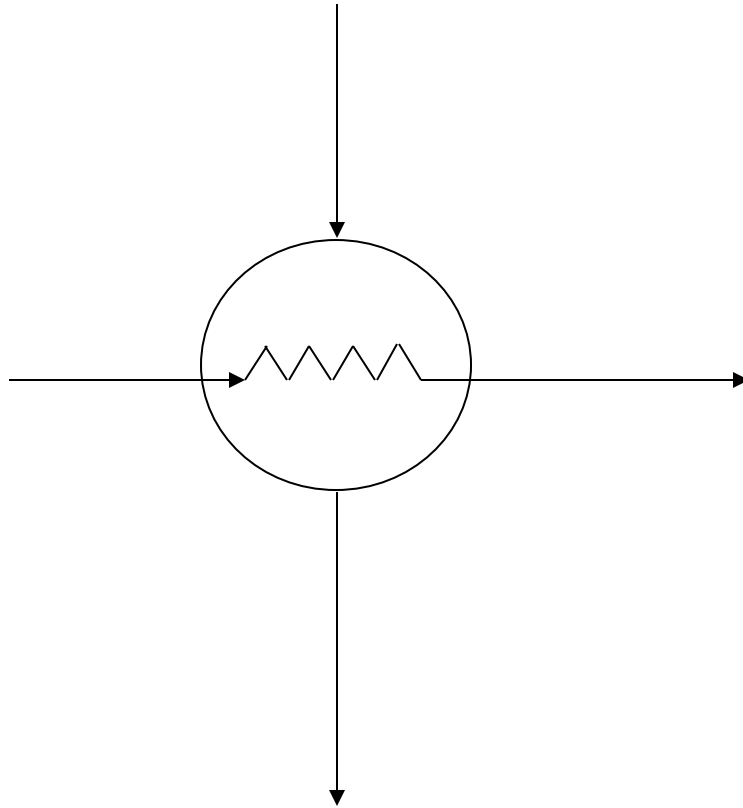


Controlling a Heating medium



Controlling a Cooling Medium

Exercise



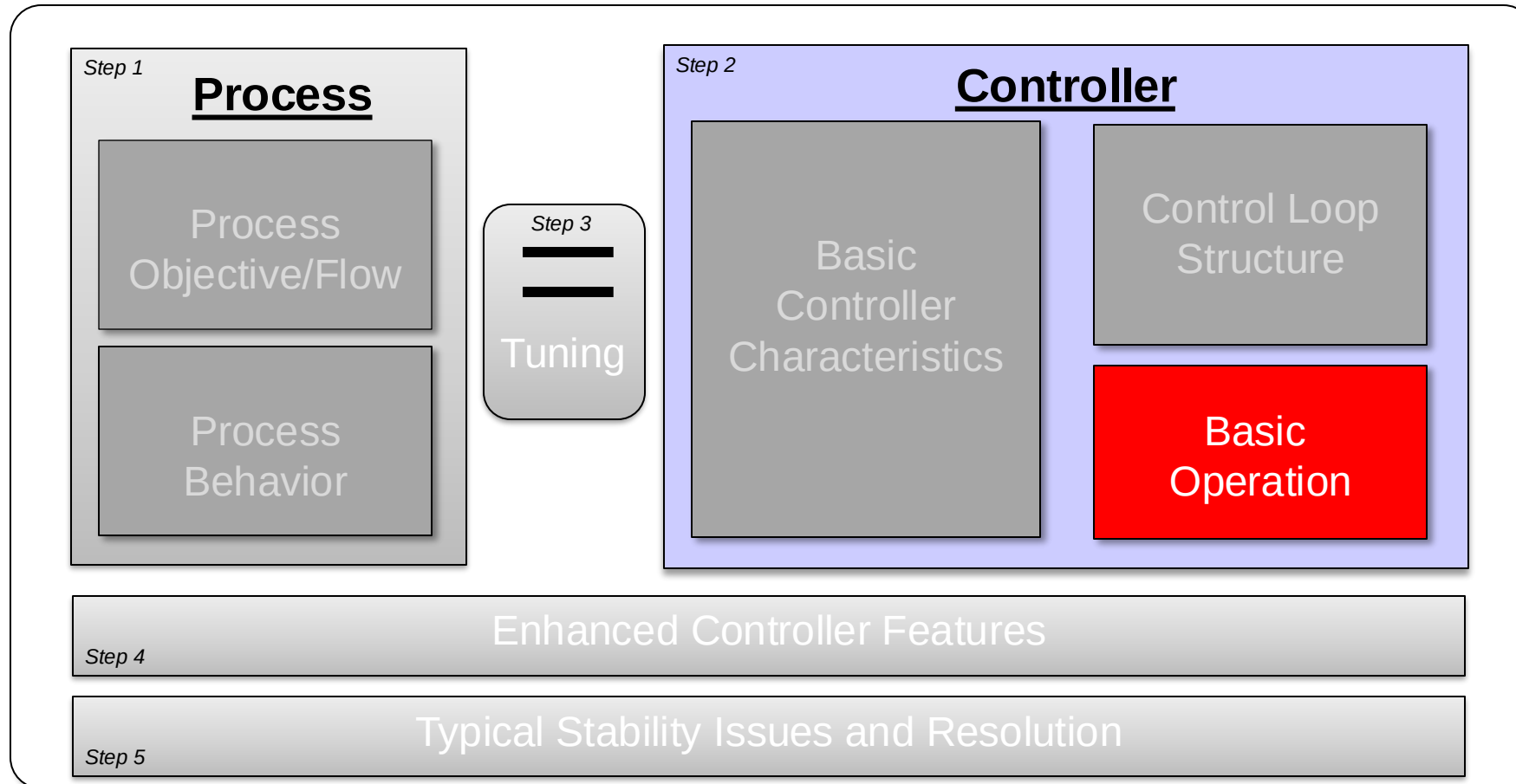
Objective

To design a single loop temperature controller. Reboiler uses steam that will fully condense. The process fluid is in the tubes.

Base Layer Control

Step 2 - Understanding the Controller

Basic Operation



Basic Operation

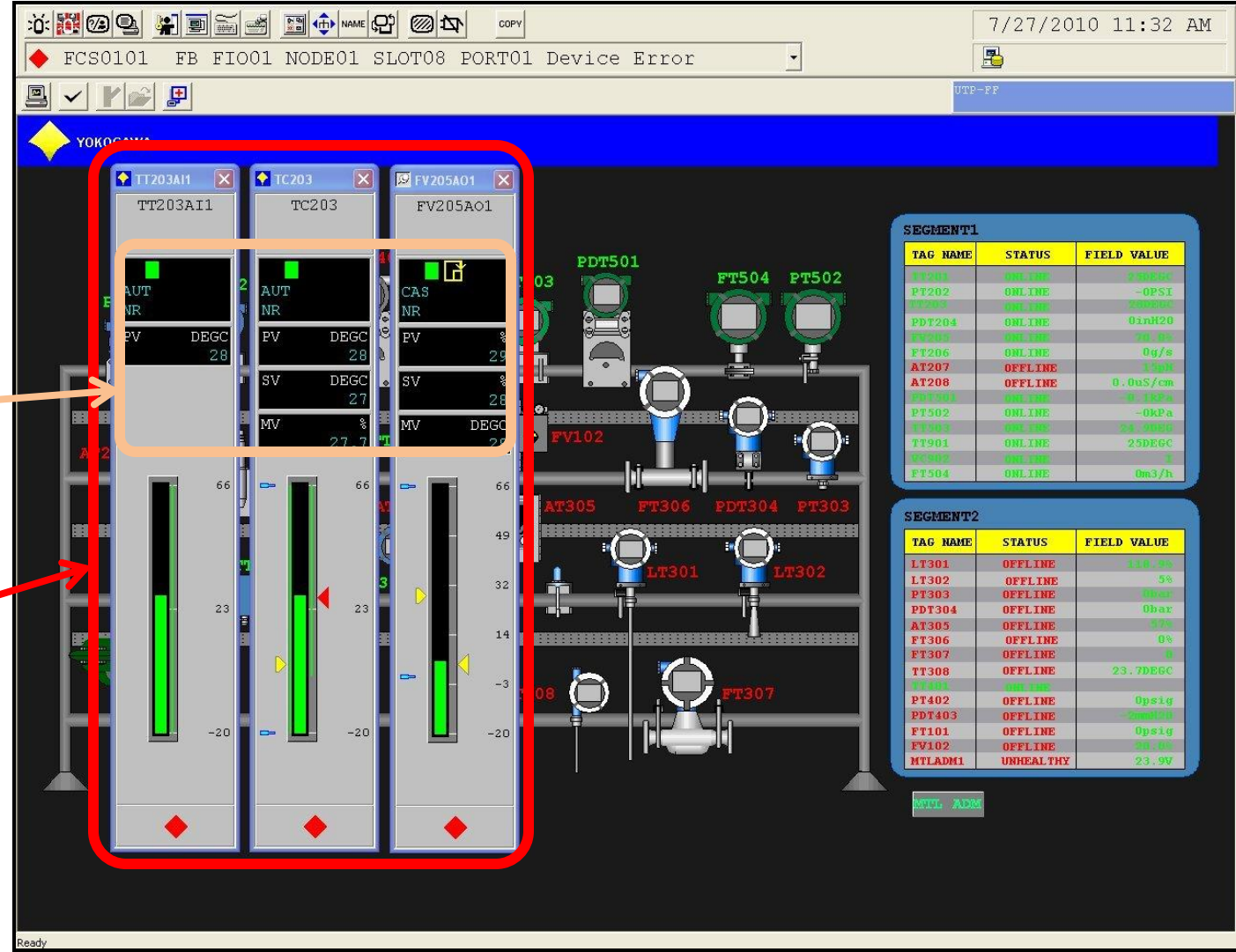
- Operation of a controller is by the following
 - In Automatic
 - Operator can manipulate the setpoint
 - Operator can switch the controller from Automatic to Manual
 - In Manual
 - Operator can manipulate the output
 - Operator can switch the controller from Manual to Automatic

Sample Controller Faceplate: Yokogawa Centum

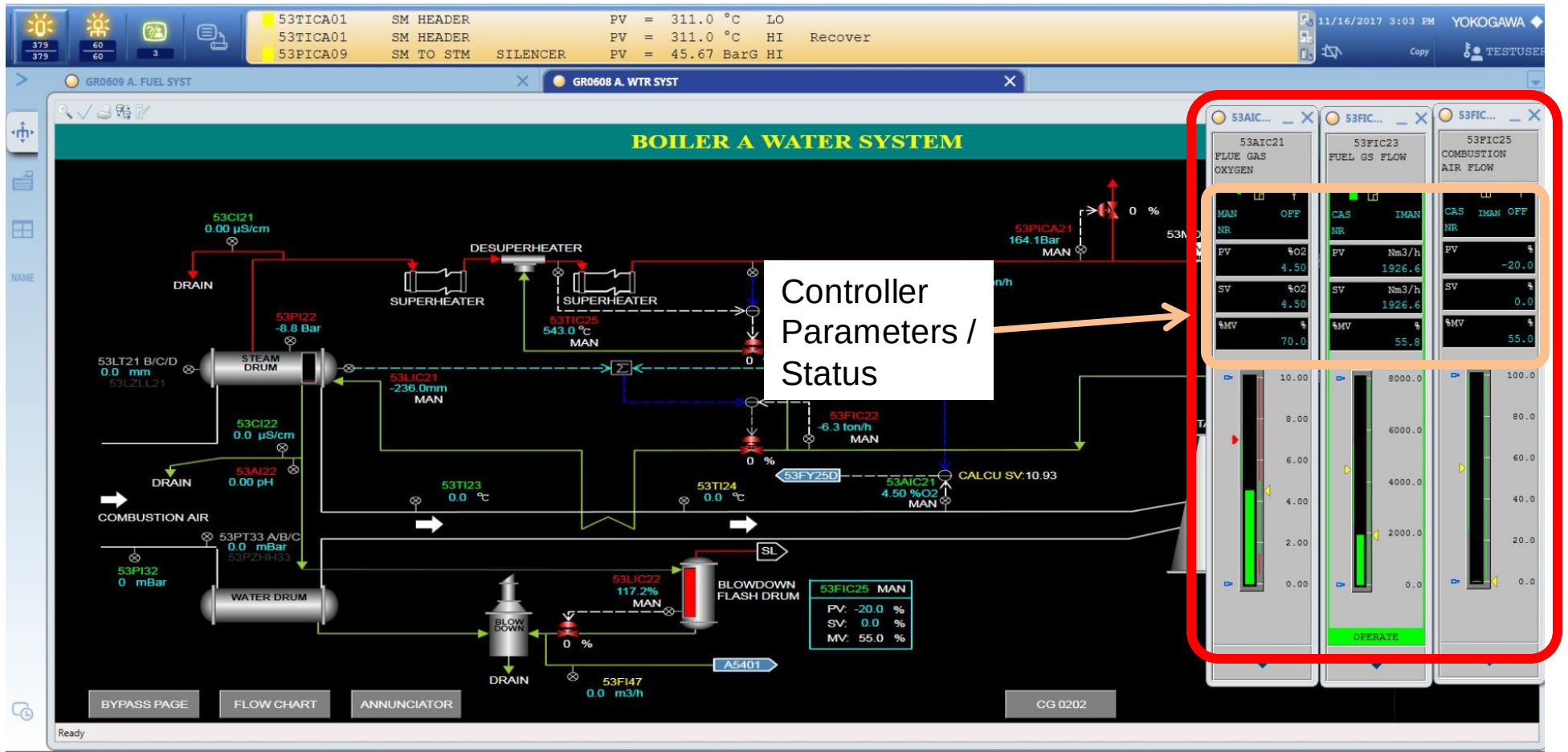


Controller
Parameters /
Status

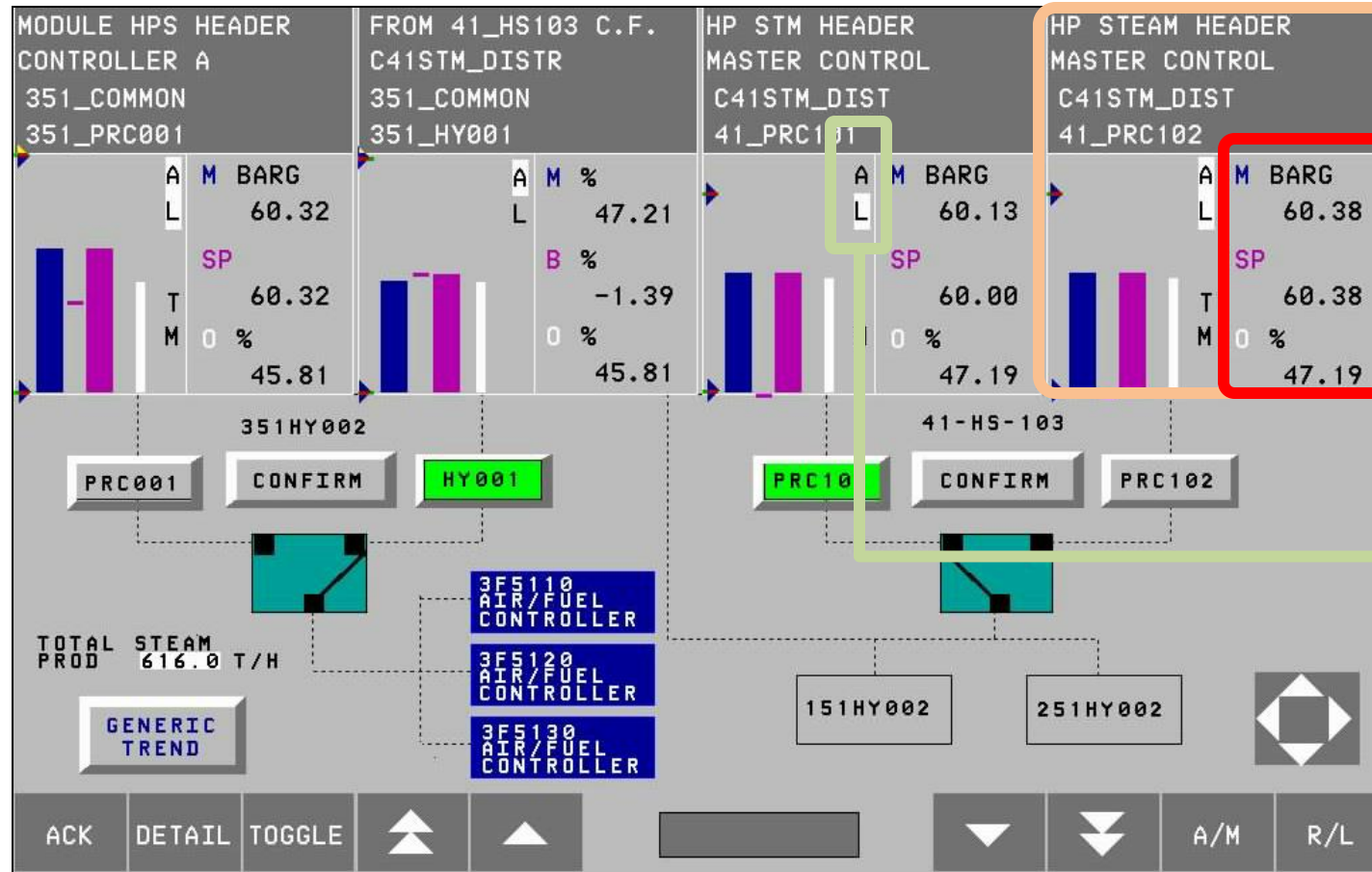
Cascade
Controllers



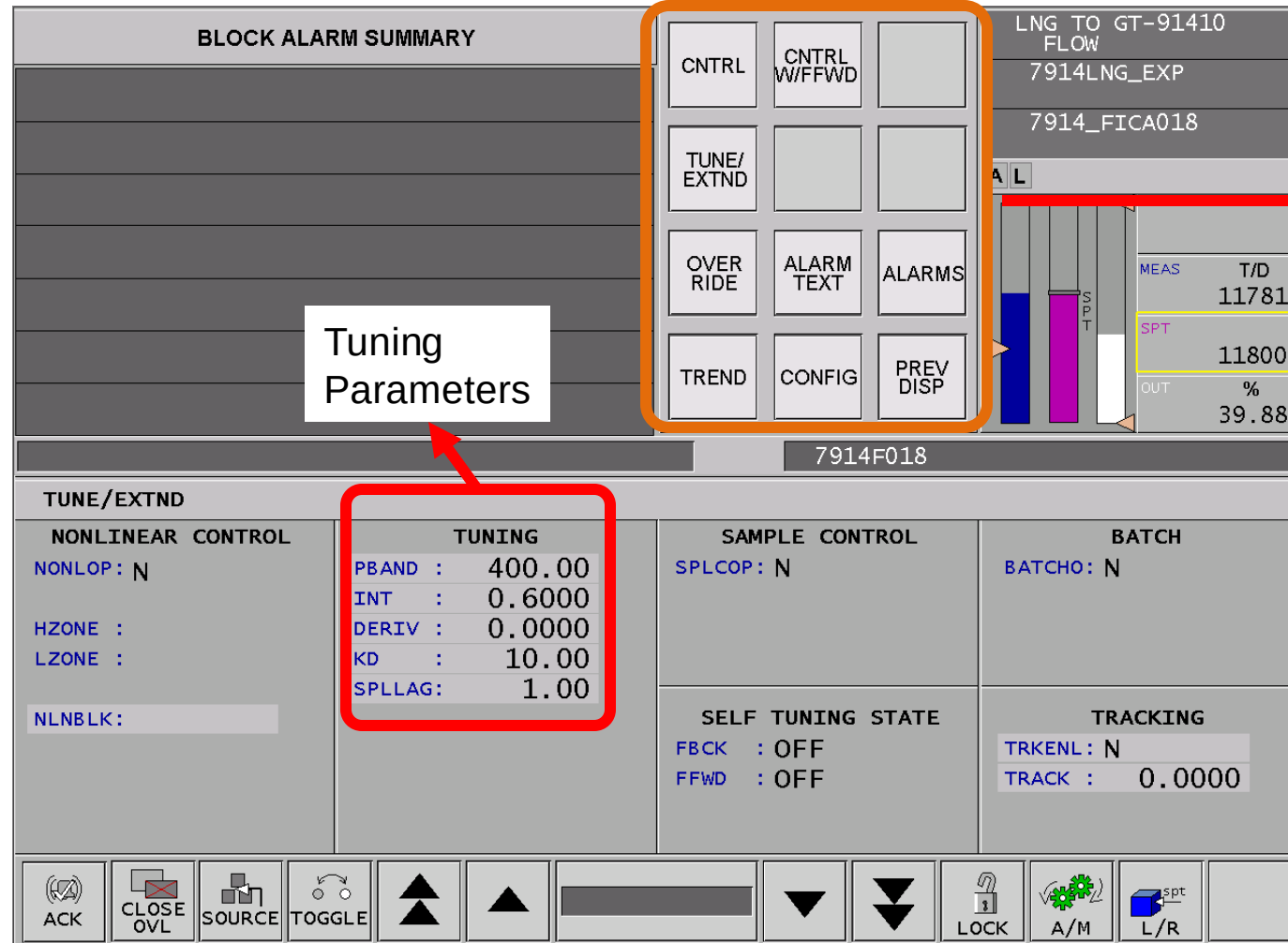
Sample Controller Faceplate: Yokogawa Centum VP



Sample Controller Faceplate Foxboro Controller Display



Sample Faceplate Foxboro Controller Configuration



Configuration Options

Sample Controller Faceplate Honeywell Experion

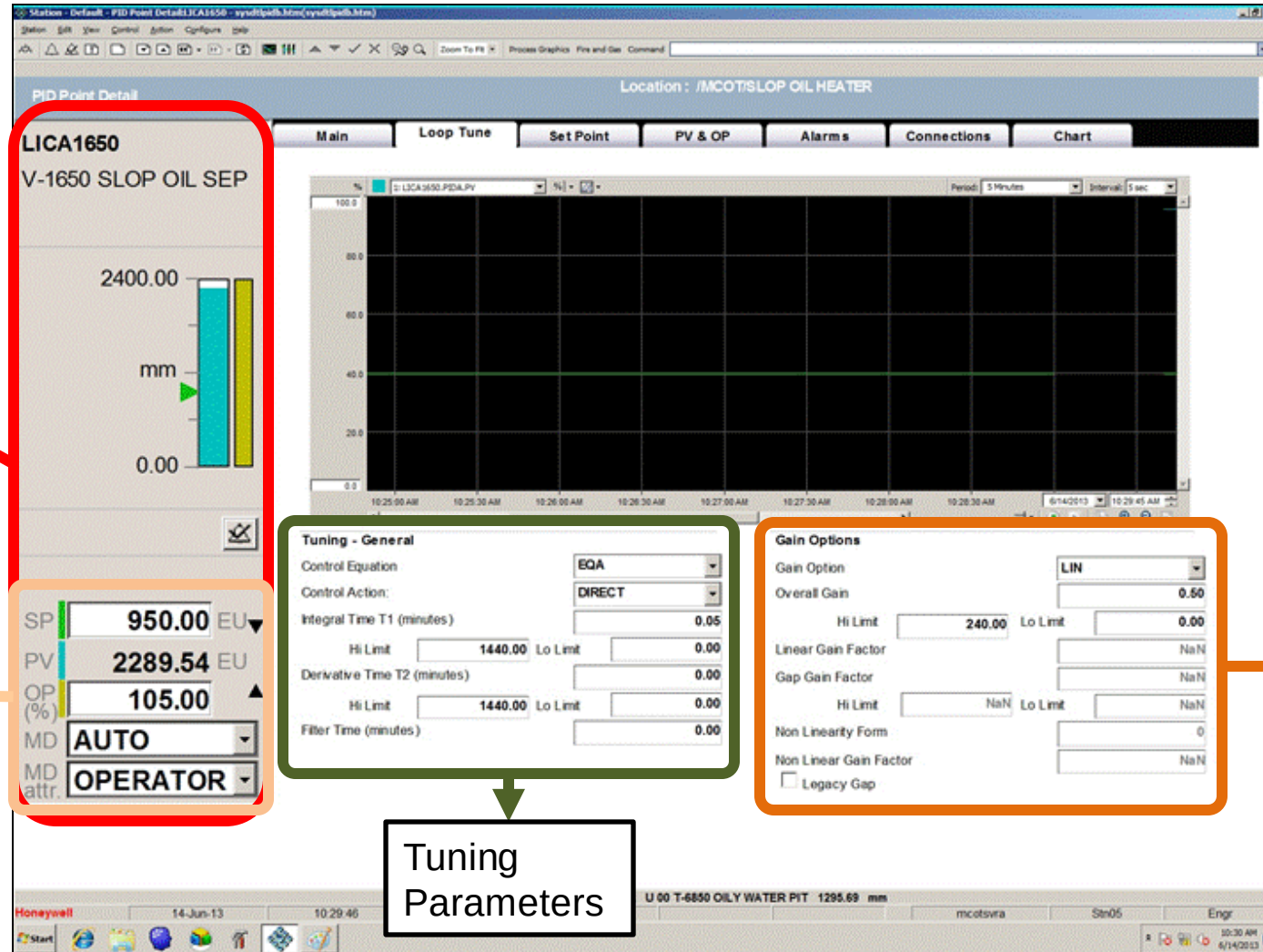
Honeywell

Honeywell
Controller
template

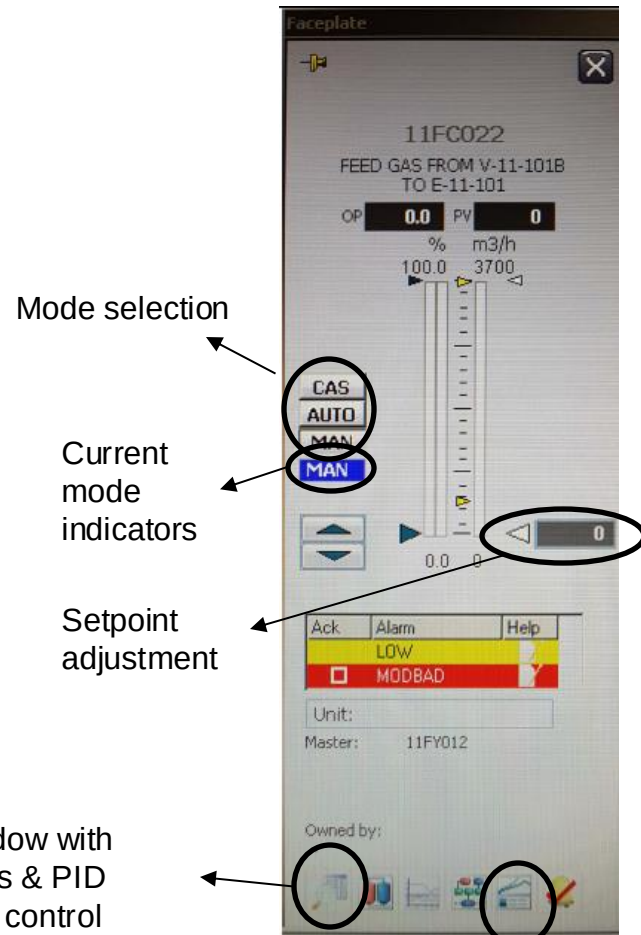
Controller
Status

Tuning
Parameters

Part of
Configuration
Parameters



Sample Controller Faceplate: Emerson

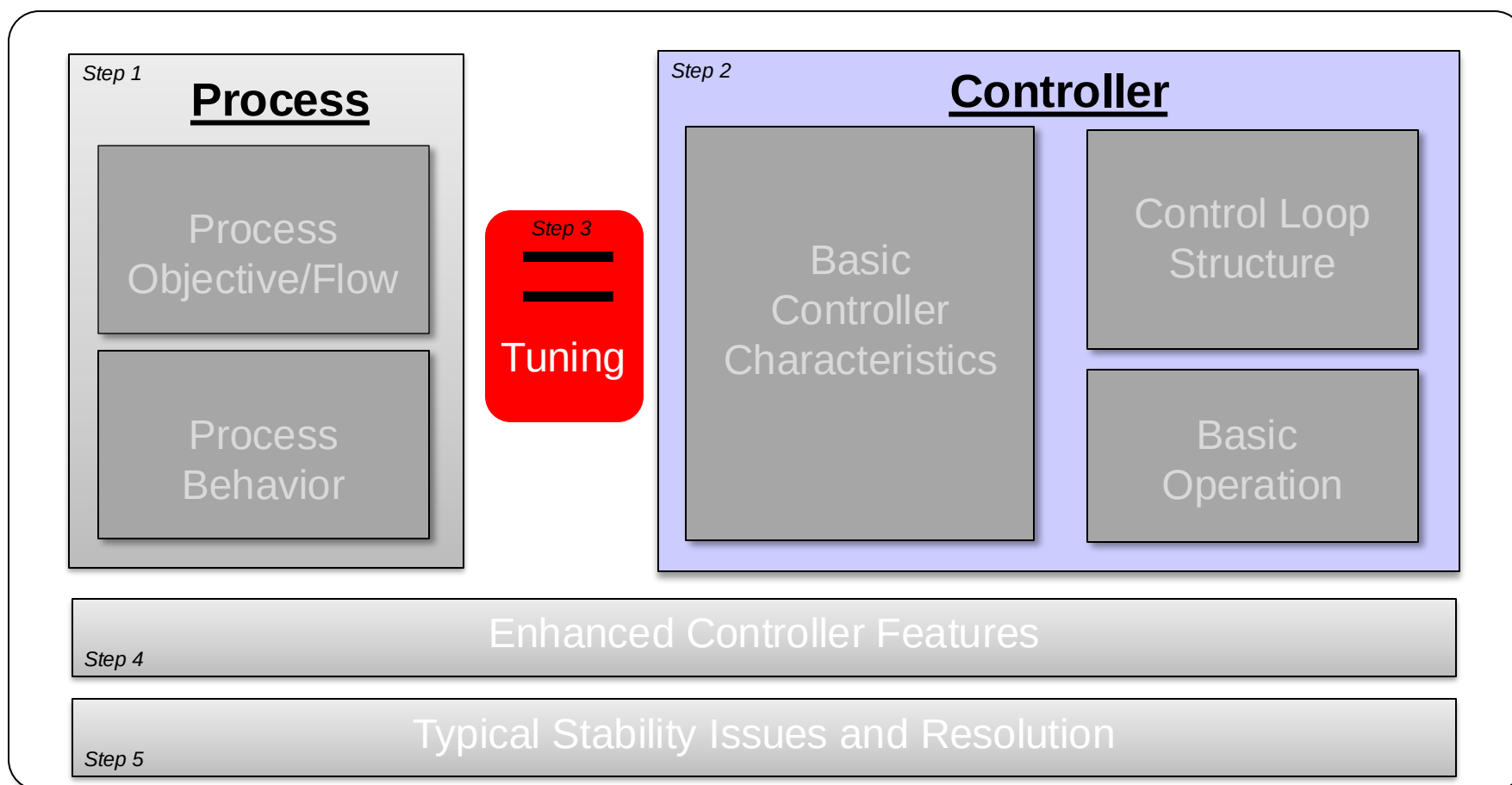


Trends

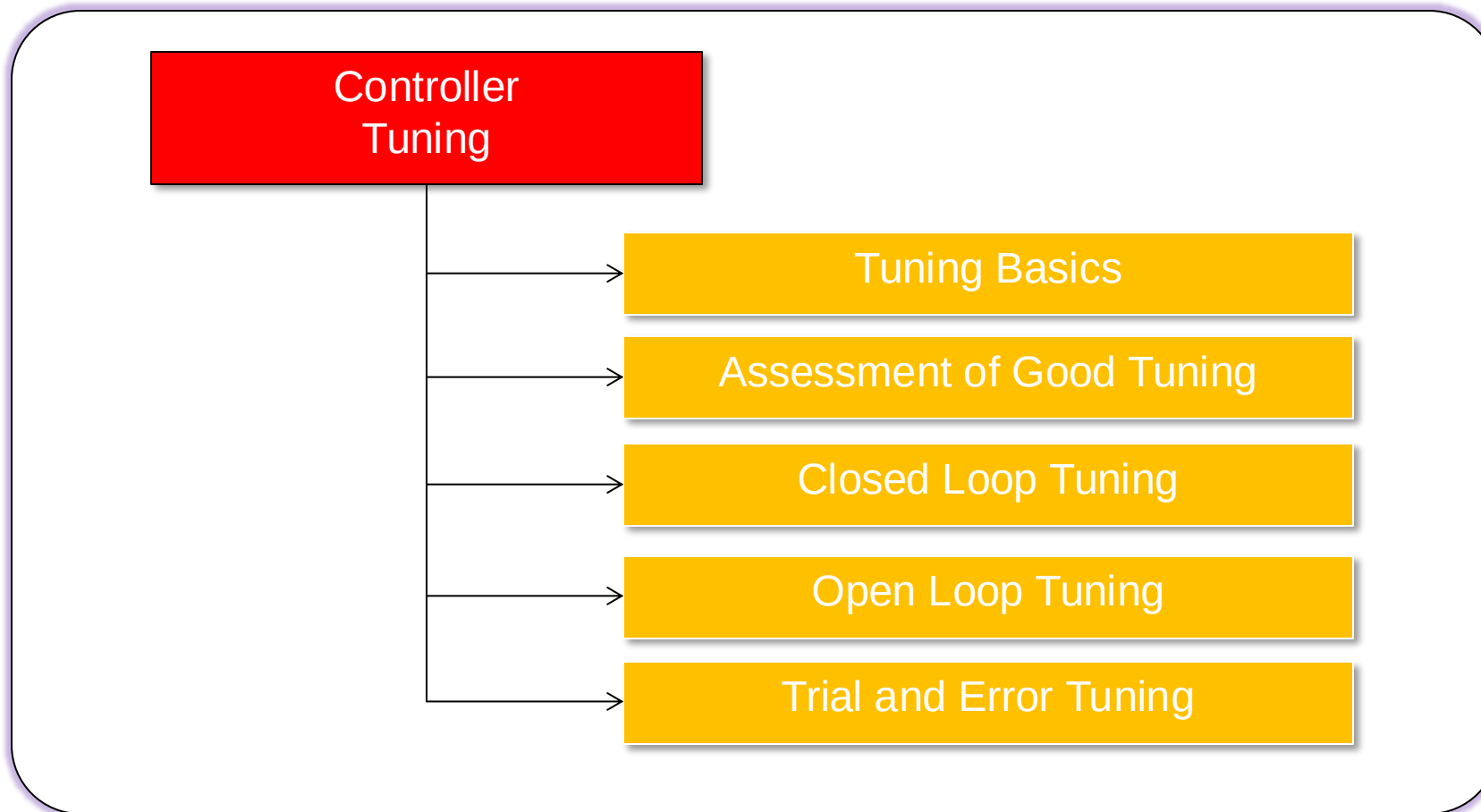
Base Layer Control

Step 3 – Matching the Process with the Controller

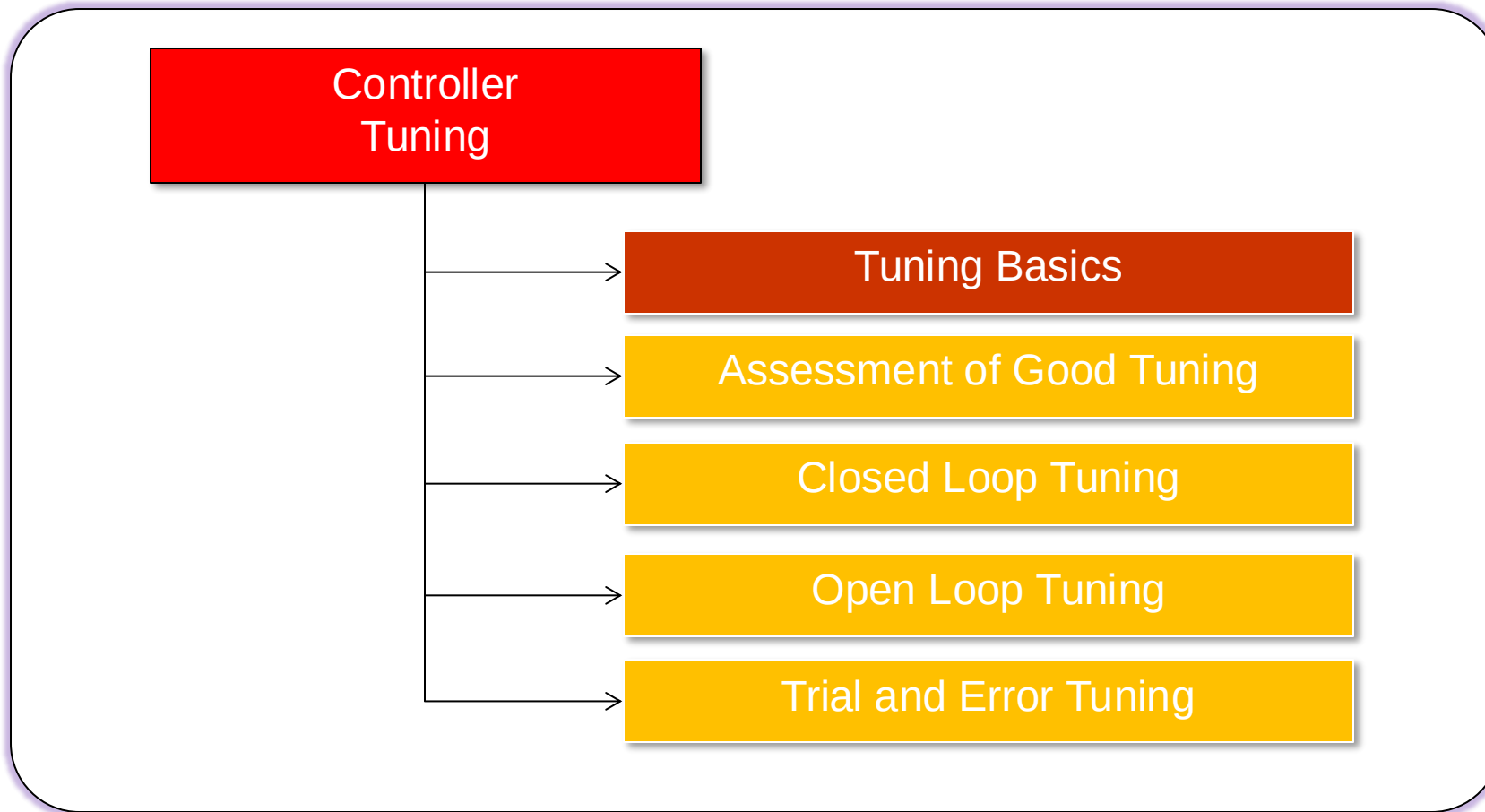
Tuning



Step 3 - Controller Tuning



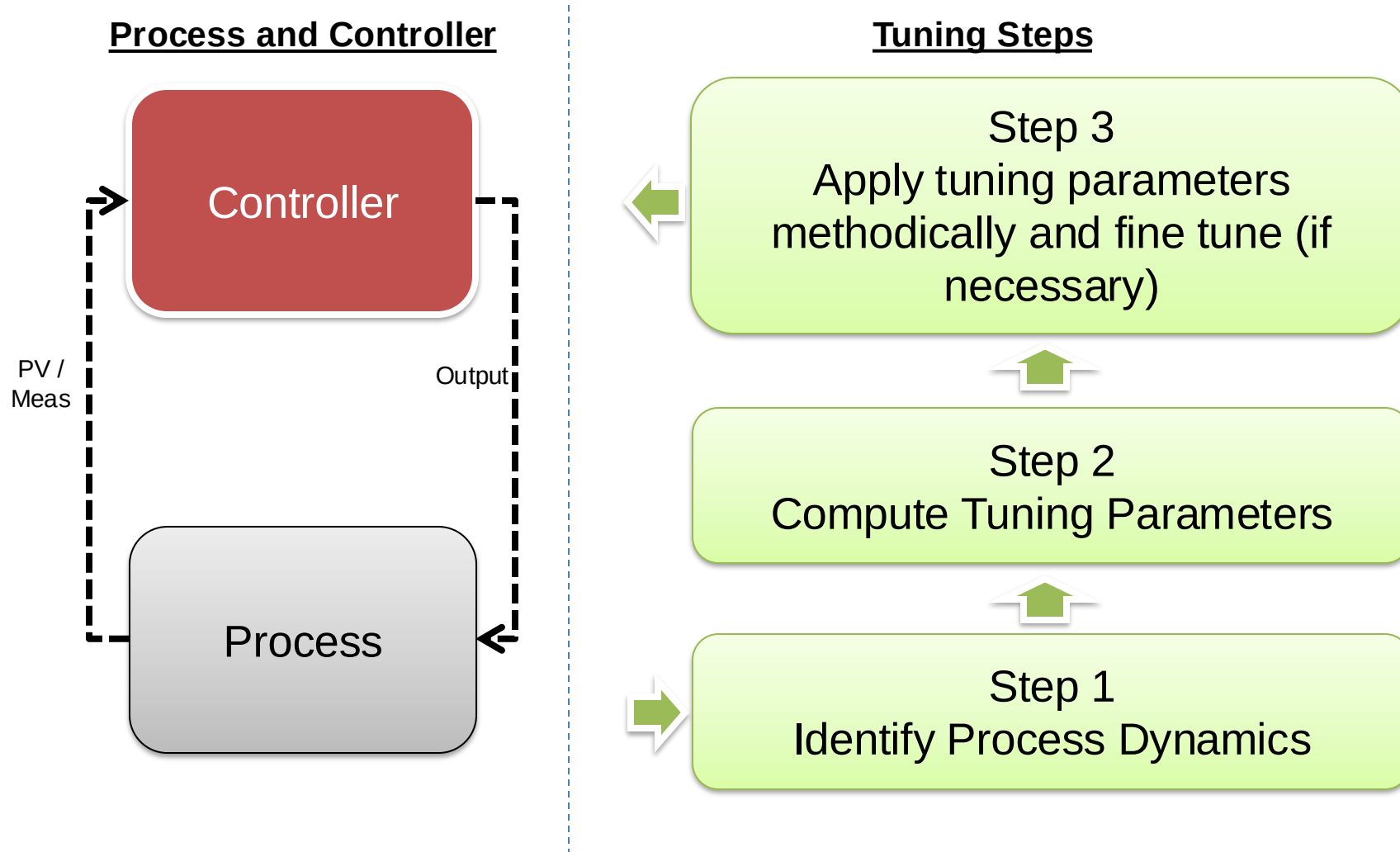
Controller Tuning



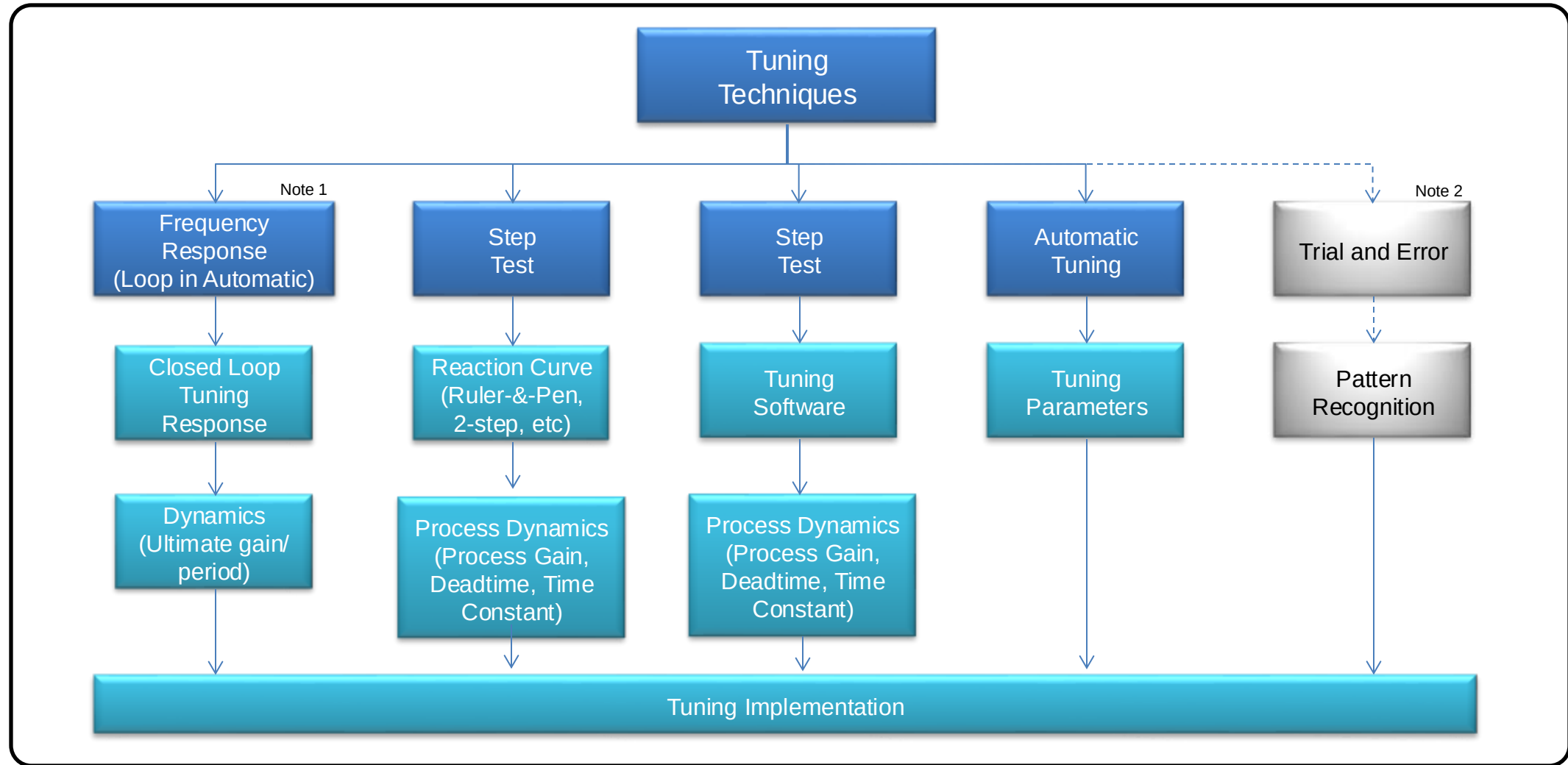
Tuning Basics

- The objective of tuning is to:
 - **“Match Controller Characteristics to Process Characteristics”**
- Process Characteristics are defined by
 - Process Gain
 - Process Deadtime
 - Process Time Constant
- Controller Characteristics is defined by
 - Proportional Setting (Proportional Gain or Band)
 - Integral Setting (Integral Time or Reset Action)
 - Derivative Setting (Derivative Time)

Basic Tuning Steps



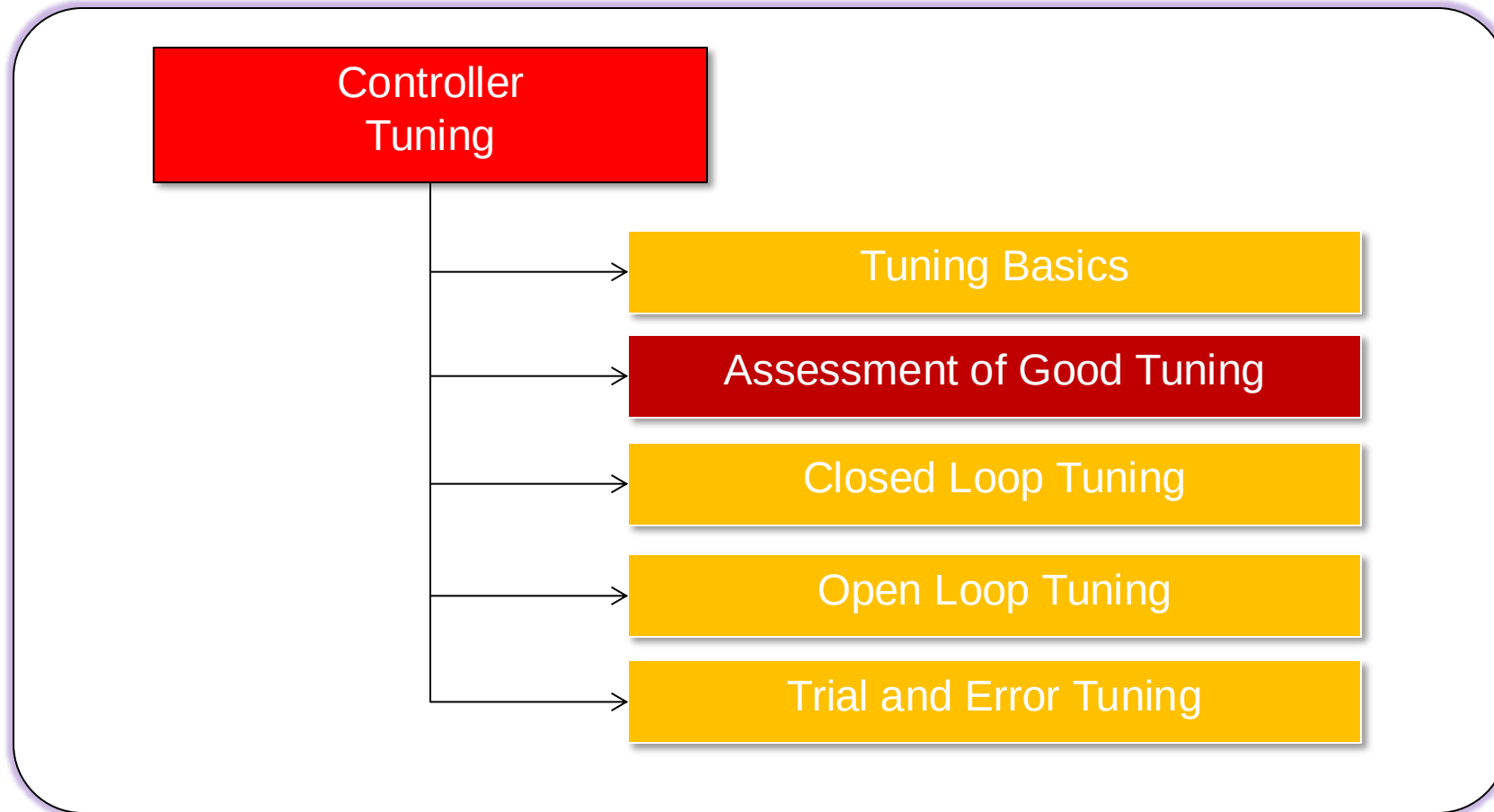
Tuning Techniques



Note 1 : Typically, not used due to the destabilising impact on the process

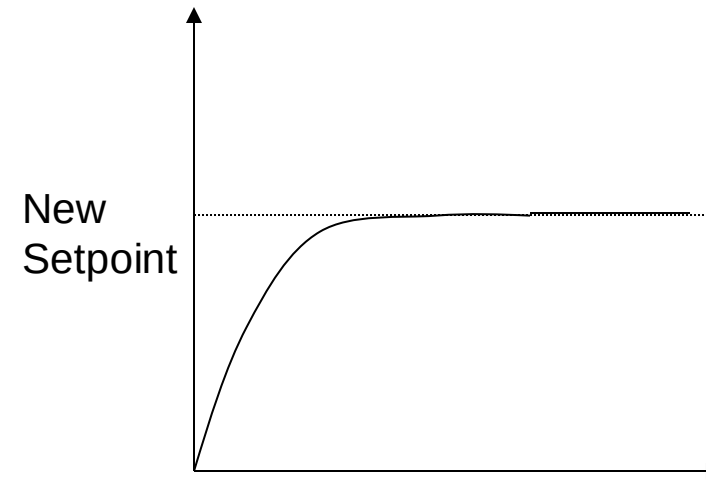
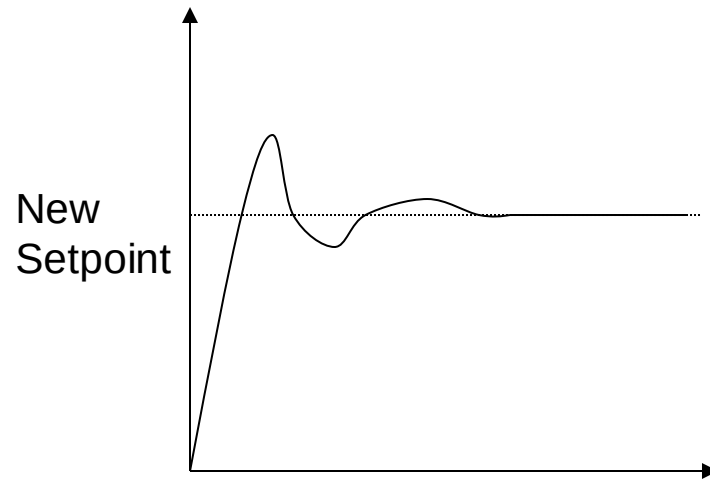
Note 2 : Not recommended

Controller Tuning



Criteria for Good tuning

- Which of the following trends represents a better tuned process?

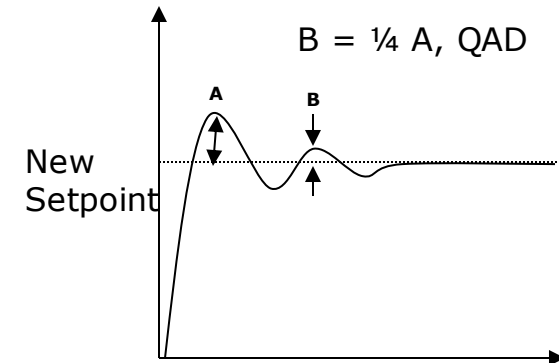
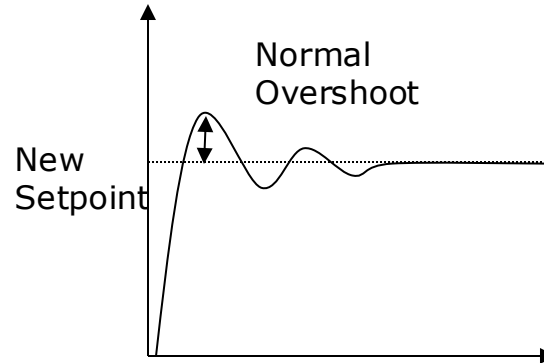
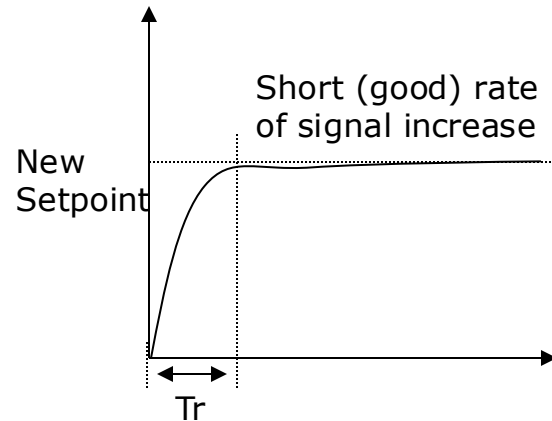
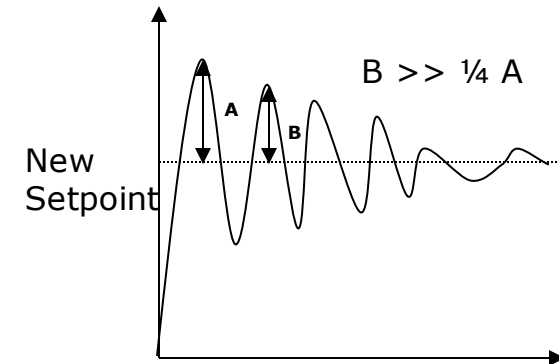
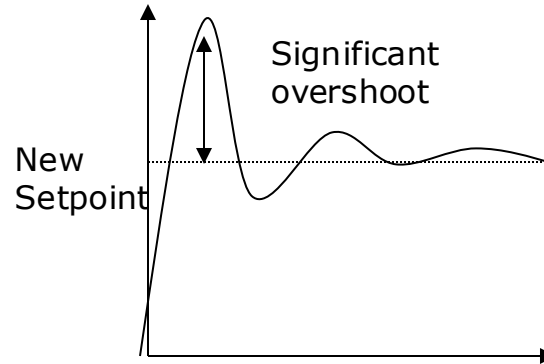
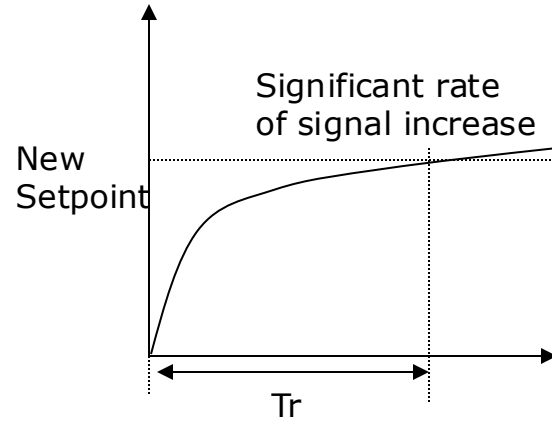


Typical Characteristics

The typical characteristic that an engineer looks for when assessing the 'goodness of control' are :

- Rate of signal increase (to the setpoint)
- Overshoot value
- Ratio of Decay

Typical Characteristics



RATE OF SIGNAL INCREASE

OVERSHOOT VALUE

DECAY RATIO

Recognised Criteria

	Criteria	Detail
1	Decay Ratio of 1/4 (most common)	$Decay Ratio = \frac{2nd\ peak\ overshoot}{1st\ peak\ overshoot}$
2	Minimum Integral Of Square of Error (ISE)	$ISE = \int_0^{\infty} E^2 dt$
3	Minimum Integral of Absolute Error (IAE)	$IAE = \int_0^{\infty} E dt$
4	Minimum Integral over time of Absolute Error (ITAE)	$ITAE = \int_0^{\infty} E t dt$
5	Minimum Integral over Time of Square of Error (ITSE)	$ITSE = \int_0^{\infty} E^2 t dt$

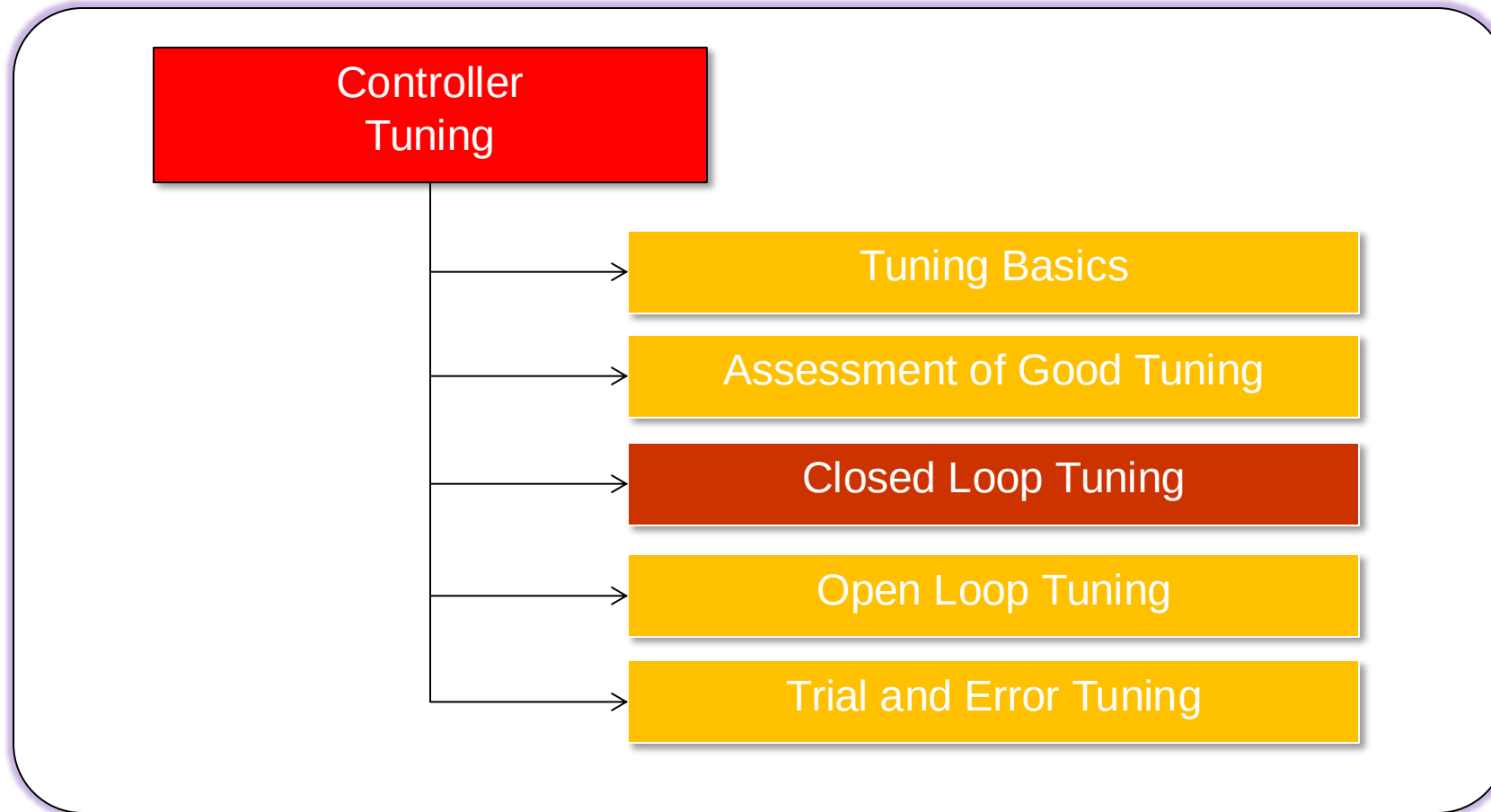
Notes

The QAD (Quarter Amplitude Damping) is the most common criteria utilized in tuning. Its basis is that for good tuning, the amplitude of each overshoot should be $\frac{1}{4}$ the size of the preceding overshoot.

Note that the QAD criterion is no longer valid in PETRONAS as the preferred response of the process is not longer with an overshoot.

The integral criteria (IAE/ISE/ITAE/ITSE) are more accurate, but however these computations need to be installed and/or available prior to the tuning exercise.

Controller Tuning



Closed Loop Tuning

As the name implies, closed loop tuning is carried out with the controller in automatic (i.e. the control loop is closed)

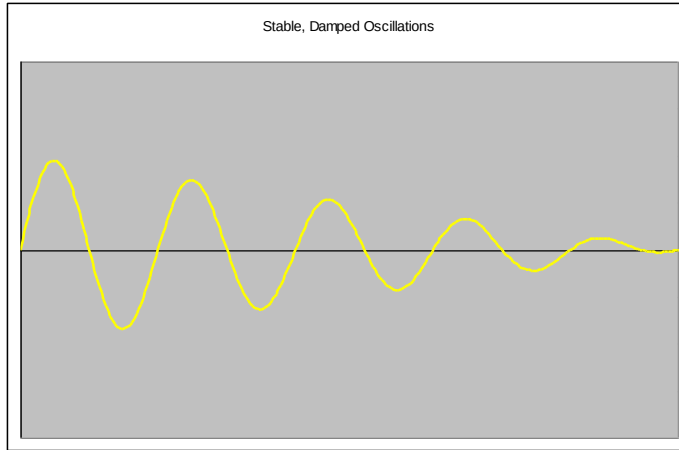
This method of tuning was one of the first methods of tuning. It was reported by Zeigler and Nichols in 1942.

It is also called the Ultimate Method because it requires the determination of the Ultimate Gain and the Ultimate period.

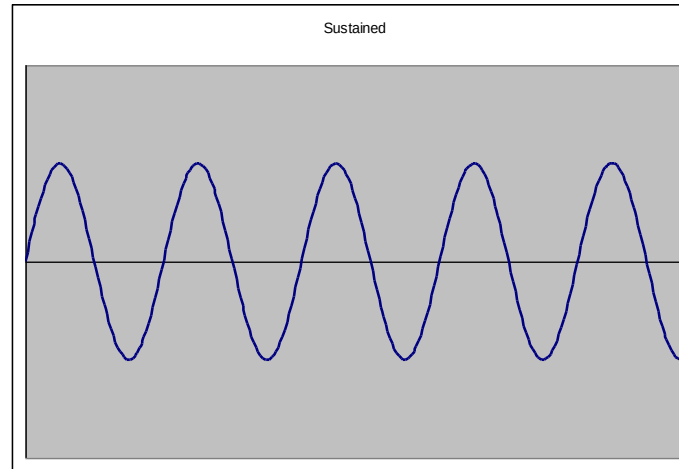
Closed Loop Steps

- Step 1 : Place control loop in manual
- Step 2 : Place gain at a small value
- Step 3 : Place integral time to maximum
- Step 4 : Place Derivative time to zero
- Step 5 : Place control loop back to Auto
- Step 6 : Change setpoint
- Step 7 : Slowly increase the gain
- Step 8 : After each gain change, initiate a setpoint change
- Step 9 : Watch the process as it starts to oscillate and stop increasing the gain when the process is perfectly oscillatory

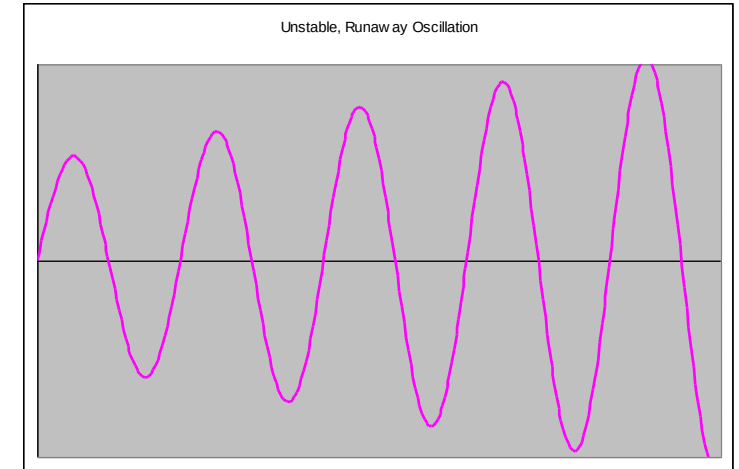
Typical Response



Small Increase in Gain,
Oscillations decay



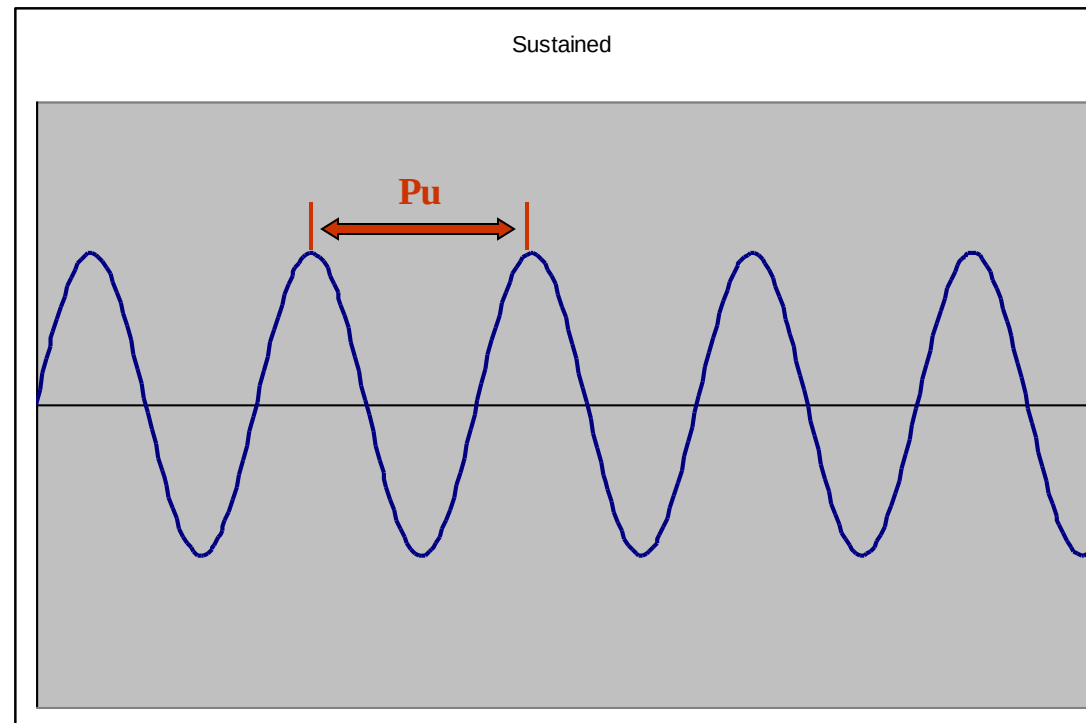
Ultimate Gain, Sustained Oscillations



Excessive Gain, Runaway Process

Obtaining the Ultimate Parameters

- Ultimate Gain
 - The Gain utilized on the controller to obtain the sustained oscillation is the Ultimate Gain (K_u).
- Ultimate Period
 - The Ultimate Period (P_u) is the period of the sustained oscillation



Obtaining the Tuning Constants

Type of Controller	Gain	Integral time	Derivative time
P	$0.5 K_u$		
PI	$0.45 K_u$	$P_u / 1.2$	
PID	$0.6 K_u$	$P_u / 2$	$P_u / 8$

For and Against

There are various advantages and disadvantages of the closed loop tuning.

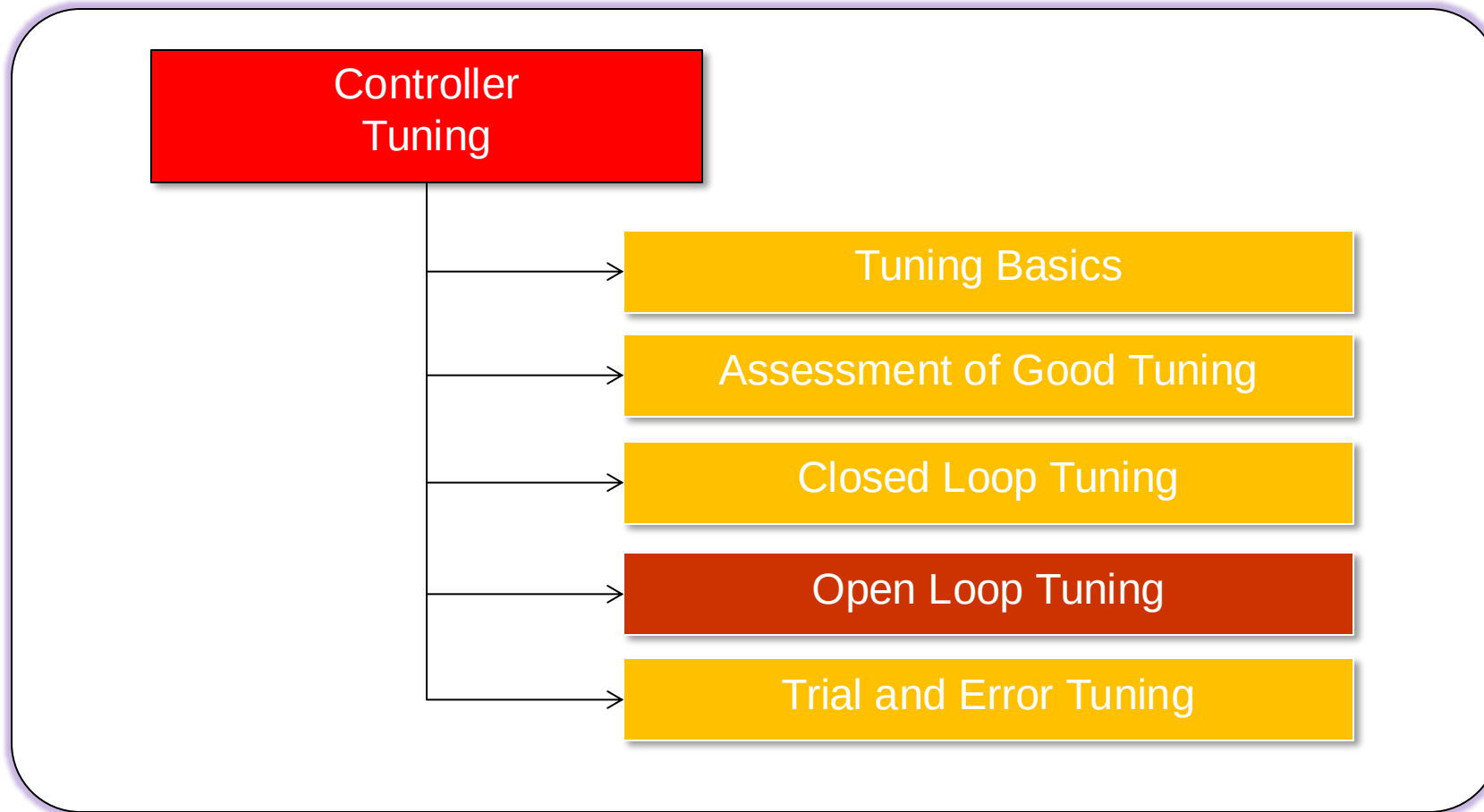
For

- Primary advantage is that this type of tuning considers the dynamics of all components in the loop
- The readings of PB_u (or K_u) and P_u are easy to read off

Against

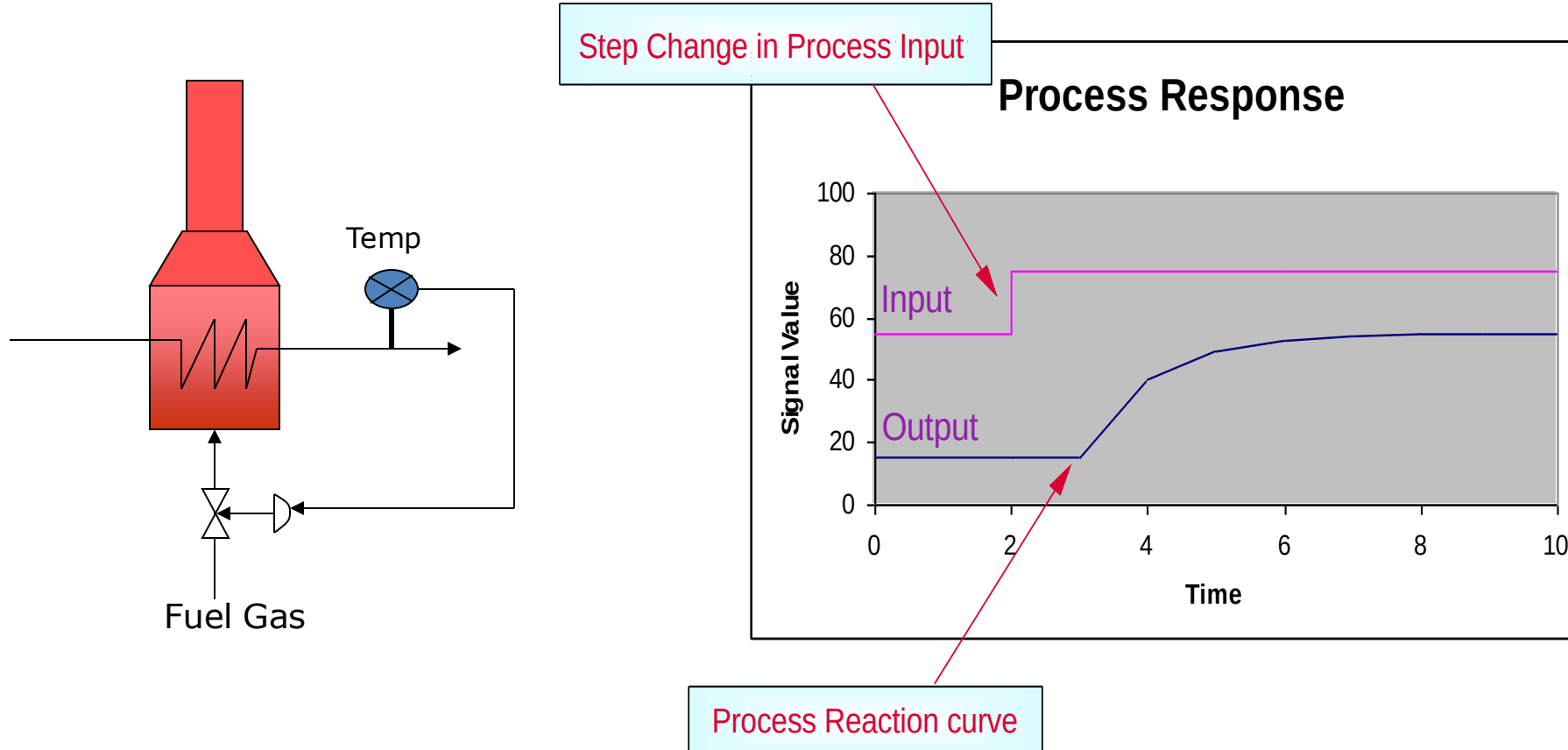
- Primary issue with closed loop tuning is that the process is made to become oscillatory affecting that process and most likely all interconnected processes
- It is also brought close to a runaway zone
- Tuning can be slow as the basis is almost like trial and error

Controller Tuning



Open Loop Tuning

- In open loop tuning (as the name implies), the control loop is placed in manual.
- The response of a process that is upset very often follows the characteristic S-shape.

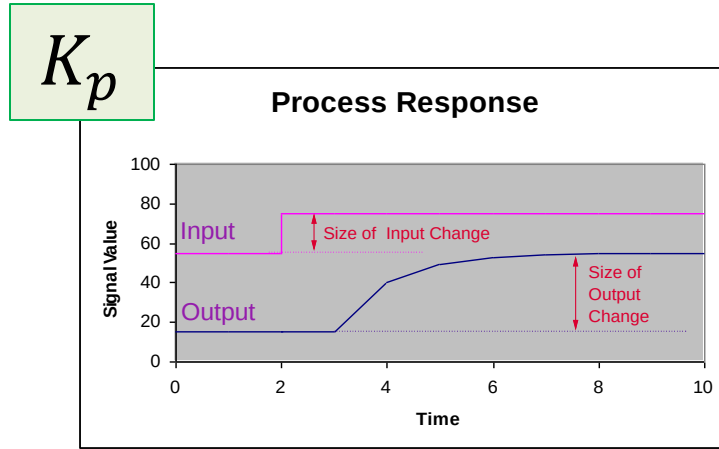


Defining the curve

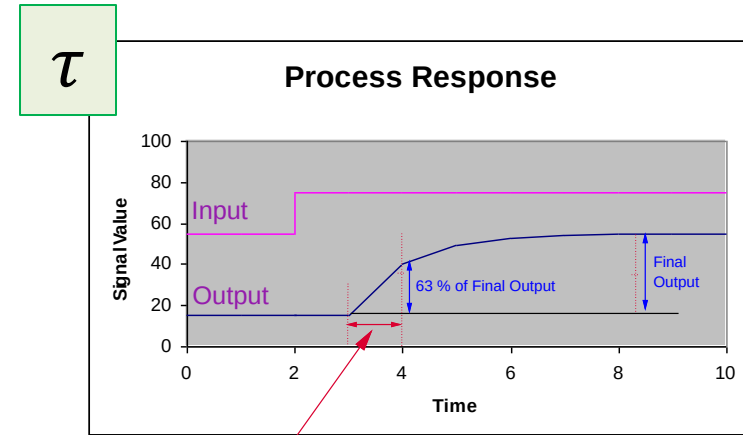
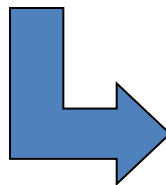
As stated earlier the process response can be defined by three components:

- A gain
- A deadtime and
- A time-constant.

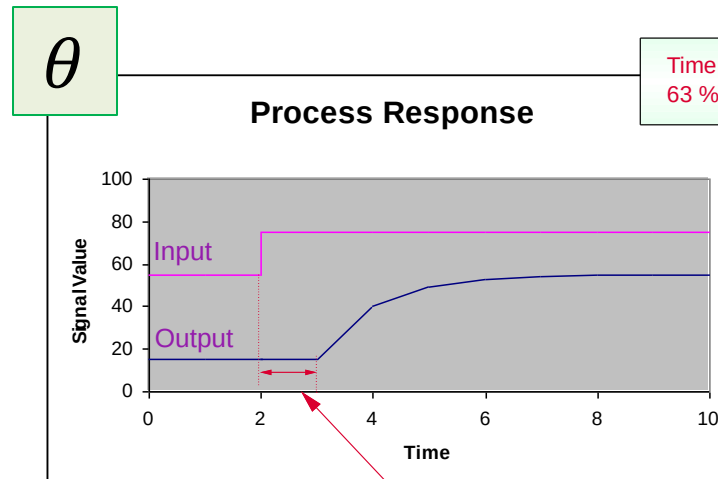
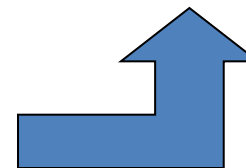
Curve Parameters



Process Gain = $\frac{\text{Size of Output Change}}{\text{Size of Input Change}}$

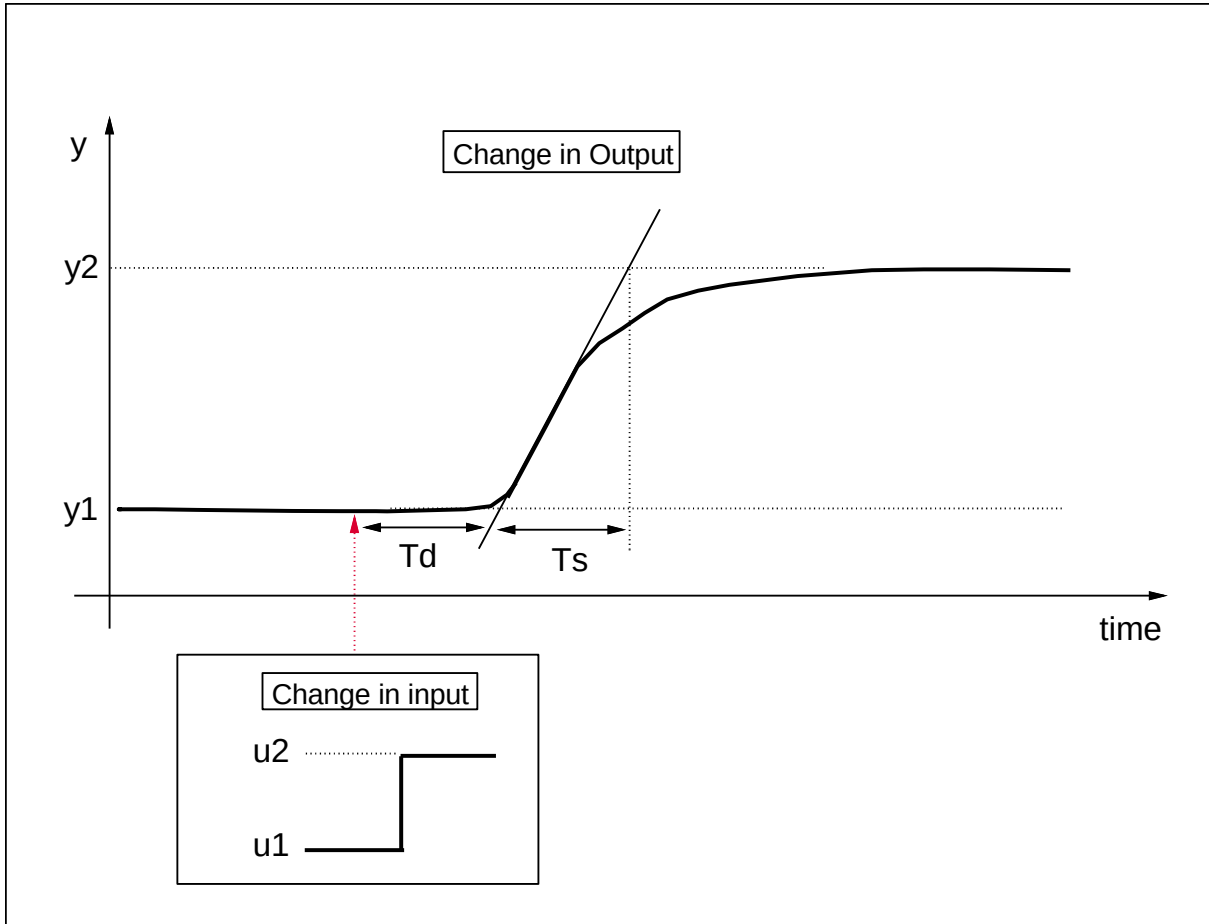


Time taken for process (Output) to reach 63 % of its final value, i.e. Process Time Constant



Time taken for process to respond, i.e. Deadtime

Putting it all together



Determine the following parameters:

- Change in input i.e.,

$$U = u_2 - u_1$$
- Change in output i.e.,

$$Y = y_2 - y_1$$

$$\Rightarrow K_p = Y/U$$
- Deadtime, T_d or θ
- Timeconstant, T_s or τ

Ziegler-Nichols Open Loop Tuning Parameters

Type of Controller	Gain	Integral time	Derivative time
P	$\frac{\tau}{\theta} \frac{1}{K_p}$		
PI	$0.9 \times \frac{\tau}{\theta} \frac{1}{K_p}$	3.3θ	
PID	$1.2 \times \frac{\tau}{\theta} \frac{1}{K_p}$	2.0θ	0.5θ

Shinskey's Open Loop Tuning Parameters

Type of Controller	Gain	Integral time	Derivative time
P	$\frac{\tau}{\theta} \frac{1}{K_p}$		
PI	$0.95 \times \frac{\tau}{\theta} \frac{1}{K_p}$	4θ	
PID	$0.85 \times \frac{\tau}{\theta} \frac{1}{K_p}$	1.6θ	0.6θ

Cohen-Coon

- Several other equations are available for the calculation of the PID values. They vary from the very simple to the very complex.

	Proportional	Integral (minutes/repeat)	Derivative
P Only Controller	$\frac{100}{1 + \mu / 3} \tau_d R$		
PI Controller	$\frac{100}{1 + \mu / 11} \tau_d R$	$3.33 \tau_d \left[\frac{1 + \mu / 11}{1 + 11 \mu / 5} \right]$	
PID Controller	$\frac{100}{1.35 (1 + \mu / 5)} \tau_d R$	$2.5 \tau_d \left[\frac{1 + \mu / 5}{1 + 3 \mu / 5} \right]$	$\frac{0.37 \tau_d}{1 + \mu / 5}$

Cohen-Coon Settings

where

τ_d = Process Dead Time

μ = Uncontrollability parameter

R = Reaction Rate

Note : For a list of other equations refer website by Abdul Aziz Ishak of Dept of Chem Eng, UTM

IMC/Lambda Tuning Equation

Type of Controller	Gain	Integral time	Derivative time
PI	$\frac{1}{K_p} \frac{\tau}{\lambda + \theta}$	τ	
PID (non-interactive)	$\frac{1}{K_p} \frac{\tau + \frac{\theta}{2}}{\lambda + \theta}$	$\tau + \frac{\theta}{2}$	$\frac{\tau\theta}{2\tau + \theta}$

Note:

1. The above tuning equation is for self-regulating processes
2. For 15% MV overshoot, the relationship between λ with τ & θ is

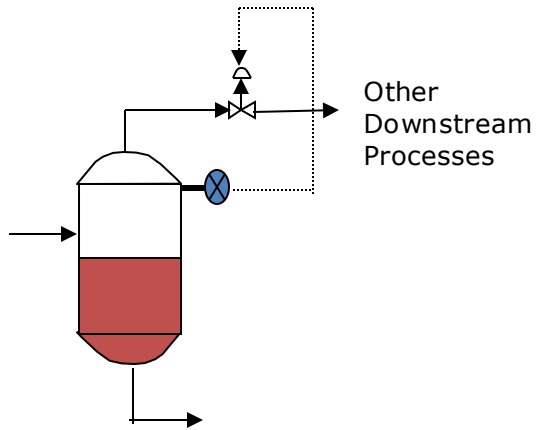
$$\lambda = 0.31\theta + 0.88\tau$$

θ = deadtime

τ = time constant

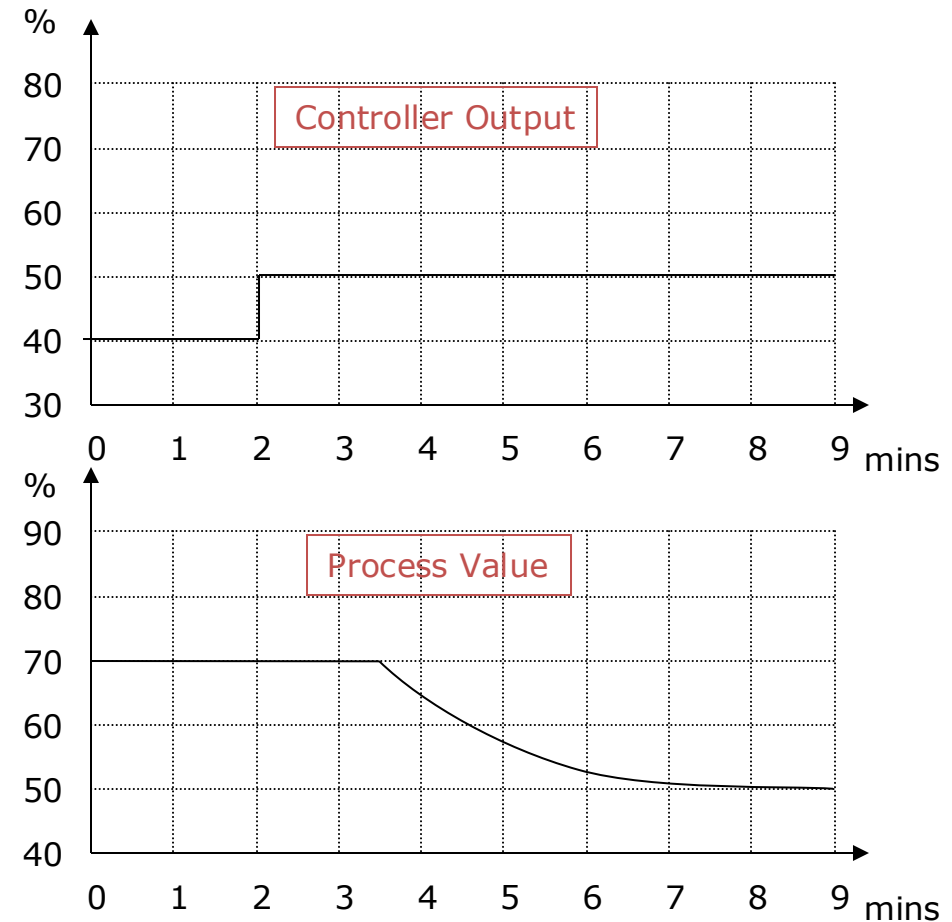


Worked Example

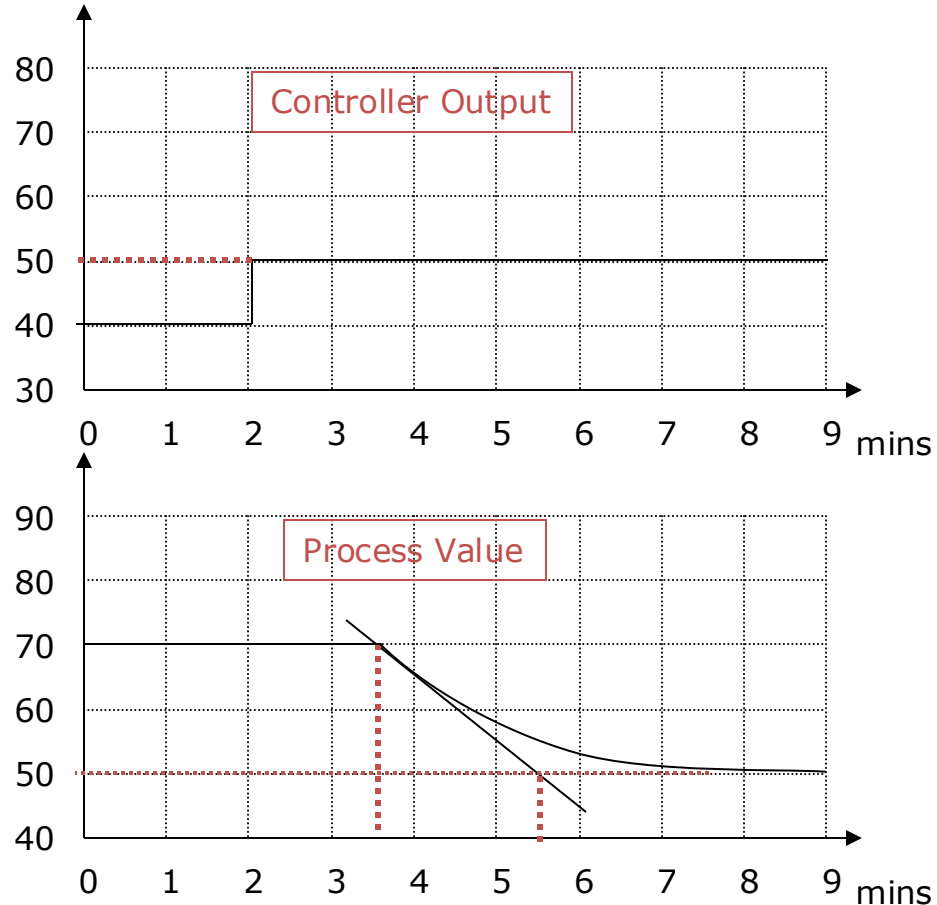


A step change was carried out on a pressure process as above, the results are indicated in the charts.

- Recommend a type of controller (P, PI or PID)
- Determine the tuning settings via
 - Zeigler and Nichols
 - Shinskey



Solution

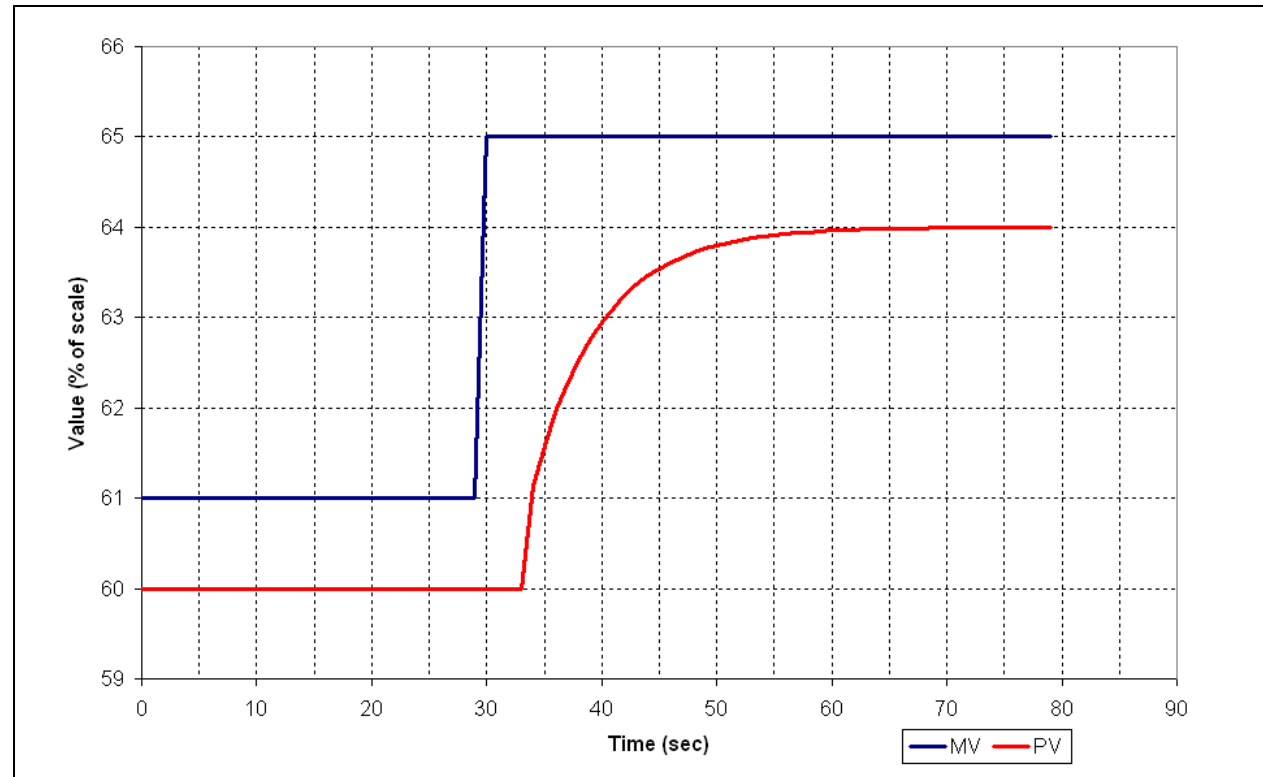


From the Plots

- | | |
|---|--|
| i) Change in Input, U
$U = (50 - 40) = 10$ | ii) Deadtime, θ
$\theta = (3.5 - 2) = 1.5$ |
| Change in Output, Y
$Y = (70 - 50) = 20$ | iii) Time constant, τ
$\tau = (5.5 - 3.5) = 2$ |
| $K_p = 2$ | |

- i) Recommended controller type: PI
- ii) Ziegler & Nichols recommended settings
 - P, Gain (PBand) = 0.67 (167)
 - I = 5.0 minutes
- Shinskey recommended settings
 - P, Gain (PBand) = 0.63 (158)
 - I = 6.0 minutes

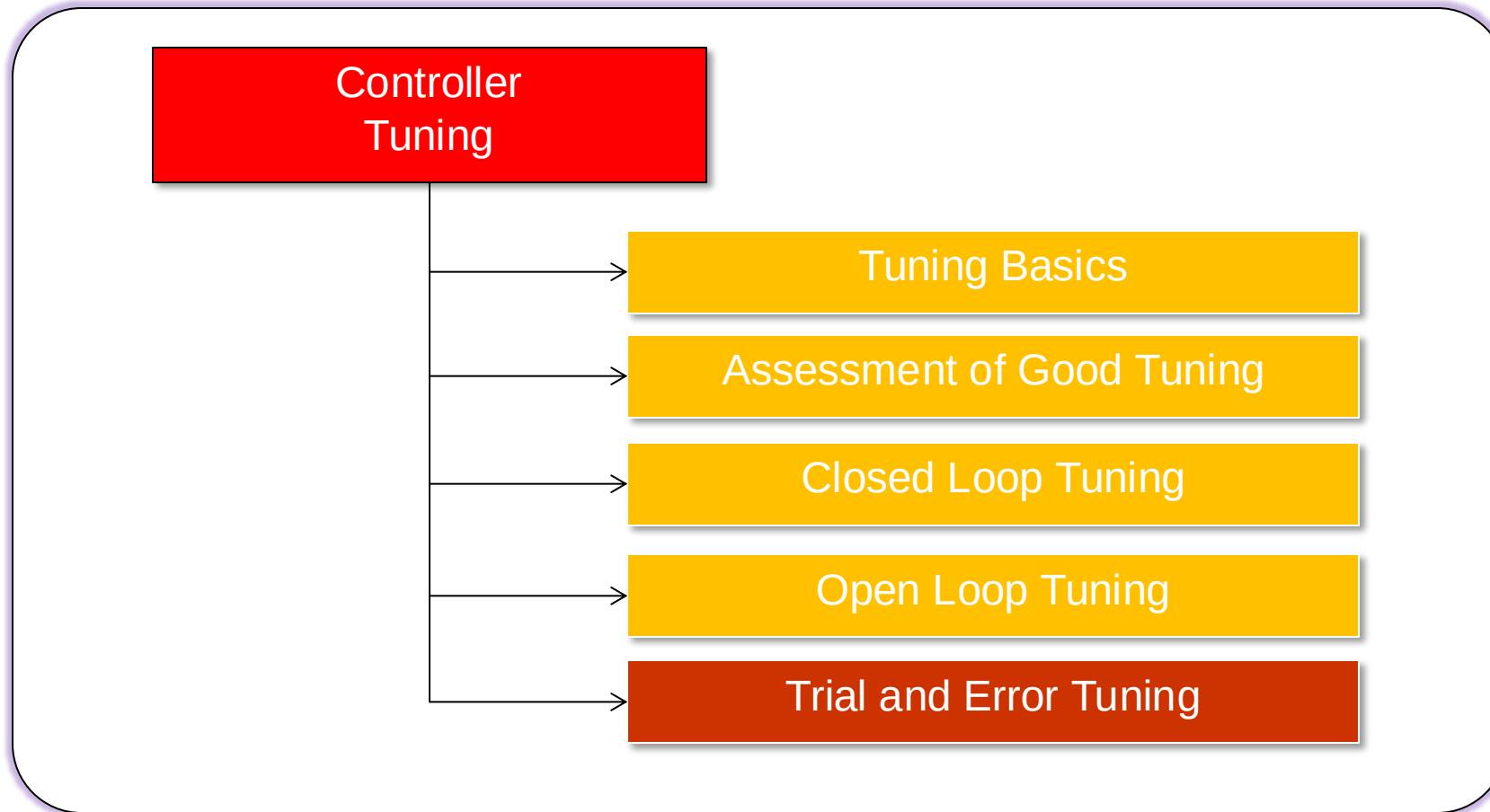
Exercise



A step change was carried out on a process and the results are indicated in the charts. Assuming this is a PI controller, determine the tuning settings via

- Zeigler and Nichols
- IMC/Lambda

Controller Tuning



Types of Trial & Error methods

Although the “Trial and Error method’ implies no specific technique, two general methods are often quoted:

- Systematic (closed loop)
- Pattern Recognition

Systematic

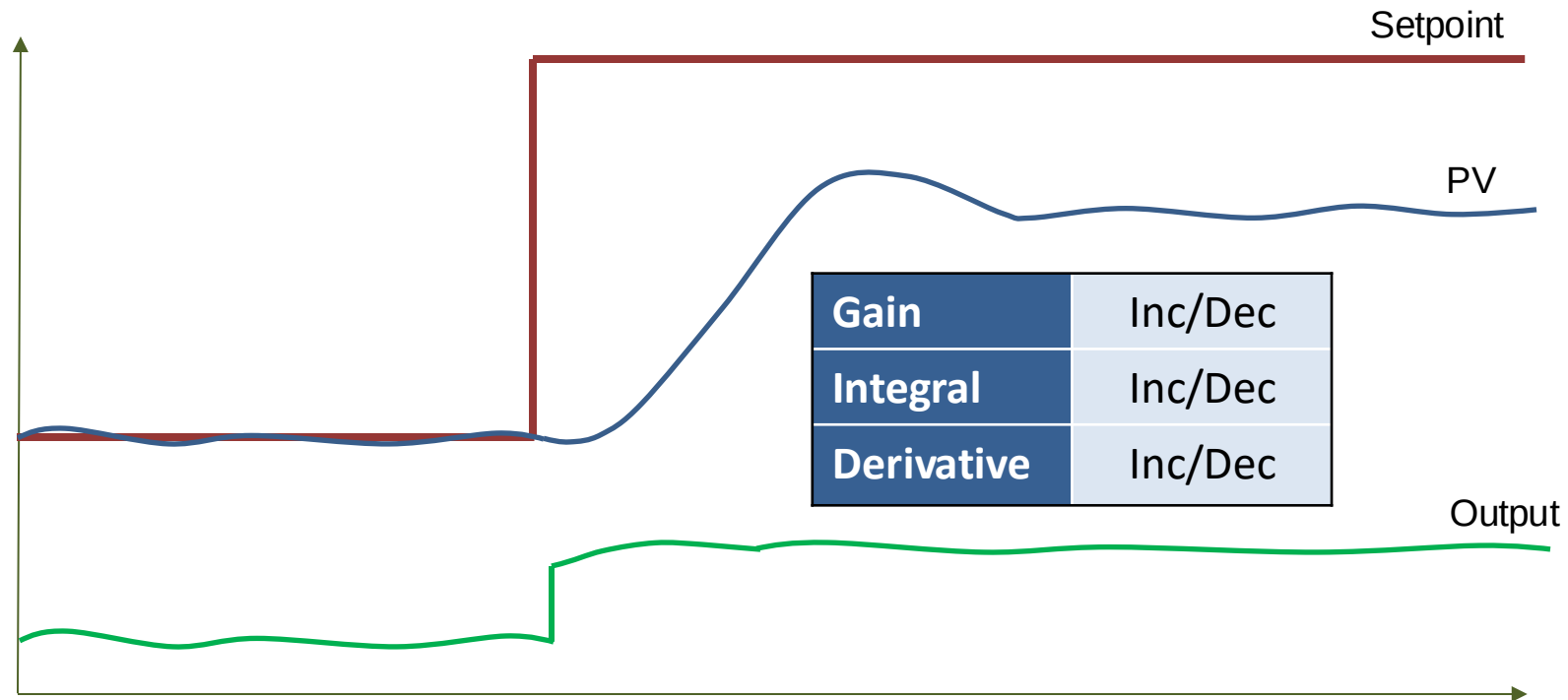
The systematic trial and error method is actually the technique for closed loop tuning developed by Zeigler and Nichols. It is considered Trial and Error because the means to obtain a sustained oscillation is purely by “Trial & Error”.

Pattern Recognition

Pattern Recognition techniques are techniques that tend to make tuning a 'black art'

It relies on the user's intuitive knowledge of the working of the PID algorithm to determine the correct 'Pattern' for process CLOSED LOOP response.

Example 1

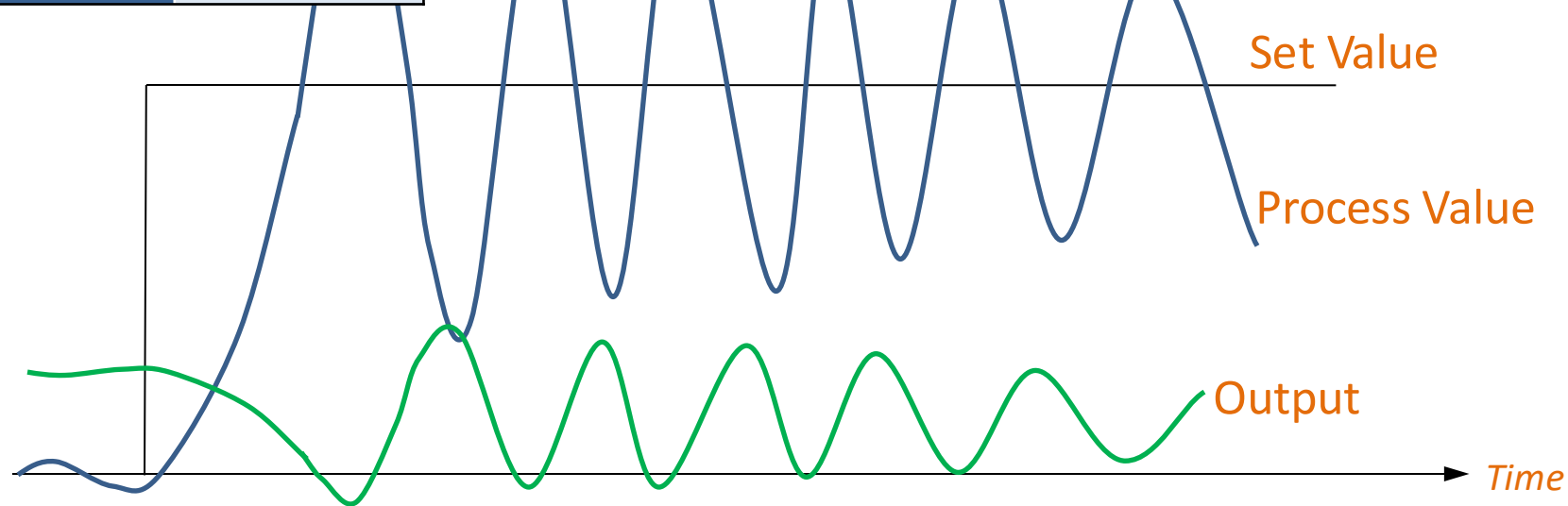


A Setpoint increase was carried out on a Flow Controller. The result of the step increase is indicated above.

- What do you think is the reason for this trend?
- What can be done to improve this 'pattern'?

Example 2

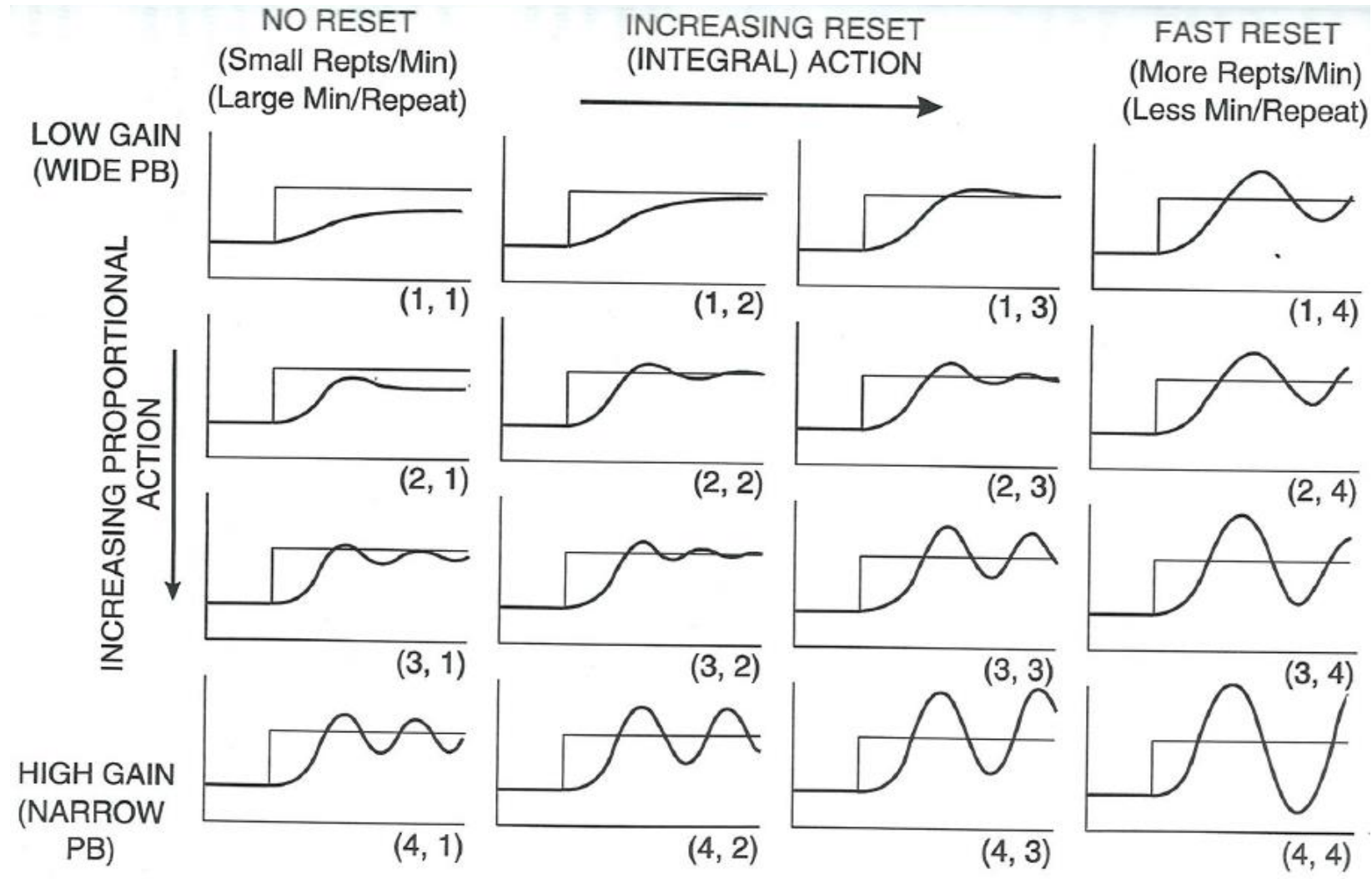
Gain	Inc/Dec
Integral	Inc/Dec
Derivative	Inc/Dec



A Setpoint increase was carried out on a pressure controller in a vessel (PI action only). The result of the step increase is indicated above.

- What do you think is the cause for this?
- What can be done to improve this 'pattern'?

Tuning on the Run



When to tune & when not to?

Controllers can be 'faulty' due to

- Instrumentation
 - Faulty/noisy transmitter
 - Faulty final control element (valve)
 - Control valve limitation
- Turndown ratio (process)
- Wrong tuning setting
- Incorrect strategy chosen

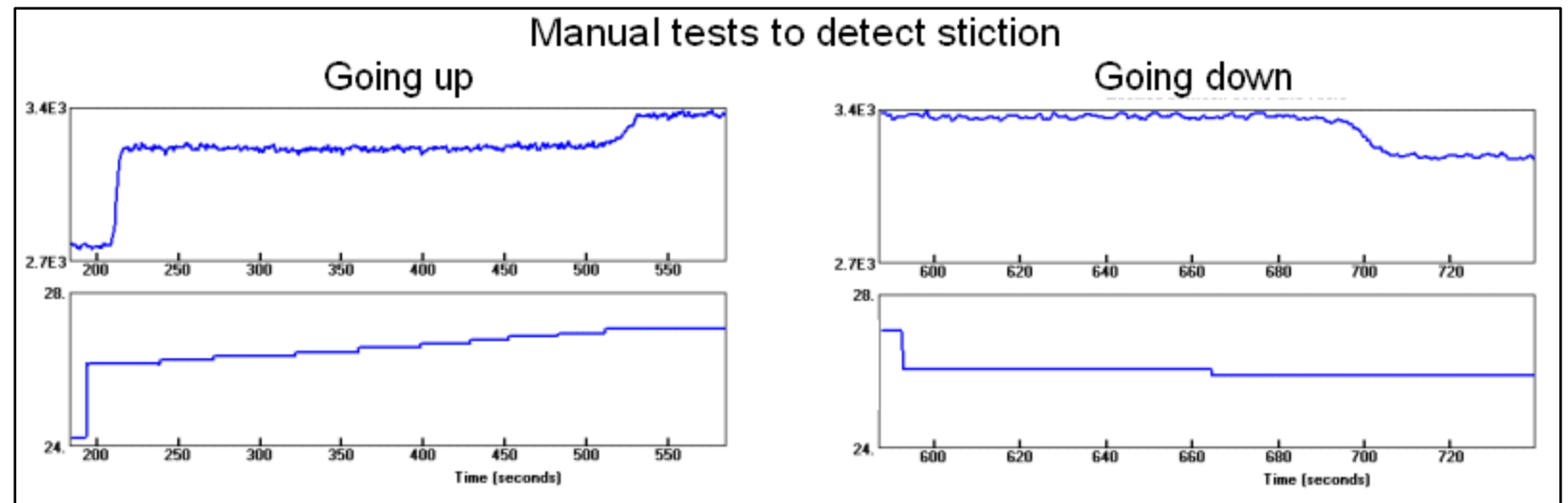
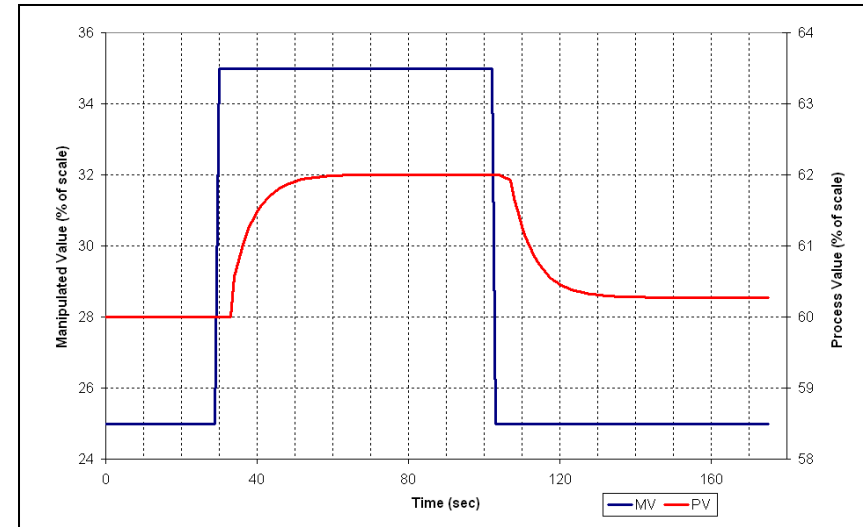
Problem Identification

Use one of the following techniques:

1. Check if PV is severely unstable
 - In Manual, if PV is not oscillating, go to Step 2.
 - If PV oscillates, try to identify source of disturbance & rectify if possible
 - Process upstream the controller can cause the oscillation
 - Check whether it's the process or the controller (elimination process)
2. Step with controller in Manual
 - Check for hysteresis, stiction
 - If present, rectify the valve problem if possible. Else, tuning needs to be changed. Note that tuning will not completely remove oscillations.
3. Carry out open or closed loop tuning
 - Sometimes reducing gain 1st can have a significant effect.

Hysteresis & Stiction

- Hysteresis
 - Do open loop test. Step both direction by same amount & see if it returns back to original PV.
- Stiction
 - Open loop test & see if valve responds well.



References: *ExpertTune*

Don't retune when . . .

- Measurement or valve is limiting
 - Retuning only helps a little bit
- Valve issues
 - Stiction > 1%

Can retune when . . .

- Controller response as per the charts in Tuning Section
- In slight cases of stiction or hysteresis, retuning can still help
 - Expect control valves to have up to 0.3% of span stiction and 0.1% span hysteresis problems

Typical Tuning Settings

Type of Measurement	Fluid/Equip	Proportional Band range	Gain Range	Integral time, (mins)	Integral Action, (repeats/min)	Basis
Flow	Liquid	200 ~ 400	0.5 ~ 0.25	< 0.5	>2	The tuning range for flow is on the basis that in an ideal flow control loop, the measurement range of the flow transmitter and the range of the flow valve is equal. This means that in an open loop response, a change of CV output by 1 % will cause the flow to respond by 1% of range giving a process gain of 1. Since (for ideal conditions), $G_p * G_I = 0.5$, then the starting point for pband is 200. However, typical CV's are oversized to varying degrees and a range of 200 ~ 400 is more reasonable. Flow systems are naturally stable and therefore tight integral times are recommended to speed up the response. Exceptions to this rule include when the valve response has deteriorated in which case tight integrals will cause responses from sustained oscillations to instability.
	Gas	150 ~ 250	0.7 ~ 0.4	0.2 ~ 1.0	1 ~ 4	Gas flow is compressible and therefore the process response is slower. For this reason a tighter Pband is needed while integral time is set larger to prevent instability.
Level		< 100	>1	> 10	< 0.1	Two important factors affect level control. One is that level process is a naturally integrating process, two is the fallacy that level control needs to be tight. In most plants, the available buffer in a vessel is never utilised as the level control is tuned to be kept at setpoint (which in most cases is meaningless as flow perturbations from upstream are immediately transferred downstream). As such, integral action in true sense is not required. The only exceptions are when level control needs to be tight (exceptional case of very small buffer volumes). Pbands are usually set below 100 to ensure full closure of an outlet valve on emptying and full opening on filling up.
Pressure	Vessel	< 50	>2	> 5	<0.2	Pressure control of a vessel is in almost all cases, the control of a large volume by utilising the flow (small volume flow) through a feed or overhead discharge valve. As such, the process tends to integrate and once again strong integral action will push the process into instability. To make up for this, a relatively tighter Pband is utilised.
	Pipe	Refer flow				Pipe pressure control has the same process dynamics as flow through the pipe.
Temperature		< 50	>2	> 5	<0.2	Temperature changes are usually a function of not just the change in input heat energy but the heat capacities of the metal in the region of the temperature probe. As such, temperature controls are typically slow. Once again, this involves the use of tight Pbands and long integral times.

Notes:

- 1) The tuning settings above are based on a well designed system. By definition, in a well designed system there is a generally a very limited amount of over-design in either the measurement range or in the range of the final element.
- 2) The tuning settings above are to be treated as INITIAL settings. Actual verification of system performance must be carried out after the tuning parameters are set. It is often necessary to fine tune the settings in order to achieve a reasonably stable system.

Retrieval Practice

1. The basic sequence with **closed** loop tuning are as follows:
 - a) Obtain the tuning parameters and apply to the controller and fine tune
 - b) Obtain the process parameters, compute the tuning parameters, apply and fine tune
 - c) Obtain the tuning parameters, compute the tuning parameters, apply to the controller and fine tune
 - d) Obtain the open loop parameters, apply to the controller and fine tune

2. The issue with classical tuning parameters such as Zeigler and Nichols or Cohen-Coon are that:
 - a) The tuning parameters obtained are too sluggish for most processes
 - b) The tuning parameters obtained are too aggressive for most processes

3. An engineer is tuning a boiler drum level and at the point of tuning the level, he only had the Z&N Tuning parameters. He decides to apply these parameters to the level controller: When you find out, you would
 - a) Monitor the process, and even if the process is stable, to retune the process using IMC/Lambda technique
 - b) Monitor the process and if the process is stable, to retain the existing tuning parameters

Key Take Aways

- The many different types of tuning
 - Closed Loop
 - Open Loop
 - ❖ Several equations exist to obtain the gain, integral and derivative time constants
 - Trial-and-Error Tuning
 - ❖ Relies heavily on working knowledge of the PID algorithm and experience
- The most common tuning methodology to obtain an accurate tuning parameter is open loop tuning
- Open Loop Tuning Methodology is always based on getting the following steps
 - Getting the Process Dynamics
 - Computing the tuning parameters using the Tuning Rules
 - Applying the Tuning Rules to the process

Control Theory and Application

Group Exercise

Tuning



- Skill Group 16.1
- Advanced Process Control

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Revision History

Rev	Date	Prepared/ Modified By	Notes
1	2005	Prakash K.K.	Original
2	2009	Andriny R.	With solutions
3	22 nd Aug 2012	Prakash K.K.	Reformatted
4	28 Aug 2013	Andriny R.	Change of tuning equation

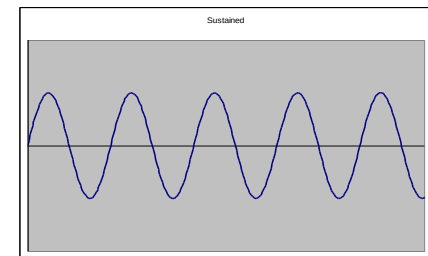
Closed Loop Tuning Query

A pressure controller is in Automatic. Closed loop tuning is to be carried out on the controller and the I action has been made very small.

The gain of the controller is increased until a gain of 6.5 until the plot below is observed. The period of oscillation is 1.5 minutes.

Determine the controller gain and the integral setting if a P controller or a P& I controller or a P&I&D controller was selected.

Reply on the next sheet



Results

Controller type	K_c , Controller Gain	T_i , Integral Time (mins)	T_d , Derivative Time (mins)
P			
P + I			
P + I + D			

Open Loop tuning

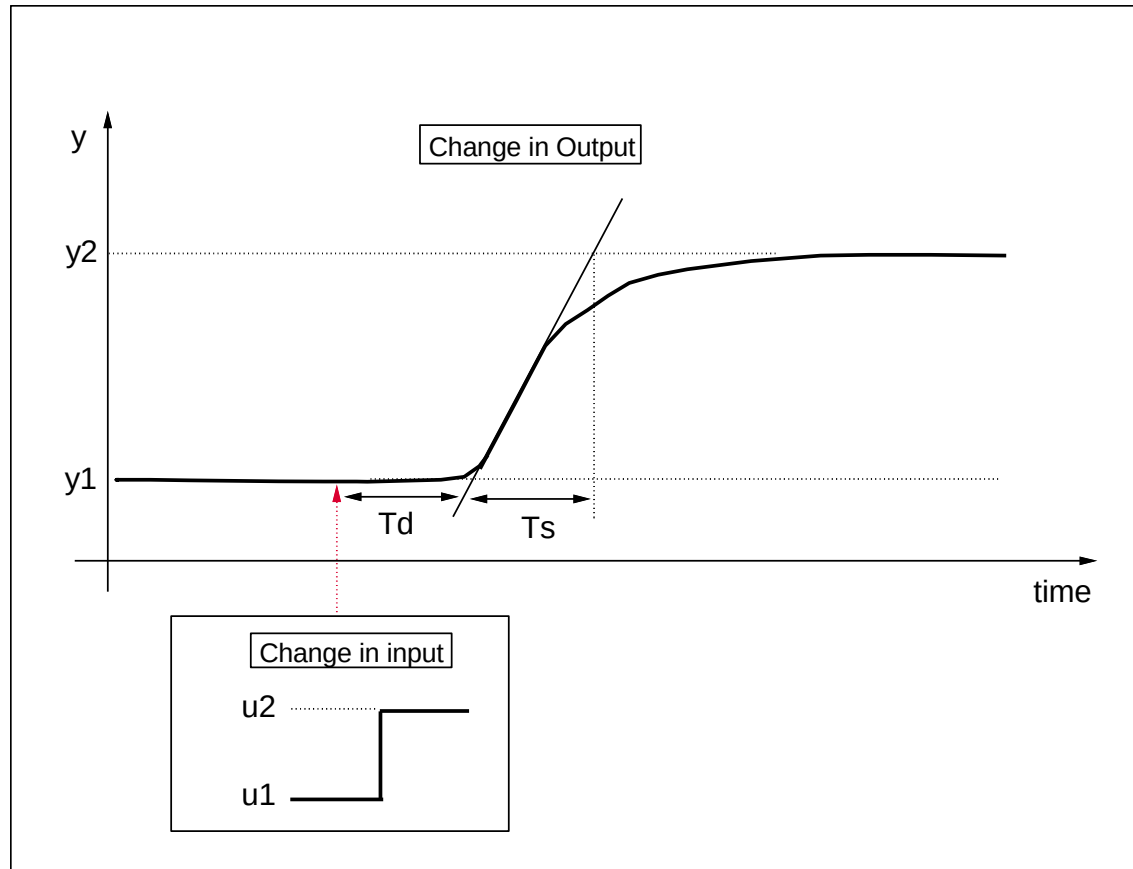
- Two control loops are tuned utilising open loop tuning.
- Loop 1
 - A pressure control loop with a pressure range 0 ~ 10 barg while the controller output is from 0~100%.
 - Loop is PID controller
- Loop 2
 - A flow control loop with a flow range of 0 ~ 3500t/d while the controller output is from 0 ~ 100%.
 - Loop is PI controller

Action Required

- Determine the tuning parameters utilising the rules of
 - Zeigler and Nichols and
 - IMC/Lambda technique
- Reply in the following answer sheet

Results (1)

Loop 1



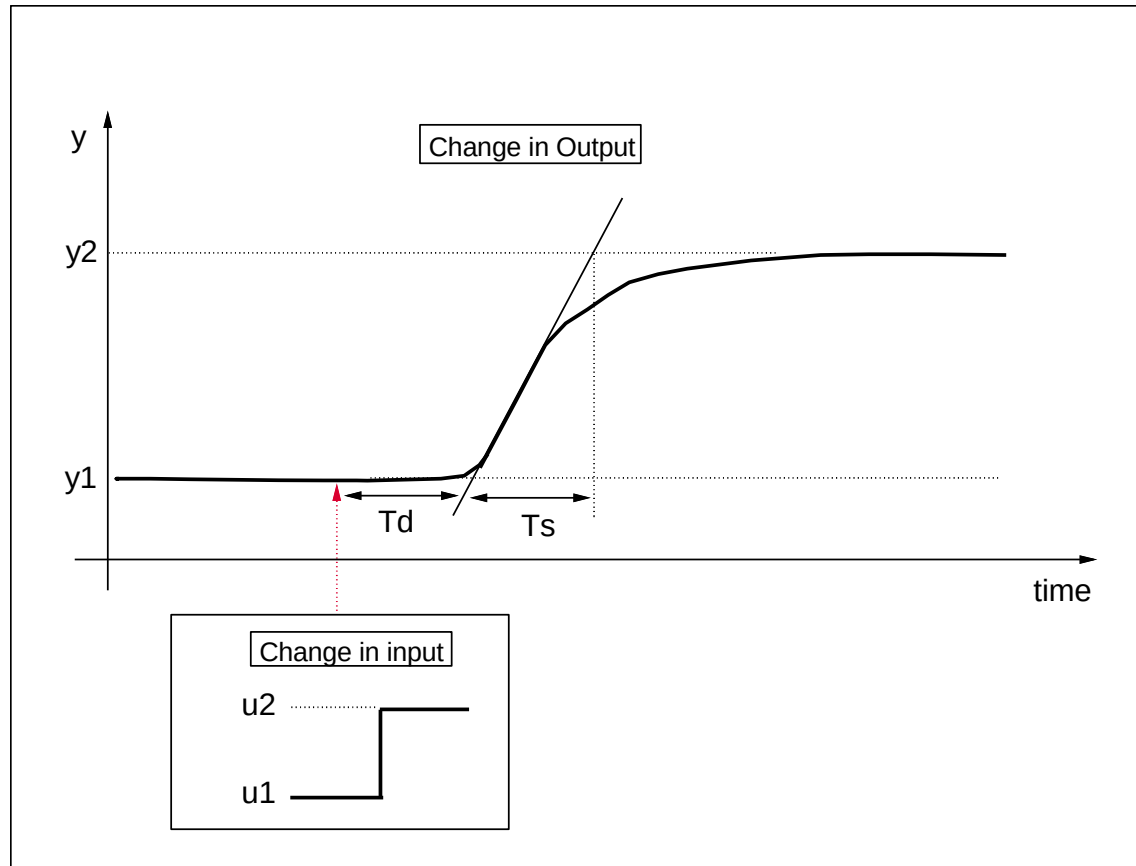
Results:
 $Y1 = 5.0$
 $Y2 = 6.0$

$U1 = 50$
 $U2 = 60$

$\theta = 10 \text{ secs}$
 $\tau = 40 \text{ secs}$

Results (1)

Loop 2



Results:
 $Y1 = 2800$
 $Y2 = 3150$
 $U1 = 45$
 $U2 = 55$
 $\theta = 2 \text{ secs}$
 $\tau = 9 \text{ secs}$

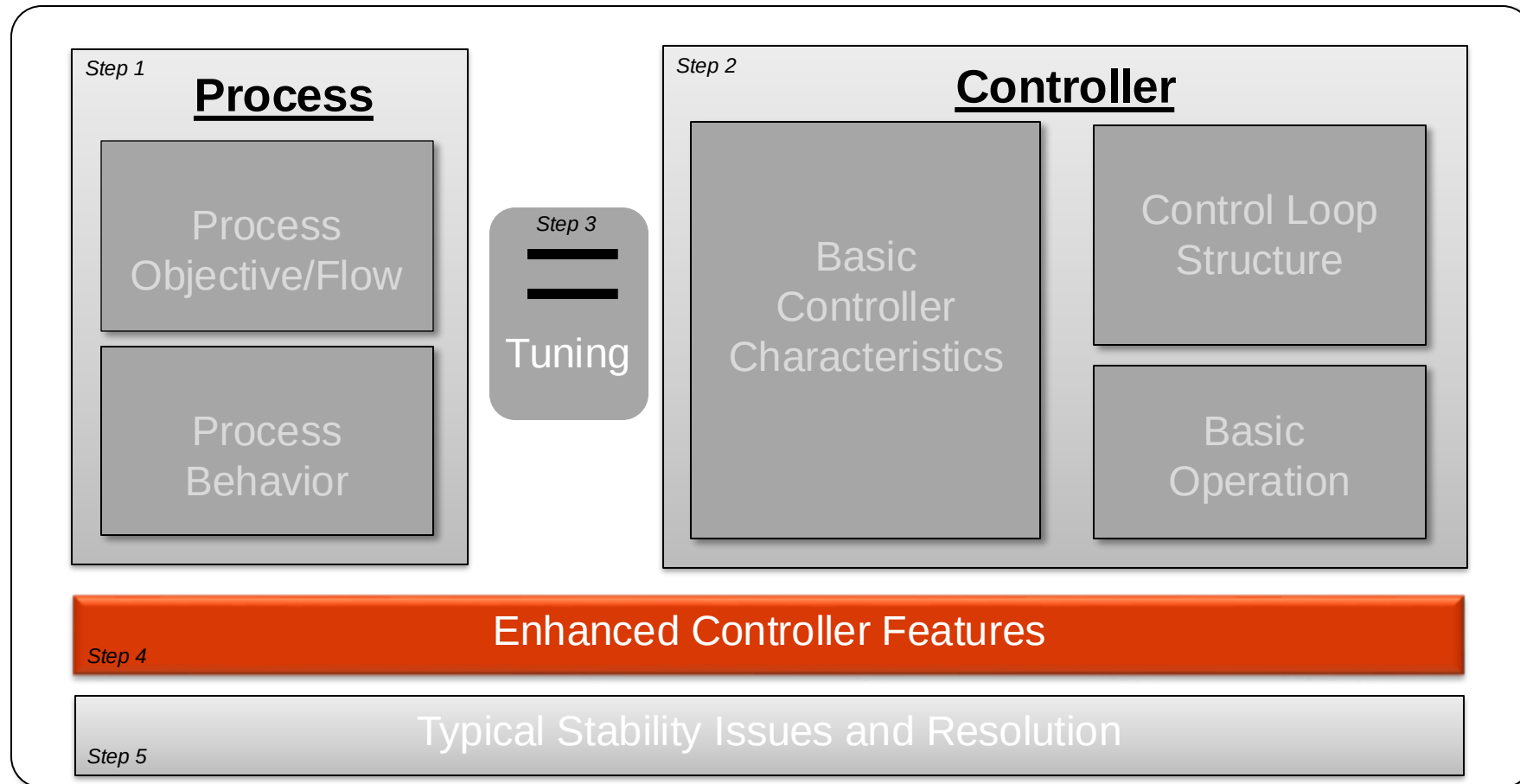
Results

For a PID (Loop 1) & PI (Loop2) Controller,

Method	Loop 1			Loop 2	
	K_c	T_i , sec	T_d , sec	K_c	T_i , sec
Z & N					
IMC Lambda					

Base Layer Control

Step 4 – Enhancing the Controller

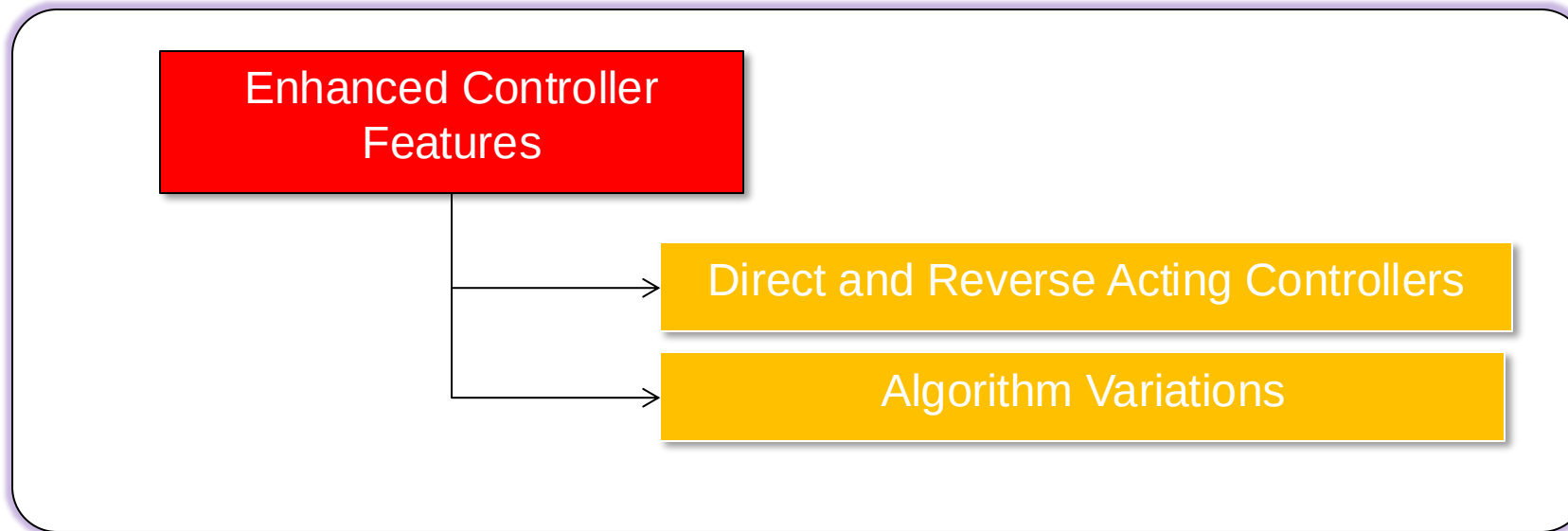


Additional Options

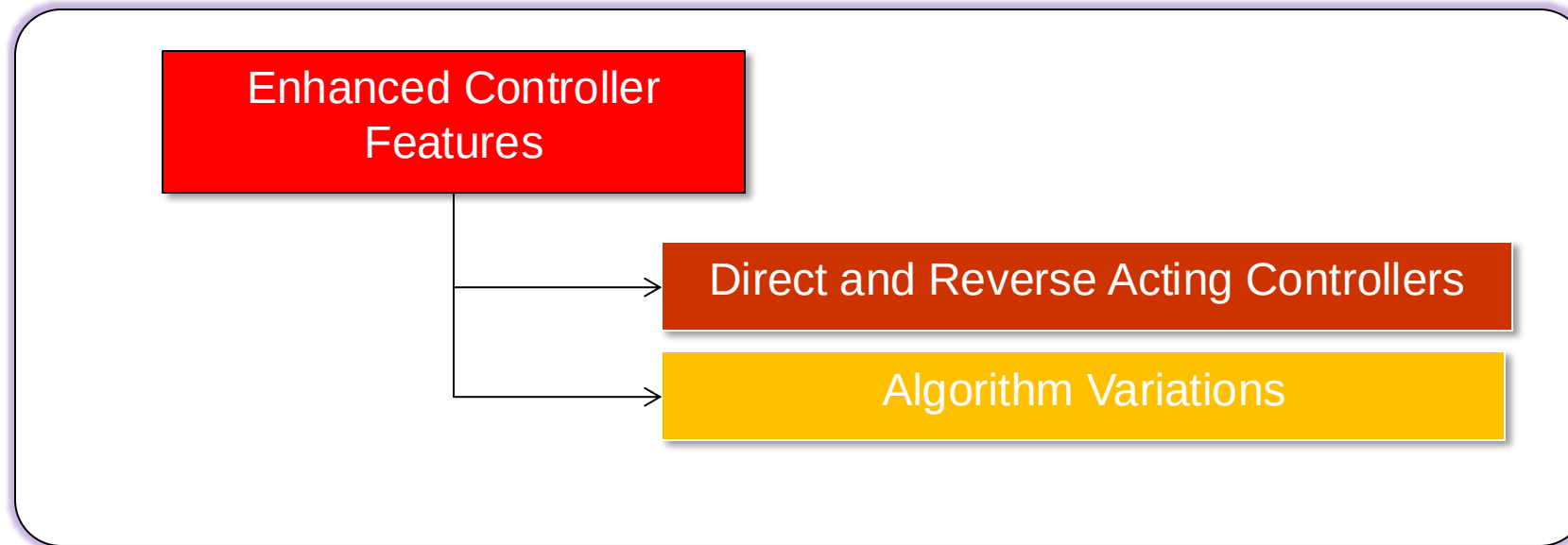
There are several options and parameters that play a part in the final controller output. This include (and are not limited to):

- Direct and Reverse acting control
- Algorithm Variations
- Digital Implementations (not covered under this course)

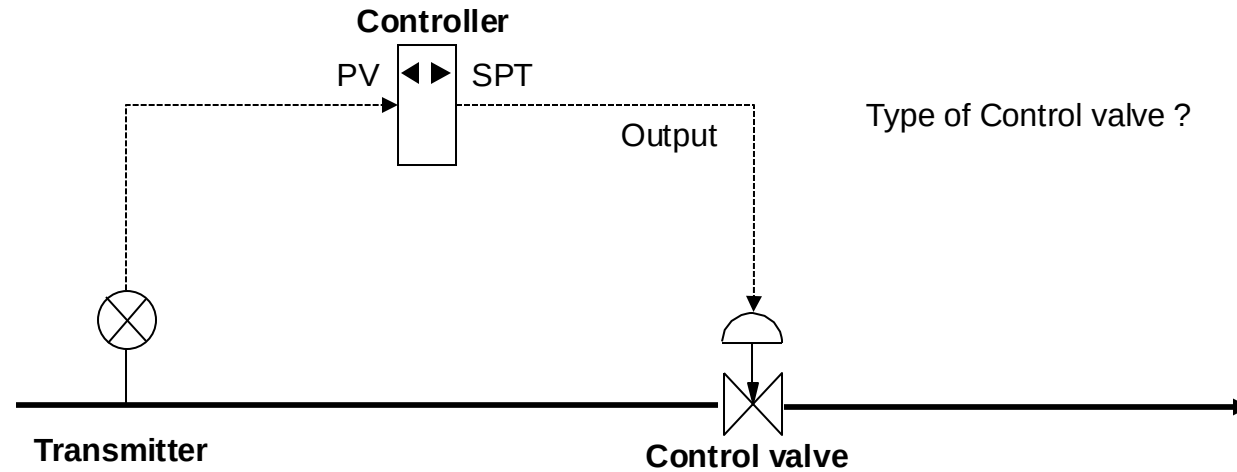
Step 4 – Enhanced Controller Features



Step 4 – Enhanced Controller Features



Direct and Reverse Acting Control



For the Process

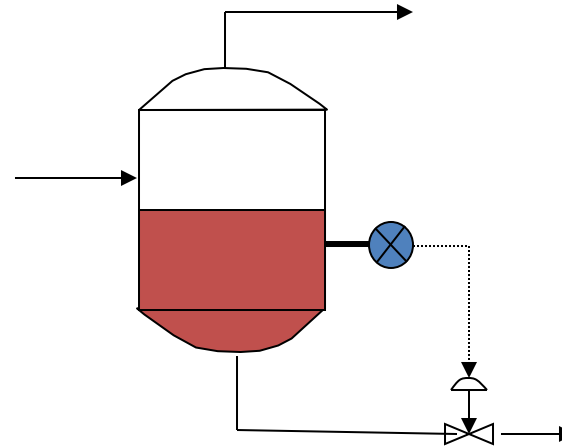
- As the measured flow increases, the valve should _____ (open/close)
- As the measured flow decreases, the valve should _____ (open/close)

For the Valve

- As the signal to the valve increases, the valve will _____ (open/close)
- As the signal to the valve decreases, the valve will _____ (open/close)

Therefore, as the measured flow decreases, the controller output should _____ (increase/decrease)

Direct and Reverse Acting Controller



Type of Control valve ?

For the Process

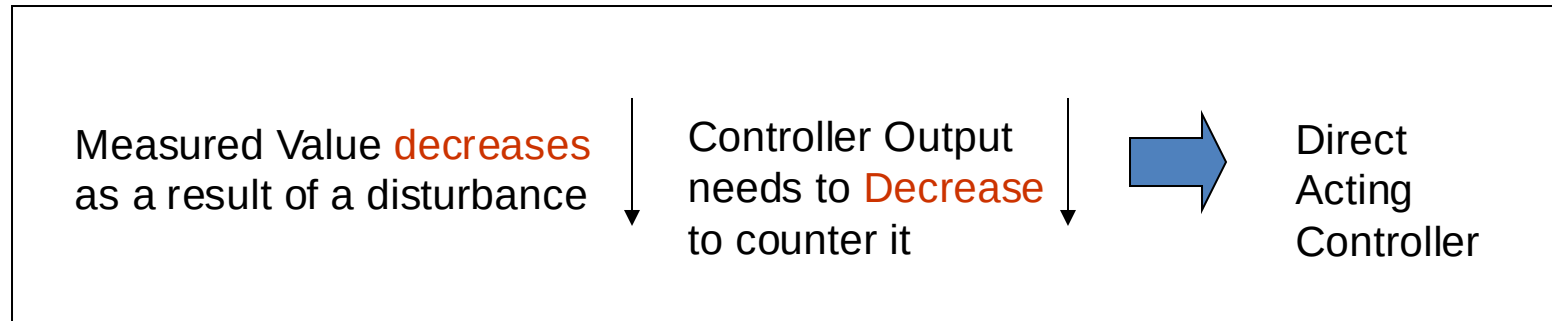
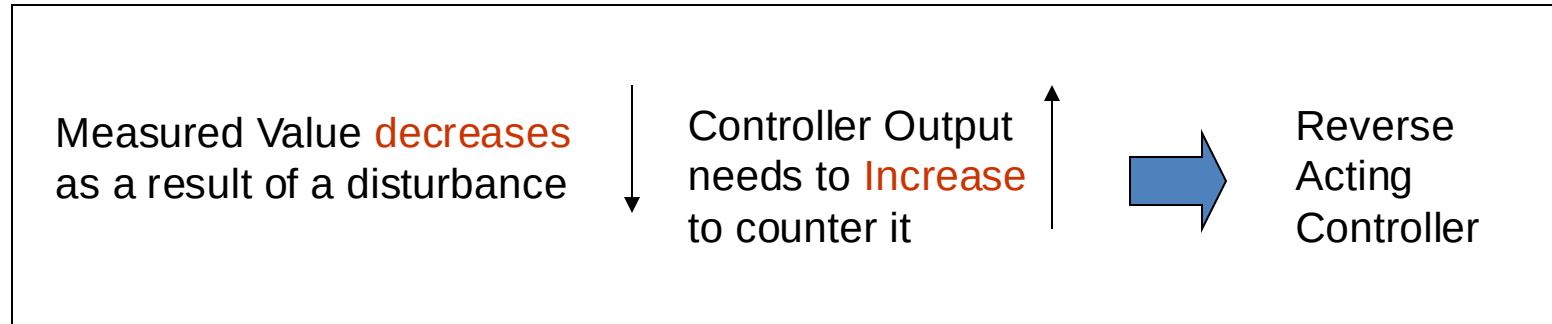
- As the level increases, the valve should _____ (open/close)
- As the level decreases, the valve should _____ (open/close)

For the Valve

- As the signal to the valve increases, the valve will _____ (open/close)
- As the signal to the valve decreases, the valve will _____ (open/close)

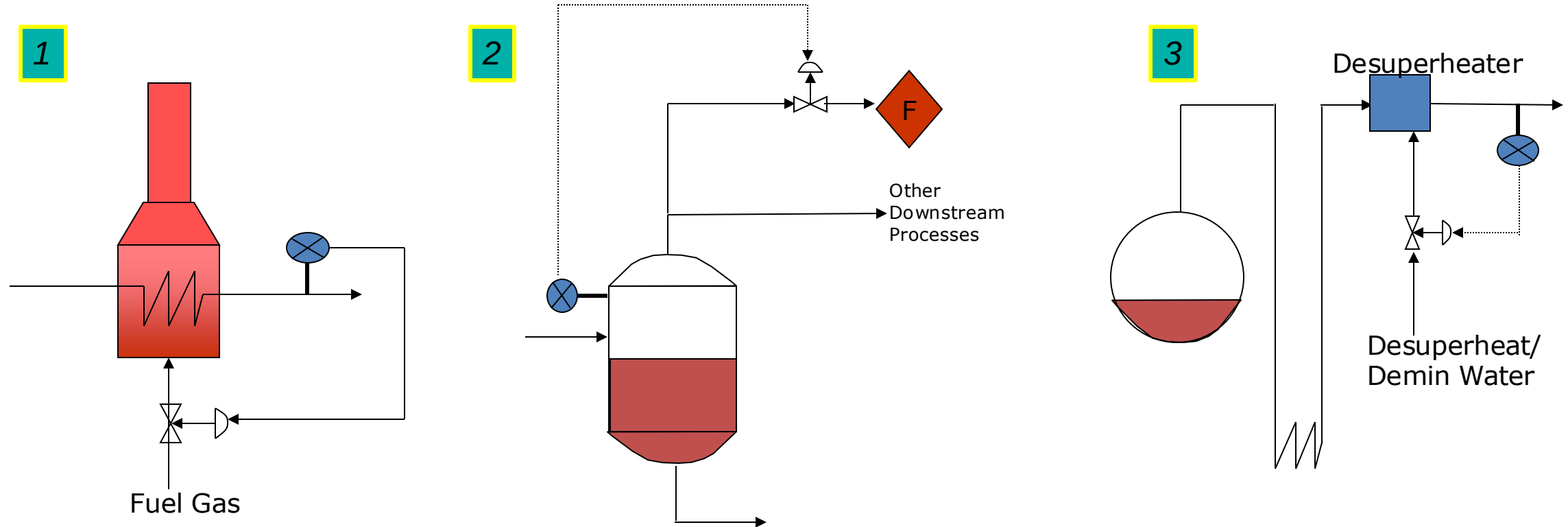
Therefore, as the level decreases, the controller output should _____ (increase/decrease)

Direct and Reverse Acting Controller



Action Of Controller

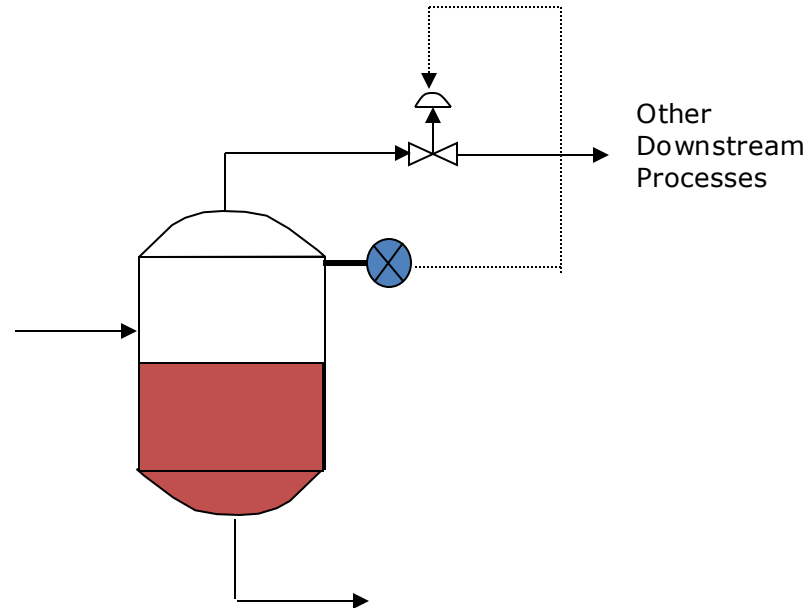
Determine Action of Controller (Direct/Reverse acting)



DCS Standardisation

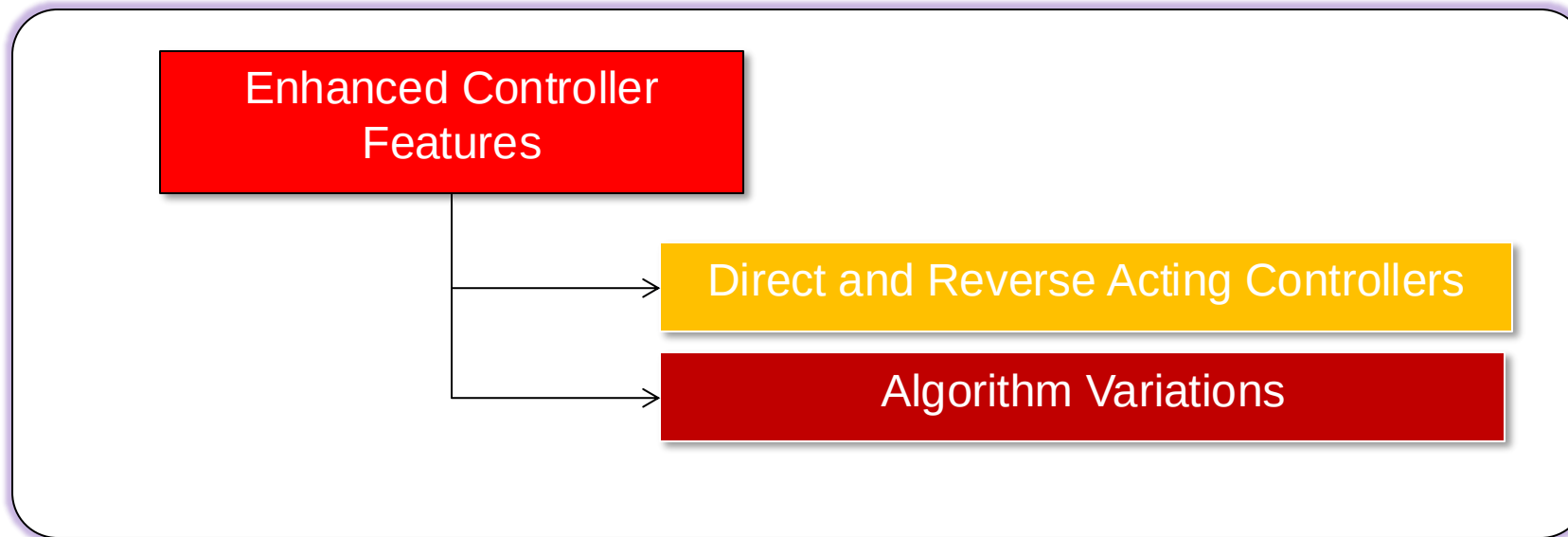
- Note that in PETRONAS and generally as a standard, it has been decided that the output of a controller shall be configured in the following manner to avoid confusion among operators:
 - 0% - valve fully shut
 - 100% - valve fully open
- To cater for this, signal reversing is done at the output control block for specific control loops.

The need for standardisation



- Revisiting the controller actions, determine action of controller with standardisation and without standardisation of output signals?
- What is the impact of standardisation?

Step 4 – Enhanced Controller Features



Algorithm Variations

Equation Form : Interacting Controllers

$$\text{Controller Output} = K * \left[E + \frac{1}{T_i} \int E \, dt \right] \times \left[1 + T_d * \frac{dE}{dt} \right]$$

Equation Form : Non-Interacting Controllers

$$\text{Controller Output} = K * \left[E + \frac{1}{T_i} \int E \, dt + T_d * \frac{dE}{dt} \right]$$

Equation Form : Parallel Controllers

$$\text{Controller Output} = K_p * E + K_i \int E \, dt + K_d * \frac{dE}{dt}$$

Other Algorithm Variations

- Proportional action
 - Most DCS vendors offer different options for P-action
 - P-action on PV
 - P-action on Error
- Derivative action
 - Again, DCS vendors offer different options for D-action
 - D-action on PV
 - D-action on Error

Difference in Equation

Proportional & Derivative Action on Error

$$\text{Output} = K * \left[E + \frac{1}{T_i} \int E \, dt + T_d * \frac{dE}{dt} \right]$$

Proportional & Derivative Action on PV

$$\text{Output} = K * \left[\Delta PV + \frac{1}{T_i} \int E \, dt + T_d * \frac{d(PV)}{dt} \right]$$

Normally used in the DCS

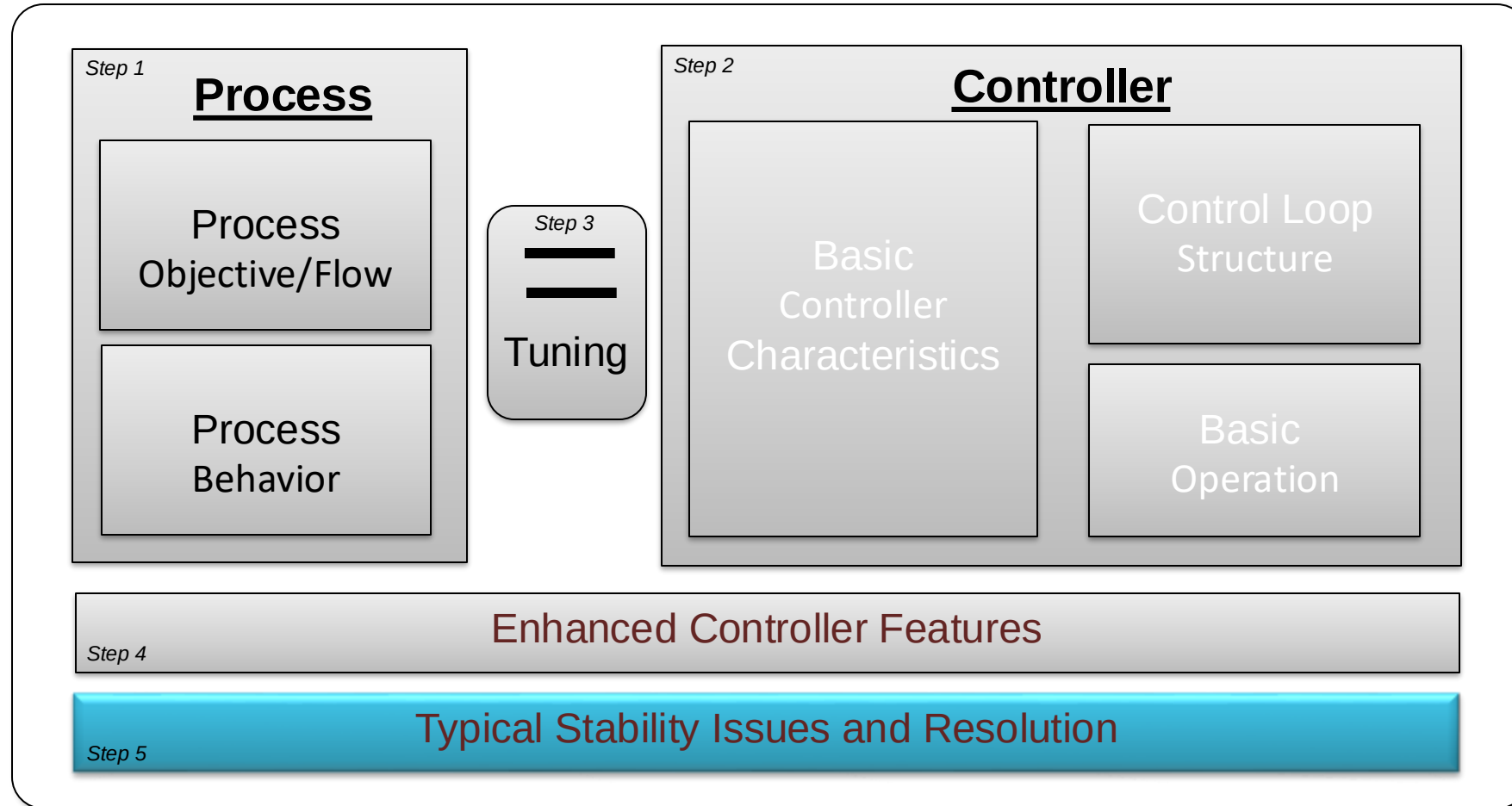
$$\text{Output} = K * \left[E + \frac{1}{T_i} \int E \, dt + T_d * \frac{d(PV)}{dt} \right]$$

Summary of Step 4

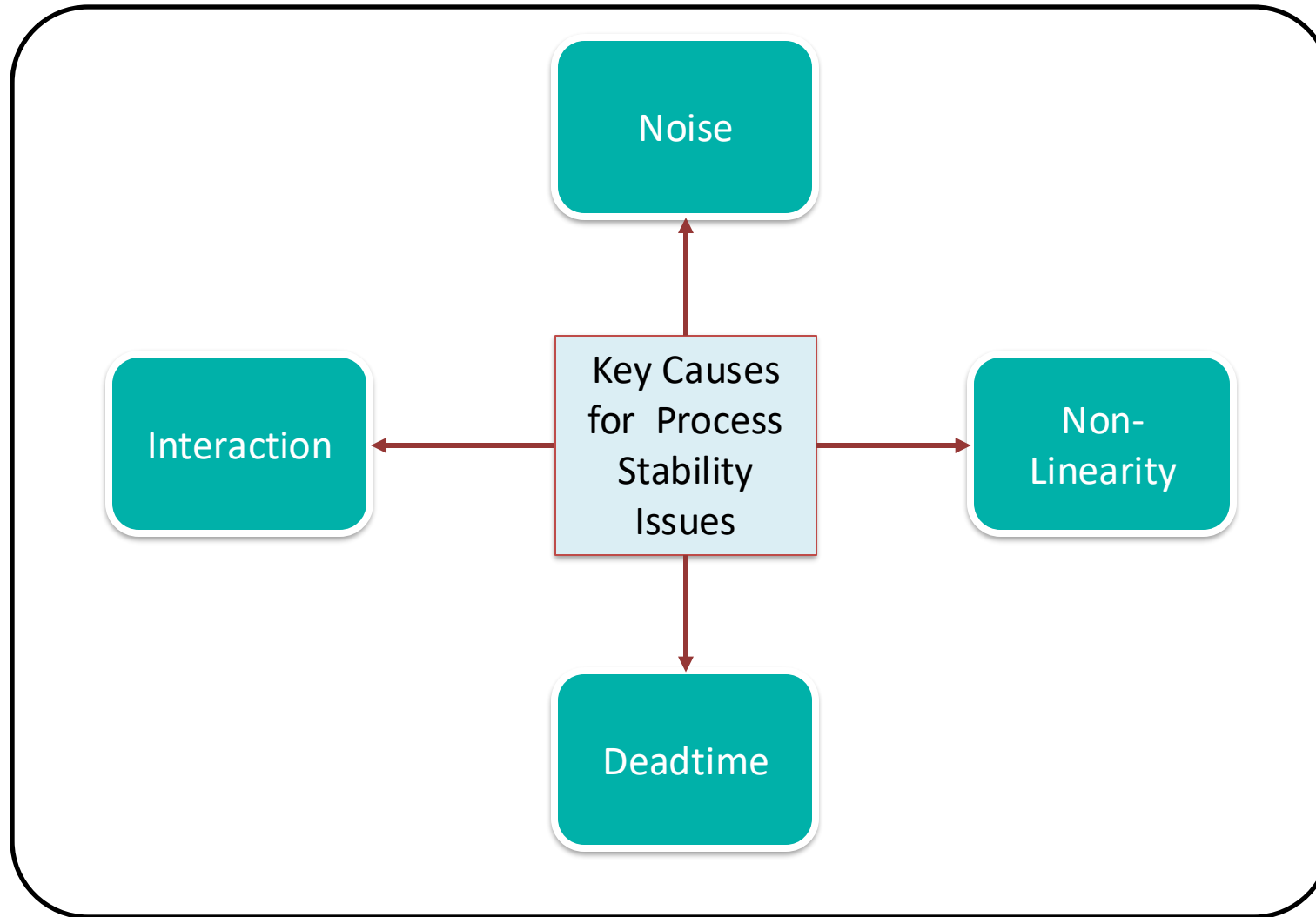
Key Take Aways

- Understand Control Direction
 - Direct & Reverse Acting
 - Look at PV move direction
 - If need to close the valve as the PV rises then it is reverse acting
 - If valve needs to open as PV rises, it is a direct acting controller
- Algorithm variations
 - Proportional and Derivative actions: on Error or PV

Scope of Base Layer Controls



Key Stability Issues



Typical Stability Issues

- Noise
 - These are signals generated either by the measuring element itself (e.g. flow orifice) or induced into the electrical signal lines THAT ARE NOT related to the process signal itself
- Non-linearity
 - Most processes are only linear in a small range of operation. Outside of this range, dynamics will start to change. The extent of the change in dynamics defines how nonlinear the system is.
- Deadtime
 - The difference when a process is manipulated is different between a flow process and a temperature process. The key difference is the time taken for the process to respond. The time the process takes to respond is the process deadtime. The longer it takes to respond, the more difficult it is to control.
- Interaction
 - Some processes such as distillation are inherently interactive. This means changing one parameter (say reflux) will change top product quality BUT at the same time change bottom product quality. The same can be said for bottoms reboiler. The manipulation of one handle will usually result in the manipulation of another handle.

Solutions

Noise

- Rectification of instrumentation
- Installation of Filter (beware)

Non-Linearity

- Not resolvable at single loop level
- Gain scheduling

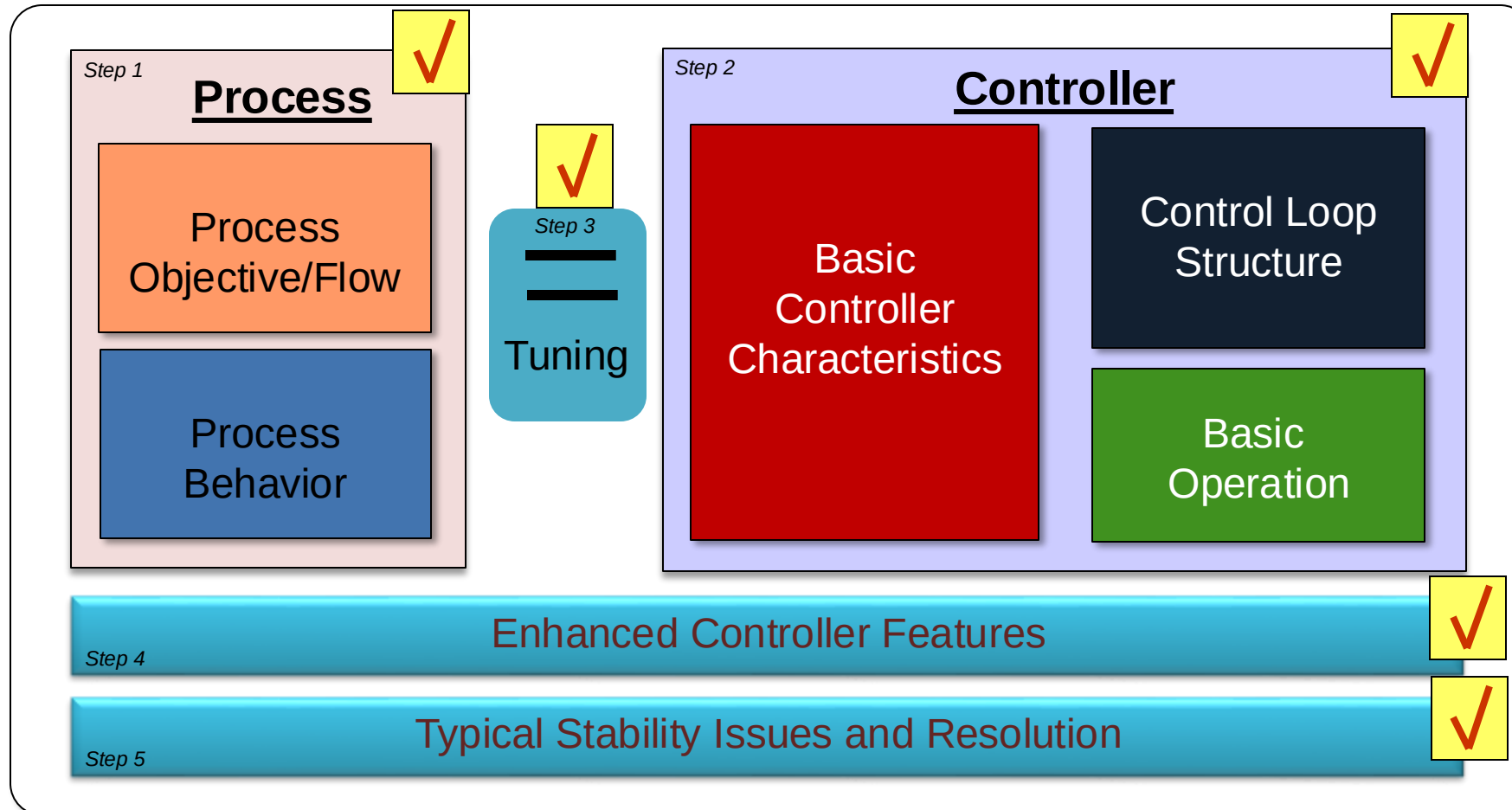
Deadtime

- Not resolvable at single (basic) loop level
- More complex applications such as “Smith Predictors” provide deadtime compensation
- Alternatively, APC may be used

Interaction

- Not resolvable at single (basic) loop level
- Static decouplers may be applied
- More complex applications such as APC may be used

Scope of Base Layer Controls



LAYER 2

Intermediate Layer Controls

Intermediate Layer Controls (ILC)

Base Layer Controls refer to feedback control loops that have a single input and a single output.

However, a number of processes may rely on one or more process inputs and one or more process outputs.

Definition

Intermediate Layer Controls refer to control loops that are more complex than Base Layer Controls but simple enough, that a comprehensive Control Narrative is not required to explain its functionality

Types of ILC

There are four primary types of loops in the Intermediate Layer:

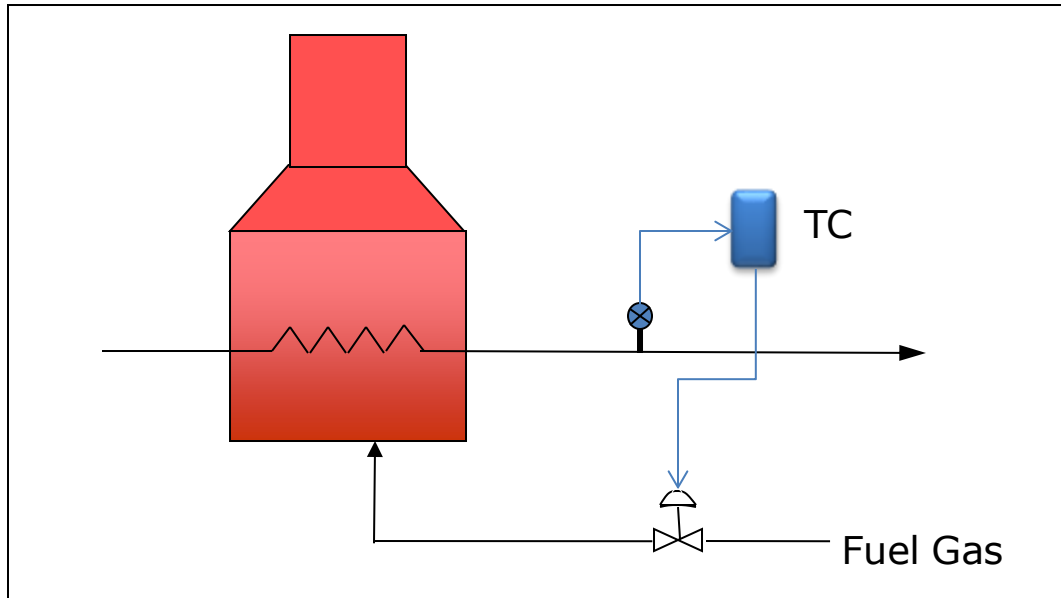
- Cascade Control
- Selective or Override Controls
- Split-range controls
- Ratio Controls

Cascade Control

Cascade control is also called Master-Slave control.

It is often used when an intermediate process variable can be utilised and control to prevent instability to a more primary process variable.

Furnace Process



- i) Consider this furnace process.
- ii) Feed flow is natural gas at a flow of 100 t/d
- iii) The outlet temperature needs to be controlled to 350C for the downstream process
- iv) The fuel gas header pressure is constant at 1.8 barg
- v) The temperature control is in automatic and stable

CASE 1

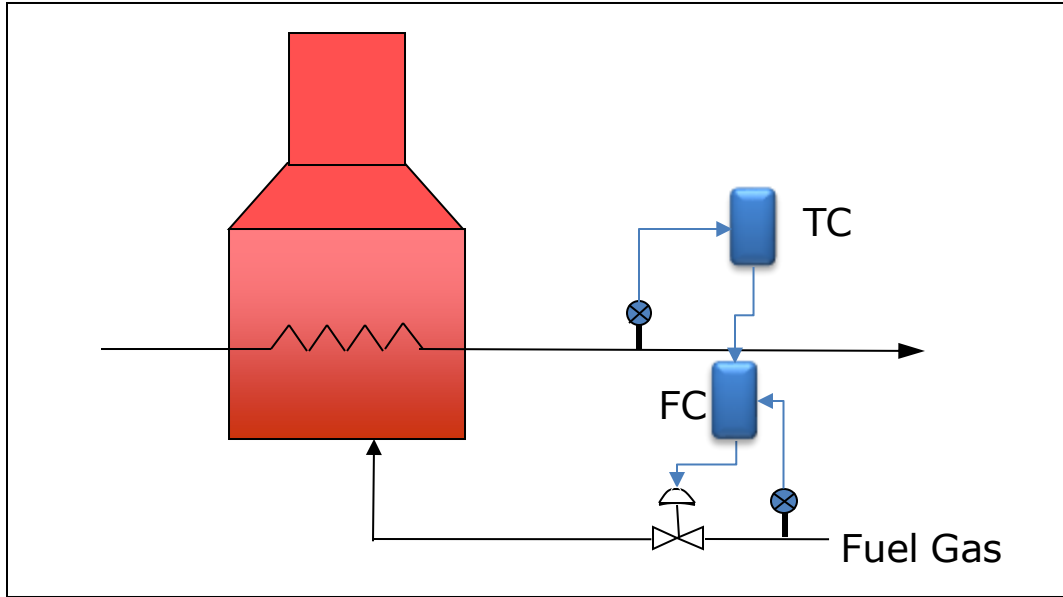
- i) Consider what happens step by step when there is a sudden feed increase by 10%.

CASE 2

- i) Consider what happens step by step when there is a sudden increase in the fuel gas pressure (increase to 2.2 barg by sudden loss of major consumer)

Which of the above cases is less acceptable?

Furnace Process Solution



- i) Consider this furnace process.
- ii) Feed flow is natural gas at a flow of 100 t/d
- iii) The outlet temperature needs to be controlled to 350C for the downstream process
- iv) The fuel gas header pressure is constant at 1.8 barg
- v) The temperature control is in automatic and stable

By installing a Flow Control, disturbances in the Fuel Gas header is immediately corrected at the SLAVE flow controller. In this case, disturbances in the slave process are prevented from being transmitted to the MASTER process.

Is there an alternative loop that can be utilised?

Terminologies

- A single loop controller can be placed in two modes:
 - Automatic
 - Manual
- With cascade functionality, the controller can be placed in two further modes pertaining to where it receives its setpoint:
 - Local (setpoint is set locally by the operator)
 - Remote/Cascade (the setpoint is received automatically from a remote source)

Cascade Concept

There are two distinct advantages gained with cascade control:

- Disturbances affecting the secondary variable can be corrected by the secondary controller before it affects the primary variable greatly
- Closing the control loop around the secondary part of the process increases the speed of response (by reducing the phase lag seen by the primary controller)

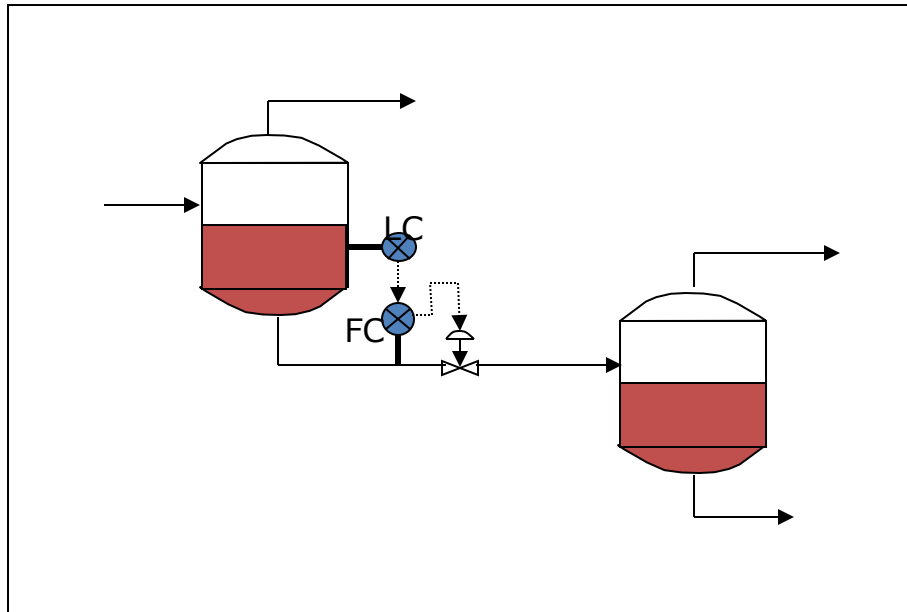
Important Factor

- The primary dynamics in a cascade function must be in the primary loop (i.e., the master loop must always be slower than the slave).
- Which of the following cascades will NOT work:
 - A) Level (Master)-Flow (Slave)
 - B) Flow (Master)-Level (Slave)
 - C) Temperature (Master) – Flow (Slave)
 - D) Flow (Master)- Temperature (Slave)
 - E) Flow (Master)- Pressure (Slave)
 - F) Temperature (Master)-Temperature (Slave)

Design Notes

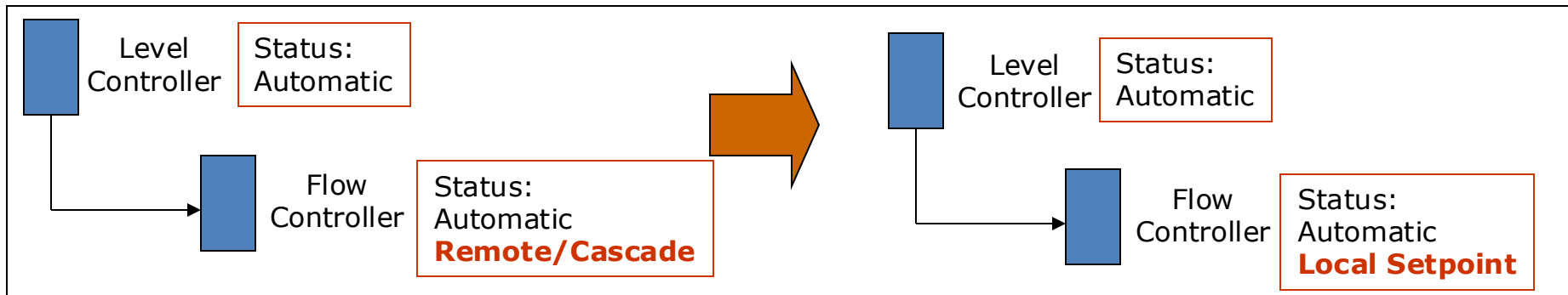
The output of the Master (Primary) controller is connected to the Remote Setpoint or Cascade Setpoint of the Slave (Secondary) controller.

Concern



Consider the process shown.

- The level in Column 1 (C1) is at 50 %, While the flow is at 50 t/d. System is full automatic and the cascade is connected.
- The cascade link is broken and the controller is placed on Local setpoint control.
- The setpoint of the flow controller is increased from 50 t/d to 60 t/d.
- Consider what will happen.



Typical Design

In typical design for cascade loop implementation in DCS systems, the following rules are followed:

- If the secondary (slave) controller is placed on local (cascade broken), the primary controller is immediately told to stop computing its PID algorithm.
- The setpoint of the primary (Master) is immediately made to track the measured value
- The output of the primary (Master) is immediately made to track the measured value (or setpoint) of the slave.

Tuning a Cascade Loop

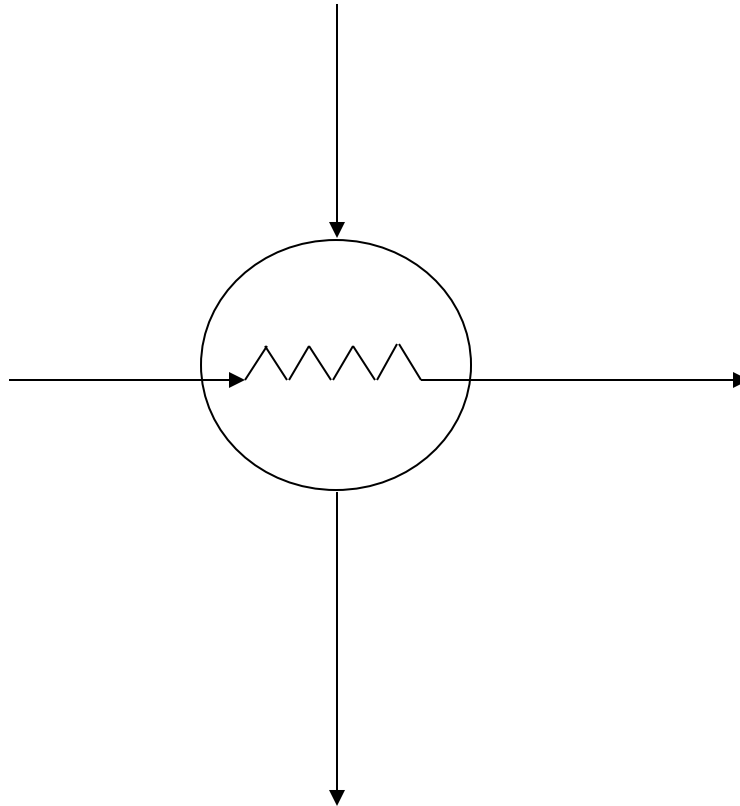
As seen earlier, the slave controller is always faster in total response than compared to master.

As a result, in all cases, the basic rule for tuning a cascade loop is to tune the slave (cascade) loop first, followed by the Master.

Steps for tuning Cascade Loop

- Tune secondary first
 - Tune as per single loop controller
- Put secondary on automatic and step its SP to obtain response curve for tuning primary
- Tuning constants in secondary are “seen” as part of the process by the primary - if they are changed then it will be necessary to retune primary

Exercise



Objective

To improve on the single loop temperature controller. Reboiler uses steam that will fully condense. The process fluid is in the tubes.

Retrieval Practice

1. The purpose of a cascade or master-slave loop is:
 - a) Ensure the disturbances in the slave loop remains in the slave loop
 - b) Ensure disturbances in the master loop does not impact the slave loop
 - c) Ensure disturbances in the slave loop does not impact the master loop
 - d) Ensure stability of the slave loop

2. There are a few configuration details associated with master slave loops. One of the items below is not one of them:
 - a) The master's output needs to be configured to the slave PV range
 - b) The master's output needs to track slave's setpoint when the slave is in manual
 - c) The master controller will need to ensure that its setpoint is tracking its PV (even though it is in Automatic. Remember, master has only Auto or Manual mode)
 - d) When the slave is in manual, the output of the master needs to track the PV of the slave

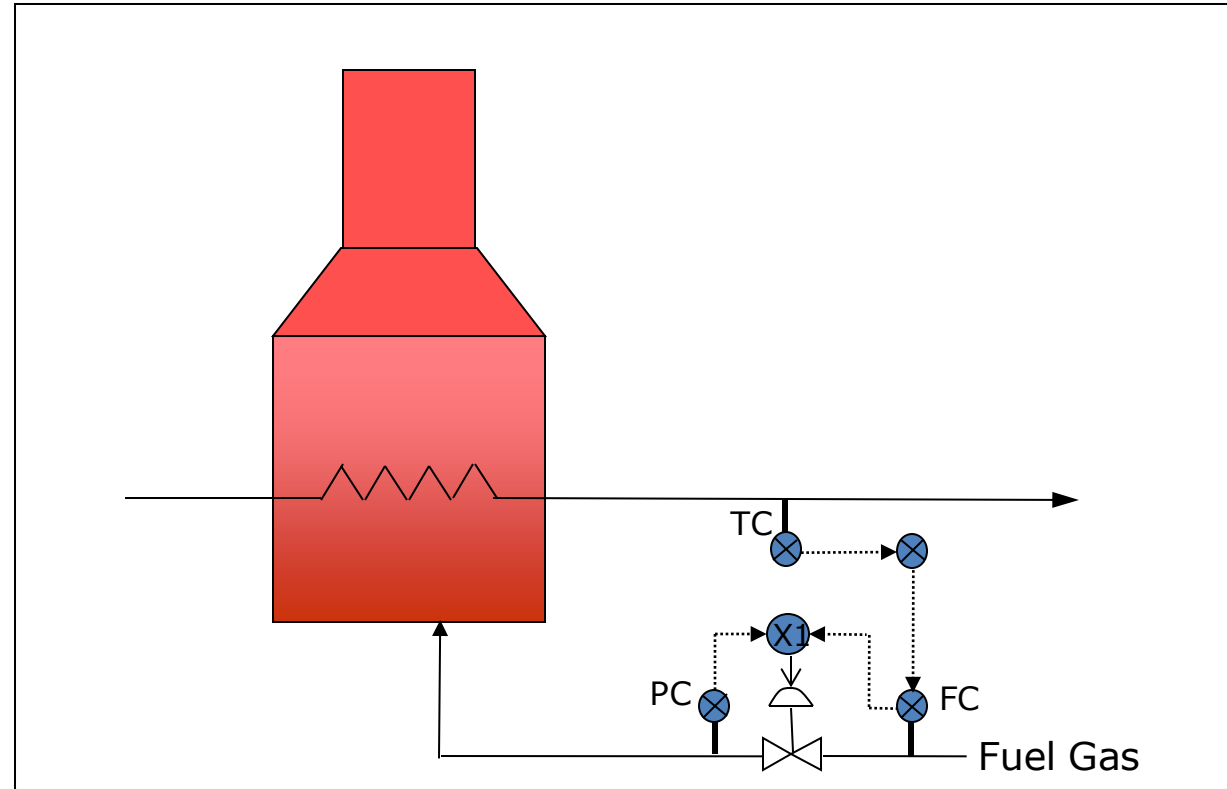
Retrieval Practice

3. While tuning the Master-Slave loop, what important concept needs to be understood?
- a) The Slave recognises the existence of the Master – the master needs to be tuned first
 - b) The Master does not recognise the existence of a slave controller and will consider the slave loop as a final element – as such the slave needs to be tuned first
 - c) The Slave does not recognise the existence of a Master Controller and will consider the Master loop as a final element – as such the slave needs to be tuned first
 - d) The Master recognises the existence of the slave – the master needs to be tuned first

Selective or Override Controls

Selective or override controls are utilised when only one of the parameters is normally in control while the other controller acts in an emergency.

Example



The objective of the scheme is to ensure process temperature is maintained at specification, while ensuring the integrity of the burner in the furnace.

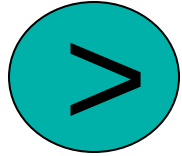
- i) What is the function of the PC?
- ii) What is the Control Action of the PC (Reverse/Direct)
- iii) What type of selector is X1

Basic Guidelines

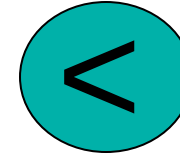
In designing a selective or override loop, a few factors need to be considered:

- Define clearly which is the normal controlling parameter and the override control
- Determine the control action of the override controller to take the process to safety
- Identify the type of signal selector, high signal selector/low signal selector to be utilised

Notation



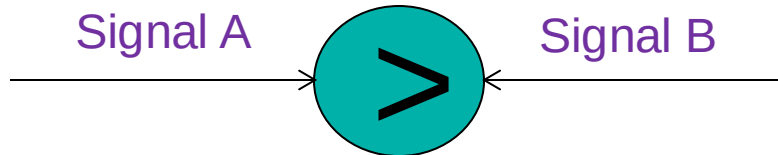
This is a **H**igh **S**ignal **S**elector
(acronym: HSS).



This is a **L**ow **S**ignal **S**elector
(acronym: LSS).

Note: Do not take it as a mathematical equation, where the symbol in the middle compares the signals coming into it from the different controllers, e.g.

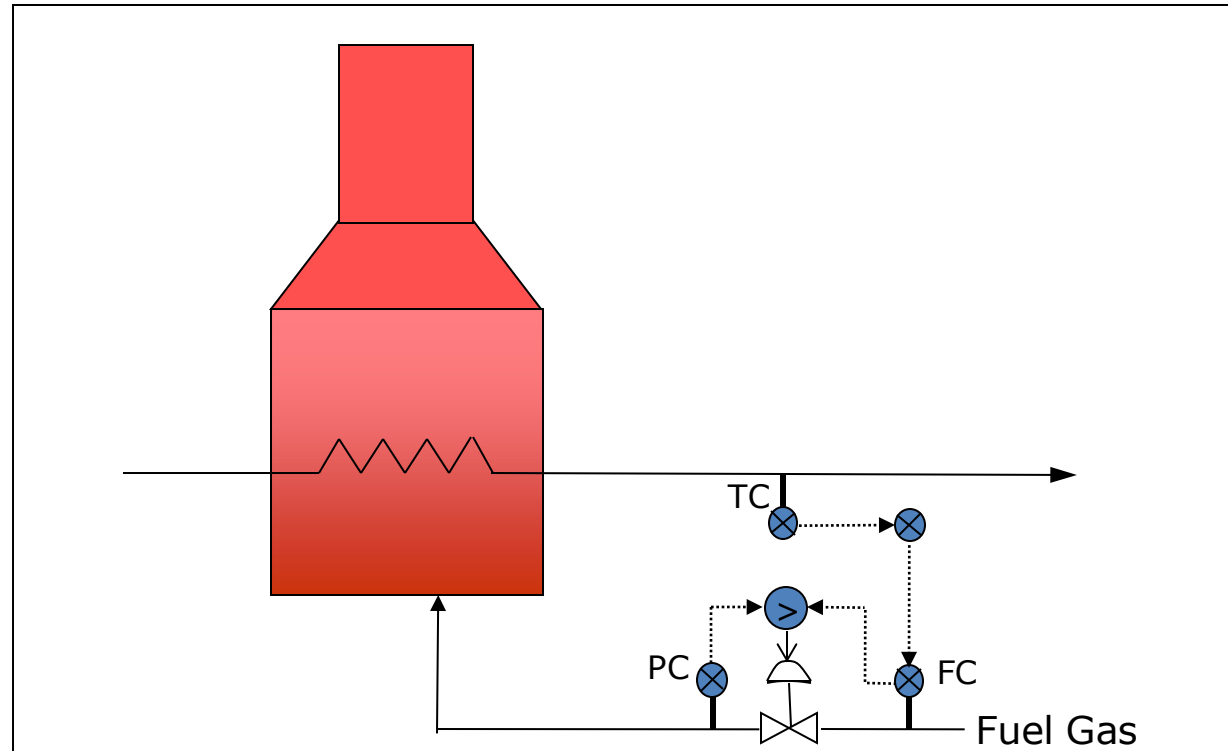
Signal A



Signal B

DOES NOT mean Signal A > Signal B!!

Saturation



Consider the above process with a low-pressure PC override:

The flow controller is normally in control. The pressure in FG line is normally at 1 barg. The setpoint of the PC is at 0.1 barg, below which it will increase the flow through the valve.

- Knowing that the flow controller is in Auto and the pressure is healthy, what is the impact on the Pressure Controller (PI controller)?

Reset Wind-up

- In all cases of selective controller, where the non-selected controller is a PI controller, the non-selected controllers' output will tend to saturate to 0 or 100%
- This process of saturation is called Reset/Integral Wind-up.
- Once saturation occurs, it takes a long time for the “protective” controller to take control of the process in an abnormal scenario.

Preventing Reset Wind-up

- To prevent reset wind-up (or to achieve anti-reset windup), the final output of the selector is used as an external feedback to both the controllers.
- This external feedback will tell the non-selected controller that its output is blocked and will thereafter limit the amount of saturation.

Design Considerations – Operational mode changes

What will happen if the override controller is put to Manual?

Retrieval Practice

1. The method to identify an override control loop is as follows:
 - a) 2 control loops and one ratio block
 - b) 2 control loops and one conversion block
 - c) 2 control loops and one selector block
 - d) 2 control loops and one calculation block

2. The configuration of an override loop is always as follows:
 - a) 2 control loops and one is the master, the other being the slave
 - b) 2 control loops and one is normally in control, while the other will cut in under abnormal conditions
 - c) 2 control loops and the primary control loop will always decide which loop is in control
 - d) 2 control loops and the loop that is always in control is called the override control loop and must be tuned very differently

Retrieval Practice

3. The method to select a high or low signal selector is as follows:
 - a) If the normal control requires to close the valve, the selector shall be low signal selector
 - b) If the normal control requires to open the valve, the selector shall be high signal selector
 - c) If the abnormal control is required to close the valve, the selector shall be a high signal selector
 - d) If the abnormal control is required to open the valve, the selector shall be a high signal selector

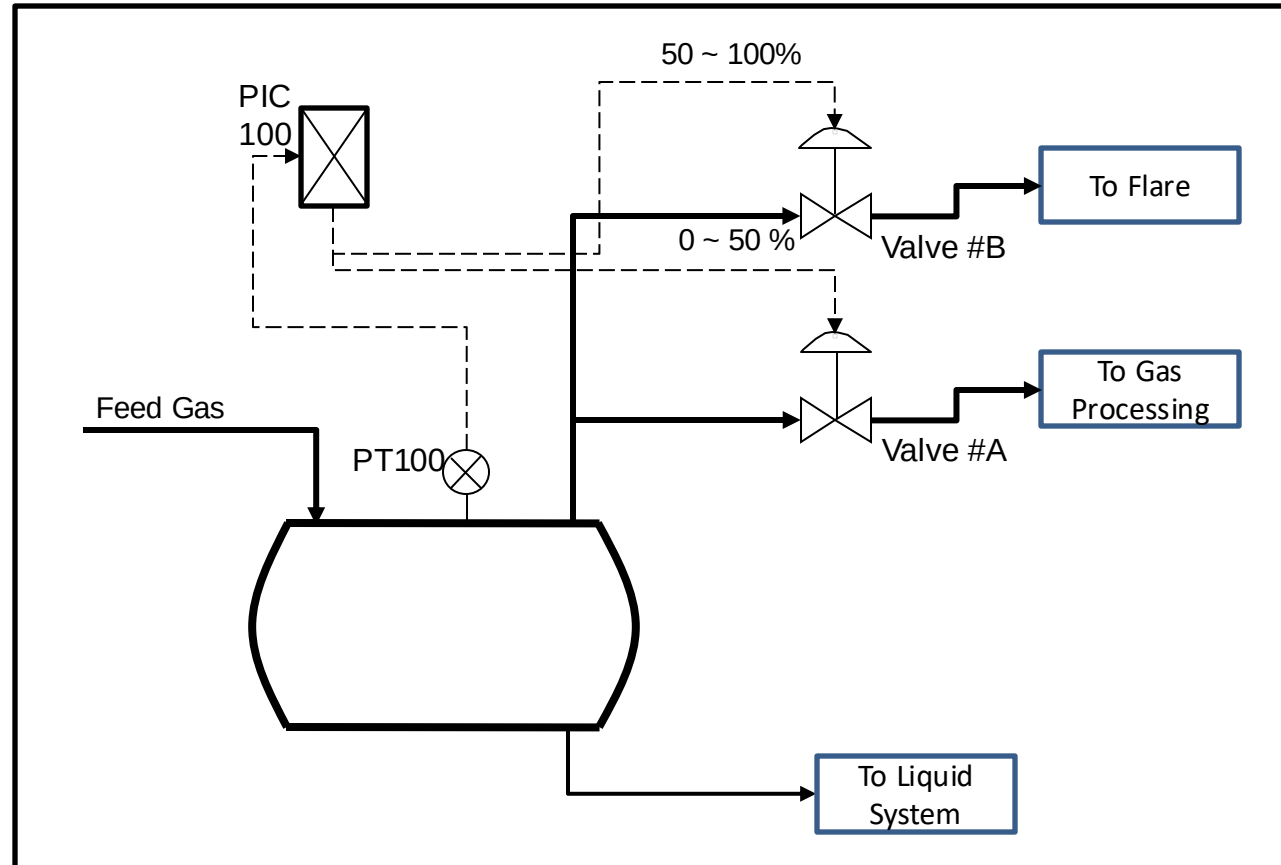
4. The primary issue in an override control is:
 - a) When one of the controller is the active controller, it will tend to saturate – this will always happen to the emergency controller
 - b) When one of the controller is the active controller, the other controller will tend to saturate – this will always happen to the emergency controller
 - c) When one of the controller is the active controller, the other controller will tend to saturate – this will happen to both controllers
 - d) What issue?

Split Range Control

In both Cascade Control and Selective Control, there are two controlled parameters and one manipulated handle.

In the case of Split range control, there is one controller and two (or more) manipulated handles.

Split-Range Control



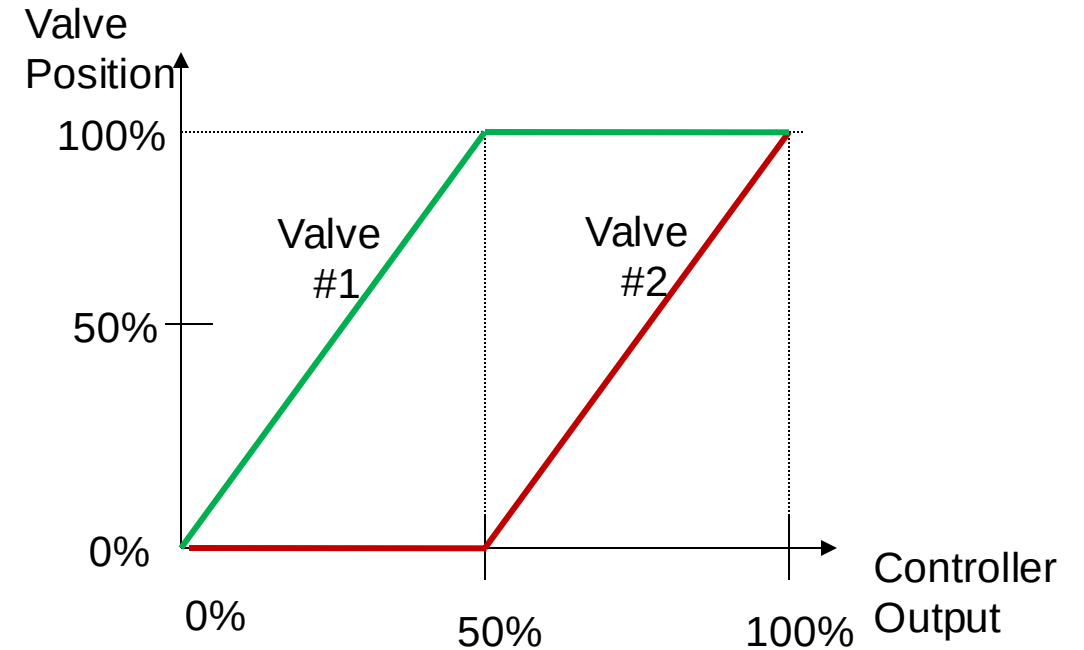
Control Objective:

- Normal pressure control is handled by controlling flow to Gas Processing.
- If pressure in the vessel rises too high, the valve to flare will open.

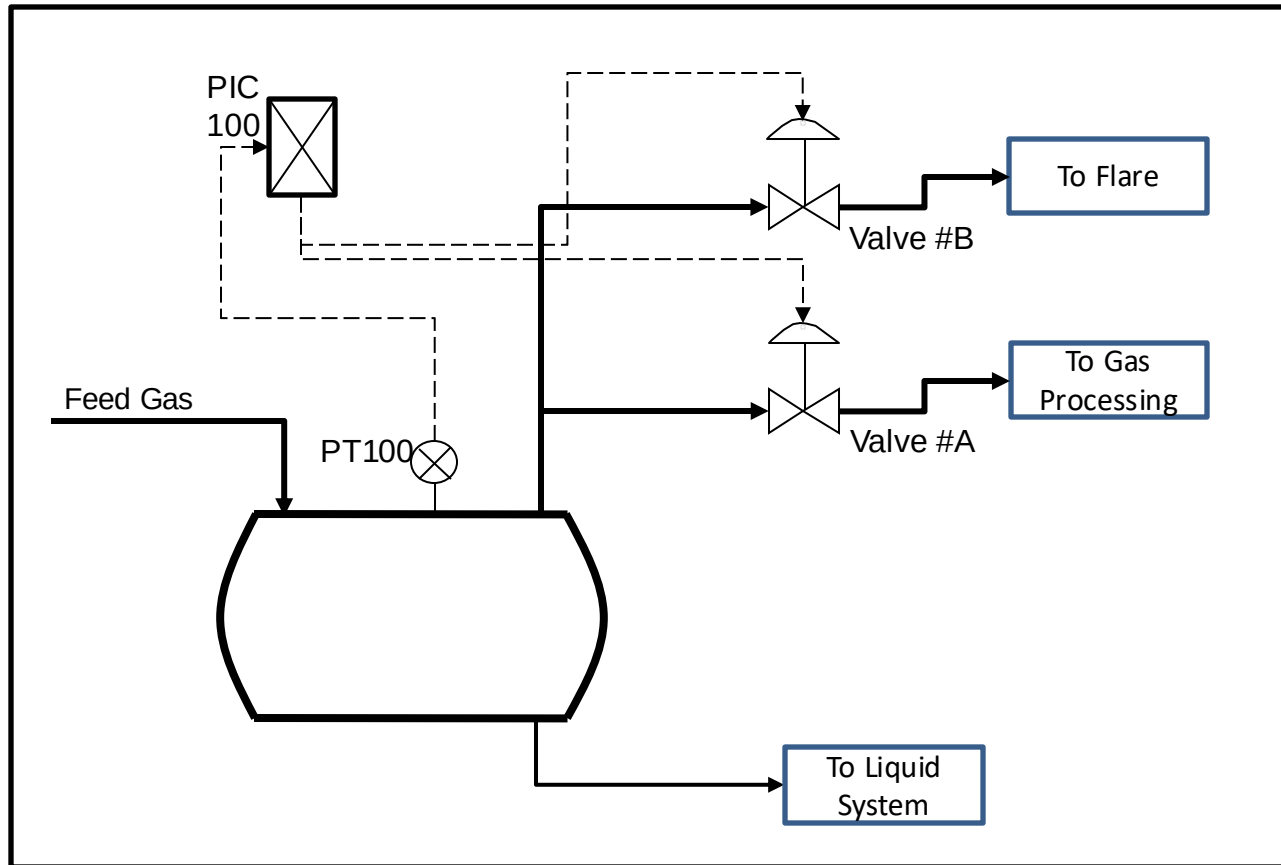
What is the action of the controller – direct or reverse?

Design Notes

- The split refers to the percentage of control signal that will affect each valve.
- At the DCS level, each of the signal is converted to 0 ~ 100 % signal to the valve.
 - As such, a 25% signal from the controller in the previous diagram will equate to :
 - 50 % signal to Valve #1
 - 0 % signal to Valve #2
- The split of the control shall ensure that over the full range of the controller output, each % change in controller output shall result in equal value changes in the process parameter.



Worked example . . . 1/2

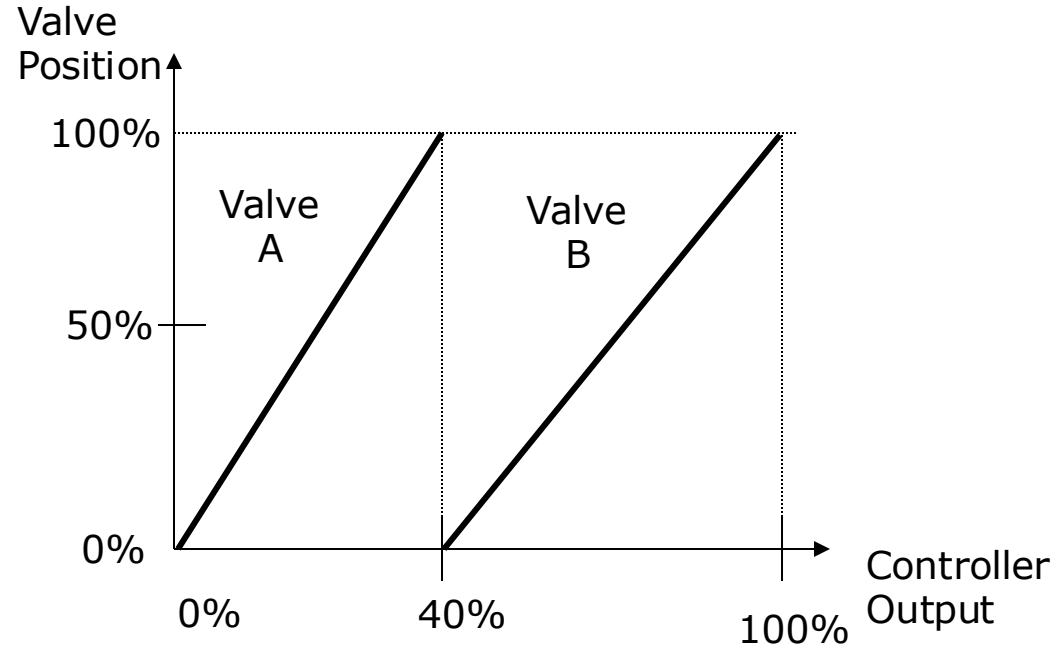


Control Objective:

- Normal pressure control is handled by controlling flow to Gas Processing.
- If pressure in the vessel rises too high, the valve to flare will open.
- Valve A max flow is 200t/d
- Valve B max flow is 300t/d

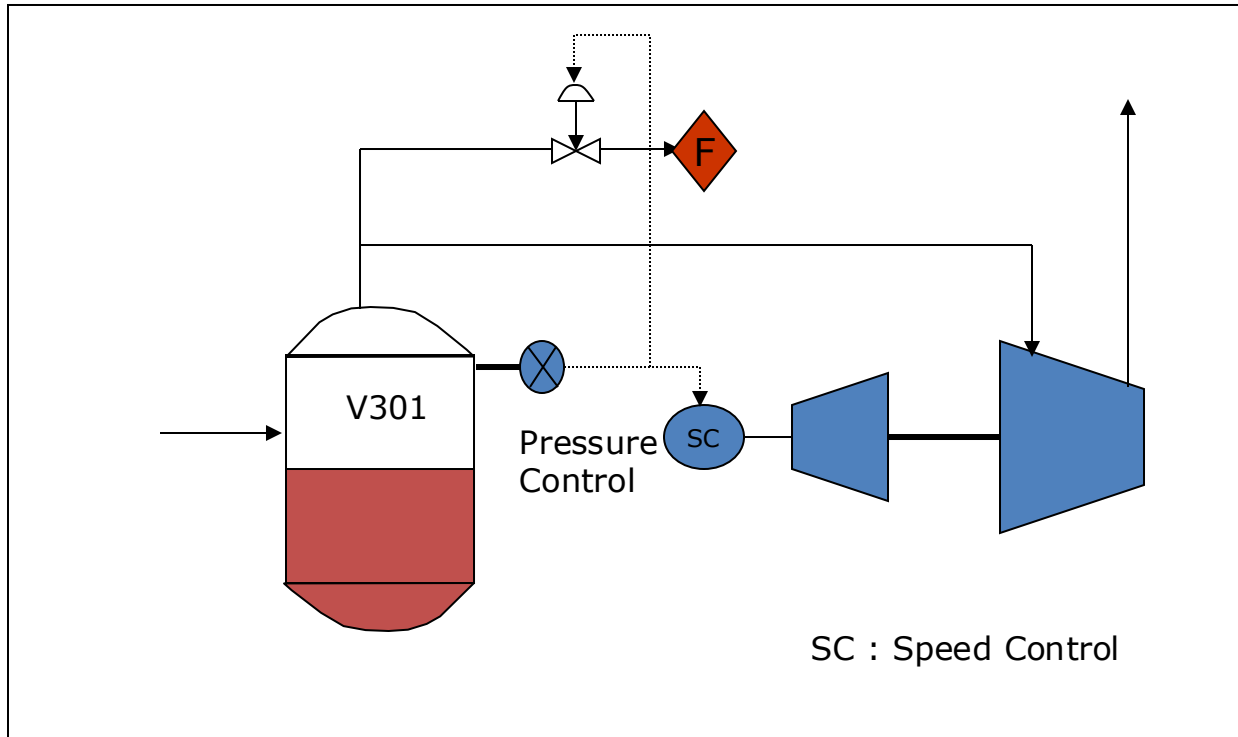
What is the action of the controller – direct or reverse?

Worked example . . . 2/2



1. Total flow = $200 + 300 = 500$.
2. Valve A = $200/500 = 40\%$.
3. Valve B = $300/500 = 60\%$.
4. Since valve B is 1.5x bigger than valve A, this means that valve B must open over a wider range by the same ratio.

Exercise

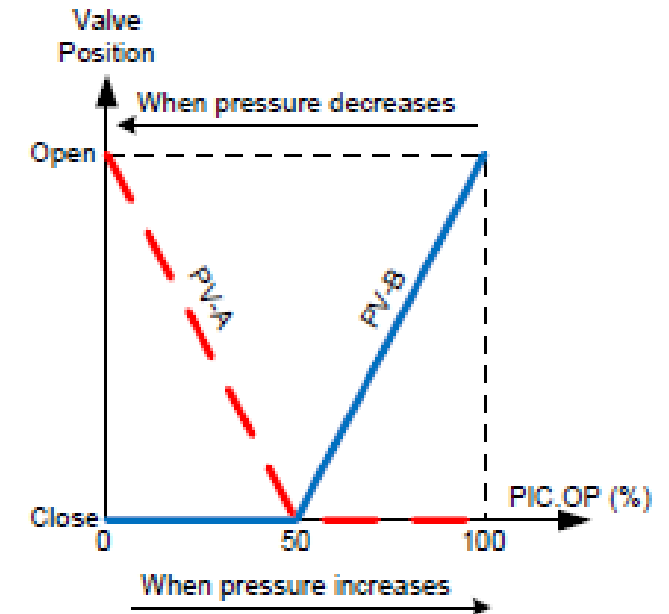
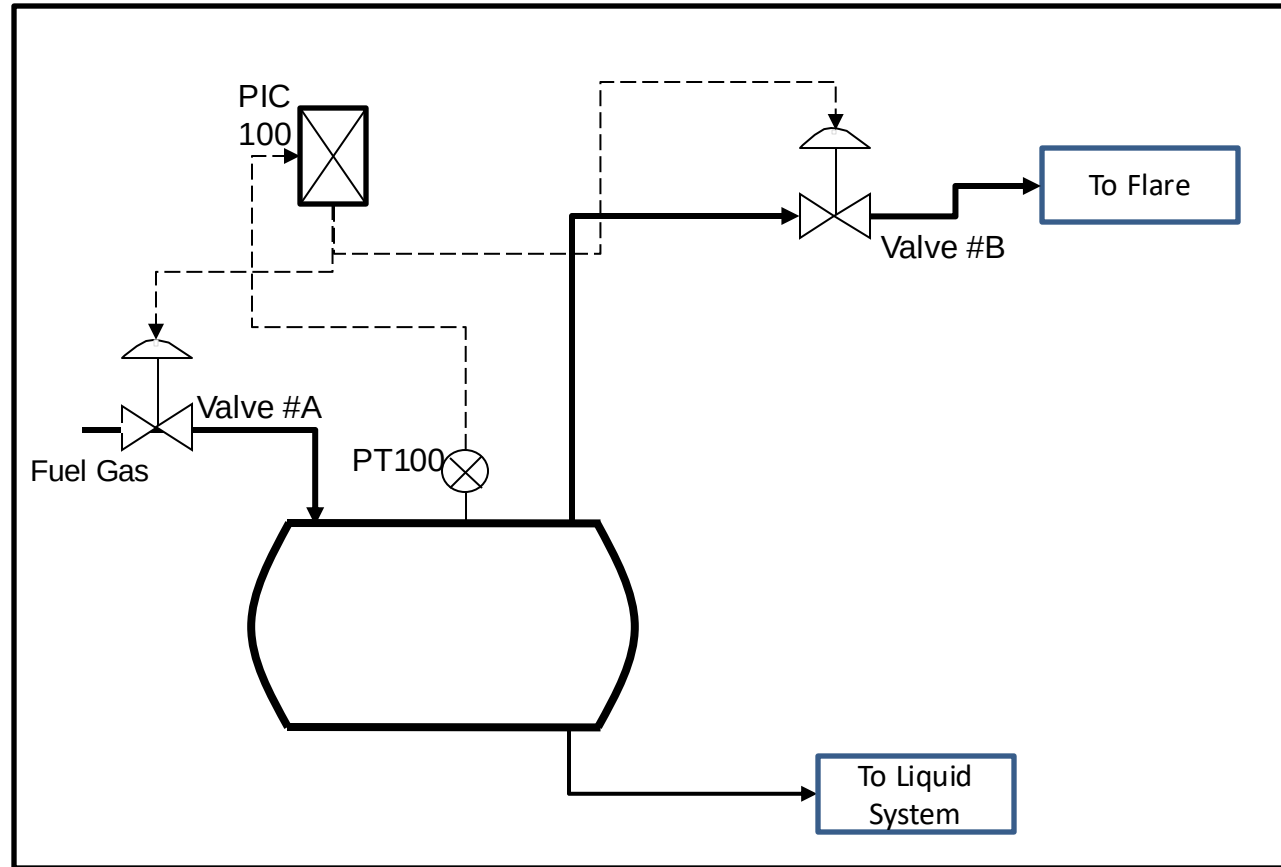


Objective: The compressor shall pressurise the hydrocarbon vapor to the Fuel Gas system.

If the pressure gets too high, the pressure control system shall open to flare. The capacity of the machine at maximum governor speed is 150 t/d while the flow capacity of the valve is 450 t/d.

What should be the split?
Draw the Valve position (lift) to Controller output chart.

Other examples of splitrange



This example is a direct acting Pressure Controller.

Retrieval Practice

1. The objective of split-range is:
 - a) To connect more than one valve to a control loop
 - b) To enhance the control of a process parameter by increasing the number of handles that has an impact on the parameter
 - c) To enhance the control of a process parameter by increasing the number of process parameters that has an impact on the first parameter
 - d) To enhance the control of a handle by increasing the number of process parameters

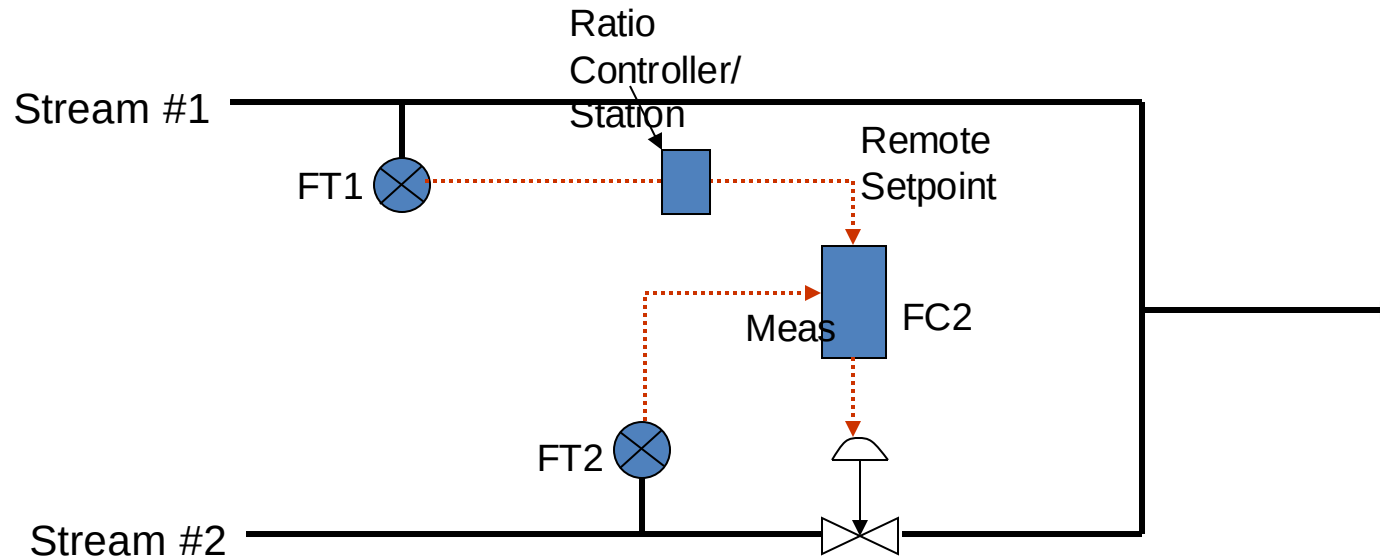
2. In a split range control loop where a controller has to manipulate two or more very differently sized control valves, the KEY ISSUE associated with the design of the system is:
 - a) The complexity of calculation with 2 valves
 - b) The inability of the controller to handle 2 different process gains
 - c) The complexity to the operator to manage 2 or more valves
 - d) The inability of the controller to handle the variability in the process

Ratio Controls

- Ratio control is utilised when the process requires the ratio between two process streams to remain constant.
- The control is achieved
 - by the manipulation of one of the process stream and
 - The monitoring of another 'wild' stream

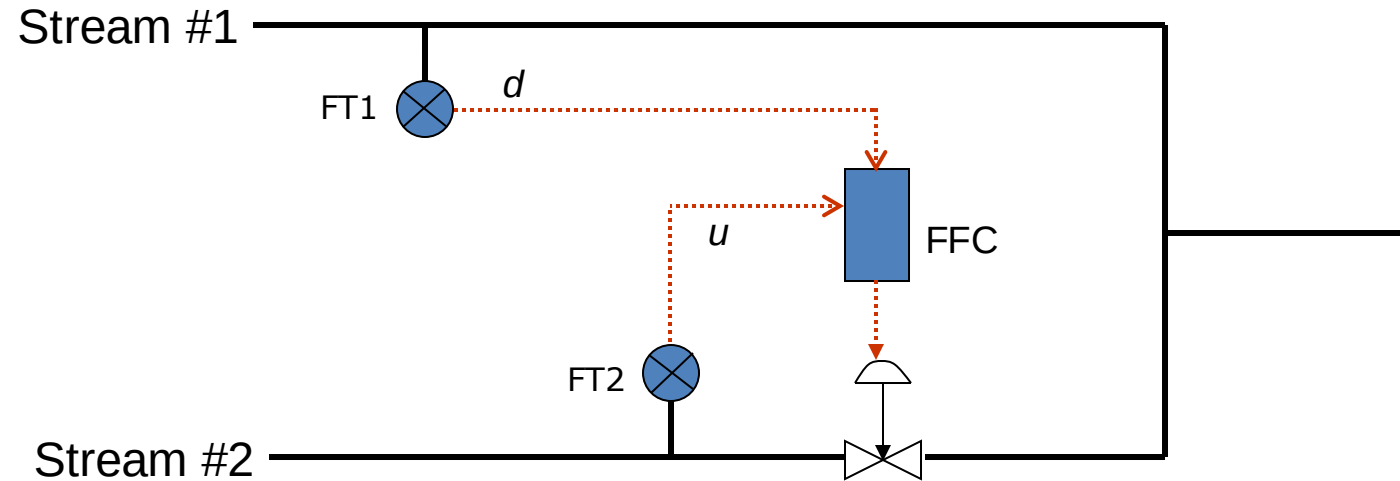
Note : The 'wild' stream maybe uncontrolled, controlled independently or manipulated by another controller

Example of Ratio Control



What is the equation in the ratio block?

Pure Ratio Control



In this example of ratio control,

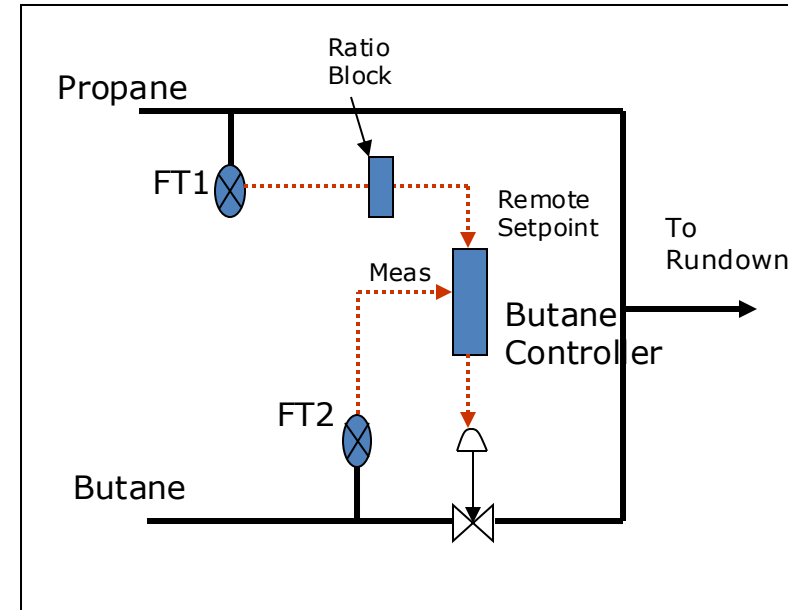
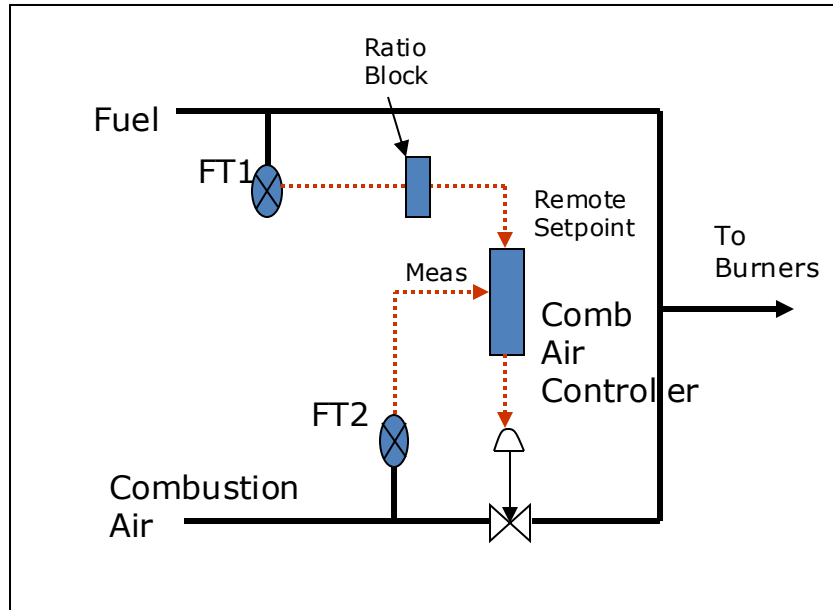
- If Ratio = d/u , then $K_p = \Delta \text{Output} / \Delta \text{input}$
 - $K_p = \frac{\Delta \text{ratio}}{\Delta u} = \frac{\Delta \left(\frac{d}{u} \right)}{\Delta u}$, because FFC is in manual, u changes but at fixed d
 - The above indicated that the Process Gain will not be a linear relationship when the ratio is changed, impacting tuning

Primary Components

The primary components of a ratio block are as follows:

- A control loop (transmitter, controller, control valve) on one pre-selected stream
- A transmitter on the 'wild' or uncontrolled stream
- A Ratio block (controller) which inputs the wild flow and provides an output to the remote setpoint/cascade input of the controller on the controlled stream
 - To prevent the ratio being changed, need to design the right access levels

Simplified Examples



- Note burner control systems are never as simple as above due to the critical nature of combustion systems. This will be covered in Level 3 Combustion Control

Control Theory and Application

Group Exercise

Intermediate Layer Controls

Skill Group 16.1 /16.4

Advanced Process Control

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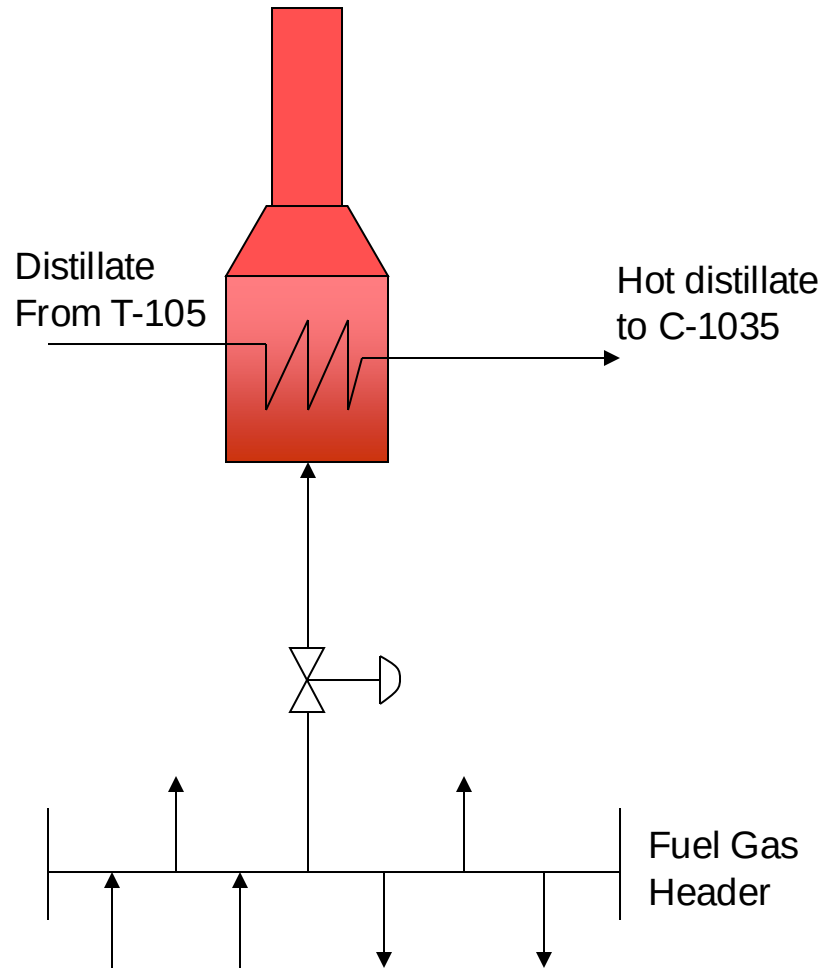
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Revision History

Rev	Date	Prepared/ Modified By	Notes
1	2005	Prakash K.K.	Original
2	2009	Andriny R.	With solutions
3	22 nd Aug 2012	Prakash K.K.	Reformatted
4	11 th July 2013	Prakash K.K.	Major Revamp, all slides inclusive addition of split range case studies and modification of all case study write-ups

Intermediate Control Case Study #1

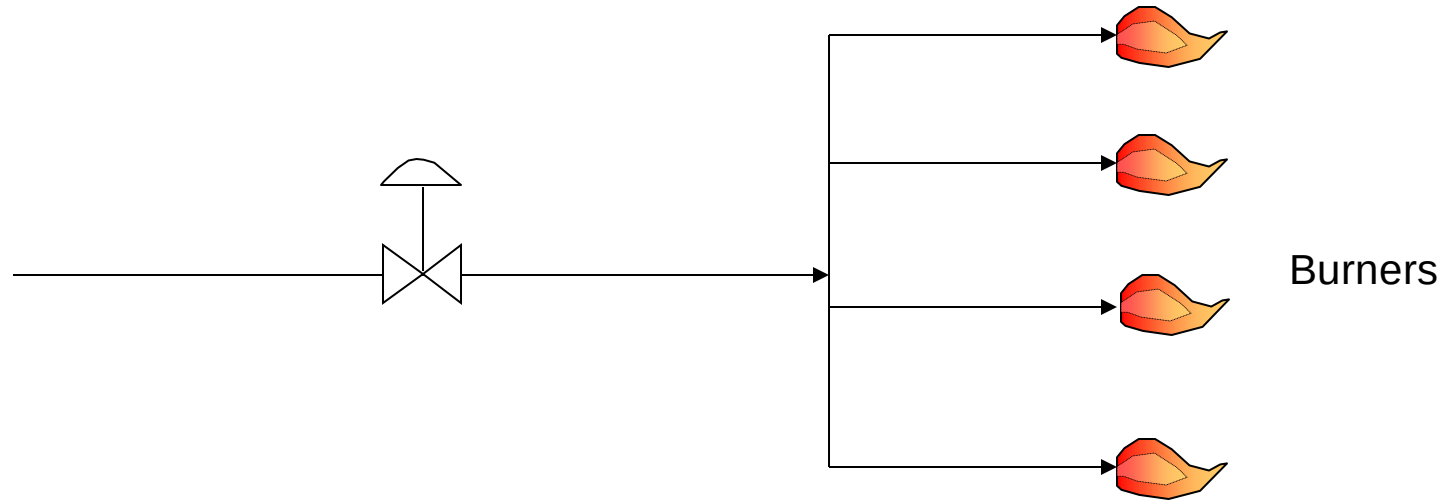
Furnace Outlet Temperature Control



- i) The furnace is heating a heavy distillate stream prior to entering a distillation column
- ii) The fuel gas header has a number of consumers and may thereafter see occasional pressure variations
- iii) Design the required control loop to achieve tight temperature control of the furnace outlet.

Intermediate Control Case Study #2

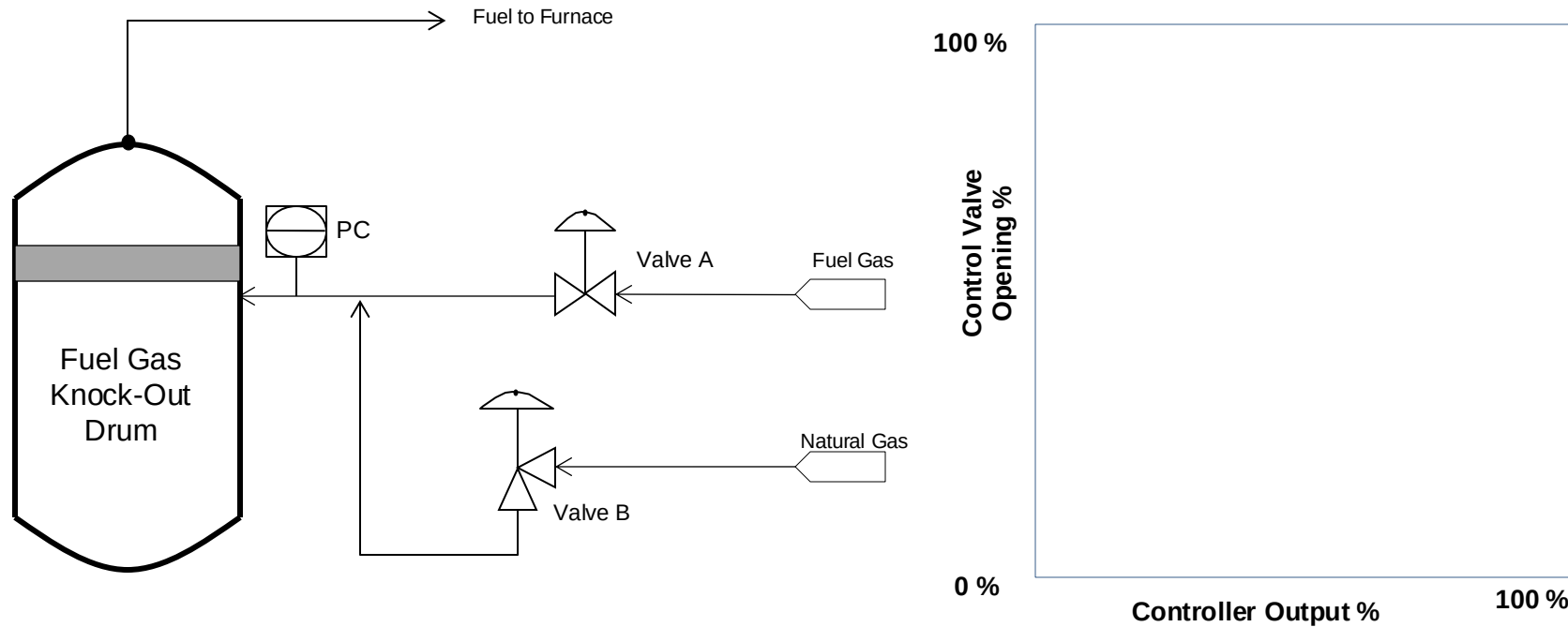
Burner Control



- i. The above process supplies fuel to burners. Fuel (of constant quality) is supplied from a natural gas header.
- ii. Flow control is utilised to control the flow to the burners
- iii. However, it is important to ensure that pressure does not drop below a minimum of 0.1 bar (kg/cm²) to prevent flame instability and flame-off. In addition, there is a maximum heat flux limit imposed by the burner manufacturer which correlates to a maximum pressure of 1.3 bar.
- iv. Develop the control scheme to achieve safe and stable control at the burners.

Intermediate Control Case Study #3

Fuel Gas Control

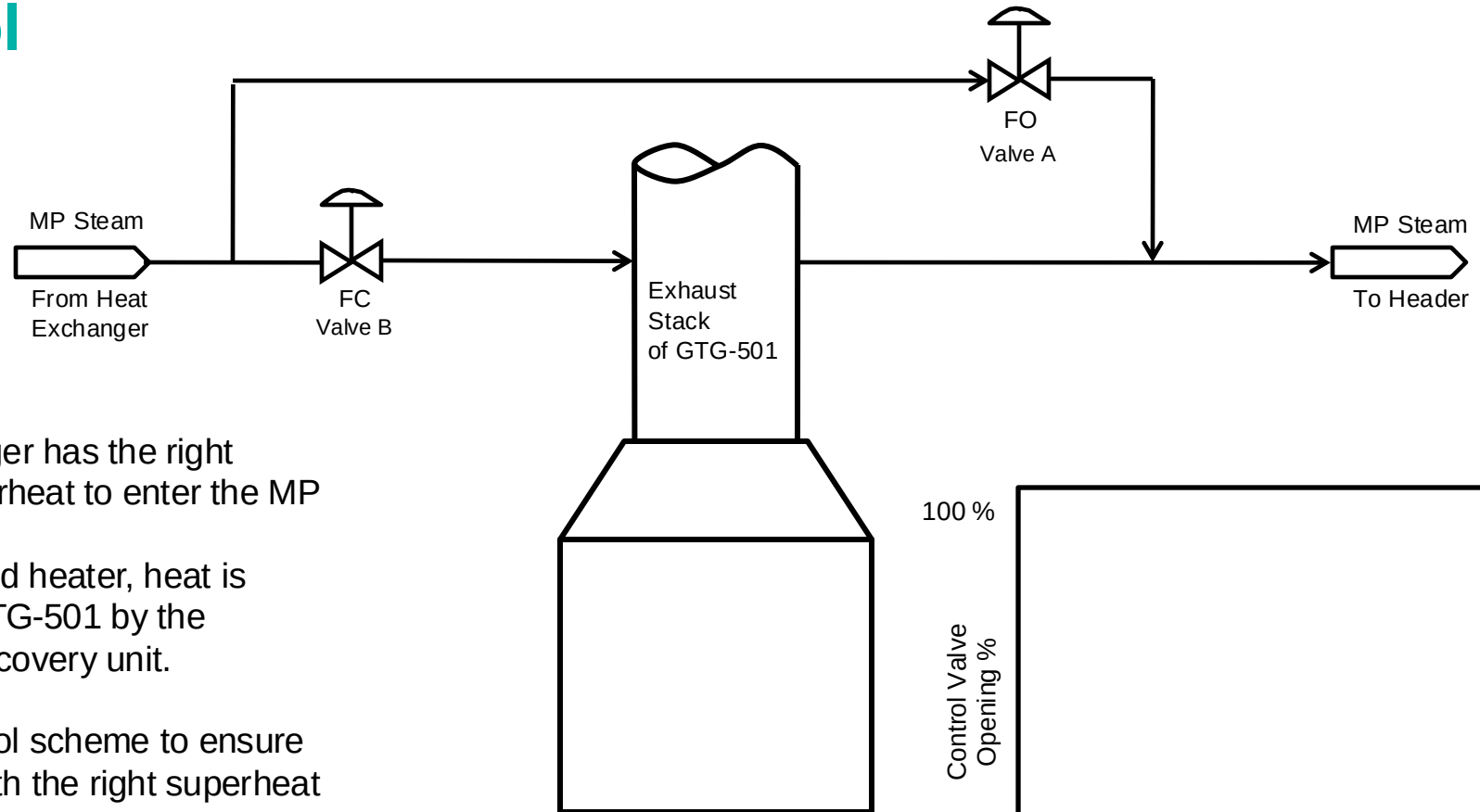


Notes:

- i) The Fuel Gas KO drum provides fuel gas to a furnace in a refinery lubes section.
- ii) Due to the critical nature of the processing in the furnace, fuel of a constant heat content is preferred.
- iii) Develop a control scheme to ensure that heat content to the furnace remains constant during normal operation.
- iv) Under abnormal conditions, fuel of different (and potentially varying) heat content will be allowed to the furnace.

Intermediate Control Case Study #4

MP Steam Control



Notes:

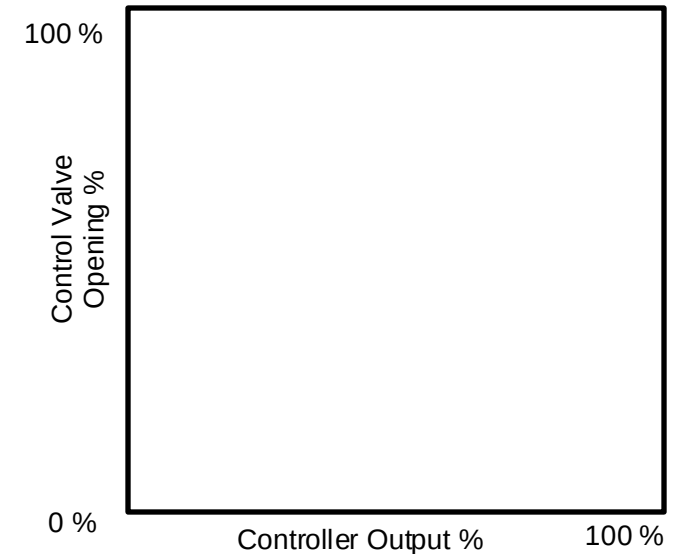
- Steam exiting a heat exchanger has the right pressure but insufficient superheat to enter the MP steam header.
- To eliminate the need of a fired heater, heat is recovered from exhaust of GTG-501 by the installation of a waste heat recovery unit.

Design Activity

- Design the appropriate control scheme to ensure that the steam is delivered with the right superheat prior to entering the header

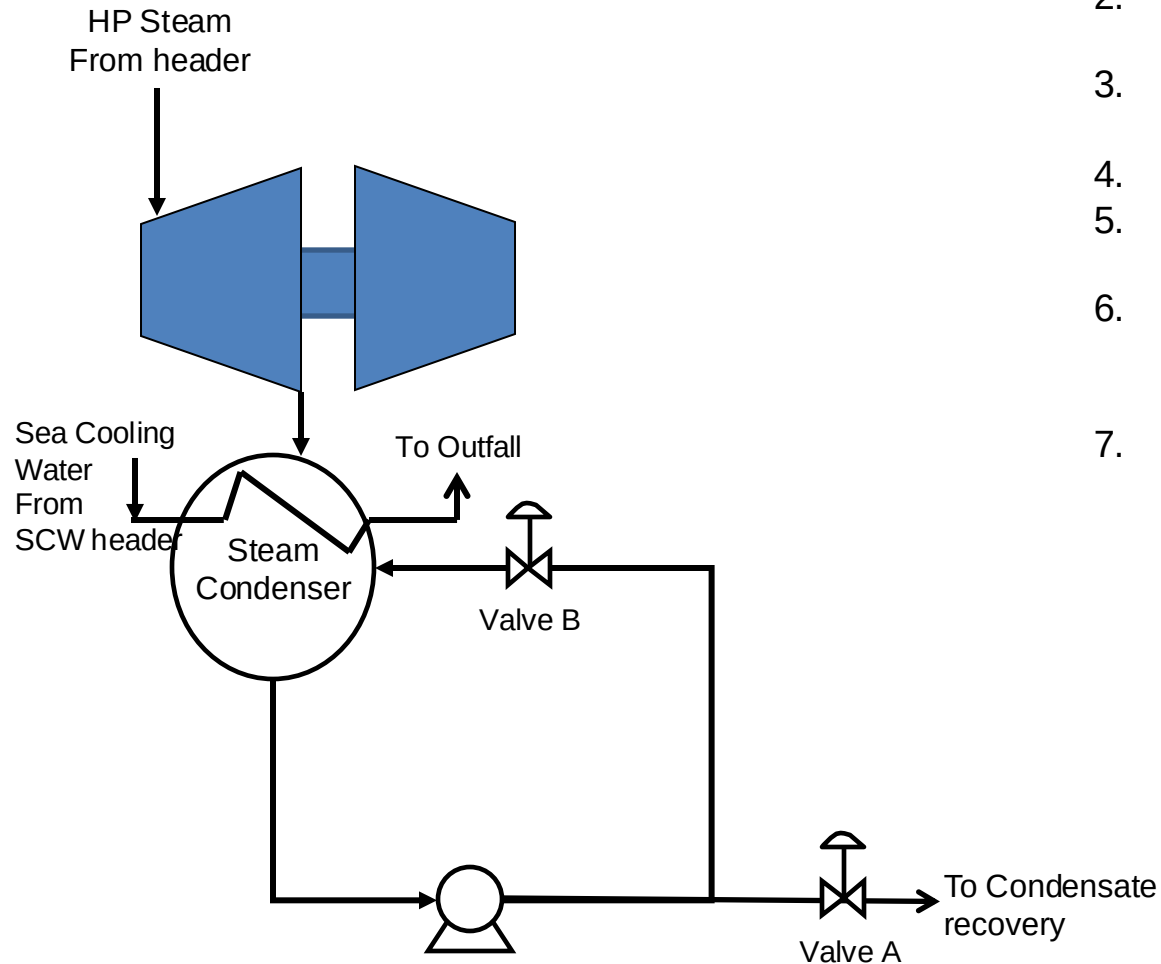
Additional Activity:

- Discuss any further modifications to cover potential fouling of the heat exchanger at low flows.

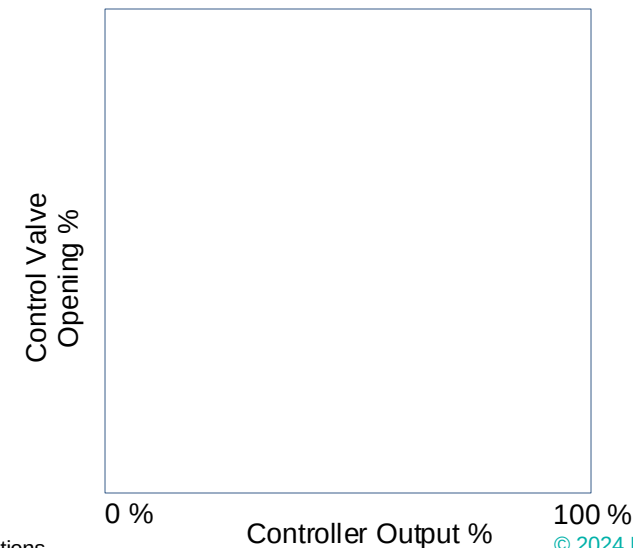


Intermediate Control Case Study #5

Condensate Level Control



1. A condensing steam turbine drives a propane compressor
2. The condensing steam turbine receives steam at 65 barg and condenses the steam in a steam condenser.
3. Steam is condensed by Sea Cooling Water (tube side) in a total condenser.
4. The condensate is pumped to a condensate recovery system.
5. Due to space limitations, it is not possible to install a minimum flow protection for the pump
6. Develop the split range chart for the condenser level control scheme to incorporate both level control and pump minimum protection.
7. Assume the condensate outlet valve is sized at 100 t/hr (valve A) & valve B is sized at 25 t/hr. Pump minimum flow is 30 t/hr.



Intermediate Layer Controls



- Cascade Control 
- Selective or Override Controls 
- Split Range Controls 
- Ratio Controls 

**For clarifications, please feel revert to:
Andriny Rusman, Principal Engineer, Process Control, GTS (andriny@petronas.com)**

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